

NATIONAL RESEARCH UNIVERSITY HIGHER SCHOOL OF
ECONOMICS

as a manuscript

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POSSESSION AND DETERMINACY IN NORTHERN KHANTY

Dissertation

for the purpose of obtaining academic degree

Doctor of Philosophy in Philology and Linguistics

Academic supervisor:

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Moscow 2025

Abstract

This thesis investigates the determinacy system of the Kazym dialect of Northern Khanty, focusing on how possessive agreement suffixes have grammaticalized into four distinct UNPOSSESSIVE determiners: the Associative Possessive, the Partitive Article, the Salient Article, and the Proprial Article. These markers, while homonymous with the Proper Possessive, operate independently from it outside prototypical possessive contexts, encoding semantic and pragmatic properties such as familiarity, uniqueness, specificity, and salience.

I analyze the semantics of Kazym Khanty Unpossessive determiners and situate them within the broader crosslinguistic picture, building on the Schwarz-Löbner typology of definites (Schwarz 2009, 2019; Löbner 1985, 2011), which categorizes definite determiners along dimensions of salience-familiarity and concept type.

I also analyze their syntax and morphophonology in Distributed Morphology (DM). Using the DM mechanism of allosemy, I account for their divergent behaviors while addressing potential weaknesses through insights from grammaticalization theory.

Key empirical findings of the study for determiner typology include the following. The Associative Possessive is a free possessive definite which presupposes a contextual associative relation between referents (cf. Beaver & Coppock 2016). The Partitive Article is an indefinite determiner which presupposes proper partitive specificity. The Salient Article presupposes relative maximal salience, with additional morphosyntactic parsing constraints imperfectly resembling proximate uniqueness (Aissen 2001). Lastly, the Proprial Article derives intensionally rigid DPs from human names. I provide a formal demonstration of its intensional rigidity in Situation Semantics. I also develop a separate dynamic analysis of the Salient Article based on Stojnić's (2021) Attention-Coherence approach to model its distribution—including its use with deictic demonstratives.

Together with demonstratives and bare nouns, these determiners form a rich and typologically unusual system. The thesis concludes by outlining directions for future research, including the distribution of bare nouns and the non-possessive functions of possessives in other Uralic languages.

Acknowledgements

First and foremost, I owe my deepest gratitude to our Khanty friends in Kazym, who have generously devoted their time, knowledge, and patience to working with me and our team. Since 2018, they have not only taught me their language but also inspired me with their wisdom and kindness. Their unwavering support and linguistic insights form the foundation of this thesis, and I can only hope to have done justice to their contributions. I am profoundly indebted to them.

I thank my scientific advisor, Natalia Ivlieva (Natalia Viktorovna). Natalia Viktorovna has played an instrumental role in my becoming a formal semanticist. Since we started working together in 2020 on this project, she has always provided a lot of encouragement and useful feedback to me. Thank you!

Apart from my Kazym Khanty teachers, I've had many teachers in linguistics that significantly impacted this study. Some of them taught classes as part of my formal education, some taught classes at summer, fall, and winter schools, some gave educating talks, and all of them pointed me to key literature and theoretical frameworks that became foundational to this study. I thank them all: Valentina Apresjan, Rajesh Bhatt, Michael Daniel, Amy Rose Deal, Kai von Fintel, Martin Haspelmath, Lisa Hofmann, Sabine Iatridou, Natalia Ivlieva, Daniar Kasenov, Alexey Kozlov, Yury Lander, Timur Maisak, Zahra Mirazzi, Elise Newman, Alexander Podobryayev, Ekaterina Rakhilina, Tatiana Reznikova, Pavel Rudnev, Yael Sharvit, Svetlana Toldova, Oleg Volkov.

I am also grateful to the faculty of CreteLing 2022, 2023; each winter V-NYI starting from 2022; Fall School for Formal Syntax and Formal Semantics (HSE, 2019); and Yerevan Academy for Linguistic and Philosophy 2024. Their classes and informal discussions with them were played a formative role for me as a formal linguist. This thesis would not be what it is without them.

A special thank you goes to my colleagues who discussed my data and ideas with me, offering encouragement and support, Rajesh Bhatt, Kai von Fintel, Enrico Flor, Lisa Hofmann, Natalia Ivlieva, Daniar Kasenov, Alexey Kozlov, Yury Lander, Ekaterina Rakhilina, Polina Pleshak, Alexander Podobryaev, Maria Polinsky, Starr Sandoval, Alexandra Simonenko, Alexey Starchenko. I am also deeply grateful to the audiences at various conferences and seminars for their insightful comments and suggestions, particularly those at: the Conference on Typology and Grammar for Young Scholars (ILS

RAS, Saint-Petersburg) over many years, the Typology of Morphosyntactic Parameters conference (IL RAS, Moscow) over several years, Sinn und Bedeutung 28 and 29, the Uralic Information Centre (Hungarian Research Centre for Linguistics), the seminars of the Northern Khanty field project and of the Laboratory of Formal Models in linguistics.

I am deeply grateful to my colleagues who generously read and provided feedback on parts of this thesis and the papers it builds upon: Pavel Astafiev, Lisa Hofmann, Ksenia Kashintseva, Alexey Kozlov, Kate Kozlova, Lera Marinina, Nikita Muravyev, Tatiana Philippova, Alina Russkikh, Alexandra Simonenko, Daniel Tiskin, Varvara Tiutiunnikova. Their insights were invaluable, and I have done my best to incorporate their advice. I also extend my thanks to the reviewers at *Rodnoy Yazyk*, *Ural-Altaic Studies*, *Voprosy Jazykoznanija*, and *Language* for their constructive comments, which significantly strengthened the foundational work for this thesis.

Over the years, I have been overwhelmed by the generosity of my colleagues—their advice, critiques, and ideas have been nothing short of extraordinary. This experience has really proved to me how much science is a collective endeavor. While I have tried to honor their contributions to the best of my ability, any remaining shortcomings in this work are solely my own.

Thanks are also due to my colleagues at the HSE University's Laboratory of Formal Models in Linguistics, at the School of Linguistics, and to my colleagues from the Northern Khanty field project. Your passion for linguistics has made this journey exhilarating and has reminded me daily that we are part of something meaningful. It is an honor to work alongside you.

I thank Adrienne Lenker and Maynard James Keenan for inspiration, and Nikita Muravyev and Alexey Yurov for introducing me to their music. And I thank Alexander Murashov for discussions of Formula One drivers.

Many friends have at various stages given me useful writing advice, were there to co-work with, to discuss writing plans, and to write collectively. You made the writing of this thesis that much less scary, despite all the complicating circumstances. For this, I'm grateful to Konnie Florinski, Fedya Golosov, Ke Lin, Dasha Mityakina, Renata Perelman, Aisha Ramazanova, Ksenia Romanova, Roma Ronko, Alina Russkikh, Varya Tiutiunnikova, Lera Vinogradova. Many more friends were there with me at different

points of my educational journey, and I received an immesurable ammount of support and joy from them. Thank you to everyone!

My family has lead me on this path since the beginning, always helping me, always praising me, always caring, always being there for me. Thank you, Maxim, mama, papa, Varya, Mura, Semion, Babochka, Lady. Thank you, Mikhailovs and Plavinskys. You mean the world to me.

*Вечно любимому Максиму,
Моим бабушкам и дедушкам: Бабочке, Баське, Боре, Диме,
а также Леди*

*To Maxim, who is forever loved,
To my grandparents: Babochka, Bas'ka, Borya, Dima,
and to Lady*

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1. Introduction

Many Uralic languages have adnominal possessive agreement markers (possessives) that index the person and number of the possessor on the possessed nominal. Such markers are well-known to exhibit non-possessive, ‘extended’ or ‘discourse’ uses that go beyond typical possessive contexts (Fraurud 2001; Kuznetsova 2003; Nikolaeva 2003; Simonenko 2017; Halm 2018; É. Kiss 2018; É. Kiss & Tánčzos 2018; Janda 2019, among others; Serdobolskaya et al. 2019). In such non-possessive functions, the markers do not seem to express any possessive meaning: they cannot be said to denote a relation between a possessor and a possessee that corresponds to the possessive prototype of ownership, parthood, or other inalienable or inherent possession, as in (Taylor 1996; Nikolaeva 2003) or (Vikner & Jensen 2002; Partee & Borschev 2003; Karvovskaya 2018). The markers rather seem to function similarly to determinacy markers, for example, the European DEFINITE ARTICLES (Schwarz 2019) or the Turkish Accusative Case conditioned on partitive specificity (Farkas & Brasoveanu 2019: §2.3).¹

I understand determinacy as a cluster of semantic and pragmatic properties of nominals such as novelty, familiarity, uniqueness, specificity, scope, etc. (Roberts 2003; Löbner 2011; Dayal 2018; Farkas & Brasoveanu 2019; Becker 2021). Intuitively speaking, determinacy is responsible for how the reference of a nominal is established and what properties its discourse referent is required to have.² Determinacy distinctions are encoded in determiners. See below for definitions of determinacy distinctions relevant to this thesis.

The Kazym dialect of Northern Khanty (< Khantyic < Uralic) presents a case where a second-person singular possessive suffix *-en/-an* [POSS.2SG] figures in as many as three frequent non-possessive determiner-like functions and a POSS.3SG *-əλ/-aλ* features another

¹ Throughout this thesis I adopt the convention of capitalizing names of language-particular categories to highlight their idiosyncrasies (Haspelmath 2010: 674), e.g., writing “(Proper) Possessive” instead of “possessive” implies that the category in question need not correspond to some universal category type but may diverge from it in different ways. Crosslinguistically valid category types, ‘crosslinguistic gram types’ in Dahl’s (2016) terminology, are rendered in small caps, e.g., STRONG DEFINITE ARTICLE. Where the distinction is of little importance, I write in lowercase.

² See §3.1.1, fn. 85 for a further terminological clarification.

determiner-like function.³ These functions are exemplified below along with the names I will be using for them in the remainder of the thesis.

Apart from the proper possessive function that describes a prototypical possessive relation (1), the Kazym Khanty POSS.2SG is also used with familiar referents that stand in an **associative relation** (Nikolaeva 2003) to another activated referent (2). Thus, in (2) the only cup available in the context is associated with the addressee, presumably, because of physical proximity, which motivates POSS.2SG marking. Note that there is no proper possessive relation between the addressee and the cup here.

(1) PROPER POSSESSIVE

năŋ kăt'-en moś-λ
 you.SG cat-POSS.2SG purr-NPST[3SG]
 'Your cat purrs.'

(2) ASSOCIATIVE POSSESSIVE

[A friend is over at the speaker's place. There's one cup on the table.]
an-en mij-e
 cup-POSS.2SG give-IMP.SG>SG
 'Give me the cup.'

Another POSS.2SG function is observed with **salient referents** like the dog in (3a) that was introduced in the preceding utterance and stands in the subject position in the target sentence.⁴ Apart from a basic anaphoric use, this function is also found in 'salience without uniqueness' contexts (3b) which are typically unavailable for canonical definite determiners due to a violation of their uniqueness presupposition (Roberts 2003), and with deictic demonstratives which occur with a pointing gesture making the referent salient (Diessel 2006).

³ I use *function* as a noncommittal term that does not distinguish between separate senses of a morpheme or merely contextualized uses of a monosemic marker (cf. Haspelmath 2003). I invite the reader to understand the functions exemplified in (1)-(5) in this way until more explicit definitions are given later in this work.

⁴ I understand salience as a function of the interlocutors' attention, following (Roberts 2011; Barlew 2014). A salient referent is at the center of attention. The referents that are part of the current question under discussion (QUD) are necessarily salient.

Barlew (2014) is the first work that describes and analyzes a SALIENT ARTICLE, working with Bulu data (< Bantu, Cameroon). The Kazym Khanty Salient Article differs in important ways from the Bulu Salient Article.

(3) SALIENT ARTICLE

- a. [“I was walking along the street when I saw a dog.”]⁵

amp-en ma pɛλ-am-a χurət-ti pit-əs
 dog-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]
 ‘The dog started barking at me.’

- b. [A theatrical play for children: “A girl_i appeared from the room on the left. Another girl_k appeared from the room on the right.”]

λɥw_{k/*i} / ^{OK}**ewen**_{k/*i} još-ən lipət təj-əs
 (s)he / girl.POSS.2SG hand-LOC flower have-PST[3SG]
 ‘She_{k/*i} [lit.: the girl] was holding flowers.’

- c. [“Who hit you?”]

ma **ši** **aj_iken-ən** χăts-əs-ij-əm
 I DEM boy.POSS.2SG-LOC hit-PST-PASS-1SG
 ‘I was hit by that boy.’

The last POSS.2SG function is found in its obligatory use **with human names** in argument positions (4) like a prototypical PROPRIAL ARTICLE (Muñoz 2019).

(4) PROPRIAL ARTICLE

wontr-en jɥχ šöp sewr-əs
 A.-POSS.2SG wood piece cleave-PST[3SG]
 ‘Andrej cleaved a log.’

One final peculiar function is found with POSS.3SG used to mark that a partitive referent, which is a part of a familiar referent from the preceding context (Enç 1991; Farkas 2002a).

(5) PARTITIVE ARTICLE

- [“I was walking down the street when I saw two dogs.”]

i **amp-əλ** ma pɛλ-am-a
 one dog-POSS.3SG I at-POSS.1SG-DAT
 χurət-ti pit-əs

⁵ Contexts given in quotes (“”) were presented as stimuli and translated into Northern Khanty along with the target sentence. Contexts without quotes were described in Russian which is the second language of all my consultants.

bark-NFIN.NPST become-PST[3SG]
'One of the dogs started barking at me.'

The existing sketches of the Kazym dialect do not describe these uses of possessive markers (Kaksin 2010; Sipos 2022), and while other dialects of Northern Khanty do feature similar phenomena the distribution of non-possessive functions of possessives described by Nikolaeva (1999; 2003) for the Obdorsk dialect and by Kashkin (Kashkin 2010) for the Tegi dialect is almost completely different from the Kazym dialect data. In particular, non-possessive possessives are largely optional in Nikolaeva's and Kashkin's data which is not the case in the Kazym dialect. Therefore, the empirical picture exemplified in (1)-(5) needs to be documented, especially as it appears to be rather unique for the Uralic family in terms of the number of non-possessive functions and their level of grammaticalization.⁶

Once confronted with such polyfunctional possessive markers, the natural question to ask is whether it is possible to unify these different functions under a single **monosemic** lexical entry. Alternatively, should the non-possessive functions be treated **polysemically**, as markers that may be homonymous with the proper possessive but are synchronically independent?⁷

Anticipating the answer to come, this thesis argues for a polysemic approach to the four non-possessive functions presented above. I will claim that they represent determinacy markers that are synchronically distinct from the Proper Possessive in (1) and from each other. The central concern of this study is with the semantics of these four markers against the background of the broader definite determiner typology that formal semantic research of the past two decades has accumulated (Farkas & Brasoveanu 2019; Schwarz 2019; Becker 2021; Reisinger et al. 2021). (Note that despite the polysemic stance adopted in this thesis, each function of Kazym Khanty possessives, possessive or

⁶ Smirnova's unpublished field report (Smirnova 2019) should be mentioned also as the first description of the Kazym Khanty Proper Possessive. The focus of that report is on the conditions under which possessive marking is obligatory with lexical possessors, see §3.2.

⁷ I use the term *marker* loosely to pick out a three-way correspondence between a phonological exponent, a semantic denotation, and a syntactic terminal, see §4.1 for theoretical motivation. The distinction between *polysemy* and *homonymy* is, of course, a thorny and theory-dependent issue. Moreover, these two notions do not play a role in the formal analysis developed in §4.7. Informally, it is safe to understand *polysemic* as "requiring more than one sense / lexical entry" and *homonymous* as "same in form, but not in meaning".

non-possessive, will be glossed as the possessive that it is (homonymous with), as in the above examples. This is because all five functions use the same exponents, see Chapter 4.)

Chapter 2 is dedicated to an overview of the current typology of definite determiners (definites) that arises from language-specific semantic studies with roots in the work of Heim (1982), Roberts (2003), Schwarz (2009), and Löbner (1985; 2011). There, I also introduce the tools of semantic analysis in research on definites and attempt unifying some of the results of my predecessors in a more comprehensive picture. Chapter 2 forms the necessary background to the three other chapters, excluding Chapter 6 which concludes this study and indicates some directions for future research. Chapters 3, 4, and 5 are relatively independent and can be read in any order. I provide cross-references between chapters where needed.

Chapter 3 presents a semantic investigation of the four Kazym Khanty determiners introduced above against the background of Chapter 2. The four determiners are contrasted with the Kazym Khanty Proper Possessive and with bare nouns (Tiutiunnikova et al. 2024; Tiutiunnikova 2024; Golosov 2024a). There, I argue that the Proper Possessive is semantically a simple possessive modifier and is not inherently definite, following Coppock and Beaver's (2015) arguments for the English Saxon Genitive. The Associative Possessive is a genuine possessive definite, which combines free possessive semantics (Partee & Borschev 2003) with definiteness. The Partitive Article is an existential quantificational determiner like the English Indefinite Article except it additionally presupposes partitivity. The Salient Article is a definite article that presupposes relative maximal salience, this explains its use in contexts like (3b) following Roberts (2003). The Proprial Article derives intensionally rigid DPs from human names. Building on (Muñoz 2019), I present a situation-semantic analysis in the style of (Schwarz 2009; Elbourne 2013) to demonstrate this. This chapter is based on (Mikhailov 2023) but it substantially expands the narrative of that paper with explicit semantic analyses and extended argumentation.

Chapter 4 is dedicated to the question of monosemic vs. polysemic treatment of polyfunctional possessive agreement markers. It is worthwhile to introduce it and my answer to it for the case of Kazym Khanty in more detail here.

The question of monosemic vs. polysemic treatment of polyfunctional possessive agreement markers has been addressed in several studies investigating similar data from other Uralic varieties. Several authors have argued for a monosemic approach, maintaining that (at least some) non-possessive functions can be subsumed under a basic, albeit underspecified, possessive meaning of establishing a relation between two referents (Fraurud 2001; Nikolaeva 2003; Gerland 2014; Janda 2015; Simonenko 2017). More recently, however, five studies (É. Kiss & Tánzos 2018; É. Kiss 2018; Halm 2018; Serdobolskaya et al. 2019; and Logvinova 2019 on Chuvash (< Turkic)) have taken a polysemic stance, arguing that the non-possessive functions they investigated are in fact **distinct markers grammaticalized from PROPER POSSESSIVE SUFFIXES**. Although, except for (Halm 2018), these authors did not formulate explicit theoretically grounded analyses.

In Chapter 4, I answer this question for a language that features four, quite distinct non-possessive functions of this sort, the Kazym dialect of Northern Khanty. I will argue that the four non-possessive functions of Kazym Khanty possessives are UNPOSSESSIVE MARKERS (UNPOSSESSIVES for short) that are homonymous with the Proper Possessive but are synchronically distinct from it, as defined in (7) below. I introduce the term ‘unpossessive’ to refer to markers like this in the hope that a dedicated name will lead to broader recognition of this phenomenon and its importance for Uralic studies and for determiner typology.

In (7), I provide a general definition of an UNPOSSESSIVE MARKER based on the general definition of a PROPER POSSESSIVE AGREEMENT MARKER formulated in (6). I formulate both definitions in the style of Haspelmath’s (2010) crosslinguistic comparative concepts, aiming to achieve applicability beyond the specific profile of Uralic possessive agreement markers. Nevertheless, this thesis is only concerned with the Uralic unpossessives,⁸ so the relevance of these definitions to the broader crosslinguistic picture will not be tested here.

⁸ With the Unpossessives of Chuvash (< Turkic; Logvinova 2019) making a brief appearance as well in Chapter 4.

- (6) A PROPER POSSESSIVE AGREEMENT MARKER is a marker which indexes the ϕ -features of a possessor nominal (cf. Dryer 2013) in an adnominal possessive construction describing inherent possessive relations.
- i. ϕ -features are any features of the set {person, number, gender}.
 - ii. An adnominal possessive construction is a nominal modifier which introduces a possessor dependent to the head nominal and describes a relation between the possessor and the head nominal referent (cf. Karvovskaya 2018). As opposed to a predicative possessive construction (Stassen 2013; Myler 2016).
 - iii. An inherent possessive relation is a stereotypical relation derived from the denotation of the head noun (Vikner & Jensen 2002; Karvovskaya 2018; cf. the possessive prototype of Taylor 1996). Inherent relations are contrasted with ‘free’ relations which are not restricted in this way and are derived from the context (Partee 1983/2011).
- (7) An UNPOSSESSIVE MARKER is a marker related to a PROPER POSSESSIVE AGREEMENT MARKER formally and, likely, related diachronically, but synchronically independent from it. It is not PROPER POSSESSIVE according to the definition in (7), i.e., semantically it is not restricted to inherent relations. It may also carry additional semantic content, and it may differ in its pragmatic behavior as a result. It may differ in morphosyntactic properties. (See Chapter 4 for a full list of unpossessive properties.)

I situate the argument for unpossessivehood of the four non-possessive functions of Kazym Khanty possessive in an explicit model of the syntax, morphophonology, and semantics of possessive agreement markers formulated in the frameworks of current mainstream generative grammar (e.g., Preminger 2014; Newman 2024), Distributed Morphology (DM; Halle & Marantz 1993), and formal semantics (Heim & Kratzer 1998), with additional support for the proposed analyses from a grammaticalization-based argument (Bisang & Malchukov et al. 2020a). The model will be presented in two steps starting with the semantics part in Chapter 3.

The morphosyntactic part of the model is presented in Chapter 4, where I develop diagnostics that aim to test whether a non-possessive function shows the same morphophonological, morphosyntactic, semantic, and pragmatic behavior as the proper possessive function, as would be expected under a monosemic approach. Using these

diagnostics, I show that the Associative Possessive (2), the Salient Article (3), the Proprial Article (4), and the Partitive Article (5) **observe different behavior than the Proper Possessive** (1) with respect to several parameters. I develop an analysis in DM (Halle & Marantz 1993) to model this behavior making crucial use of the mechanism of allosemy from the recent work by Wood (2015), Myler (2016), and others. This allows us to cover the differences between these markers while also accounting for their similarities. I address some potential weaknesses of this analysis, specifically, the stipulative nature of allosemy, by justifying it with an appeal to how grammaticalization processes normally work (Bisang & Malchukov et al. 2020a).

Finally, Chapter 5 is concerned with figuring out the conditions of use of the Salient Article and, specifically, with the question of what factors do **not** play a role, contrary to previously voiced suggestions. It presents a broader and more detailed inquiry into the distribution of the Salient Article, as compared to Chapter 3. I argue against the Salient Article being sensitive to information-structural distinctions (contra Mikhailov 2021b), and I also critically investigate the hypothesis first proposed in the report (Muravyev 2022) that the Salient Article is a PROXIMATE MARKER, signaling prominent third person participants and subject to the proximate uniqueness constraint (Aissen 1997; 2001): there may be at most one proximate third person per obviation span (e.g., per clause). I show that even though with subjects and direct objects there may be at most one Salient Article per clause (a proximate uniqueness signature), this is not so for indirect objects, and with demonstrative. The full pattern is quite unlike what we see in canonical proximate markers.

Chapters 4 and 5 reproduce the narratives of (Mikhailov 2024a; Mikhailov 2024c) in shortened form (due to overlap with Chapter 3), and additionally, Chapter 4 includes arguments for unpossessivehood of the Partitive Article from (Mikhailov 2021b) that were not discussed in (Mikhailov 2024c), while the last sections of Chapter 5 (§§5.4.2, 5.5) argue for relaxing the analysis of the Salient Article from (Mikhailov 2024a), as well as for a mixed approach to its distribution (also contra Mikhailov 2024a). There, I also implement the resulting semantic analysis of the Salient Article in a dynamic semantic system, based on Stojnić's (2021) Attention-Coherence approach, which allows me to model the Salient Article's use with demonstratives (3c) and in 'salience without uniqueness' contexts (3b).

Overall, this thesis is the first case study of Uralic UNPOSSESSIVES that provides an explicit comprehensive model for their quirky properties and sets them against the background of determinacy markers crosslinguistically. The model and the unpossessive diagnostics proposed here are designed to be applicable to other languages with non-possessive uses of possessives. The semantic analyses are purposefully framed in an explicit formal semantic framework which allows for high degree of crosslinguistic comparability as attested by the studies surveyed in Chapter 2.

The main empirical findings of the study are as follows. The Kazym Khanty determiner system features four Unpossessive determiners. The Associative Possessive is a genuine possessive definite, which combines free possessive semantics (Partee & Borschev 2003) with definiteness. The Partitive Article is an existential quantificational determiner which presupposes that the DP referent is part of another strongly familiar referent. The Salient Article is a relative maximal salience article which imposes additional idiosyncratic restrictions similar to proximate uniqueness of languages with obviation systems (Aissen 2001). The Proprial Article derives rigid DPs from human names (Kripke 1980; Muñoz 2019).

Taking into account the findings of Tiutiunnikova and colleagues (2024; and other works), who investigated the bare nouns of Kazym Khanty (as discussed in §3.3.1), and of Marinina (2024), who investigated the three demonstrative determiners of Kazym Khanty (as discussed in §2.4.3), the Kazym Khanty determinacy system appears to be an unusually rich and diverse system which incorporates at least five uncommon types of definites (three of the Unpossessives and two of the Demonstratives), a Partitive Article, and bare nouns that are interpreted existentially across the board, contrary to the predictions of (Chierchia 1998; Dayal 2004) and much like the ‘indefinite’ determiners of Salish languages (see §2.1).

The added twist of the system is that four of its determiners are grammaticalized from possessives, as I argue in Chapter 4, which also makes the system stand out among the determiner systems found crosslinguistically (König 2018; Becker 2021).

I anticipate that the application of the unpossessive diagnostics developed here to other languages with non-possessively used possessives will also uncover unusual determinacy markers and systems. The semantic framework developed here provides a solid foundation for their investigation.

The rest of this introduction presents the language under study (§1.1) and the basics of the extensional truth-conditional semantic system that the following chapters use (§1.2).

1.1. Kazym dialect of Northern Khanty

Northern Khanty is a Uralic language of the Khantyic branch spoken in areas by the Ob' river and its tributaries (Kazym, Kunovat, Synja, etc.) in the Yamal-Nenets and Khanty-Mansi Autonomous Regions of the Russian Federation (Kaksin 2010; Sipos 2022). It is spoken by about 8865 speakers which makes 37% of the ethnic Khanty according to the Russian Census of 2010 and it is undergoing language shift with less than 10,000 speakers remaining⁹. Semistructured interviews with some 50 Khanty respondents from the Kazym and Juil'sk villages (Beloyarsky district), Khanty-Mansiysk, and Beloyarsky show that only people born before the 1980s speak the language, while the younger generations generally do not (Aristova 2023).

The data presented in this thesis come from my fieldwork with speakers of the Kazym dialect residing in Kazym, conducted from 2018 to 2024, during joint field trips of the School of Linguistics (HSE University) and the Department of Theoretical and Applied Linguistics (Lomonosov Moscow State University) in 2018-2019 and 2021-2024 under Svetlana Toldova and Alexey Kozlov, as well as online with the same speakers in 2020-2021.

The data were collected via context-based semantic elicitation (Kibrik 1972; Matthewson 2004) using Russian and Northern Khanty stimuli with Russian as the metalanguage. The consultants were presented with contextualized stimuli and asked for acceptability judgements regarding particular forms in contexts. In most cases the context was also translated. Sometimes the speakers' comments were used to create further stimuli to test the hypotheses suggested by their comments.

Each judgement reported in this thesis was taken from at least three and at most ten consultants. Overall, thirteen consultants participated in the elicitation. Eight consultants were women, five were men, and they ranged from fifty to seventy-seven years old. All consultants were sequential bilinguals with Russian acquired in school or shortly prior to

⁹ The Russian Census of 2010 data are from (Koshkareva 2016).

school. Two consultants acquired Northern Khanty simultaneously with Komi Zyrian. All consultants were born and raised in Kazym dialect-speaking areas, except for one consultant native in the Obdorsk dialect who learned the Kazym dialect in school. All consultants were aware of the goals of this study and were paid for their participation.

1.1.1. Abbreviations used in Kazym Khanty examples

1, 2, 3	first, second, third person
ACC	accusative
ADD	additive
APPR	approximative
CVB	converb
DAT	dative
DEC	numeral suffix for 11–17
DEM	demonstrative
DU	dual
EMPH	emphatic
LOC	locative
NEG	negation
NFIN	general non-finite form
NOM	nominative
NPST	non-past
NSG	non-singular number
OPT	optative
PASS	passive
PL	plural
POSS	possessive marker
PROP	propriative
PST	past
PTCL	particle
SG	singular

1.2. Semantic framework

This section provides the basics of the semantic framework that I will employ in what follows. The readers familiar with (Heim & Kratzer 1998) are free to skip ahead to

Chapter 2. The details of the syntax of possessive agreement constructions and the spell-out rules assumed for the Kazym Khanty Unpossessives won't matter until Chapter 4, so I delay introducing them until then.

I assume a simple static extensional framework based on (Heim & Kratzer 1998). Some additional semantic and pragmatic assumptions will be introduced in Chapter 2 when necessary, along with brief mentions of Dynamic Semantics and Situation Semantics. In two places in Chapters 3 and 5, I will make use, respectively, of a situation-semantic extension of the system presented here and of a different dynamic system, but these will be presented when needed.

In the current framework, natural language expressions are directly interpreted by the interpretation function $\llbracket \cdot \rrbracket$ with respect to two parameters of interpretation, an assignment function g and an utterance context c . Table 1 shows the semantic types of denotations that we will be working with in what follows. These include simple types, e and t , and functional types such as the type of predicates, $\langle e, t \rangle$.¹⁰

SEMANTIC TYPE OF DENOTATION	CONVENTIONAL NAME	EXAMPLES
e	individual	Proper names (<i>Zahra, Serge, Eduardo</i>), definite descriptions (<i>the dog</i>), personal pronouns (<i>she, he</i>)
t	truth-value	Declarative sentences (<i>Zahra smiles, She pets the dog</i>)
$\langle e, t \rangle$	predicate	Intransitive verbs (<i>smiles</i>), common nouns (<i>dog</i>), verb phrases (<i>pets the dog</i>)
$\langle e, \langle e, t \rangle \rangle$	relation (binary predicate)	Transitive verbs (<i>pets</i>)
$\langle \langle e, t \rangle, e \rangle$	definite determiner	<i>the</i>
$\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$	quantificational determiner	<i>some, a/an, every</i>
$\langle \langle e, t \rangle, t \rangle$	generalized quantifier	<i>somebody, every dog</i>

Table 1. Semantic types of some expressions

The denotations of simple expressions are specified in the lexicon, as shown in (8) for English along with formal notational abbreviations. I will be assuming that the same denotations are appropriate for similar expressions in other languages discussed here

¹⁰ For any two types α and β , a function of type $\langle \alpha, \beta \rangle$ maps objects of type α to objects of type β .

unless stated otherwise. (The types of variables in the denotations will sometimes be omitted.)

(8) SAMPLE LEXICAL DENOTATIONS FOR SOME ENGLISH EXPRESSIONS

- a. $\llbracket \text{Zahra} \rrbracket^{g,c} = \text{the individual Zahra} = z$
- b. $\llbracket \text{smiles} \rrbracket^{g,c} = \text{the function which maps every individual } x \text{ to truth if and only if (iff) } x \text{ smiles, and to falsity otherwise} = \lambda x_e. \text{smile}'(x)$
- c. $\llbracket \text{pets} \rrbracket^{g,c} = \lambda y_e \lambda x_e. \text{pet}'(x, y)$
- d. $\llbracket \text{dog} \rrbracket^{g,c} = \lambda x_e. \text{dog}'(x)$
- e. $\llbracket \text{the} \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle}$: defined iff there is a unique x that satisfies P , when defined, this function returns $x = \lambda P_{\langle e,t \rangle}. \iota x_e. P(x)$
- f. $\llbracket a/\text{some} \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle}$. [there is an individual x that satisfies both P and Q] $= \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle}. \exists x_e. P(x) \wedge Q(x)$
- g. $\llbracket I \rrbracket^{g,c} = \text{the speaker of the utterance context } c = s_c$
- h. $\llbracket \text{she}_i \rrbracket^{g,c} = \text{defined iff the individual assigned to the index } i \text{ by the assignment function } g \text{ is female, when defined, it denotes this individual} = \text{female}'(g(i)). g(i)^{11}$

Table 2 summarizes the formal notation used in the denotations in (8) and throughout this thesis.

ABBREVIATION	ITS CONTENT
$\lambda x_e. \dots$	The function which maps an individual (type e) to ...
$\iota x_e. P(x)$	The unique individual that satisfies P if there is one, undefined (#) otherwise
$\exists x. \dots$	'there is an x such that ...'
$\exists! x. \dots$	'there is only one x such that ...'
$\forall x. \dots \rightarrow \dots$	'for all x , if ..., then ...'
$\subset, \subseteq$	proper and improper subset relations between sets; $\subseteq$ also holds in the case of identity between two sets
$<, \leq$	proper 'part-of' and improper 'part-of' relations between individuals / situations; $\leq$ also holds in the case of identity
$\wedge$	'and'
i	An index representing a discourse referent; an index is mapped to a specific real individual by the assignment function g
$\text{smile}', \text{pet}', \text{dog}'$	The predicates denoted by the corresponding expressions

¹¹ Here I'm using Hackl's (2019) notation for presuppositions.

$p.q$	Defined if p holds, undefined (#) otherwise; when defined, the denotation is q
-------	--

Table 2. Abbreviations used in denotations

Parameters of interpretation, g and c , will sometimes be omitted where they don't play a role. The context c plays a role in the interpretation of first and second person pronouns, as well as of salience-based determiners discussed in the following chapters. Third person pronouns and familiarity-based determiners (Chapter 2) are represented as (including) variables or indices like i which receive their meaning from the assignment function g . These indices should be understood as discourse referents in the tradition of Dynamic Semantics (Karttunen 1976; Heim 1982; Roberts 2003), see Chapter 2, section 2.1.¹²

Complex expressions are interpreted by combining the denotations of their constituents according to semantic composition rules. The only rule we will be using is Functional Application (Heim & Kratzer 1998: 94–95): expressions with functional denotations take other expressions as arguments.

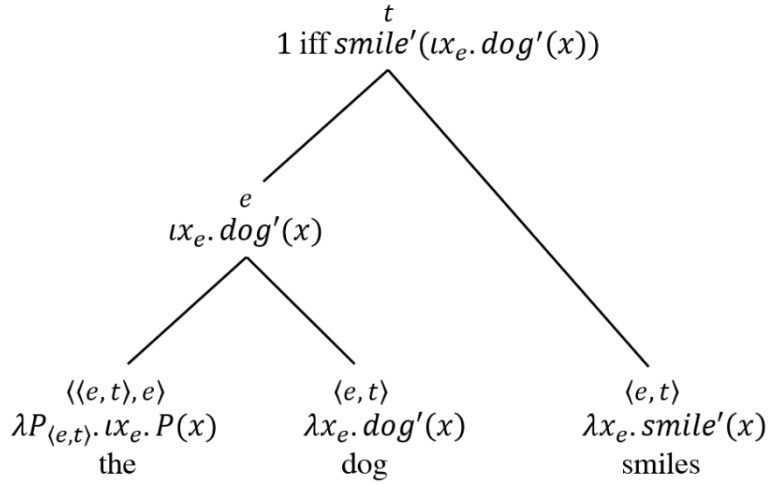
(9) FUNCTIONAL APPLICATION (FA)

If C is a complex expression with daughters A and B , $\llbracket A \rrbracket^{g,c}$ has the type $\langle \alpha, \beta \rangle$ and $\llbracket B \rrbracket^{g,c}$ has the type α and is in the domain of the function denoted by A , $\llbracket B \rrbracket^{g,c} \in \text{Dom}(\llbracket A \rrbracket^{g,c})$, then $\llbracket C \rrbracket^{g,c} = \llbracket A \rrbracket^{g,c}(\llbracket B \rrbracket^{g,c})$ and $\llbracket C \rrbracket^{g,c}$ has the type β .

Here is an illustration (10) in the form of an annotated tree where each node has a denotation and a type. For the current purposes, a representation which depicts the constituency of a complex expression to derive its denotation can be called its Logical Form (LF). The two mother nodes, representing the determiner phrase and the sentence as a whole, receive their denotation by combining their daughter nodes via FA.

¹² In Chapter 3, we will also add free possessives (in the sense of Partee & Borschev 2003) to the list of expressions that are interpreted with appeal to the assignment function.

(10)



In prose, the sentence in (10) is defined iff there is a unique dog (this is the presupposition contributed by ι), when defined, it is true iff the unique dog smiles.

In what follows I will make use of simpler LFs presented in bracket notation along with the semantics of the expression, e.g., as in (11)-(12). (I will usually assume that a nominal with a determiner is syntactically a DP and a clause is a TP or a CP. The syntactic categories of clauses will not play a role.)

$$(11) \quad [_{TP} [_{DP} \text{the}_{\langle\langle e, t \rangle, e \rangle} \text{dog}_{\langle e, t \rangle}]_e \text{smiles}_{\langle e, t \rangle}]_t$$

$$(12) \quad \llbracket \text{the dog smiles} \rrbracket^{g, c} = 1 \text{ iff } \text{smile}'(\iota x_e. \text{dog}'(x))$$

2. Many types of definites crosslinguistically

The goal of this chapter is to introduce the core concepts in the study of definiteness and determiner semantics more broadly and at the same time to give an overview of recent developments in crosslinguistic research on definite articles (definites). The focus will be on in-depth language-particular semantic research, stemming from (Schwarz 2009), since this is essentially what the current work is about. I will mostly overlook the broader typological studies based on secondary data such as the recent functionally-oriented monograph by (Becker 2021).

This chapter is structured as follows. In §2.1, based on Roberts' (2003) work on English definites, I define definiteness as consisting in two presuppositions: weak familiarity and informational uniqueness. All definite expressions presuppose at least these two requirements. Here I also discuss how Heim's (1991) principle *Maximize presupposition!* allows us to account for the competition between definites and indefinites.

There are more specialized types of definites crosslinguistically, especially when we look at determiners (and not pronouns or proper names). In §2.2, I introduce STRONG ARTICLES which require strong familiarity together with informational uniqueness, and in §2.3, I introduce SALIENT ARTICLES which additionally require salience. In the course of these sections, I discuss how these requirements may be represented in our simple extensional framework from §1.2 for better comparability with my discussion of the Northern Khanty data in Chapter 3.

I also try to clarify how the two types of familiarity and salience relate to each other and how the two types of familiarity are to be distinguished in §2.3.2, using a more fine-grained notion of context in the spirit of Dynamic Semantics. I argue that the two types of familiarity together with salience and (relative) maximal salience, proposed by Roberts for personal pronouns, make up an implicational scale of constraints on the discourse referent in question, the Salience-Familiarity Scale (§2.3.2). A brief comparison with Gundel and colleagues' Givenness Hierarchy (Gundel & Hedberg & Zacharski 1993) is provided in §2.3.3 which also lets us clarify the difference between relative maximal salience and absolute maximal salience, conceptualized in degree-based terms in §2.3.3.1.

In §2.4, we look beyond the Saliency-Familiarity Scale at definites selecting different Löbnerian concept types (Löbner 2011) and at even more specialized types of definites, including demonstrative determiners. Section 2.5 concludes the chapter.

2.1. Definiteness and Maximize presupposition!

Following (Roberts 2003) among many others (see Heim 2019), I understand definiteness to consist in two requirements: familiarity and uniqueness. The other kinds of requirements discussed in further sections, such as saliency, are not a part of the necessary and sufficient conditions of definiteness in my view.

To illustrate Roberts' analysis let us consider the distribution of the English Definite Article. Both uniqueness and familiarity are notions that have been proposed as the core of the English Definite Article's semantics in the long-standing tradition prior to Roberts.¹³

Uniqueness requires there to be one and only one referent satisfying the descriptive content of the determiner phrase (DP) in the model / the world / the topic situation.¹⁴ Thus, in (13) the Definite Article is used because the Queen of England is a globally unique entity. In (14), upon hearing the sentence the addressee will be justified in assuming that there was one and only one locally unique button in the box, i.e., the uniqueness requirement is accommodated, on which see (21) below.¹⁵

(13) **The Queen of England** had a bad year in 1993. (Roberts 2003: 209)

(14) I found a box in my attic the other day. I opened the lid and pushed **the button I found inside**. You won't believe what happened. (adapted from (Ibid.))

Following Roberts' (2003) proposal to be presented below, we will not be making use of 'semantic' uniqueness (uniqueness in the world) but rather we will talk about

¹³ For uniqueness, see, e.g., (Russell 1905; Link 1983; Elbourne 2013; Coppock & Beaver 2015), and for familiarity, (Christophersen 1939; Heim 1982; Schwarz 2009; Arkoh & Matthewson 2013; Beaver & Coppock 2015), among many others, as well as the recent critical discussion in (Coppock 2022).

¹⁴ For the purposes of convenience, let us assume that all nominals with determiners are determiner phrases (DPs) syntactically and that the complement of a determiner is a noun phrase (NP). I will use 'the descriptive content of the DP', 'the DP description', and 'the NP denotation' interchangeably to refer to the semantic object denoted by the NP complement of the given determiner which provides the descriptive information conveyed by this DP.

¹⁵ For plural DPs, uniqueness is understood as maximality: plural definites require the referent to be the unique maximal entity satisfying the descriptive content of the DP (Link 1983).

‘informational’ uniqueness which is uniqueness among (weakly) familiar discourse referents.

Familiarity requires there to be a discourse referent introduced in prior context that satisfies the definite DP’s descriptive content. A discourse referent (Karttunen 1976) is a virtual entity assumed to exist in the current context by the interlocutors for the purposes of the current information exchange. The interlocutors mutually accept a body of descriptive information regarding this entity, although there might be no corresponding individual in the actual world of which this information holds. Discourse referents are introduced into the context by indefinite DPs and are picked up by anaphoric expressions like personal pronouns and other definite DPs. In formal notation, discourse referents are represented as indices such as i or 8. These indices are mapped by the assignment function g to individuals in the actual world.¹⁶

A related notion is the common ground (Stalnaker 1974): the body of information that the interlocutors mutually accept and coordinate their communication with respect to it. The discourse referents of a given context are entailed to exist by the common ground. Familiarity, thus, entails existence in the worlds under discussion (another commonly discussed implication of definite DPs).

According to Heim’s seminal thesis (1982), the use of definite DPs (including personal pronouns) and indefinite DPs in English is governed by the Extended Novelty-Familiarity Condition (Heim 1982: 238). Under this Condition, the use of a definite DP presupposes the existence of a corresponding discourse referent in the context (familiarity), while the use of an indefinite DP presupposes the absence of a corresponding discourse referent in the context, so that indefinite DPs introduce novel discourse referents (novelty).

¹⁶ The cases where there is no actual individual corresponding to the given discourse referent are cases where a discourse referent is introduced in the scope of negation, modal quantifiers, and other expressions which limit the ‘lifespan’ of a discourse referent. For instance, the sentence *Bill doesn’t have a car₅* cannot be followed up with *It₅ is black* because the discourse referent 5 was introduced in the scope of negation and does not ‘survive’ outside its scope (Heim 1982: 165ff). We will mostly not be concerned with cases like this in what follows.

Sometimes, anaphoric pick-up for such a discourse referent is possible, e.g., with a continuation like *He wouldn’t have a parking space for it₅*, where the pronoun is in the scope of the modal auxiliary *would*. Thus, the above picture is a simplification. See (Hofmann 2022) and references therein for dynamic-semantic analyses of anaphora of this kind.

Thus, in (15) the Definite Article is obligatorily used upon the second mention of *the glass* since the context does provide an appropriate discourse referent, which has been introduced by the first mention with the Indefinite Article. On the other hand, the Indefinite Article is infelicitous in the second clause since that would violate its novelty presupposition, whereas the Definite Article cannot be used upon the first mention of the glass since it is not familiar in any sense at that point.¹⁷

(15) A / #**the wine glass** broke last night. **The** / #**a glass** had been very expensive. (based on (Roberts 2003: 293))

Note also that this example presents a problem for theories of uniqueness since the Definite Article here does not seem to require uniqueness of the glass in the world or even in the relevant household. This is resolved below via Roberts' notion of informational uniqueness.

Roberts points to the fact that discourse referents are not only introduced by linguistic material (which is the case of strong familiarity / anaphoricity) but may also be introduced via entailment from the common ground or from the extralinguistic context of the utterance situation. This observation is captured under her notion of weak familiarity.¹⁸

(16) WEAK FAMILIARITY OF DISCOURSE REFERENTS (Roberts 2003: 298, 304, 306)

A discourse referent is weakly familiar iff the existence of the entity in question is entailed by the interlocutors' common ground.

Both uniqueness and familiarity effects may be observed in cases involving bridging (Clark 1975) such as (17). In (17a), the introduction of a unicycle into the context entails the existence of its wheel, which licenses the Definite Article since the wheel is familiar via entailment from the common ground. However, the Definite Article is barred in (17b) even though, presumably, the tires of a car also become familiar as soon as a car

¹⁷ Here and throughout this thesis, * marks ungrammaticality and # marks infelicity. % indicates that some speakers accepted the marked expression, but others did not. An asterisk (*) outside of brackets *(X) indicates that omission of the bracketed material results in ungrammaticality and vice versa for (*X). The same holds for other acceptability signs.

¹⁸ See (55) for a potential refinement of the weak vs. strong familiarity distinction.

is introduced. The reason for that is that uniqueness is not satisfied here since a car has four tires unlike a unicycle which only has one wheel.

- (17) a. Every unicycle had a spoke missing from **the wheel**.
 b. #Every car had a puncture in **the tire**. (adapted from (Roberts 2003: 291))

But what about the glass example (15) above, in which uniqueness in the strict sense does not hold? This problem dissolves if one adopts, following Roberts, **informational uniqueness**. Unlike semantic uniqueness, standardly assumed in works prior to Roberts, informational uniqueness does not require uniqueness in the world but is only concerned with uniqueness among weakly familiar discourse referents. Roberts' analysis of definite DPs is summarized below.

- (18) WEAK FAMILIARITY AND INFORMATIONAL UNIQUENESS OF DEFINITE DPs (informal)
 (Roberts 2003: 308)¹⁹

Given a context C , use of a definite DP_i presupposes that it has as antecedent²⁰ a discourse referent x_i which is:

- a. weakly familiar in C , and
 b. unique among discourse referents in C in being contextually entailed to satisfy the descriptive content of DP_i .

In Chapters 3 and 5, I will propose an analysis for the Kazym Khanty Salient Article that is based on Roberts' analysis in (18). But until then, it will also be useful to have a simpler, more traditional static analysis for the English Definite Article (Partee 1986; Coppock & Beaver 2015: 396, 408).

- (19) $\llbracket the \rrbracket = \lambda P_{\langle e,t \rangle} . \iota x_e . P(x)$

Given a predicate P and some individual d ,

$$[\iota x_e . P(x)] = d,$$

iff the extension of P contains only d , $\{x: P(x)\} = \{d\}$, undefined otherwise.

¹⁹ The reader interested in the formal version is referred to (Roberts 2003: 310).

²⁰ Since, as we saw above, the discourse referent need not be introduced by any explicit linguistic material, the term 'antecedent' should be taken to refer to the corresponding discourse referent here, and not to any explicit DP.

In prose, the Definite Article takes a predicate (type $\langle e, t \rangle$) and returns the unique individual that satisfies this predicate, if there is one, but yields undefinedness otherwise. Following (Partee 1986; Coppock & Beaver 2015; Coppock 2022: fn. 6), the (semantic) uniqueness and existence presuppositions are encoded in ι and, for this reason, not explicitly represented in (19).

It is easy to see that the analysis in (19) has a hard time dealing with the glass example (15) where uniqueness is so blatantly violated. This is one of the reasons for adopting more intricate frameworks such as Dynamic Semantics or Situation Semantics. The dynamic option is the one Roberts (2003) pursues (18) and, as we saw, it tackles non-global uniqueness via the notion of informational uniqueness. With informational uniqueness, we are no longer speaking of actual individuals but rather we require uniqueness among familiar discourse referents. It is reasonable to assume that in (15) there is only one ‘glass’ discourse referent and, thus, informational uniqueness is satisfied.

Below, where simplification is not harmful, we will work with the simpler analysis in (19).

There are cases in which it seems that Roberts’ conditions in (18) are not satisfied but the Definite Article is felicitously used. For instance, consider (20) below, repeated from (14). Clearly, neither weak familiarity nor informational uniqueness are satisfied in this context, as no familiar and unique ‘button’ referent has been introduced in (or is entailed by) the prior discourse. What happens here, however, is that the addressee **accommodates** semantic uniqueness and existence (“there was one and only one button that the speaker found inside the box”) and this entails informational uniqueness. As Roberts puts it, “since the presupposition of informational uniqueness in definite NPs can be satisfied by semantic uniqueness plus weak familiarity, speakers actually use this fact and the expectation that a hearer will cooperatively accommodate (when retrievable and plausible [see below — *S. M.*]) to convey the information that a definite NP’s intended denotation [exists and] is semantically unique” (Roberts 2003: 319).

(20) I found a box in my attic the other day. I opened the lid and pushed **the button I found inside**. You won’t believe what happened. (adapted from Roberts 2003: 209)

This analysis crucially relies on the notion of **presupposition accommodation** (Lewis, D. 1979; Beaver & Zeevat 2007; von Stechow 2008), a mechanism whereby the

addressee adds information to her beliefs in order to make sense of an apparent presupposition failure on the speaker's part. Accommodation is commonly used by speakers to background newly introduced information. It is generally available given that the following conditions obtain:

(21) NECESSARY CONDITIONS ON PRESUPPOSITION ACCOMMODATION (Roberts 2003: 302–303)

- a. Retrievalability: what the hearer is to accommodate is easily inferable, by virtue of its salience²¹ and relevance to the immediate context, and
- b. Plausibility: the accommodated material is unobjectionable, i.e., it does not contradict the common ground or the addressee's beliefs and is not unlikely.

Another type of example where Roberts appeals to accommodation is provided by superlatives as in (22) (see her discussion of similar examples in (Roberts 2003: 302–303)). With superlatives, the Definite Article is the only possible option, even though the DP referent is often novel in such cases.

(22) Last weekend we climbed **the / #a biggest mountain in West Virginia**. (adapted from (Roberts 2003: 294))

Under standard assumptions (Heim 1999), the superlative introduces a uniqueness requirement that the DP referent have the highest degree of the property denoted by the gradable adjective with respect to its comparison class. This ensures semantic uniqueness of the NP denotation and, by Roberts' reasoning above, licenses accommodation of weak familiarity of the DP referent. Semantic uniqueness with weak familiarity gives us informational uniqueness and, thus, the Definite Article is used with superlatives (22).

Note that the simplified static analysis in (19) also predicts that the Definite Article will be used in such cases given the pragmatic principle of *Maximize presupposition!* (MP; Heim 1991; 2019: §1.7; Coppock & Beaver 2015).²²

²¹ Roberts does not clarify in what sense the material to be accommodated has to be salient. I believe that the retrievalability condition can be stated by appealing to relevance only, but I do not explore it further here. See the above cited works.

²² Compare also Magri's (2009: 253) formulation of MP which might be clearer: "A sentence ϕ sounds odd if there exists an alternative sentence ψ which conveys the same information as ϕ but triggers stronger presuppositions than ψ ".

(23) MAXIMIZE PRESUPPOSITION! (Heim 1991)

Make your contribution presuppose as much as possible!

Informally speaking, MP forces the Definite Article to be used in all cases where its uniqueness presupposition is satisfied in the context. The use of the Indefinite Article leads to an anti-uniqueness inference. The addressee infers that the speaker used an Indefinite Article where a Definite Article was possible because the latter's requirements are not satisfied in the context: there is more than one referent satisfying the DP's descriptive content. With superlatives, the anti-uniqueness inference gives rise to a contradiction, hence the infelicity in (23). (See also (Hawkins 1991; Gundel et al. 1993).)

Assuming this account allows us to abandon Heim's (1982) Extended Novelty-Familiarity Condition and similar analyses of the Indefinite Article which assume that it presupposes anti-uniqueness or novelty. Instead, anti-uniqueness and novelty inferences associated with (some) uses of the Indefinite Article arise as 'antipresuppositions' (Percus 2006) via pragmatic reasoning driven by MP.^{23,24} The Indefinite Article itself can then be assigned the simple, presupposition-less denotation in (24). According to (24), *a/an* is an existential quantifier which takes the NP denotation (*P*) and the denotation of the rest of the sentence (*Q*) and yields truth iff there is an individual *x* that satisfies both *P* and *Q*.²⁵

$$(24) \llbracket a/an \rrbracket = \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle} \cdot \exists x_e. P(x) \wedge Q(x)$$

With this denotation at hand, a sentence like *A wine glass broke* (15) receives the Logical Form in (25) and the truth conditions in (26). Note that (26) does not presuppose

²³ See (Percus 2006; Magri 2009; Coppock & Beaver 2015; Sudo 2024) for specific implementations.

²⁴ Pavel Astafiev (p.c.) contends that this does not necessarily work as desired in a dynamic framework with some standard assumptions. For instance, if there is a familiar discourse referent 4, a predicate like *dog'*, and it is not common ground that $dog'(g(4)) = 1$, then the current suggestion predicts that it would be felicitous to utter *a^t dog* (as part of a larger sentence) to pick up the familiar discourse referent and convey the new information that it is a dog. But, as Astafiev (2025) discusses, anaphoric DPs normally presuppose their descriptive content, and so this prediction is unwelcome. Rather we expect (i) such utterances to be infelicitous both with the Indefinite and with the Definite Article, and (ii) that the Definite Article is used with a familiar discourse referent. The solution proposed by (Beaver & Coppock 2015) for English seems to solve this issue, although their assumptions are nonstandard. The interested reader is referred to cited works. I will maintain the (simplifying) proposal from the main text in the current work.

²⁵ Alternatively, if one wishes indefinite DPs to yield a denotation of type *e*, we may use the choice-functional analysis which assigns the Indefinite Article a denotation of type $\langle \langle e, t \rangle, e \rangle$, see (Reinhart 1997; Winter 1997; Matthewson 1999) and most recently (Mirazzi 2024).

novelty or anti-uniqueness since the Indefinite Article is not a presupposition trigger under this analysis.

(25) $[TP [DP a_{\langle\langle e,t \rangle, \langle\langle e,t \rangle, t \rangle\rangle} \text{ wine glass}_{\langle e,t \rangle}]_{\langle\langle e,t \rangle, t \rangle} \text{ broke}_{\langle e,t \rangle}]_t$

(26) $\llbracket \text{A wine glass broke} \rrbracket = 1 \text{ iff } \exists x. \text{wineglass}'(x) \wedge \text{broke}'(x)$
 true iff there is a wine glass and it broke

Thus, Heim's (1991; 2019) *Maximize presupposition!* leads to the view of the English determiner system schematized in Figure 1, adapted from (Dayal 2018: 3). The Indefinite Article could in principle be used in the contexts where *the* actually appears, but its use is restricted by MP since the Definite Article carries stronger presuppositions and thus wins over in contexts where its presuppositions are satisfied.

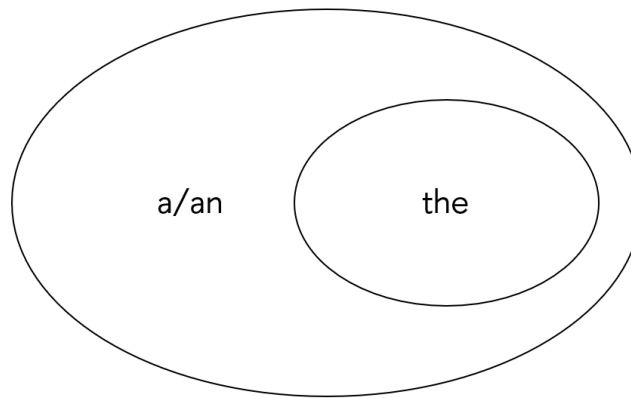


Figure 1. Relation between the Definite and the Indefinite Article in English (Dayal 2018: 3)

This view receives support from determiner systems where there's only an INDEFINITE ARTICLE (or a paradigm of INDEFINITE ARTICLES) with underspecified semantics as in (24). This is the case, for instance, with Salish languages (Matthewson 1996; 1999; Mattina 2006; Gillon 2009; 2013; Lyon 2011; 2013; 2015; Reisinger et al. 2021).²⁶ The determiner systems of Salish languages can get quite elaborate; however, they do not appear to track the definite/indefinite contrast.

²⁶ Although, whether the semantics is indeed underspecified as in (15) is a controversial matter, see (Matthewson 2008; Reisinger et al. 2021). Demonstrative determiners in Salish languages are also (more) freely used with novel referents than expected (Reisinger & Huijsmans 2021; Huijsmans & Reisinger 2022).

The empirical picture reported here, however, is uncontroversial and is largely the same across the determiner system of the Salish family according to these studies.

This is illustrated with examples (27)-(28) from St’át’imcets (< Northern Interior Salish; British Columbia, Canada), a language that features at least seven distinct determiners (six encoding distinctions in number and spatio-temporal distance from the speaker and one narrow-scope “polarity” determiner). The important fact for us is that these determiners do not encode definiteness distinctions. Thus, in example (27) the Singular Present Determiner *ta ...-a* is used with a novel non-unique entity ‘star’, and example (28) provides a minimal pair where the same determiner is used with the unique and familiar ‘sun’.²⁷

(27) ka hál’h-a [ta nkakúsent-a]
 OOC show-OOC DET star-DET
 ‘A star appeared.’ (Matthewson 1999: (58))

(28) ka hál’h-a [ta snéqwem-a]
 OOC show-OOC DET sun-DET
 ‘The sun appeared.’ (Matthewson 1999: (64))

As Matthewson (1999: 108) puts it: “St’át’imcets DPs behave precisely as we would expect for indefinites in a language which lacks definites.” Absent a contrasting DEFINITE ARTICLE, indefinite DPs are used in canonical definite contexts (see Matthewson 1996; 1999 and the works cited above for more examples).

This concludes our discussion of the English Definite Article and of MP effects in DEFINITE-INDEFINITE determiner systems, and we now turn to other types of definite determiners found crosslinguistically.

2.2. Familiarity

In recent years, starting with Schwarz’s (2009) dissertation, there has been a rise in crosslinguistic research on definite determiner semantics. (See also (Wespel 2008).)

Schwarz investigates a phenomenon found in several Germanic varieties where two morphological paradigms of definite articles are distinguished. This is illustrated in (29) with data from the Fering dialect of Northern Frisian (< West Germanic) from (Ebert, K. 1971) (cited after (Schwarz 2009: 13)).

²⁷ OOC in the glosses stands for ‘out of control’.

(29) FERING WEAK VS. STRONG ARTICLES (Schwarz 2009: (10))

a. Ik skal deel tu **a** / ***di** **kuupmaan**.

I must down to the^{weak} / the^{strong} grocer

‘I have to go down to the grocer.’

b. Oki hee an hingst keeft. ***A** / **Di** **hingst** haaltet.

Oki has a horse bought. the^{weak} / the^{strong} horse limps

‘Oki has bought a horse. The horse limps.’

The difference between (29a) and (29b) is that in (29a) the grocer is familiar in the context in virtue of the interlocutors’ common knowledge: the grocer is unique and familiar in some restricted situation, perhaps, in the interlocutors’ town. Thus, the grocer need not be previously mentioned to license a definite article, in this case, Ebert’s A-form which Schwarz calls ‘the Weak Definite Article’. In (29b), however, a horse is explicitly introduced in the preceding context which makes it strongly familiar in Roberts’ (2003) terminology. This licenses the use of the Fering D-form, which Schwarz calls ‘the Strong Definite Article’. The Strong Definite Article is infelicitous with weakly familiar discourse referents, as (29a) shows. At the same time, the Weak Definite Article is infelicitous with strongly familiar discourse referents (29b).

Schwarz argues that the same semantic contrast is found in Standard German contracted vs. non-contracted preposition-article combinations (30) and, thus, that two different articles are involved in (30a)-(30b), although this distinction is obscured by their homonymy in environments without prepositions.

(30) STANDARD GERMAN CONTRACTED FORM (WEAK ARTICLE) AND NON-CONTRACTED FORM (STRONG ARTICLE) (Schwarz 2009: (8))

a. Hans ging **zu dem** **Haus**.

Hans went to the^{strong} house.

‘Hans went to the house.’

b. Hans ging **zum** **Haus**.

Hans went to the^{weak} house.

‘Hans went to the house.’

Working in a situation-semantic framework, Schwarz (2009) proposes that the difference between the two types of definite articles is that the WEAK DEFINITE ARTICLE

only presupposes uniqueness of the referent in the contextually given ‘resource’ situation (31b), while the STRONG DEFINITE ARTICLE additionally requires familiarity by means of a silent pronoun/index introduced as a syntactic argument of the article (32a). (See (Schwarz 2009: Chapter 3) for details of the framework.) Consider the Logical Forms and denotations provided for the two articles in (31)-(32).^{28,29}

(31) WEAK ARTICLE (UNIQUENESS-BASED) (Schwarz 2019: 12)

a. $[_{DP} [the^{weak} s_r] NP]$

b. $\llbracket the^{wk} \rrbracket^g = \lambda s_s \lambda P_{\langle e, st \rangle}. \iota x. P(x)(s)$

(32) STRONG ARTICLE (FAMILIARITY-BASED) (Ibid.)

a. $[_{DP} i [the^{strong} s_r] NP]$

b. $\llbracket the^{str} \rrbracket^g = \lambda s_s \lambda P_{\langle e, st \rangle} \lambda y. \iota x. [P(x)(s) \wedge x = y]$

c. $\llbracket [_{DP} i [the^{str} s_r] NP] \rrbracket^g = \iota x. [\llbracket NP \rrbracket^g(x)(\llbracket s_r \rrbracket^g) \wedge x = g(i)]$

An important role is played by the assignment function g in this analysis. To appreciate its role, it’s useful to consider the denotation of the whole DP in (32a), once the STRONG ARTICLE has composed with all three of its arguments (32c). In prose, this DP denotes the unique individual which fits the DP description in the contextually given resource situation s_r such that this individual is identical to the individual assigned by g to index i .

The role of the assignment function g is to supply the meanings of the contextually dependent pronouns, the resource situation pronoun s_r and the index i , to the denotation of the DP. A complex expression that contains pronouns such as (32) is only felicitous in a context c if c provides an assignment function g_c which assigns some values to the pronouns. (This is Heim and Kratzer’s (1998: 243) Appropriateness Condition on contexts.) Intuitively speaking, this amounts to a familiarity presupposition for pronouns, and so, the difference between the WEAK ARTICLE and the STRONG ARTICLE is that the

²⁸ Recall from the discussion of the simple denotation for *the* in (19) that ι presupposes uniqueness and existence.

²⁹ I follow Schwarz’s convention of representing the Weak and the Strong Articles as *the* with a superscript “str(ong)/w(ea)k” (Schwarz uses a subscript). These two denotations are proposed as crosslinguistic gram types (Dahl 2016) which justifies using the same arbitrary label for both the Northern Frisian A- and D-forms and the Standard German contracted and non-contracted article-preposition complexes, as well as for analogous constructions in other languages (Schwarz 2019).

STRONG ARTICLE requires the DP referent to be familiar (via the anaphoric index), while the WEAK ARTICLE does not.

The interesting question is how to account for the infelicity of the Fering Weak Article in anaphoric cases such as (33) repeated from (29b). One could think of an account in terms of *Maximize presupposition!* (23), along the lines of the competition between English Definite and Indefinite Articles discussed in §2.1. The Strong Article appears to carry a stronger presupposition than the Weak Article and so, by MP, the Strong Article is obligatory where its presuppositions are satisfied, and the Weak Article is infelicitous there.

- (33) Oki hee an hingst keeft. *A / Di hingst haaltet.
Oki has a horse bought. the^{weak} / the^{strong} horse limps
'Oki has bought a horse. The horse limps.' (Schwarz 2009: (10))

However, this only appears to be the case.³⁰ First of all, Schwarz (2009: 44ff., 281ff.) discusses a range of contexts where the Weak Article can be used apparently anaphorically. A part of these contexts shows that the two Articles distributions overlap which should not be possible under MP-based competition (see the original work for examples). Second, if we consider the analyses proposed by Schwarz (31)-(32) (or the simplified versions below (34)-(35)), it is apparent that the Strong Article's presupposition is not properly stronger than the Weak Article's one, i.e., the Strong Article's presupposition does not **entail** the Weak Article's presupposition. This is so because the Strong Article presupposes uniqueness among individuals that satisfy the NP denotation **and** that are equivalent to the familiar discourse referent, provided by the silent pronoun. This makes the Strong Article's uniqueness requirement very different from the Weak Article's plain semantic uniqueness presupposition.

On the other hand, the Strong Article's presupposition clearly delimits a narrower range of contexts of use than the Weak Article's one. Specifically, it implies that the discourse referent for the DP is to be found among the familiar discourse referents in the preceding text. Thus, the use of a Strong Article will inform the addressee to interpret the

³⁰ I thank Pavel Astafiev (p.c.) for discussions on this point. See (Coppock 2022: 123ff) for important critical discussion of the WEAK vs. STRONG competition, and see also (Lyons 1999: 166ff) for another discussion of Fering facts.

DP anaphorically, while the Weak Article merely implies uniqueness, of which familiarity is a special case under the current analysis. This suggests the possibility of a Gricean pragmatic analysis (Grice 1975) of the competition pattern along the lines of (Gundel et al. 1993), who discuss how the distribution of uniqueness-signaling determiners and familiarity-signaling determiners is regulated via Gricean quantity implicatures (see §2.3.3). Figure 2 below schematizes this difference in the breadth of available contexts of the two Articles.

Before that, a note on the formal details. While situations are a useful tool to account for a wide range of properties of definites (as (Schwarz 2009; Elbourne 2013) attest, among many others), it will be enough for our purposes to work with a simplified analysis for STRONG ARTICLE DEFINITES without situations (34). I assume that the anaphoric index is carried by the article itself, in the spirit of (Heim 1982; Roberts 2003), and more recently (Hackl 2019; Ahn 2024; Sudo 2024).

$$(34) \llbracket the_i^{str} \rrbracket^g = \lambda P_{\langle e, t \rangle} : \exists! x_e [P(x) \wedge x = g(i)]. g(i)$$

In prose, given an NP denotation, the STRONG ARTICLE presupposes that there is only one individual that satisfies the NP denotation and that it is identical to the familiar discourse referent i .³¹ If this presupposition is satisfied, the STRONG ARTICLE returns the individual assigned to i by the assignment function g .

Compare this to our simple denotation for *the* from (19) repeated below. This is equivalent to a simplified version of Schwarz’s WEAK ARTICLE (31b).

$$(35) \llbracket the^{wk} \rrbracket = \lambda P_{\langle e, t \rangle} . \iota x_e . P(x)$$

Note that the presupposition imposed by the STRONG ARTICLE in (34) makes use of discourse referents by incorporating an index which gives us a familiarity presupposition, while the WEAK ARTICLE (35) is just about actual individuals and only presupposes semantic uniqueness (and existence).³² I suggest that the STRONG ARTICLE essentially

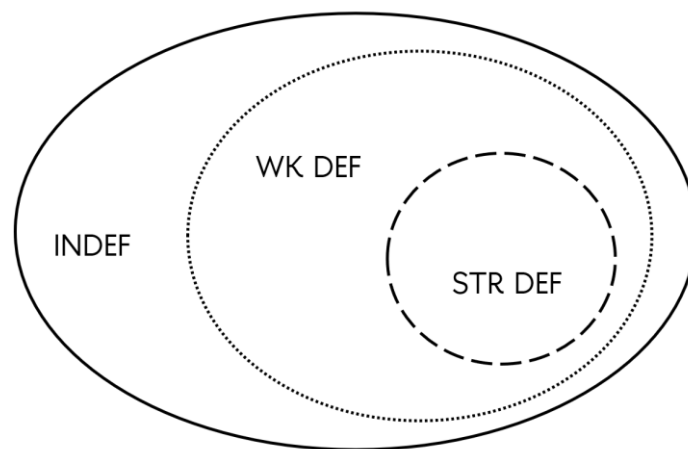
³¹ Since (34) does not feature ι in the denotation, I introduce the uniqueness presupposition explicitly as a domain condition (the part after the colon and before the dot) on the function denoted by the STRONG ARTICLE, following the notational convention of (Heim & Kratzer 1998). Thus, $\exists! x. \dots$ is to be read as ‘there is only one x such that ...’, which is different from $\iota x. \dots$ itself a shorthand for ‘the unique x such that ...’.

³² Compare the proposal by (Lewis, K. S. 2022). I thank Pavel Astafiev (p.c.) for bringing Lewis’s work to my attention.

presupposes informational uniqueness (18) among strongly familiar referents (to be discussed below).

We may still adopt Roberts' analysis of *the* as a WEAK DEFINITE ARTICLE which requires weak familiarity and informational uniqueness and distinguish it from the STRONG DEFINITE ARTICLE by saying that the STRONG DEFINITE ARTICLE requires strong familiarity (Roberts 2003: 304-305). This is not what Schwarz and other work such as (Arkoh & Matthewson 2013) do, but this is the move I think we ought to make if we want to maintain a meaningful difference between WEAK and STRONG ARTICLES (cf. Coppock 2022; 2024), see §2.3.2.³³

The denotations in (34)-(35) suggest the distribution for WEAK and STRONG ARTICLES schematized in Figure 2. STRONG ARTICLE DPs have a narrower distribution, and their contexts of use are a subset of the contexts that are in principle available to WEAK ARTICLE DPs. However, in languages where both are present, WEAK ARTICLES are dispreferred at least in some STRONG ARTICLE contexts due to being less informative for the purposes of finding a suitable referent along the lines of (Gundel et al. 1993), see §2.3.3.³⁴



³³ Although, this move, I believe, would predict WEAK and STRONG articles to be MP-competitors contrary to fact (see above), since the set of weakly familiar discourse referents is a proper superset of the set of strongly familiar discourse referents. It is important to rectify this matter in the future.

³⁴ However, see (Coppock 2022; 2024) for an extended criticism of Schwarz's analysis and the idea that the two types of definites are in complementary distribution. Taking Coppock's arguments into account is beyond the scope of this thesis, although, as mentioned above, an attempt at refining the distinction is made in §2.3.2.

Figure 2. Relation between INDEFINITE ARTICLES (INDEF), WEAK DEFINITE ARTICLES (WK DEF), and STRONG DEFINITE ARTICLES (STR DEF), based on (Heim 1991; Gundel et al. 1993; Schwarz 2009)

In a minimally different determiner system without a strong article, the weak article covers all definite article contexts (Coppock 2022; 2024), both uniqueness-based and familiarity-based, which is the case of English (19), as noted above. Note also that the relationship between different pairs of circles in Figure 2 is different. While the INDEFINITE ARTICLE is excluded in definite contexts due to MP, the WEAK DEFINITE ARTICLE is not excluded in STRONG DEFINITE contexts across the board because they do not compete for MP purposes, but rather in terms of more general (and more lax) pragmatic principles along the lines of (Gundel et al. 1993).

It is worthwhile to discuss in some more detail the idea that the STRONG ARTICLE (34) does not presuppose semantic uniqueness in Roberts' terms, but rather triggers an informational uniqueness-like presupposition (see (18)) with strong familiarity.³⁵ We can see this if we consider a case where more than one individual meets the DP description. Say there are two individuals y and x such that, for some NP denotation P , $P(y)$, $P(x)$, and $x \neq y$ all hold. In this case, $\llbracket the_i^{str} \rrbracket^g(P)$ may still be felicitous as long as $x = g(i)$, but $y \neq g(i)$, which informally boils down to x being strongly familiar while y is not.

This also requires the natural assumption that $\llbracket the_i^{str} \rrbracket^g(P)$ is infelicitous in a context where more than one (strongly familiar) discourse referent meets the DP description. Here's why. When a speaker utters a DP of the form $the_i^{str} NP$, the addressee is required to infer which discourse referent/index is intended. If there is more than one (strongly familiar) discourse referent meeting the DP description, this task is critically inhibited which leads to infelicity due to a violation of Grice's (1975) Maxim of Manner

³⁵ The discourse referent x_i is required to be unique among **strongly familiar** discourse referents in C in being contextually entailed to satisfy the descriptive content of DP_i , cf. (18).

(Heim 1982), specifically, the second submaxim “Avoid ambiguity”.^{36,37} This is summarized in (36).

(36) PRAGMATICALLY DERIVED UNIQUENESS (STRONG DEFINITE version)

The use of a STRONG ARTICLE DEFINITE DP [*the*_{*i*}^{*str*} NP], which requires strong familiarity, is infelicitous in a context *c* if *c* contains another strongly familiar discourse referent *k*, *k* ≠ *i*, which satisfies the DP description, due to a violation of Grice’s (1975) Maxim of Manner: the addressee cannot assign a unique index to the DP during comprehension.

Of course, we could also encode informational uniqueness among strongly familiar referents directly as a presupposition of STRONG DEFINITE DPs in the style of Roberts’ (18), but it is preferable to not hard-code in the semantics what could be derived pragmatically, unless, of course, this leads to undesired predictions.

The meaning we assigned to the STRONG ARTICLE in (34) reflects the intuition pursued in (Ahn 2024) that the difference between definite article-based descriptions and pronouns is that descriptions carry additional information provided by their NP restriction, while pronouns carry at most their ϕ -feature information.³⁸ An important difference is that what Ahn proposes for English *the* (Ahn 2024: (26b)), which is a weak article, is equivalent to our STRONG ARTICLE denotation. If the presence of an index

³⁶ Compare the same effect with pronouns in (i). (Knowing that Eduardo and Serge are male names,) the addressee who hears (i) would be hard-pressed to assign a denotation for the pronoun since both discourse referents, *i* and *k*, fit the conditions imposed by the pronoun.

(i) I met Eduardo_{*i*} and Serge_{*k*}. #He_{*i/k*} gave me a cake.

See (Roberts 2003: §4.2) for pertinent discussion.

³⁷ In a way, the reasoning given above echoes Roberts’ reasoning discussed in §2.1. There, we derived informational uniqueness from semantic uniqueness and an accommodation of weak familiarity for cases like superlative descriptions. Here, we derive informational uniqueness among strongly familiar referents from an informational uniqueness-like presupposition of (34) and a presupposition of strong familiarity. An important difference from Heim’s (1982) conversational uniqueness based on Gricean reasoning is that Heim eschews the uniqueness presupposition altogether, which is what Roberts (2003: §4.2) criticizes.

Although, see also Lewis’s (2022: §2.2) pertinent criticisms of pragmatically-derived uniqueness in general.

³⁸ Compare a pronoun denotation like (i) in the spirit of (Heim & Kratzer 1998).

(i) $\llbracket she_k \rrbracket^g = g(k)$, defined if *g(k)* is female and singular, undefined otherwise

Using Hackl’s (2019) notation for presuppositions we can shrink (i) to (ii) which makes the similarity between pronouns and the STRONG ARTICLE (34) even clearer.

(ii) $\llbracket she_k \rrbracket^g = female(g(k)) \wedge singular(g(k)).g(k)$

(The presupposition comes before the dot.)

interpreted as $g(i)$ is equivalent to a presupposition of strong familiarity, as I argued above, Ahn's denotation cannot be right for *the*, since *the* clearly does not require strong familiarity.

2.2.1. Akan Definite Article

Another interesting example of a STRONG ARTICLE is discussed by Arkoh and Matthewson (2013) who show that determiners like the German Strong Article are attested crosslinguistically, drawing on data from Akan (< Kwa < Niger-Congo). Importantly, Akan differs from Standard German in that it only has a strong article, with no weak counterpart to it (and no indefinite article, for that matter).³⁹

Much like *the* in English, the Akan Definite Article *nó* is not available upon the first mention of a novel referent such as *èkùtú* 'orange' in (37) and it is obligatory upon the second mention where the referent is already familiar. (See fn. 17 for acceptability sign notation.)

(37) [Beginning of conversation.]

Mò-tó-ò èkùtú (*nó). Èkùtú *(nó) yè dèw pápá

1SG.SUBJ-buy-PAST orange (*FAM) orange *(FAM) be nice good

'I bought an orange. The orange was really tasty.' (Arkoh & Matthewson 2013: 2)

Unlike *the*, however, the Akan Definite Article cannot be used with novel unique referents. Thus, in (38) neither *nyímpá* 'person' nor *àsìrán* 'moon' can be marked with *nó*, despite both of them being unique.⁴⁰ Cf. the English translation of (38) where omitting *the* in either case is impossible.

(38) Ámstrònyí **nyímpá** áà ó-dzí-ì kán tú-ù kó-ò àsìrán dò

A. is person REL 3SG.SUBJ-eat-PAST first fly-PAST go-PAST moon TOP

³⁹ Note that Bombi (2018) has argued against Arkoh and Matthewson's analysis and for analyzing the Akan Definite Article as a UNIQUENESS-BASED/WEAK DEFINITE like English *the*. (See also Owusu (2022) for yet another, more elaborate view of the Akan data, discussed in §2.4; and Coppock (2024).)

I, however, present the data as they are given in (Arkoh & Matthewson 2013) since this is an influential reference and my goal is simply to illustrate the fact that the uniqueness vs. familiarity distinction has been found relevant for determiner semantics crosslinguistically. The reader interested in less controversial instances of the ANAPHORIC ARTICLE is referred to the works cited in (Schwarz 2019).

⁴⁰ In German, this sentence is rendered using the Weak Article (Schwarz 2009: (43)), which corresponds with the Akan bare form according to Arkoh and Matthewson. The Strong Article is excluded here, just as expected under their analysis.

‘Armstrong was the first person to fly to the moon.’ (Ibid.)

Another pair of contexts that distinguishes STRONG and WEAK ARTICLES, highlighted by Schwarz (2009: 52ff) is ‘part-whole’ bridging (the relation between *fridge* and *freezer* in *The fridge was so big that the pumpkin could easily be stowed in **the** freezer*) vs. ‘producer-product’ bridging (the relation between *play* and *author* in *The play displeased the critic so much that he tore **the** author to pieces in his review*). **Part-whole bridging** is based on uniqueness in the same situation (a fridge has a unique freezer) and requires the WEAK ARTICLE in the many languages that feature this distinction (Schwarz 2019: §3). **Producer-product bridging** comes about with relational nouns and is based on familiarity (the author of **this** play), while presence of the antecedent in the same situation is not guaranteed, which is why the STRONG ARTICLE is used in such contexts crosslinguistically (Ibid.). As expected, the Akan Definite Article can only be used in producer-product bridging contexts, and not in part-whole bridging contexts (Arkoh & Matthewson 2013: §2.3.1).

Just as discussed above (36), a STRONG DEFINITE ARTICLE such as the Akan Definite Article is infelicitous in a context where there is more than one strongly familiar referent satisfying the DP description. This is illustrated in (39) where the only feasible option to refer to one of the two strongly familiar children is to use the personal pronoun (39b), while the definite description is ruled out (39a). (Cf. also the English gloss (39) and example (41) discussed in the next section.)

(39) (adapted from Arkoh & Matthewson 2013: (80))⁴¹

[“A (certain) child came from Winneba and another child_i came from Swedru.”]

a. #Àbòfrá nŏ_i bá-ì árá nà kyirèkyirènyí nŏ hwí-ì àbòfrá nŏ_i.

child FAM come-PST just FOC teacher FAM cane-PST child FAM

Intended: Immediately the child_i came, the teacher caned the child_i.

b. Ó_i-bá-ì árá nà kyirèkyirènyí nŏ hwí-ì nŏ_i.

3SG.SUBJ-come-PST just FOC teacher FAM cane-PST 3SG.ANIM.OBJ

‘Immediately s/he_i came, the teacher caned him/her_i.

⁴¹ I made a correction to the (b) part of the example that was marked with the infelicity sign ‘#’ in Arkoh and Matthewson’s paper by mistake which is clear from their discussion of the example (39b) being felicitous in contrast to (39a).

The Akan and the Germanic data show that a clear empirical difference exists between two types of definite determiners, so it makes sense to ask, when investigating a definite determiner, which type it resembles the most.⁴² As I will show in Chapter 3, the Kazym Khanty Unpossessives have an even narrower distribution than STRONG and WEAK ARTICLE DEFINITES in virtue of presupposing, for instance, an associative relation (the Associative Possessive) or relative maximal salience (the Salient Article; this restriction is discussed in the next section).

There are also narrower types of definites in other languages which are worth mentioning here.

2.3. *Salience*

Salience has been appealed to in the analysis of definites before (e.g., Lewis, D. 1979; von Stechow 2013). In particular, Roberts (Roberts 2003: 330) argues that personal pronouns are essentially definites, triggering similar presuppositions to definite descriptions (18), with the additional requirement that the antecedent be salient in the context and that it be the most salient discourse referent satisfying the descriptive content of the pronoun (i.e., person, number, and gender).

(40) PRESUPPOSITIONS OF PRONOUNS (informal) (Roberts 2003: (49))

Given a context C , use of pronoun Pro_i presupposes that it has as antecedent a discourse referent x_i which is:

- a. weakly familiar in C ,
- b. salient in C , and
- c. unique in being the most salient discourse referent in C which is contextually entailed to satisfy the descriptive content suggested by the person, number and gender of Pro_i .

To illustrate this analysis, consider (41). In this example, two female referents are introduced. Using the definite DP *the woman* to refer to either of them is infelicitous because it would violate informational uniqueness: neither woman is unique among the

⁴² See (Serdobolskaya 2017) for a conclusion along the same lines based on the differential object marking pattern in Beserman Udmurt (< Permic < Uralic) which involves a distinction between Accusative-Marked and Possessive Accusative-Marked Objects which is based on the uniqueness-familiarity distinction.

(Schwarz 2019: 15) collects further references to languages exhibiting the STRONG-WEAK split.

discourse referents in satisfying the DP content.⁴³ On the other hand, using the pronoun *she* is clearly felicitous, and the pronoun would refer to the latter of the two women since she is the most salient one due to her most recent mention. Thus, Roberts (2003: 323) suggests that “[a]naphora resolution for pronouns effectively screens out non-salient portions of the discourse and the discourse referents therein, so that only salient discourse referents are accessible to serve as antecedents”.⁴⁴ Let us call this a ‘**salience without uniqueness**’ context (since uniqueness simpliciter is not satisfied here).

- (41) A woman entered from stage left. Another woman entered from stage right. **She** / **#the woman** was carrying a basket of flowers. (adapted from (Roberts 2003: 324))

We can spell out Roberts’ analysis in the current framework as (42), where the pronoun denotes the individual assigned to i by the assignment, given that the presuppositions of the pronoun hold (relative maximal salience, as well as gender and number).⁴⁵

$$(42) \llbracket she_i \rrbracket^{g,c} = fem(g(i)) \wedge sg(g(i)) \wedge sal(i, c) \wedge \forall j \in Dom(g) [fem(g(j)) \wedge sg(g(j)) \wedge j \geq_{sal(c)} i \rightarrow j = i]. g(i)$$

Following Roberts (2003: (50)), I formalize (relative) maximal salience using the set of salient discourse referents Sal_c , which is a subset of the set of weakly familiar discourse referents $Dom(g)$, and the partial order $\geq_{sal(c)}$ over $Sal_c \times Sal_c$ (pairs of salient discourse referents): $j \geq_{sal(c)} i$ is true just in case j is at least as salient in c as i .

Drawing on data from the definite article of Bulu (< Bantu, Cameroon), Barlew (2014) has further argued that the notion of salience is relevant to definite determiners more broadly. But first, some definitions are needed. I explicate the notion of salience following (Barlew 2014) as follows.

Building on Roberts’ later work (Roberts 2011) (see also Gundel et al. 1993; Grosz & Joshi & Weinstein 1995), Barlew (2014: 623–624) takes **salience** to be a function of

⁴³ Recall from previous section that STRONG DEFINITE ARTICLES (39) are infelicitous in contexts like this just like the English Definite Article. Further see (47) below for the Bulu Salient Article in this context.

⁴⁴ See the original article for references to experimental and corpus-based work supporting this conclusion. Cf. also (Gundel et al. 1993) on the same point with the Givenness Hierarchy framework discussed in §2.3.2.

⁴⁵ Presuppositions are given before the dot, cf. §1.2 and fn. 38.

attention capture. A referent that is salient to a given individual is a referent that she is paying attention to. Just like attention, salience is a gradient notion so that a referent may be salient to a higher or lesser degree.⁴⁶

Based on results from psychology (see the references in the original paper, as well as recent discussion in Luck et al. 2021), Barlew distinguishes two types of attention capture. **Top-down attention capture** is driven by an individual's intentions: an individual pays attention to entities relevant to her goals. **Bottom-up attention capture** is stimuli-driven: an individual pays attention to perceptually prominent entities be that due to their movement, sudden change, standing out from the surroundings, or other factors. Crucially, the two kinds of attention interact (Anderson 2021; Luck et al. 2021) and are both relevant for discourse salience.

Assuming that information sharing is a goal of the interlocutors (Grosz et al. 1995; Roberts 2011; 2012; 2022), the interlocutors will pay attention to discourse referents that are relevant to addressing the current **question(s) under discussion** (QUD).⁴⁷ Roughly speaking, questions under discussion are questions that need to be addressed by the interlocutors in order to share information on the current topic under discussion or to achieve their other relevant goals. Following Roberts (Roberts 2012; 2019), I assume that each utterance in a discourse answers a (possibly implicit) QUD. Given this, the discourse referents that figure in the QUD are guaranteed to be relevant and, therefore, salient. Specifically, a discourse referent is salient due to the QUD if some property of this discourse referent needs to be settled in order to address this QUD.⁴⁸ Traditionally, such discourse referents are called topical.⁴⁹ This is the kind of top-down attention-based salience that will be most relevant to us in what follows.

Assuming a structured model of discourse like (Asher & Lascarides 2003) where different utterances/QUDs are connected via rhetorical relations, this also lets us understand the phenomenon of higher-order topics which may not be a part of the

⁴⁶ This fact will become important below in §2.3.3, where I distinguish relative and absolute salience. See also the discussion of Khanty data in §5.2.3.

⁴⁷ For an introduction to Roberts' model of the information structure of discourse structured by QUDs and its applications to anaphora resolution, English Focus, and other phenomena, see (Roberts 2012; 2022).

⁴⁸ Thank you to Lisa Hofmann (p.c.) for discussion.

⁴⁹ This gives us a modern definition of the notion of 'aboutness' topic (Reinhart 1981; Strawson 1964; Lambrecht 1994) that I will make use of in Chapter 5.

immediate QUD but at the same time are a part of a superordinate QUD that the immediate QUD helps address. On this use of QUD-based discourse salience, see (Roberts 2022).

Another source of salience relevant for this thesis is found with deictic demonstratives. Following general consensus in the work on demonstratives, in particular (Diessel 2006),⁵⁰ I assume that demonstratives serve to coordinate the interlocutors' joint attention. Speakers use demonstratives and deictic pointing gestures to focus the addressee's attention on a particular referent, making it salient (see also Reisinger & Huijsmans 2021; Stojnić 2021: Ch. 4; Huijsmans & Reisinger 2022; Marinina 2024).

Finally, plain introduction of a novel referent makes it salient at least to a small degree (as well as mention of non-salient familiar referents), since all mentioned entities come into the addressee's attention upon mention (bottom-up attention capture; cf. von Heusinger 2013; Stojnić 2021: Ch. 4) and are relevant to the ongoing conversation (top-down attention capture), at least with respect to the immediate QUD.

Also, it must be noted that for determining salience only the addressee's attentional state is relevant (Prince 1992; Gundel et al. 1993). The speaker's attentional state may be assumed, since the speaker necessarily pays attention to any discourse referent that she refers to. Otherwise, it would be impossible to refer to it.

The different cases of attention capture relevant for salience are summarized in (43).

(43) SOURCES OF SALIENCE

a. **Top-down attention capture:**

- The referent is relevant to the interlocutors' current goals, and so they attend to it;
- The referent is a part of the current question under discussion, and so the interlocutors attend to it;

b. **Bottom-up attention capture:**

- The referent is perceptually prominent in the speech situation due to movement, sudden change, standing out from the surroundings, or other factors, and so the interlocutors attend to it;

⁵⁰ I thank Lera Marinina (p.c.) for introducing me to Diessel's paper.

- The speaker directs the addressee’s attention to the referent via a demonstrative accompanied by a pointing gesture;
- The referent is mentioned by the speaker and thus comes to the addressee’s attention.

This gives us a clear enough informal understanding of how salience comes about. In Chapter 5, §5.5 I will make use of Stojnić’s (2021) dynamic system which formalizes bottom-up salience (43b). The formalization of top-down salience must be left for another occasion.

With this in place, we can give a formal definition of discourse referent salience in the current context, notated as $sal(i, c)$ in (42), based on (Barlew 2014). Barlew uses a dynamic framework like (Roberts 2003), augmented with a rich context parameter c which includes the common ground C_c , the interlocutors s_c and a_c , the utterance time t_c , and the set of discourse referents D_c . Salience is then defined as in (44) in terms of an attention relation: $att(i, a_c, t_c) \leftrightarrow a_c$ is attending to i at t_c .⁵¹

(44) Given a context c with speaker s_c and addressee a_c and a discourse referent $i \in D_c$:
 $sal(i, c) \leftrightarrow att(i, s_c, t_c) \wedge C_c$ entails that $att(i, a_c, t_c)$

- i. A discourse referent i is salient in c iff the speaker is attending to i at t_c and the common ground entails that the addressee is too attending to i at t_c . (Barlew 2014: 632–633)
- ii. Attention capture occurs due to the factors listed in (43).

(Salience is given an extended interpretation in degree-semantic terms in §2.3.3.1.)

2.3.1. Bulu Salient Article

Barlew shows that the Bulu Salient Article *-tè* is restricted precisely to discourse referents that the addressee attends to.

Thus, in (45a) where the Salient Article is used, the referent of *kàlàtā tè* ‘the book’ is important to both speaker and addressee, and this ensures that it is attended to in the context involving a robber, making it salient (top-down attention capture). (The context

⁵¹ From this we may also define the set of salient discourse referents:

(i) $Sal_c =_{def} \{i \in Dom(g) : sal(i, c)\}$

It is clear that $Sal_c \subseteq Dom(g)$, i.e., it is a subset of the set of familiar discourse referents.

suggests an implicit QUD “Is the book safe?” which, based on the discussion above, guarantees the discourse referent’s salience.) In (45b), on the other hand, the addressee is uninterested in the book even though we may assume that the book is familiar to him, and the Salient Article is infelicitous since the addressee cannot be assumed to attend to the discourse referent. Note that in both contexts, the target sentence is the first sentence in the discourse, there is no immediately preceding linguistic context.

- (45) a. [Andung and Abondo have a special book that is a family heirloom passed down from Andung’s mother. It has family genealogy written inside it. They always keep it on the nightstand beside their bed. One day when they come home, they find their house has been broken into. When they come to the nightstand, they see that the book is gone. They exchange a glance, and then Andung says:]

kálàtà tè à nè ndzáńán

book TE PN₁ COP missing

‘The book is missing.’ (Barlew 2014: 626)

- b. [Minimally different from (a), except that Andung is speaking to her teenage son, who does not care about genealogy or family heirlooms.]

#kálàtà tè à nè ndzáńán

book TE PN₁ COP missing

Intended: ‘The book is missing.’ (Barlew 2014: 626)

In (46), the Salient Article may only be used in the context where the referent is made perceptually prominent (46b) by suddenly entering the addressee’s vision (bottom-up attention capture). It is inappropriate in the context where there is no particular perceptual prominence to the referent (46a).

- (46) a. [Abondo is sitting on a bus when a man he does not know sits down beside him. The man says:]

#vjǎn tè wó fài dón

sun TE PN₁₁ shine today

Intended: ‘The sun is bright today.’

- b. [Identical to (46a), except that the stranger first opens the window shade on the bus, letting in sunlight.]

vjǎn tè wó fài dón

sun TE PN₁₁ shine today

Intended: ‘The sun is bright today.’

Barlew further argues that the Salient Article does not require maximal salience, as personal pronouns do on Roberts’ account, but rather requires any (positive) level of salience with informational uniqueness. His argument is based on examples modelled after Roberts’ ‘salience without uniqueness’ context (41). The Salient Article, as it turns out, is infelicitous in this kind of a context (47a), just like the English Definite Article and the Akan Definite Article (39), but unlike personal pronouns (Barlew 2014: 630–631). Naturally, the issue disappears if one of the men is more salient as in (47b)

(47) (adapted from Barlew 2014: (9))⁵²

- a. [I say to you: “Yesterday I saw two men and a woman at the farm. One of the man had black hair. The other man had white hair. The man that had black hair was white. The man that had white hair was black.” Then I continue:]

#**fām tè** è mbó è kóbò fùlàsí.

man TE PN₉ AUX PN₉ speak French

Intended: The [black haired] man was speaking French.

- b. [I say to you: “Yesterday I saw two men and a woman at the farm. One of the man had black hair.” Then I continue:]

fām tè è mbó è kóbò fùlàsí.

man TE PN₉ AUX PN₉ speak French

‘The [black haired] man was speaking French.’

According to Barlew, this is because both referents in (47a) are salient due to being recently introduced, which gives rise to a violation of informational uniqueness among salient referents. On the other hand, personal pronouns require maximal salience, so no problem of this kind arises for them since the more recently mentioned discourse referent is the most salient in this kind of example. (In Chapter 5, I will argue that the Kazym Khanty Salient Article differs crucially from the Salient Article of Bulu in that the Kazym Khanty Article marks the maximally salient referent which is apparent in particular in ‘salience without uniqueness’ contexts.)

⁵² I corrected minor inconsistencies in glosses between (47a) and (47b).

Finally, Barlew argues that the Bulu Salient Article is distinct from the ANAPHORIC ARTICLES described by Schwarz and Arkoh and Matthewson (Barlew 2014: 634–636), although he does not provide an in-depth comparison. Part of his argument consists in the observation that neither assumption about what kind of familiarity (weak or strong) an anaphoric article requires would yield the correct distribution of the Salient Article in his examples. For example, if the Salient Article merely required weak familiarity, it should presumably be available in both (45a) and (45b). On the other hand, if it required strong familiarity, it should be barred in both cases, Barlew suggests. Thus, only a salience requirement correctly captures the observed distribution.

Using the definition of salience from (44) above, Barlew’s analysis of the Salient Article is presented in (48), reformulated in a static manner for the sake of comparability with the other article types discussed above. The Salient Article takes a predicate P , presupposes salience and uniqueness among salient discourse referents that satisfy P , and returns the unique salient discourse referent i that satisfies P (Barlew 2014: (13)).

$$(48) \quad \llbracket -t\dot{e}_i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} : \\ \begin{array}{ll} \text{i.} & \text{sal}(i, c) \wedge P(g(i)) \wedge \\ \text{ii.} & \forall k \in D_c [\text{sal}(k, c) \wedge P(g(k)) \rightarrow k = i]. \end{array} \\ g(i)$$

According to (48), the Bulu Salient Article presupposes (i) that the index (representing a discourse referent) that it carries is salient and that the individual corresponding to this index satisfies the NP denotation and (ii) that there is no other salient index with these properties. If this presupposition is satisfied, (48) returns the individual corresponding to this discourse referent.

I believe the same result can be achieved with a simpler semantics that is parallel to the STRONG ARTICLE denotation from (34), namely with the semantics in (49).

(49) SIMPLIFIED DENOTATION FOR THE BULU SALIENT ARTICLE

$$\llbracket -t\dot{e}_i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} : \text{sal}(i, c) \wedge \exists! x_e [P(x) \wedge x = g(i)]. g(i)$$

The only difference from the STRONG ARTICLE here is the salience presupposition. The ‘uniqueness-among-the-salient’ requirement can be derived pragmatically from Grice’s Maxim of Manner in the same way as we did for the STRONG ARTICLE (36). The difference here would be that we are only concerned with salient discourse referents that

satisfy the NP denotation now. Non-salient strongly familiar discourse referents are, in Roberts' words, 'screened out', because the Salient Article presupposes salience of the discourse referent; analogously to how weakly familiar discourse referents were screened out for the purposes of STRONG ARTICLE anaphora resolution

2.3.2. *Definite typology so far and weak vs. strong familiarity*

The benefit of the alternative denotation in (49) is that it highlights how the requirements imposed by the Salient Article are essentially a subset of the STRONG ARTICLE requirements, which are themselves a subset of the WEAK ARTICLE requirements. This is represented in Figure 3.

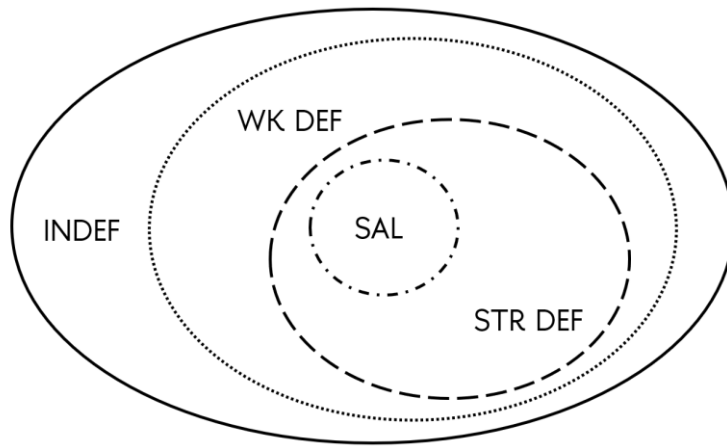


Figure 3. Relation between the SALIENT ARTICLE (SAL) and the other definites (based on Figure 2)

Interestingly, if we do adopt the modified SALIENT ARTICLE denotation in (49), its presupposition is strictly stronger than the simplified STRONG ARTICLE's presupposition in (34). So, following Heim (1991), we predict that in a single determinacy system a SALIENT ARTICLE would outcompete a STRONG ARTICLE in contexts with a salient discourse referent due to *Maximize presupposition!*. I believe that the same is true under Barlew's original analysis (48). Recall that the WEAK ARTICLE and the STRONG ARTICLE are not subject to MP, as discussed under (33). I believe that the same considerations hold if we consider the SALIENT ARTICLE vs. the WEAK ARTICLE: they should not be competitors for the purposes of MP, since the SALIENT ARTICLE's presupposition is too different from the presupposition of semantic uniqueness. However, they may be expected to be in a laxer pragmatic competition in style of (Gundel et al. 1993; see below). Finally, the

INDEFINITE ARTICLE, having no presuppositions, is theoretically an MP competitor for the SALIENT ARTICLE, just as for the two other definite types.

Whether these considerations are indeed the case in the world's languages remains to be seen in future work. As far as I am aware, there are no other languages besides Bulu which realize the SALIENT ARTICLE gram type in the current determiner typology, and Bulu does not have other articles (Barlew 2014).

We may also add personal pronouns to Figure 3 by adding a smaller circle inside SAL, since personal pronouns presuppose (relative) maximal salience among possible referents for this pronoun (40) which entails proper salience.⁵³ It will be important to distinguish relative maximal salience, which pronouns require, from absolute maximal salience, which I define formally in §2.3.3.1.

As discussed just now, salience entails strong familiarity which itself entails weak familiarity. Thus, the requirements imposed by the definites surveyed so far form the scale in (50), based on (Roberts 2003; Barlew 2014). ' $x > y$ ' should be read as ' x delineates a set of contexts which is a superset of the contexts delineated by y '. Assuming the Roberts-style approach to WEAK and STRONG definites, as sketched above and discussed below, and assuming that each position in (50) also presupposes uniqueness in the relevant sense (possibly, pragmatically derived (36)), we get the dynamic equivalent of Figure 3 with maximal salience added (presented graphically as Figure 6 at the end of the chapter)⁵⁴.

(50) THE SALIENCE-FAMILIARITY SCALE (first version)

weak familiarity > strong familiarity > salience > maximal salience

(In a more detailed scale, 'maximal salience' should be divided into 'relative maximal salience > absolute maximal salience', see (61).)

I would like to suggest that the SFS is typologically relevant, showing how the distributions expected from idealized restrictions on discourse referents at different points of the scale relate to each other. Testing this, however, is a task for future work, since the

⁵³ This is done in Figure 6 at the end of this chapter.

⁵⁴ Although, the Roberts-style approach may problematically make WEAK and STRONG ARTICLES MP-competitors, see fn. 33.

current work is primarily concerned with Northern Khanty data which does not allow us to do this.

Note that I am assuming here that the Bulu Salient Article requires strong familiarity together with salience, not weak familiarity. And I tacitly assumed the same for the Akan Definite Article in §2.2.1., contrary to the assumptions of Barlew, Arkoh and Matthewson, and Schwarz.

For the Akan Definite Article (or its STRONG ARTICLE brethren in Germanic) it is easy to see why weak familiarity by itself is not appropriate. As Roberts (2003) has shown for English (see §2.1 above), weak familiarity, taken together with informational uniqueness, covers a wide range of uses including global uniqueness and ‘novelty + semantic uniqueness’ uses (e.g., with superlatives). However, in Akan, Standard German, and Fering, strong article definites cannot occur in these contexts, precisely because the referents there are ‘not familiar enough’. This is not what we would expect if STRONG ARTICLES required mere weak familiarity (or ‘hearer-oldness’ in Prince’s (1992) roughly equivalent terms, which Arkoh and Matthewson use).

The reason that Arkoh and Matthewson argue for a weak familiarity requirement is that there are cases, in both Akan and Standard German, where there is no linguistic antecedent but a STRONG ARTICLE is used or at least preferred (see the discussion in (Arkoh & Matthewson 2013: §2.4)). What I would like to suggest instead is that in these cases strong familiarity is accommodated due to the entity in question being known to the interlocutors and/or relevant in the context. This is parallel to how existence of an appropriate discourse referent is accommodated with novel unique DPs yielding weak familiarity, as Roberts has argued (2003: 302ff.). Exactly what it means to accommodate strong familiarity depends on its definition. I suggest a slightly broader definition below based on the distinction between the context prior to the utterance and the context of interpretation of the utterance (Stalnaker 1998; von Stechow 2008). In my proposal, a discourse referent is strongly familiar if it is represented in prior context. Accommodation of strong familiarity is accommodation of existence to the prior context. (This is different from accommodation for weak familiarity, since weak familiarity is about the context of interpretation, not the prior context; see below.)

While I forgo a demonstration of how Roberts-style accommodation of strong familiarity could handle the problematic cases discussed by Arkoh and Matthewson,⁵⁵ I do believe that this is the way to go for research on STRONG ARTICLES. Roberts has independently shown that accommodation plays a significant role in the actual use of definites.⁵⁶ At the same time, weak familiarity massively overgenerates for STRONG ARTICLES (cf. Barlew 2014: 636). So, it seems that we have no choice but to maintain that STRONG ARTICLES require strong familiarity and that a pragmatic rescue operation, accommodation, applies where strong familiarity is not met yet a STRONG ARTICLE is used. Future work is required to demonstrate how this happens and how such accommodation is constrained. At the very least, it seems clear that in contexts where even weak familiarity of the discourse referent in question requires accommodation of existence (e.g., with novel superlatives), strong familiarity cannot be accommodated, since that would require ‘stretching’ our context even more than for weak familiarity.

Ditto for the Bulu Salient Article. Barlew frames his discussion in terms of weak familiarity. In particular, his set of discourse referents D_c is really the set of referents that are weakly familiar in c . The reason for this seems to be that in most of the examples he discusses the target DP does not have a proper linguistic antecedent (e.g., the book example (45) above). Yet the attention-based definition of salience that Barlew provides (44), repeated in shortened form in (51), requires the speaker to attend to a discourse referent and the common ground to entail that the addressee also attends to it.

(51) $sal(i, c) \leftrightarrow att(i, s_c, t_c) \wedge C_c$ entails that $att(i, a_c, t_c)$

Surely, if the interlocutors are attending to a discourse referent, it is strongly familiar. It is hard to see how it could be otherwise. (The same point in slightly different terms is made by Gundel et al. (1993) based on their Givenness Hierarchy, discussed below in §2.3.3.)

The issue here, I believe, is the idea that strong familiarity requires there to be a proper linguistic antecedent (Roberts 2003: 297, 304). The cases where this is so provide

⁵⁵ In large part because I don’t have access to primary data. Moreover, this is beyond the scope of this thesis which is first and foremost a study of the Kazym Khanty determiner system where these issues are not directly relevant.

⁵⁶ See (Beaver & Zeevat 2007; von Stechow 2008) on the necessity of accommodation more generally.

the clearest examples of strong familiarity. And these are the canonical cases where STRONG DEFINITES and personal pronouns are licensed. However, it is also clear that neither of these two types of definites are restricted to such cases.

This observation is already found in (Karttunen 1968; Heim 1982; cf. Roberts 2003). In Heim's (1982: 201) words:⁵⁷

Does the [context] also reflect what is familiar by contextual salience [*sic*]? So far we have not assumed it does, but let us make the assumption now. Let us assume that anything that is familiar for either reason [previous mention or contextual salience] is represented by a [discourse referent in the context]. ... For instance, if halfway through a conversation between A and B a dog comes running up to them and draws their attention, then that event presumably makes the [context assignment g_c] increase by a [discourse referent]. ... Suppose now that ... A says [(52)]. This sentence has a reading under which "it" carries the referential index 7.

(52) It₇ is going to bite. (Heim 1982: 201)

Thus, it seems that if we are willing to allow strong familiarity of the discourse referent 7 in (52), we ought to also allow strong familiarity for discourse referents in the Bulu examples where the Salient Article is used. Thus, I contend that STRONG ARTICLES and SALIENT ARTICLES require (at least) strong familiarity. (Incidentally, (52) also shows that (relative maximal) salience may too result from extra-linguistic events since the pronoun *it* is felicitous in this example.)

An important difference between weak familiarity and strong familiarity is that weak familiarity, as defined by Roberts (16), needs only entailment by the common ground. This means that a weakly familiar discourse referent isn't necessarily included in the domain (*Dom*) of the assignment function g prior to the utterance in question, but it is introduced once its existence is relevant to the conversation.⁵⁸

⁵⁷ Corrections and comments in square brackets are mine — SM.

⁵⁸ Cf. von Fintel's (2008: §2) elucidation of Stalnaker's (1998) discussion of the timing of accommodation, specifically in the case of first-person pronouns. To quote Stalnaker (1998: 8) with von Fintel's emphasis:

Suppose Phoebe says "I saw an interesting movie last night." To determine the content of her remark, one needs to know who is speaking, and so Phoebe, if she is speaking appropriately, must be presuming that the information that she is speaking is available to her audience – that it is shared information. But she need not presume that this information was available before she began to speak. *The prior context that is relevant to the interpretation of a speech act is the context as it is changed by the fact that the speech act was made, but prior to the acceptance or rejection of the speech act.*

For strong familiarity, Roberts proposed that the discourse referent needs to be introduced by overt linguistic means, an antecedent DP. I shed doubt on this idea above. Instead, I propose that strong familiarity amounts to the requirement that the discourse referent is included in the domain of g (55) prior to the utterance in question. This may be due to prior mention (53a) or to events in the utterance situation (53b), as in the dog example from Heim (52).

(53) REFERENT INTRODUCTION

Referents may be introduced:

- a. linguistically,
- b. by non-linguistic events in the utterance situation,
- c. by entailment from the common ground when relevant (Roberts 2003: 305), or
- d. by accommodation.

Ideally, we would want to eliminate accommodation from the list in (53), subsuming all such cases under (53c). This requires us to have a theory of discourse referent relevance which I currently do not have. For now, I stick with my speculative remarks about accommodation, trying to clarify what is possible given the scope of this thesis.

To spell out the proposed familiarity distinction, it would be useful to distinguish three types of contexts in the style of dynamic semantics as in (54). (The output context is added for the sake of illustration.)

(54) THREE TYPES OF CONTEXTS IN A DYNAMIC INTERPRETATION OF AN UTTERANCE

Let a context c be a tuple that includes the common ground C_c , the interlocutors s_c and a_c , the utterance time t_c , and the assignment function g_c : $c = \langle C_c, s_c, a_c, t_c, g_c \rangle$. When an utterance is interpreted, three types of contexts may be distinguished:

- 1) Context prior to the current utterance: c'
- 2) Context relative to which the utterance is interpreted and to which the utterance's denotation is applied (in a dynamic semantics, fn. 59): c_0
 - a. simple case: $c_0 = c'$
 - b. cases of accommodation: c_0 is the result of updating c' with all information that is relevant and/or needs to be and can be accommodated for interpretation to proceed smoothly

- in dynamic semantic notation:⁵⁹

$$c_0 = c'[\text{accommodated information}]$$

- 3) Output context, the result of updating c_0 with the utterance's denotation ϕ : $c_2 = c_0[\phi]$

This allows us to distinguish weakly and strongly familiar discourse referents along the lines above, using the notation for contexts from (54).

(55) WEAK VS. STRONG FAMILIARITY

- a. A strongly familiar discourse referent i is included in the domain of the assignment function prior to its mention in the current utterance.

For a given assignment $g_{c'}$, i is strongly familiar iff $i \in \text{Dom}(g_{c'})$

- b. A weakly familiar discourse referent i is entailed to exist by the common ground but is not necessarily in $\text{Dom}(g_{c'})$ prior to its mention in the current utterance.

Thus, weak familiarity includes two cases:

- 'Narrow' weak familiarity: $i \notin \text{Dom}(g_{c'})$ and $i \in \text{Dom}(g_{c_0})$
- Strong familiarity: $i \in \text{Dom}(g_{c'})$ ⁶⁰

The case of narrow weak familiarity is found when the discourse referent has to be accommodated. This is possible if either (i) its existence already follows from the common ground but the referent only now became relevant, so it was not represented in prior context c' and a representation for it is created in c_0 (i.e., an index is added to $\text{Dom}(g_{c_0})$), or (ii) its existence follows from the common ground together with some proposition(s) to be accommodated. An important requirement is that accommodated information is retrievable and plausible as discussed by Roberts (2003), see (21), as well as (Beaver & Zeevat 2007; von Stechow 2008). As it is accommodated, an index for the referent in question is added to $\text{Dom}(g_{c_0})$, making it familiar.

⁵⁹ Here, I'm using the notation of dynamic semantics where declarative sentences are taken to denote context change potentials (CCPs), instead of truth values, and the result of uttering a sentence that denotes a CCP ϕ in a context c is the update of c with ϕ , formally: $c[\phi]$.

This puts the conversation in a new context, call it c_2 , so we have: $c_2 = c[\phi]$, c_2 is the result of updating c with ϕ .

(See Coppock & Champollion 2024 for a user-friendly introduction; relevant classical models of this kind include Heim 1982/2011; Groenendijk & Stokhof 1991; Muskens 1996.)

⁶⁰ See (Martin 2012) for another, more explicit proposal on the weak vs. strong familiarity distinction that is similar in spirit but very different in details.

The first case may be illustrated with (56). Like any protracted event, a war has a unique beginning, this will generally follow from any common ground, hence, if there is a strongly familiar war in the context, its beginning will be weakly familiar (and informationally unique) by entailment from the common ground.

(56) [Some specific war is strongly familiar.]

I remember **the beginning** of the war very well. (Roberts 2003: (20); based on Poesio & Vieira 1998)

The second case is illustrated in (57). The fact that Bill was on a date last night needn't be common ground, it may be accommodated which will make the woman he went out with weakly familiar, thus licensing the definite article according to Roberts.

(57) What's wrong with Bill? Oh, **the woman** he went out with last night was nasty to him. (Roberts 2003: (27); based on Poesio & Vieira 1998)

The difference of (55) from Roberts' proposal and with the subsequent literature is in where the line is drawn in (53) with respect to how the discourse referent needs to be introduced in order to be strongly familiar (cf. Roberts 2003: (29)). I assume that extralinguistic events may have the same effect on the assignment, and be treated the same as, proper linguistic introduction of discourse referents. The crucial difference is then spelled out as whether the discourse referent was in $Dom(g_{c'})$ prior to the utterance (strong familiarity) or not (weak familiarity). I suggest that this interpretation of strong familiarity may provide the correct boundary to delimit STRONG ARTICLE definites. (See (Schwarz 2019: §4.2) for evidence of subtle crosslinguistic variation in the use of STRONG ARTICLE definites.)

Note that (54) requires our semantic model to make use not only of the input and the output context, as in (Groenendijk & Stokhof 1991) and much other dynamic semantic work, but also of the context prior to the utterance. According to the above hypothesis, the choice between Germanic Weak and Strong Articles will hinge upon whether the discourse referent is in $Dom(g_{c'})$ or not, while the rest of the utterance is interpreted relative to a different context, namely, c_0 . This complicates our dynamic semantic system. Future work is needed to determine whether this extension is justified and whether the above hypothesis is on the right track. If this will turn out to be so, we will have answered Coppock's (2022) skepticism regarding the strong vs. weak distinction.

It is also interesting whether there are other phenomena for which the distinction between c' and c_0 is relevant. One possible candidate is the distinction between projective implications that impose a strong contextual felicity constraint and those that do not in Tonhauser et al.'s (2013) classification of projective implications. For instance, according to Tonhauser and colleagues, personal pronouns in English and Paraguayan Guaraní (< Tupí-Guaraní; Paraguay) impose a strong contextual felicity constraint with respect to their familiarity presupposition. This means that an utterance of a personal pronoun in these languages is infelicitous in a context where a suitable referent has not been introduced in prior context — a clear case of strong familiarity.

On the other hand, many other kinds of expressions do not impose a strong contextual felicity constraint with respect to their projective implications. For example, this is the case for non-restrictive relative clauses, nominal appositives, predicates meaning 'know', and constructions meaning 'to stop doing X'. While for the former two this fact is uncontroversial (Potts 2005; AnderBois & Brasoveanu & Henderson 2015), the latter two are classical examples of presupposition triggers, expressions that are canonically thought to impose something just like a strong contextual felicity constraint with respect to the presuppositions they bear. Tonhauser et al. have argued that predicates meaning 'know' and constructions meaning 'to stop doing X' in fact do not impose a strong contextual felicity constraint in English and Paraguayan Guaraní, which distinguishes them from personal pronouns in these languages.

It seems correct that the difference between weak familiarity and strong familiarity corresponds to whether the definite imposes a strong contextual felicity constraint with respect to the familiarity presupposition (strong familiarity) or not (weak familiarity). An interesting topic to explore in the future is whether the above-proposed model with three contexts (54) can also account more generally for the difference between Tonhauser et al.'s (2013) classes A and D projective implications, which impose a strong contextual felicity constraint, vs. classes B and C, which do not. (See the original paper for details.) Under my current proposal, implications from classes A and D are unaccommodatable: they need to be entailed or implied by the common ground of the prior context c' , otherwise the utterance will be infelicitous, which is what Tonhauser et al.'s data show.

2.3.3. Comparison with the Givenness Hierarchy and relative vs. absolute salience

An important precursor to the Salience-Familiarity Scale introduced above, is Gundel, Hedberg, and Zacharski's (1993) Givenness Hierarchy (GH) in (58). Below the GH, I provide English and Mandarin expressions which conventionally signal each possible cognitive status of the referent (see Table 4 below for Mandarin glosses).

(58) THE GIVENNESS HIERARCHY (Gundel et al. 1993: 275)⁶¹

in focus > activated > familiar^{GH} > uniquely identifiable > referential > type identifiable

{it}	{HE, that, this, this N}	{that N}	{the N}	{indefinite this N}	{a N}
{Ø, tā '(s)he/it'}	{TĀ '(s)he/it', zhè 'this', nèi 'that', zhè N}		{nèi N 'that N'}		{yī N 'one/a N', Ø N}

Gundel et al. suggest that different types of nominal expressions conventionally signal different cognitive status of their referent in language use, enhancing reference resolution for the addressee (cf. Roberts 2003). Each status in (58) provides the necessary and sufficient condition for the use of the corresponding form(s). Gundel et al. demonstrate this for English and four other languages based on a corpus study: Mandarin, Japanese, Russian, and Spanish (see Gundel et al. 1993: §4).⁶²

The statuses are defined as follows:

(59) GH STATUS DEFINITIONS (Gundel et al. 1993: §2)

- a. **In focus:** the referent is at the current center of the addressee's attention;
- b. **Activated:** the referent is represented in the addressee's current short-term memory;
- c. **Familiar^{GH}:** the referent is represented in the addressee's memory (either long-term or short-term if it has recently been mentioned or perceived);
- d. **Uniquely identifiable:** the addressee can identify the speaker's intended referent on the basis of the nominal alone;

⁶¹ I use "familiar^{GH}" with the superscript for the Givenness Hierarchy status to distinguish it from (formal semantic) familiarity of my framework.

⁶² See also (Gundel et al. 2010) for an investigation of four additional languages within the GH framework, that largely confirms and extends the results of the 1993 study.

- e. **Referential**: the speaker intends to refer to a particular object, and the addressee either needs to construct or to retrieve a representation of the speaker's intended referent by the time the sentence has been processed;
- f. **Type identifiable**: the addressee is able to access a representation of the **type** of object described by the expression.

What is interesting is that expressions are not restricted to the status that they conventionally signal in this theory, but they may also be used if the referent has any status to the left of the signalled one on the hierarchy, because statuses in GH stand in an implicational relation: a status on the left implies all the statuses on the right. (Cf. my discussion of the SFS in §2.3.2.)

This is illustrated in Table 3 and Table 4 which show how different types of nominals, including pronouns, NPs with demonstratives and articles, and bare nouns in Mandarin, are distributed in corpora of English and Mandarin respectively according to status on the Givenness Hierarchy.⁶³

	IN FOCUS	ACTIVATED	FAMILIAR	UNIQUE	REFERENTIAL	TYPE	TOTALS
<i>it</i>	214	1					215
<i>he</i>		1					1
<i>this</i>		15					15
<i>that</i>	1	17					18
<i>this</i> N	1	11					12
<i>that</i> N		10	7				17
<i>the</i> N	30	95	47	108			280
indefinite							
<i>this</i> N					1		1
<i>a(n)</i> N					41	55	96
TOTALS	246	150	54	108	42	55	655

Table 3. Distribution of English forms according to highest status on the GH (Gundel et al. 1993: Table 3). Statuses in columns correspond to (59). English forms are: unstressed pronoun (*it*), stressed pronoun (*HE* with caps indicating stress), substantive demonstratives *this* and *that* (without an NP complement), demonstrative descriptions

⁶³ Gundel et al. (1993) annotated corpus data from a variety of written and spoken sources and narrative film descriptions. See (Gundel et al. 1993: fns. 20, 21) for details.

with *this* and *that* (with an NP complement), definite descriptions with *the*, ‘indefinite’ *this* N, and DPs with the Indefinite Article *a(n)*.

	IN FOCUS	ACTIVATED	FAMILIAR	UNIQUE	REFERENTIAL	TYPE	TOTALS
$\emptyset$	25	1					26
<i>tā</i>	40						40
<i>zhè</i>		2					2
<i>nèi</i>							
<i>zhè</i> N	12	26	1				39
<i>nèi</i> N	1	7	2				10
<i>yī</i> N					17	2	19
N	12	17	14	49	2	10	104
TOTALS	90	53	17	49	19	12	240

Table 4. Distribution of Mandarin forms according to highest status on the GH (Gundel et al. 1993: Table 2). Statuses in columns correspond to (59). Mandarin forms are: pro-dropped zero pronominal ($\emptyset$), pronoun *tā* ‘(s)he/it’⁶⁴, substantive demonstratives *zhè* ‘this’ and *nèi* ‘that’, demonstrative descriptions with *zhè* and *nèi*, DPs with *yī* ‘one/a’, bare nominals.

Consider the English data first. We see that familiar^{GH} referents are referred to both using demonstrative descriptions with *that* (*that book*), which conventionally signal familiar^{GH} status, and using simple definite descriptions with *the* (*the book*), which conventionally signal a lower status, unique identifiability. Even more forms are used for activated referents in English. These include: unstressed personal pronouns, stressed personal pronouns (*HE* where caps indicate stress), substantive demonstratives *this* and *that* (without an NP complement), demonstrative descriptions with *this* and *that* (with an NP complement), and, again, definite descriptions with *the*. Overall, we see that there is no (grammatical) competition whereby a stronger / more semantically marked expression (say, a demonstrative description) rules out the use of weaker / less semantically marked expression (a simple definite description). There is no *Maximize presupposition!* signature here, unlike the competition of definite expressions with indefinite descriptions where the availability of a definite rules out the indefinite, exactly as in (Heim 1991).

⁶⁴ Stressed *TĀ* is mentioned but is not discussed by Gundel et al.

Roughly, the same is true of the Mandarin data modulo the differences of the Mandarin determinacy system from the English one. Thus, Mandarin has pro-drop, making zero pronominals ($\emptyset$) an option for referents in focus. Mandarin also does not have obligatory articles, so bare nouns are found with all statuses (cf. English definite descriptions). But Mandarin demonstrative descriptions with *nèi* ‘that’ and DPs with the numeral *yī* ‘one/a’ are functioning somewhat like English definite descriptions and DPs with *a(n)*, so Gundel et al. suggest that these two Mandarin forms signal unique identifiability and type identifiability respectively, just like their English counterparts.

Thus, we observe an important pattern. Expressions of a single determinacy system conventionally signalling different GH statuses are not necessarily in complementary distribution with each other. The case where we do find complementary distribution is when we consider strictly indefinite expressions with strictly definite ones. Here, as discussed in §2.1, we may appeal to *Maximize presupposition!* (Heim 1991) to explain why Mandarin *yī* N descriptions are excluded with unique referents and with referents of higher GH statuses.

For analyses of some of the patterns observed in Table 3 and Table 4, see the original paper and see (Gundel et al. 2010) for extensions. What’s interesting for us is how the GH statuses (58) correspond to the Salience-Familiarity Scale (SFS) (60), repeated from (50).

(60) THE SALIENCE-FAMILIARITY SCALE (first version)

weak familiarity > strong familiarity > salience > maximal salience

The top four positions of the GH can be argued to correspond to (the inverse of) the four positions of the SFS. There is also a near identity between the top two GH positions and the bottom two SFS positions. Gundel and colleagues define ‘in focus’ as ‘at the center of current attention’ and ‘activated’ as ‘represented in current short-term memory’. I believe it is not a stretch to say that Barlew-Roberts salience entails activation and vice versa. Both appear to be approaches to the same phenomenon of attention-modulated salience. While this notion, and its linguistic expression(s) are often discussed based on intuitive distinctions in the literature, some work towards developing an explicit and predictive operationalization would be useful. As a first step, I attempt such a clarification in the next section, but more work is needed.

As for referents that are in focus, they are maximally salient. In the next section, I lay the groundwork for distinguishing two different types of maximal salience.

2.3.3.1. *Degree-based relative vs. absolute salience*

Recall that according to Roberts, personal pronouns in English presuppose **relative** maximal salience (42). Gundel et al. make a distinction, not found in Roberts' work, between stressed pronouns and unstressed pronouns and assign them different statuses. Stressed pronouns, just like substantive demonstratives, signal mere activation (salience in our terms), while unstressed pronouns (and zero pronominals in languages with pro-drop) signal that the referent is in focus. It is not entirely clear to me how to cash out this distinction in terms of our framework. And at the same time, whether stressed personal pronouns really presuppose the same discourse referent status as substantive demonstratives and demonstrative descriptions with *this* in English (all of them signal activation in terms of the GH).

Intuitively speaking, the 'in focus' status of unstressed personal pronouns and zero pronouns (the referent is at the current center of attention) seems to be stronger than relative maximal salience, since the latter only requires that the referent be more salient than other potential referents of this DP. Relative maximal salience is compatible with it not being at the very center of the current attention, i.e., not being maximally salient in an absolute sense.

I would like to suggest that we ought to distinguish relative maximal salience and absolute maximal salience, as in (61), and that the GH 'in focus' status corresponds to absolute maximal salience.

(61) THE SALIENCE-FAMILIARITY SCALE (main version)⁶⁵

weak familiarity > strong familiarity > salience > relative maximal salience >
absolute maximal salience

To formalize absolute maximal salience, let us conceptualize salience as a totally closed scale in the sense of (Kennedy & McNally 2005) in analogy with adjective like *full/empty*, *open/closed*, *visible/invisible*.⁶⁶ Degrees of salience are assumed to be

⁶⁵ The reason why (61) is called the main version of the SFS is that I suggest a possible extension including partitive specificity in §3.1.2, which, however, is not very important for this study.

⁶⁶ I thank Kate Kozlova (p.c.) for introducing me to Kennedy and McNally's work.

isomorphic to the real numbers between 0 and 1. 1 corresponds to maximal salience, meaning that the referent is at the current center of attention (GH ‘in focus’), and 0 corresponds to no salience, the referent is not in the field of attention. If a referent is salient, then its salience degree is larger than 0 (thus, salience is a minimum-standard gradable property like *awake*, *visible*, *open*, *bent*). This is formalized in (62) using the degree-semantic framework of (Kennedy & McNally 2005).

(62) DEGREE-BASED SALIENCE

Given a context c with speaker s_c and addressee a_c and a discourse referent $i \in Dom(g_c)$:

a. **Salience measure function:** $m_{sal(c)}(i) = d$, the degree of salience of discourse referent i in context c is equal to d , where d is a real number in $[0; 1]$.

b. **Salience:**

$$sal(i, c) = [m_{sal(c)}(i) > 0],$$

which is equivalent to $[att(i, s_c, t_c) \wedge C_c \text{ entails that } att(i, a_c, t_c)]^{67}$

c. **Relative maximal salience** for given descriptive content P (e.g., features of a pronoun):

$$rel.max.sal(P, i, c) = \forall j \in Dom(g_c) [P(g_c(j)) \wedge m_{sal(c)}(j) \geq m_{sal(c)}(i) \rightarrow j = i]^{68}$$

d. **Absolute maximal salience:**

$$abs.max.sal(i, c) = [m_{sal(c)}(i) = 1] \quad (\text{equivalent to GH ‘in focus’})$$

Based on these definitions, we can illustrate simple cases where the three bottom statuses of the Salience-Familiarity Scale (61) come apart.

Figure 4 presents the degree of salience of three discourse referents i, k, l in two scenarios. Let’s assume that i and k are dogs and l is a cat.

⁶⁷ Based on (44).

⁶⁸ The difference with the Roberts-based semantics of personal pronouns (42) is that I’m using here the salience measure function instead of Roberts’ relation $j \geq_{sal(c)} i$.

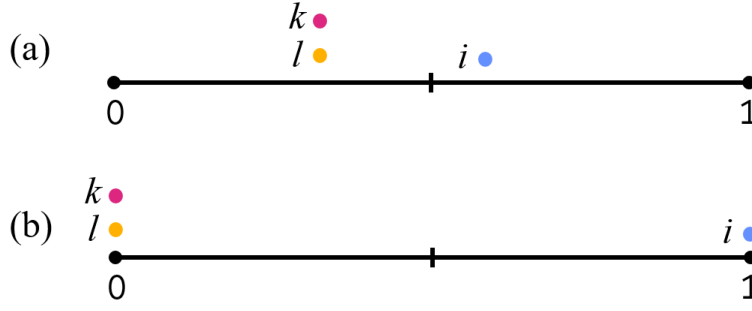


Figure 4. Degrees of salience for three discourse referents i, k, l in two hypothetical scenarios on the scale of salience

Consider first scenario (a). What do we predict in terms of which definite expressions may be used to pick up these referents? All three referents are salient (because their degree of salience exceeds 0). i is maximally salient relative to dog' (and relative to whatever descriptive content the personal pronouns of the language in question carry), although i is not absolutely maximally salient since $m_{Sal(c)}(i) < 1$. Relative maximal salience entails that a personal pronoun may be used to pick up i , as well as substantive demonstratives and demonstrative and definite descriptions, following Gundel et al.'s observations discussed above. Assuming that zero pronouns in Mandarin and unstressed personal pronouns in English require 'in focus' status / absolute maximal salience, they will not be appropriate devices to pick up i in this scenario. Thus, relative maximal salience and absolute maximal salience come apart here. We expect zero pronouns and unstressed personal pronouns to be available to refer to i in scenario (b) where it has absolute maximal salience.

k is not maximally salient relative to dog' in scenario (a) and will not be accessible for anaphoric pick-up by a definite DP based on a noun meaning 'dog'. Discourse referent l , on the other hand, is the most salient cat and so can be referred to using a definite DP based on 'cat'. The availability of a personal pronoun for l depends on whether 'cats' and 'dogs' differ in gender in the pronominal system in question, otherwise the personal pronoun will only be able to pick up i as the maximally salient referent. And, again, zero pronouns and unstressed personal pronouns are predicted to be unavailable for l .

The Bulu Salient Article (Barlew 2014), which presupposes salience and informational uniqueness (49), can be used to refer to l in this scenario, but not to i or k , since informational uniqueness among referents satisfying dog' is violated for both i and

k , even though i is more salient than k . Thus, it seems that based on our discussion the Bulu Salient Article is actually expected to have less options than personal pronouns (and demonstratives) regarding discourse referents available for pick up. This is because personal pronouns, presupposing relative maximal salience, can screen out less salient discourse referents (like k in Figure 4a) to pick up the relatively maximally salient discourse referent, while the Bulu Salient Article is simply barred because informational uniqueness is violated.

This means that the superset relation on the Salience-Familiarity Scale (61) only holds, if we do not take into account informational uniqueness relative to each status on the scale, as discussed when I first introduced the SFS (50). For instance, it is generally true that *salience* > *relative maximal salience*, that is there are more salient referents than there are maximally salient referents (relative to some P). However, if we compare definite expressions which presuppose salience + informational uniqueness with those presupposing relative maximal salience, the latter are predicted to have a wider distribution. (This is ignoring other potential constraints on their use, some of which are discussed by Gundel et al., comparing definite descriptions, demonstratives, and pronouns.)

2.3.3.2. *The SFS vs. the GH*

Coming back to relative vs. absolute maximal salience distinction, I suggested that it corresponds to the difference, put forward by Gundel et al. ((58), Table 3, and Table 4), between stressed pronouns and demonstratives, on the one hand, and unstressed and zero pronouns, on the other. In Chapters 3 and 5, I will argue that the Kazym Khanty Salient Article also requires relative maximal salience, providing the first example of a (non-demonstrative) definite determiner which requires this SFS status. As far as I know, there are no definite determiners which require **absolute** maximal salience. This status (equivalent to Gundel et al.'s 'in focus') seems to only be encoded by reduced forms such as unstressed and zero pronouns, which finds a natural functional explanation in that this discourse referent status requires the least descriptive content (possibly, none) to identify the referent since it's already at the center of attention (Gundel et al. 1993: §4.2; and references there).

If relative maximal salience corresponds to the GH 'activated' status, a question arises where do merely salient referents fit in. Judging by the definition of 'activated' in

terms of short-term memory (59b) and the definition of ‘familiar^{GH}’ (59c), ‘salient’ seems to correspond more closely to ‘activated’. But we already said that GH ‘activated’ is equivalent to SFS ‘relatively maximally salient’. It seems that here the GH is conflating the two statuses that we distinguished in the SFS. ‘Activated’, meaning ‘represented in the addressee’s short-term memory’, fits both salience simpliciter and relative maximal salience. I think there is a natural reason why there is no distinction of this sort on the GH. After all, current crosslinguistic research seems to have only identified one definite expression type that requires salience simpliciter, namely, the Salient Article of Bulu. Neither English, nor Mandarin, nor the other languages investigated by Gundel et al. (1993; 2010) have expressions marking salience simpliciter. Thus, on the empirical side, the ‘salience’ point on the Salience-Familiarity Scale (61) seems to be the most poorly motivated one, with only one exemplar encoding salience as its presupposition on the discourse referent of the DP. This is unlike the other four positions, which all seem to find solid empirical justification when we look beyond simple descriptions with articles and bare nouns to include demonstratives and pronouns, as Gundel et al. have shown. In §2.4.3, I will discuss the semantics of demonstratives in a little more detail, and the other subsections of §2.4 discuss definite presuppositions that do not fit either the GH or the SFS. Before moving on to that, let me clarify the relation between the remaining two statuses of the SFS to the corresponding GH positions.

As suggested above, ‘familiar^{GH}’ and ‘uniquely identifiable’ of the GH correspond to strong familiarity and weak familiarity in SFS terms respectively. The definition of unique identifiability (59d) seems exactly like a cognitive version of Roberts’ weak familiarity (16)/(55). If the addressee can identify the speaker’s intended referent on the basis of the nominal alone (which presumably means ‘the nominal as used in the current context, including pragmatic reasoning’), its existence is entailed by the common ground. (Given informational uniqueness,) the converse also holds.

However, the issue is not so simply resolved with the ‘familiar^{GH}’ status which stands for ‘represented (at least) in long-term memory’. This seems to be closer to weak familiarity than to strong familiarity (cf. Prince’s (1992) ‘hearer-old’ status), since strong familiarity requires the discourse referent to be represented in the current context (55), a stronger requirement clearly. This is how familiarity is understood in the SFS and in semantics more broadly, i.e., in its relation to the common ground / the context (recall

Roberts' definition (16)). While this is not the place to settle the issue of which definition of familiarity has the empirical upper hand, I can offer conceptual arguments for the common ground-based, semantic definition adopted in the current thesis.

The common ground-based definition has the benefit that it is easier to operationalize in formal terms and to integrate compositionally, since the common ground has played an important role in formal semantics for at least four and a half decades (Kamp 1981; Heim 1982) and there are many approaches on the market that achieve these two desiderata. The ability to state a compositional definition of a definite's semantics is crucial, since there is a range of phenomena that heavily implicate a compositional solution. These include, but are not limited to, indeterminate uses of definites and uses with *only* (Coppock & Beaver 2015), uses with superlatives and Haddock descriptions (Bumford 2017; Hackl 2019), donkey sentences and bishop sentences (Elbourne 2013; 2016), and many more. I explore one such case for the Kazym Khanty Salient Article in Chapter 5, laying the groundwork for a compositional account of its cooccurrence with demonstratives.

On the other hand, memory has not been integrated into formal semantic models, which would significantly complicate any attempt at a semantic analysis of definite determiners. It seems to me that all that is relevant to semantics memory-wise can be stated in terms of the common ground, which is why I maintain the semantic definition of (strong) familiarity in the remainder of this thesis.⁶⁹

Some Kazym Khanty data will take us in another direction as well, and so do further extensions to the definite typology we have sketched so far, discussed in the next section.

2.4. *Beyond the Salience-Familiarity Scale*

Schwarz (2019: §5) mentions some types of definites that do not fit the Salience-Familiarity Scale as proposed above (50).

For instance, according to Wespel (2008), Haitian Creole exhibits a split between bare nouns and DPs marked with the article *la/a* (derived from French *la*). The article is

⁶⁹ Svetlana Toldova (p.c.) suggests that the two definitions of familiarity (as well as of other statuses) may complement each other, so that the semantic contribution of a definite can be stated in SFS terms, while its pragmatic communicative effect beyond the confines of compositional interpretation is as the GH has it. It is of course an empirical and a theoretical question of whether one should pursue such an integration and how to implement it, both are beyond the scope of my work. I thank Svetlana Toldova for this promising suggestion.

used in all typical definite contexts, including anaphoric uses (63b), situational uniqueness (63a), global uniqueness. (The former two are illustrated here with a part-whole, uniqueness-based bridging example (63a) and a producer-product, familiarity-based bridging example (63b), following Schwarz.)

(63) (Wespel 2008: 114)

a. WEAK ARTICLE DEFINITE context

Yè, mwen viste yon vil provens. Meri a pi
yesterday I visited one town province. town.hall DEF more
wo ke legliz la
high than church DEF

‘Yesterday I visited a town in the province. The town hall was higher than the church.’

a. STRONG ARTICLE DEFINITE context

Eli te renmen liv la, e kounye a
E. PST love book DEF and now DEF
li vle rankontre otè a.
she wan meet author DEF

‘Eli loved the book, and now she wants to meet the author.’

The context where the bare form is used is with ‘functional descriptions’ (Löbner 1985; 2011), i.e., DPs where the head noun denotes a function from possessors to unique possessums, when the possessor is explicitly introduced (Wespel 2008: §4.5.2), like ‘south department’ in (64).

(64) (Wespel 2008: (40))

Nan **depatman sid** peyi Dayiti, jis avan ou pran Tiburon,
in department south country H. just before you take T.
nou jwenn on lokalite ki rele Lèzanglè
we join INDF place REL call L.

‘In the southern district of Haiti, near Tiburon, there is a place called Les Anglais.’

Schwarz (2009: 26ff.), following Wespel, discusses a potential analysis whereby the Haitian Creole Article in fact denotes the resource situation pronoun s_r of Situation Semantics (cf. Schwarz’s situation-semantic analyses above (31)-(32)). The resource

situation s_r is the situation relative to which the definite DP is interpreted. Specifically, in Situation Semantics, the uniqueness presupposition is restricted to hold specifically in s_r , which in informal terms is a familiar situation that contains the DP referent. The resource situation pronoun is proposed to be absent syntactically in the cases where the restriction of the DP is introduced by other means, as in (64) with an overt possessor (see Schwarz 2019 and Wespel 2008 for more examples).

Regardless of the specific analysis that we settle on for the Haitian Creole Article, it is clear that this marker cuts the cake of the definite semantic space (Figure 3) rather differently than the types surveyed so far, with the relevant distinction based not on the level of salience-familiarity but on the type of the head noun and the presence of overt restrictions.

2.4.1. Réyoné concept type-based determiner system

A more recent example, also from a French-based creole, comes from the work of Albers (2024) who discusses the determiner system of Réyoné, a French-related creole of the Indian Ocean. Albers shows that Réyoné exhibits a three-way split, distinguishing bare nouns, DPs with *lo*, and DPs with *-la*. The determiner *-la* that was described as a demonstrative in previous work (e.g., Bollée 2013).

Bare nouns are used with global and larger situation uniques.⁷⁰

- (65) La travay ziska {**solèy** / **#lo** **solèy** / **#solèy-la**} la tonbé.
 PRF work until sun DET_{FUNC} sun sun-DET_{SORT} PRF fall⁷¹

‘They worked until the sun went down.’ (adapted from Albers 2024: (16))

- (66) M’i sar {**volkan** / **#lo** **volkan** / **#volkan-la**}.
 1SG.*i* go volcano DET_{FUNC} volcano volcano-DET_{SORT}

‘I’ll go to the volcano.’ (adapted from Albers 2024: (17))

DPs with *-la* are used for referents present in the utterance situation (visible situation) and in anaphoric contexts.

- (67) {The speaker wants to draw the attention to an insect on the floor:}

⁷⁰ And also, with immediate situation uniques that are not part of the ‘visible situation’ in Hawkins’ (1978) terminology. See (Albers 2024: 12-14) for discussion.

⁷¹ Réyoné-specific glosses are: DET_{FUNC} – functional determiner; DET_{SORT} – sortal determiner; FUT – future; *i* – predicate marker (Albers 2024: fn. 7); INDF – indefinite; OBJ – object; PRF – perfective; TEL – telic.

Pangar {#bébèt / #lo bébèt/ bébèt-la} va pik aou
 attention insect DET_{FUNC} insect insect-DET_{SORT} FUT sting 2SG.OBJ

‘Attention, the/that insect will bite you.’ (adapted from Albers 2024: (25))

(68) In madam la rantré. **Madam-#(la)** te i port in tant.

INDF lady PRF come.in lady-DET_{SORT} PST i carry INDF basket

‘A woman came in. The woman carried a basket.’ (adapted from Albers 2024: (27))

Thus, bare nouns and DPs with *-la* may seem to correspond to WEAK and STRONG ARTICLE DEFINITES, respectively (under the conception of strong familiarity proposed in last section (55)), although this is not how Albers chooses to analyze them as I summarize below.⁷²

DPs with *lo*, however, have an unexpected distribution. The article *lo* appears with what Albers calls functional concept nouns (following Löbner 1985; 1998; 2011), *e.g.*, (69). These are nouns such as ‘goalkeeper (of a football team)’ or ‘lid (of a pot)’ which in Löbner’s theory of concept types denote functions from possessors to unique corresponding individuals, type $\langle e, e \rangle$. Functional concepts are opposed to individual concepts (type e), sortal concepts (type $\langle e, t \rangle$), and relational concepts (type $\langle e, \langle e, t \rangle \rangle$). Unlike functional concepts, relational concepts like ‘brother’ are not inherently unique (a single person may have several brothers, but a country has only one capital). Some examples are provided in Table 5.⁷³

A. INDIVIDUAL e <i>sun, pope</i>	B. SORTAL $\langle e, t \rangle$ <i>flower, dog</i>
C. FUNCTIONAL $\langle e, e \rangle$ <i>capital, price, goalkeeper</i>	D. RELATIONAL $\langle e, \langle e, t \rangle \rangle$ <i>brother, hand, friend</i>

Table 5. Löbner’s concept types, based on (Albers 2024: 10)

⁷² There are important nuances to their mutual distribution, though, as discussed by Albers (2024: §5).

⁷³ I am simplifying slightly here. As Albers (2024: 9–10) discusses, all of these types should also include a situation argument, so individual concepts would be type $\langle s, e \rangle$ and so on. This is relevant, for instance, with larger but not global situation uniques, such as the volcano in (66). I will continue making this simplification as before.

- (69) A: Kosa i ariv **kanapé-la**?
 what i arrive sofa-DET_{SORT}
 ‘What’s up with that/the sofa?’
- B: {#**Dosyé/ Lo dosyé**} la fine kasé.
 back / DET_{FUNC} back PRF TEL break
 ‘The back of this/the sofa is broken.’ (adapted from Albers 2024: (33b))

Albers proposes that *lo* combines only with functional nouns (cell C of Table 5), while *-la* is reserved for sortal nouns (cell B), and individual nouns appear bare (cell A).⁷⁴ Thus, instead of the distinctions in levels of salience-familiarity that we used previously to differentiate definite determiners (and that I pointed at above), this theory is based on differentiating the semantic types of determiners’ complement (cf. Wespel 2008). In some cases, this requires appealing to type-shifting in order to combine a determiner that requires a certain type with an inappropriately-typed NP (Albers 2024: 10ff.).

We may tentatively suggest that bare nouns in Réyoné which denote individual concepts do not undergo any type-shifting, nor have a covert article, but, having the type *e* (cell A of Table 5), are simply used by themselves. The Sortal Determiner *-la*, requiring sortal concepts, would have the standard denotation of type $\langle\langle e, t \rangle, e\rangle$ (cf. WEAK (35) or, maybe, STRONG (34)). And the Functional Determiner *lo* may have the unconventional type $\langle\langle e, e \rangle, e\rangle$. Albers does not provide a compositional treatment of the Réyoné determiner system, so what I am now suggesting is quite speculative. Albers instead opts for a more thorough description of the distribution of each DP type, and this allows her to take into account a wide range of different constructions and to relate the different marking possibilities with Löbner’ scale of definiteness, see (Albers 2024: §7).

A lexical entry for the Réyoné Functional Determiner *lo* within the current framework could be as follows:

- (70) AN ATTEMPT AT THE SEMANTICS OF THE RÉYONÉ FUNCTIONAL DETERMINER *lo*

$$\llbracket lo_i \rrbracket^g = \lambda f_{\langle e, e \rangle}. \iota x. [f(g(i)) = x]$$

- (71) $\llbracket lo_i dosyé \rrbracket^g = \iota x. [back'(g(i)) = x]$

⁷⁴ Relational nouns may either appear bare with a possessor or be shifted to a sortal concept and appear with *-la*.

the unique individual which is the back of $g(i)$, where $g(i)$ is the aforementioned sofa

For the DP *lo dosyé* ‘the back’ from (69), this gives us the denotation in (71). The job of the Determiner under this analysis is to feed a familiar possessor to the functional concept denoted by the noun, yielding the unique individual which corresponds to this possessor according to this function.

However, there are important complications if one were to try giving a compositional treatment of the Réyoné determiners (see fn. 72). For instance (Albers 2024: 16–17), the Functional Determiner *lo* is only obligatory in case the possessor of the functional noun is left implicit (69) and in identificational sentences, which feature an explicit possessor, as well as with superlatives and other modifiers which turn a sortal NP into a ‘compositionally created functional concept’ mapping an index or a restriction of some sort to a unique individual (Albers 2024: §3.4.2).

If the possessor is present DP-internally as in (72), the determiner is **optional**, even though from the semantics in (70) we would rather expect such DPs to be bare only, just like in Haitian Creole (64).

(72) {**Dosyé** / **Lo** **dosyé**} kanapé-la la fine kasé.

back / DET_{FUNC} back sofa-DET_{SORT} PRF TEL break

‘The back of this/the sofa is broken.’ (adapted from Albers 2024: (33a))

Clearly, our simple-minded denotation from (70) would not be able to compose with a functional noun which already received a possessor denotation from the explicit possessor DP, because the resulting denotation would already be of type e instead of the base type $\langle e, e \rangle$ which the Determiner requires. So, let my presentation of the complex Réyoné data not create the wrong picture in the reader’s mind. A compositional treatment of these data deserves a separate study.

Figure 5 is a possible graphical representation of the Réyoné determiner system as compared to the three types of definites considered in §§2.1-2.3. (The size of the circles representing the distribution of each expression (type) is not informative while overlap between the different circles is.)

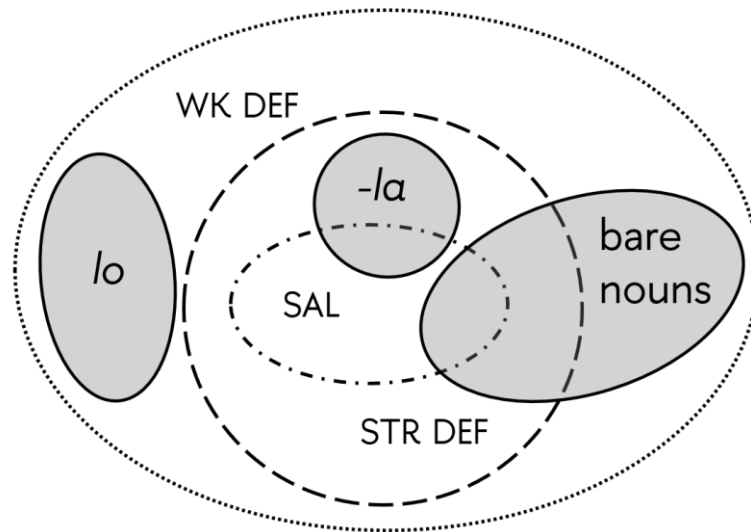


Figure 5. Relation between the Réyoné Functional Determiner *lo*, Sortal Determiner *-la*, and bare nouns, compared to WEAK, STRONG, and SALIENT ARTICLES (based on (Albers 2024) and Figure 3)

Individual concepts (type *e*) appear bare in Réyoné regardless of whether they are strongly familiar or salient in the current context, so the circle representing their distribution overlaps with the three types of salience-familiarity definites discussed previously. DPs with *lo* seem to be used only in uniqueness contexts with functional concepts (functional nouns, superlatives, etc.), therefore, the circle that represents their distribution is contained in the WEAK ARTICLE DEFINITE circle and does not overlap others. DPs with *-la* seem to appear only in STRONG ARTICLE DEFINITE contexts (as defined in (55)), so this circle is contained in the STRONG ARTICLE one.

Assuming that my understanding of the Réyoné data is largely correct, Figure 5 does make for an interesting representation of the definite typology we have seen so far. There is some degree of correspondence between the Löbner-style concept type-based approach to definites and the Schwarz-style familiarity-based one, as seen from the positions of the Functional *lo* and the Sortal *-la* in Figure 5. Still, it is clear that Figure 5 loses important information about the determiner system of Réyoné: bare nouns crosscut the whole salience-familiarity typology; the mutual distribution of the three Réyoné constructions is not represented in Figure 5; and neither are their selectional restrictions for concept type, etc. In general, the two approaches to definites seem not to be reducible to each other.

2.4.2. Schwarz-Löbner definite typology

So far, we have seen that definites vary crosslinguistically along basically two dimensions:

(73) THE SCHWARZ-LÖBNER DEFINITE TYPOLOGY⁷⁵

- a. **Salience-Familiarity dimension:** A definite determiner may presuppose one of the four statuses of the discourse referent along the Salience-Familiarity Scale (Roberts 2003; Barlew 2014; Schwarz 2019; this work, Ch. 3 and 5; among many others).
- b. **Concept type dimension:** A definite determiner may be restricted to one of the four concept types (Wespel 2008; Löbner 2011; Albers 2024, among many others).

It is clear, however, that there is more variation still. Above we mentioned the Haitian Creole Definite Article, which likely should be characterized in terms beyond its position on either dimension. It might have a wholly different theoretical status.

Another relevant determiner that requires something more than (73) is actually the Akan Definite Article, according to the recent work by Owusu (2022). (Cf. footnote 39.) Bombi (2018) and Owusu (2022) have argued that Arkoh and Matthewson's STRONG ARTICLE DEFINITE analysis is not correct for Akan data. Bombi proposed a WEAK ARTICLE DEFINITE analysis, accounting for some not strongly familiar unique uses, but Owusu went even further than that, arguing for the denotation in (74).

(74) SEMANTICS OF THE AKAN DEFINITE ARTICLE (Owusu's (2022: (70)) initial proposal)⁷⁶

$$\llbracket n\acute{o}_i \rrbracket^{g,c} = \lambda s \lambda P_{\langle e, \langle s, t \rangle \rangle}:$$

$$\text{i. } \exists! x [P(x)(s) \wedge x = g(i)] \wedge$$

$$\text{ii. } \exists s' \geq s [|\{x: P(x)(s')\}| > 1].$$

$$\iota x. P(x)(s) \wedge x = g(i)$$

⁷⁵ See the many references collected in (Schwarz 2019; Coppock 2022: fn. 1; Albers 2024: fn. 1) for other crosslinguistic work on the semantics of definite determiners in terms of these two dimension.

⁷⁶ Note that Owusu uses a different transcription than Arkoh and Matthewson (2013).

Owusu's final proposal (Owusu 2022: (93)) provides a predicative denotation for the Akan Definite Article based on arguments from (Coppock & Beaver 2015) for predicative denotations for definites as applied to Akan data, which I omit for simplicity.

The principal novelty of the proposal in (74) is the presupposition of non-uniqueness in a larger situation (74ii). This is why the Akan Definite Article is infelicitous with global situation uniques (‘sun’, ‘president’; Owusu 2022: §2.2.3), while it is acceptable with local uniques and anaphorically accessible referents, crosscutting Schwarz’s typology. In present terms, keeping in mind our definition of strong familiarity (55), the Akan Definite Article is a STRONG DEFINITE ARTICLE that additionally presupposes non-uniqueness in a larger situation.⁷⁷ The presupposition of non-uniqueness may perhaps be viewed as a Löbner-style restriction to sortal concepts bringing the two approaches together in the semantics of this determiner, but Owusu does not take Löbner’s theory into account, so I will not try to do so here either.

The Akan data show us how else definite semantics may be parametrized crosslinguistically by incorporating additional presuppositions. One of the definite determiners found in Kazym Khanty provides a similar case in that it combines WEAK DEFINITE ARTICLE semantics with an additional presupposition, see the discussion of the Associative Possessive in the next chapter.

2.4.3. Demonstratives as definite determiners

Other candidates for inclusion into the definite typology are demonstratives, as proposed most recently by Ahn (2022; 2024; 2025). Ahn argues that demonstratives are just like (WEAK ARTICLE) definites except that they have an additional restriction argument, *R* (75), which can be filled with a deictic gesture,⁷⁸ a relative clause, or an anaphoric index. The latter option brings demonstratives rather close to STRONG ARTICLE DEFINITES and some have indeed argued that demonstratives are STRONG ARTICLE DEFINITES in the determiner systems under discussion (e.g., Jenks 2018; cf. Dayal & Jiang 2022; Yu 2024).

(75) DEMONSTRATIVES’ LOGICAL FORM

$$[DP [D \cdot Dem_{\langle \langle e, t \rangle, \langle \langle e, t \rangle, e \rangle \rangle} NP_{\langle e, t \rangle}]_{\langle \langle e, t \rangle, e \rangle} R_{\langle e, t \rangle}]_e$$

(76) AHN’S UNIFIED ANALYSIS OF DEMONSTRATIVES AS DEFINITE DETERMINERS

⁷⁷ Although, I am not sure that strong familiarity indeed holds in all examples Owusu considers (e.g., Owusu 2022: (11)). Perhaps, a WEAK ARTICLE-based denotation, without an index, should be adopted instead.

⁷⁸ See (Ebert et al. 2020) on demonstratives as dimension shifters which allow to integrate information from the physical modality into linguistic at-issue content.

$$\llbracket \text{bi-sup} \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda R_{\langle e,t \rangle} \iota x_e \forall y [P(y) \wedge R(y) \rightarrow y \leq x]$$

where the possible values of R are:

- i. pointing gesture to location A :

$$\llbracket \rightarrow A \rrbracket^{g,c} = \lambda x. x \text{ is at } A \text{ at the actual world } w_0$$

- ii. anaphoric index function: $\llbracket idx \rrbracket^{g,c}(\llbracket i \rrbracket^{g,c}) = \lambda x. [x = g(i)]$

- iii. relative clause denotation

(adapted from Ahn 2022: (37), (57), (61), (64))

Demonstratives vary with respect to distance presuppositions, which Rs they admit (Marinina 2024; Yu 2024), and along other dimensions. Huijsmans and Reisinger (2021; 2022), exploring the demonstrative system of the Coast Salish language ʔayʔajuθəm (British Columbia, Canada), describe a system which distinguishes demonstratives based on distance, gender, number, evidentiality, and salience.

The ʔayʔajuθəm Gesture Demonstratives are used with a pointing gesture to establish joint attention and they only distinguish ‘current direct evidence’ (CDE) used mostly when the referent is visible and ‘evidence-neutral’ forms for non-visible referents. The Salience Demonstratives are used with referents at the center of the interlocutors’ joint attention and they distinguish CDE, ‘previous direct evidence’ and evidence-neutral forms.⁷⁹ So, in Ahn’s terms (76), the two classes of ʔayʔajuθəm demonstratives lexically distinguish what type of additional restriction R is used, a pointing gesture or an anaphoric index.

The salience presupposition of the Salient Demonstratives is reminiscent of the Bulu Salient Article and of personal pronouns on Roberts’ (2003) analysis (40). In fact, Huijsmans and Reisinger adopt Roberts’ analysis of pronouns for the ʔayʔajuθəm Salience Demonstratives, with the only difference being that the Demonstratives additionally take the NP denotation, just like other definite determiners do.

Building on the work of Ahn and Huijsmans and Reisinger, Marinina (2024) investigates the three demonstratives of Kazym Khanty: the Proximal *tām*, the Distal *tum*,

⁷⁹ Skilton (2021), investigating the demonstratives of Ticuna (isolate; Brazil, Columbia, Peru), also argues for a system distinguishing demonstratives that presuppose visibility of the referent (cf. CDE) and demonstratives that implicate invisibility of the referent (cf. evidence neutrality) as a *Maximize presupposition!*-based antipresupposition (Heim 1991; Percus 2006), along the lines of the discussion of (23) above.

and the Unmarked *śi*. All three demonstratives have exophoric (gestural) and anaphoric uses, but they are not used with relative clauses. The Unmarked Demonstrative *śi* is overwhelmingly more frequent in anaphoric uses than the other two.

Marinina proposes the Ahn-style unified denotations in (77) for the three demonstratives, with the additional proximal/distal presupposition for *tām* and *tum* based on Huijsmans and Reisinger's analyses of the Gestural-Salience distinction in ʔayʔajuθəm.⁸⁰

(77) KAZYM KHANTY DEMONSTRATIVES (adapted from Marinina 2024: (11-13))

$$a. \llbracket \acute{s}i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda R_{\langle e,t \rangle} . \iota x_e . P(x) \wedge R(x) \quad (\text{Unmarked})$$

$$b. \llbracket t\acute{a}m \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda R_{\langle e,t \rangle} . \iota x_e . P(x) \wedge R(x) \wedge PROX(x)(P) \quad (\text{Proximal})$$

$$c. \llbracket tum \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda R_{\langle e,t \rangle} . \iota x_e . P(x) \wedge R(x) \wedge DIST(x)(P) \quad (\text{Distal})$$

(78) ALLOSEMY-BASED PRESUPPOSITIONS OF *tām* (adapted from Marinina 2024: (14))

$$a. PROX = \lambda x \lambda P . x \text{ is close in space to } l_c / \text{___Gesture Phrase (R)}$$

$$b. PROX = \lambda x \lambda P . sal(x, c) \wedge \exists y [sal(y, c) \wedge P(y) \wedge x >_{sal(c)} y] \wedge \\ \forall z [sal(z, c) \wedge P(z) \wedge z \neq x \rightarrow x >_{sal(c)} z] / \text{___Index Phrase (R)}$$

x is salient in *c*, it is the most salient P-referent in *c* and there is at least one other less salient P-referent in *c*

(79) ALLOSEMY-BASED PRESUPPOSITIONS OF *tum* (adapted from Marinina 2024: (15))

$$a. DIST = \lambda x \lambda P . x \text{ is far in space from } l_c / \text{___Gesture Phrase (R)}$$

$$b. DIST = \lambda x \lambda P . sal(x, c) \wedge \exists y [sal(y, c) \wedge P(y) \wedge y >_{sal(c)} x] / \text{___Index Phrase (R)}$$

x is salient in *c* and there is another salient P-referent in *c*

These denotations are based on the mechanism of allosemy from recent work in Distributed Morphology (DM; Wood 2015; Myler 2016, among others).⁸¹ The basic idea is that the type of presupposition that the demonstrative in question triggers depends on its syntactic context, its R.

⁸⁰ I slightly adopt the notation used in (77)-(79) for consistency with the current work. $>_{sal(c)}$ stands for 'is more salient in *c* than', cf. the analysis of the Kazym Khanty Salient Article in §5.4.2.

Note that the denotations in (77)-(79) speak of salience of individuals and not of discourse referents as we did elsewhere. This discrepancy is irrelevant for our purposes and the denotations can be easily reformulated in terms of discourse referents if needed.

⁸¹ See also Chapter 4 where the DM architecture and allosemy are introduced in some more detail.

In an exophoric use, with a gestural R (represented syntactically as a Gesture Phrase; (78a)-(79a)), *PROX* and *DIST* are concerned with the distance of the referent to the location of the utterance situation l_c .

In an anaphoric use, in the context of an Index Phrase R, *PROX* and *DIST* encode ‘competitive’ salience presuppositions. An anaphorically-used Proximal *tām* presupposes relative maximal salience and the existence of another less salient individual fitting the NP denotation. An anaphorically used distal *tum* presupposes non-maximal salience compared to another salient referent fitting the NP denotation.

This is designed to capture Marinina’s observation that *tām* and *tum* are only used anaphorically in case there are two referents fitting the same description that contrast in salience. (Thus, there is a ‘competition’ in salience between the two referents.) In the more usual case, where only one referent fitting the description is salient, neither presupposition is satisfied and the Unmarked *ši* is used since it triggers no additional presuppositions (77a).

Marinina’s analysis shows how demonstratives’ polyfunctionality can be captured in a unified manner using Ahn’s R and the mechanism of allosemy, bridging the gap between gesture-based and salience-based demonstratives that was lexically encoded in ʔayʔajuθəm but is not encoded in Kazym Khanty.

We also see that in anaphoric uses the Kazym Khanty demonstratives make use of more elaborate salience presuppositions than the salience simpliciter presupposition of the Bulu Salient Article (48) and the relative maximal salience presupposition of personal pronouns (40) and ʔayʔajuθəm Salience Demonstratives. This provides another example of the semantic distinctions made by definite determiners crosslinguistically.

2.5. Summary

In this chapter, I provided an overview of the recent definite typology building on language-particular formal semantic work investigating definite determiners. Alongside this overview, I introduced the core concepts and tools needed in the semantic analysis of definite determiners. I will make use of them in the analysis of Kazym Khanty Unpossessives in the following chapters.

The principle dimension of variation in the semantics of definites for our purposes is represented by the Salience-Familiarity Scale (SFS), repeated from (61):

(80) THE SALIENCE-FAMILIARITY SCALE

weak familiarity > strong familiarity > salience > relative maximal salience > absolute maximal salience

(The SFS is based on a slight revision of how we understand the weak-strong familiarity distinction that I proposed in (55).)

The basic question that the SFS is concerned with is what degree of familiarity-salience of the DP referent is needed for it to be marked with the given determiner. The crosslinguistic work surveyed above shows that each position of the SFS is attested as a presupposition of a definite determiner in some language alongside a presupposition of uniqueness among discourse referents of the appropriate SFS status. (Excluding absolute maximal salience which so far seems to only be found with unstressed and zero pronouns.)

The canonical example of a WEAK DEFINITE ARTICLE is English *the* (Roberts 2003). STRONG DEFINITE ARTICLES (Schwarz 2009) are found in Germanic languages alongside competing WEAK ARTICLES (see also the references in fn. 75). The Bulu Salient Article (Barlew 2014) requires salience simpliciter (with uniqueness), and the ʔayʔajuθəm Salience Demonstratives (Reisinger & Huijsmans 2021; Huijsmans & Reisinger 2022) require relative maximal salience just like personal pronouns do.

We also saw definites that cannot be neatly characterized in terms of the SFS. They may select specific concept types in the sense of (Löbner 2011), like the Réyoné Functional and Sortal Determiners do (Albers 2024), or impose some other presuppositions, like the Akan Definite Article under Owusu's (2022) analysis or the demonstratives investigated by Huijsmans and Reisinger (2021; 2022), Marinina (2024), and others.

In the next chapter, I will show that Kazym Khanty features several determiners, in addition to the demonstratives discussed by Marinina (2024), that incorporate some familiar and some novel requirements. The added twist is that the Kazym Khanty Unpossessive determiners that I describe there are grammaticalized from possessive agreement markers. This makes them unique with respect to diachronic source in the context of the definite typology surveyed above. The Unpossessive determiner system of Kazym Khanty will be shown to be similar to the Réyoné determiner system in that it does not neatly fit into the SFS-based typology in Figure 6 (reworked from Figure 3).

Figure 6 includes the INDEFINITE ARTICLE and four definite determiner types discussed above, but it does not include non-determiner definites, such as zero pronouns which require the narrowest SFS status, absolute maximal salience. I suggested in §2.3.3 that no determiner may require absolute maximal salience which has a functional motivation (Gundel et al. 1993). Thus, Figure 6 includes the only types that are relevant to determiners.

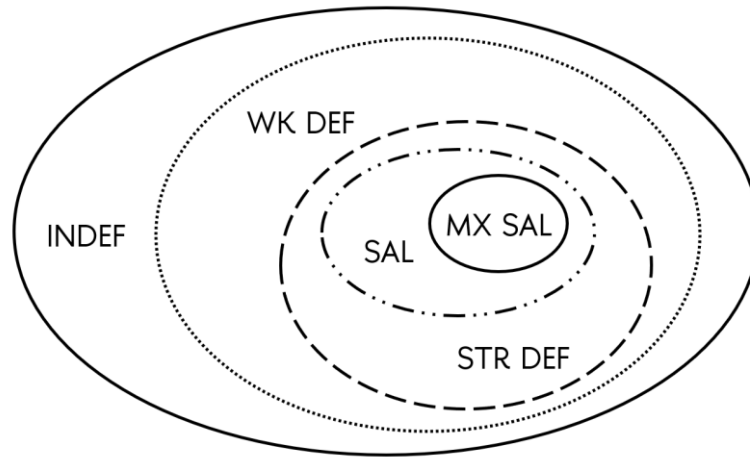


Figure 6. Relation between types of definites in the Saliency-Familiarity Scale typology.

Abbreviations: INDEF – INDEFINITE; WK DEF – WEAK DEFINITE; STR DEF – STRONG DEFINITE; SAL – SALIENT ARTICLE; MX SAL – RELATIVE MAXIMAL SALIENCE ARTICLE

3. Types of definites (and indefinites) in Kazym Khanty⁸²

This chapter is structured as follows. Section §3.1 introduces two more crosslinguistic parallels for the discussion to come, indeterminate possessives in English (Coppock & Beaver 2015) and the Turkish differential object marking conditioned on partitive specificity according to Enç (1991). Section §3.2 introduces the basic morphosyntax of the Kazym Khanty adnominal possessive construction. Sections §§3.3-3.7 examine the semantics of five Kazym Khanty (Un)possessives: the Proper Possessive, the Associative Possessive, the Partitive Article, the Salient Article, and the Proprial Article. Section §3.8 summarizes the main results of the chapter.

I start in §3.3 with the Proper Possessive. I argue that the Proper Possessive is semantically a simple possessive modifier and is not inherently definite, echoing the conclusions of (Coppock & Beaver 2015; Beaver & Coppock 2016) for English and several other languages. To account for the composition of Proper Possessive-marked nominals (Proper Possessive descriptions), I briefly touch on the interpretation of bare nouns, reviewing the results of (Tiutiunnikova et al. 2024; Tiutiunnikova 2024; Golosov 2024a).

On the other hand, in §3.4, I argue that the Associative Possessive is a genuine possessive definite, which combines free possessive semantics (Partee & Borschev 2003) with definiteness. What makes it an UNPOSSESSIVE in current terminology is that unlike the Proper Possessive, the Associative Possessive usually describes free relations picked up from the context. The relation choice is not restricted to prototypical possessive relations.

Sections §§3.5, 3.6, 3.7 present the Partitive Article, the Salient Article, and the Proprial Article respectively. Unlike the two Possessives, these Articles do not vary in ϕ -features of the supposed possessor. The Partitive Article is an existential quantificational determiner like the English Indefinite Article which additionally presupposes that the DP referent is part of another strongly familiar referent.

⁸² This chapter is a more explicit and extended version of (Mikhailov 2023) which adds formal semantic analyses for each marker and some new examples comparing the four Unpossessive determiners, alongside a brief but important mention of bare nouns and a quite involved discussion of nonrigidity in §3.7.3.

The Salient Article is a RELATIVE MAXIMAL SALIENCE ARTICLE in the sense of Chapter 2, which imposes additional idiosyncratic restrictions similar to proximate uniqueness of languages with obviation systems (Aissen 2001). The discussion of these restrictions is set aside until Chapter 5; in the confines of the current chapter I focus specifically on how the Salient Article relates to the types of definites surveyed above.

Finally, the Proprial Article derives intensionally rigid DPs from human names.⁸³ In §3.7.3, I provide a detailed discussion of how Salient Article-marked DPs and bare nouns are intensionally nonrigid in Kazym Khanty, while Proprial Article-marked DPs are rigid. I show how this is derived from the denotations proposed for the three nominal types throughout this chapter in a situation-semantic extension of the current semantic framework that combines (Schwarz 2009; Elbourne 2013) with Muñoz’s (2019) semantics for PROPRIAL ARTICLES.

Furthermore, in §§3.6.4, 3.8, I discuss if and how the four Unpossessive determiners compete. I illustrate this with examples where their distributions overlap and explain how the denotations that I propose account for this. Figure 7 at the end of this Chapter shows how the Kazym Khanty Unpossessives relate to the Saliency-Familiarity Scale typology from Chapter 2. To spoil a bit, it will be shown that the Unpossessives crosscut the SFS typology.

3.1. Two more crosslinguistic parallels

3.1.1. Indeterminate possessives in English

Coppock and Beaver (2015; C&B) discuss the English determiner system and propose that a significant role is played by type-shifting rules, IOTA and EX, a much larger role than assumed previously (Partee 1986; Chierchia 1998). IOTA derives determinate, definite denotations (81), just like *the* in traditional analyses. EX derives indeterminate, existential generalized quantifier denotations (82), just like *a/an* and *some* in traditional analyses.

$$(81) \text{ IOTA} =_{\text{def}} \lambda P_{\langle e,t \rangle}. \iota x. P(x)$$

$$(82) \text{ EX} =_{\text{def}} \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle}. \exists x. P(x) \wedge Q(x)$$

⁸³ Intensionally rigid DPs are rigid designators in the sense of (Kripke 1980): they denote the same individual under any circumstances of evaluation.

The reason for this is that C&B argue for predicative denotations for *the* and *a/an*, type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$, as opposed to standard types $\langle\langle e, t \rangle, e\rangle$ and $\langle\langle e, t \rangle, \langle\langle e, t \rangle, t \rangle\rangle$, respectively, assumed in the previous literature (e.g., (19), (24) from Chapter 2).

(83) PREDICATIVE DENOTATION FOR *the* (adapted from Coppock & Beaver 2015: (50))⁸⁴

$$\llbracket the \rrbracket = \lambda P: |\{x: P(x)\}| \leq 1. \lambda x. P(x)$$

Their argument is based on so-called indeterminate uses of definite DPs where they cannot be assumed to have the type *e*. In C&B's parlance, determinate DPs have the type *e* while indeterminate DPs have the generalized quantifier type $\langle\langle e, t \rangle, t\rangle$.⁸⁵ I refer the reader to the original paper for their arguments and analysis, as it is not directly relevant to the Kazym Khanty determiner system.

What is relevant for us here is C&B's discussion of Saxon Genitive DPs, which are traditionally also assumed to be definite (Barker 2000; Partee & Borschev 2003), as incorporating "an implicit definite article" (Vikner & Jensen 2002: 201). C&B instead argue that Saxon Genitives are not inherently definite. Rather, depending on the syntactic position in which a DP with the Saxon Genitive is used, its determinacy changes. In argument positions, such DPs behave as definite as evidenced by (84a) where a Saxon Genitive DP causes a contradiction in Löbner's (2011) 'negation' test.⁸⁶ In this example, two identical DP *Mary's pet rabbit* are attributed mutually exclusive properties, making the utterance unacceptable. (Cf. (84b), which shows that DPs with *some* do not cause a contradiction in this test and hence are indeterminate according to C&B).

(84) a. #**Mary's pet rabbit** is in the cage and **Mary's pet rabbit** is outside the cage.

⁸⁴ In prose, *the* is an identity function on predicates and it presupposes that there is at most one individual that fits the DP description (and, possibly, none). This is what C&B call a weak uniqueness presupposition. On their account the existence presupposition arises after IOTA applies to $\llbracket the NP \rrbracket$.

⁸⁵ We may also include DPs of the predicative type $\langle e, t \rangle$ in the indeterminate class, although C&B use the term narrowly to refer to DPs that underwent EX.

You may recall, that in Chapter 1, I gave a broad definition of determinacy as the cluster of semantic distinctions encoded by determiners. Coppock and Beaver's understanding of determinacy is thus much narrower: it boils down to denoting an individual. Coppock and Beaver's narrowly understood determinacy is one of the distinctions included in broad determinacy, as understood in this thesis.

⁸⁶ Also known as the 'homogeneity test' (Dayal 2018). The test is based on Löbner's (1985; 2000; 2011) observation that definite DPs behave like individual terms in predicate logic do, for instance, in terms of consistency: if $P(\llbracket DP \rrbracket)$ is true for some predicate *P*, then necessarily $\neg P(\llbracket DP \rrbracket)$ is false (the negation of the predicate does not hold of the denotation of the DP).

b. Some rabbit is in the cage and some rabbit is outside the cage. (adapted from (Coppock & Beaver 2015: 417))

A striking observation is that in predicate position, DPs with the Saxon Genitive are indeterminate (85a) much like DPs with *some* (85b) are, but unlike DPs with *the* (85c).

(85) a. The rabbit in the cage is **Mary's pet** and the rabbit sitting just outside the cage is **Mary's pet**.

b. The rabbit in the cage is **a pet Mary owns** and the rabbit sitting just outside the cage is **a pet Mary owns**.

c. #The rabbit in the cage is **the pet Mary owns** and the rabbit sitting just outside the cage is **the pet Mary owns**. (adapted from (Coppock & Beaver 2015: 418))

Based on these observations, Coppock and Beaver consider the Saxon Genitive unmarked in definiteness (86), and explain its behavior in argument position by IOTA (81) applying to it. The denotation of a Saxon Genitive DP prior to any type-shifts is as in (86), depending on whether the head noun is relational (86a) or not (86b)

(86) a. $\llbracket \text{Mary's pet} \rrbracket = \lambda x_e. \text{pet}'(m, x)$

a predicate of individuals that stand in the *pet'* relation to Mary

b. $\llbracket \text{Zahra's horse} \rrbracket = \lambda x_e. \text{horse}'(x) \wedge \text{POSS}(z, x)$

a predicate of horse-individuals that stand in an inherent possessive relation to Zahra

(87) $\text{IOTA}(\llbracket \text{Mary's pet} \rrbracket) = \iota x_e. \text{pet}'(m, x)$

the unique individual that is Mary's pet; #, if there is no unique individual that is Mary's pet

In §3.3, we will see that Kazym Khanty (proper) possessives are indeterminate even in argument positions according to the negation test.

A separate issue, prominent in work on the semantics of possessives, is the semantics of the possessive relation that a possessive construction introduces (in the case of non-relational head nouns). I will assume with (Barker 1995; Vikner & Jensen 2002; Partee & Borschev 2003; Partee 2011/1983; Karvovskaya 2018) that in the usual case, an adnominal possessive construction introduces an 'inherent' possessive relation, represented in simplified form as POSS in (86b). Inherent relations are stereotypical relations derived from the denotation of the head noun (see Karvovskaya 2018: §2.3;

Vikner & Jensen 2002 and the other papers cited for details), they are contrasted with ‘free’ relations which are not restricted in this way and are derived from the context (cf. Partee 2011/1983; Storto 2005).

As discussed in the introductory chapter, in many Uralic languages, possessives are also used to mark familiar referents (Nikolaeva 2003; Fraurud 2001, etc.). In this chapter, my goal is to investigate the semantics of the Kazym Khanty Unpossessives in the framework of determiner semantics developed in Chapter 2 and to situate the Unpossessives in the known determiner typology.

3.1.2. Partitive specificity in Turkish

Before proceeding to Kazym Khanty data, it is necessary to introduce another type of determinacy, partitive specificity. A partitive specific referent is a part of another familiar referent in a given context (Enç 1991; Farkas 2002a; von Stechow 2002; Farkas & Brasoveanu 2019). For example, in Turkish, partitive-specific direct objects are marked by the Accusative Case (88). Omission of the case marker in this example triggers the antipresupposition (Percus 2006; and §2.1) that the two girls are not part of the group of children from the preceding context.

(88) [“Several children entered my room.”]

İki **kız-#(i)** taniyordum.

two girl-ACC I:knew

‘I knew two (of the) girls.’ (adapted from (Enç 1991: 6))

Here is Enç’s definition of partitive specificity adapted to our notation.

(89) PARTITIVE SPECIFICITY (Enç’s (1991: 7) version)

A $DP_{\langle i, k \rangle}$ carries a pair of indices. The first index represents the DP referent. The second index is needed to relate the DP referent to the superset antecedent referent. A $DP_{\langle i, k \rangle}$ is partitive-specific iff $g(i) \subseteq g(k)$, the (set of) individual(s) that the assignment function maps the first index to is a subset of the set of individuals that the assignment maps the second index to.⁸⁷

⁸⁷ This is based on the assumption that plural DPs denote sets of individuals while singular DPs denote individuals proper. Accordingly, partitive specificity for a singular DP should be defined as $\{g(i)\} \subseteq g(k)$, instead (Enç 1991: (22)). Contemporary approaches to plurality mostly opt for a different, mereological definition based on Link’s work (Link 1983) which does not make a type-theoretic distinction

The close connection between partitive specificity and definiteness is revealed by the fact that typologically, differential object marking (DOM) often depends on whether the referent is definite/partitive-specific or not (Farkas 2002a; Aissen 2003; Serdobolskaya 2017). For instance, the Turkish Accusative Case is used with definite DPs and with partitive-specific DPs, while non-partitive-specific and indefinite direct objects remain unmarked. In other words, Turkish DOM combines definite and partitive-specific DPs into one class, contrasting them with other types of DPs. In §3.5, I will introduce a similar marker in Kazym Khanty, the Partitive Article.

Viewed this way (89), partitive specificity may be seen as an additional position on the Salience-Familiarity Scale proposed in Chapter 2, (61).

(90) THE SALIENCE-FAMILIARITY SCALE WITH PARTITIVE SPECIFICITY

partitive specificity > weak familiarity > strong familiarity > salience > relative maximal salience > absolute maximal salience

Under Enç's definition, familiarity is seen as a special case of partitive specificity, the case when $g(i) = g(k)$.⁸⁸ (See also Aissen's (2003) Definiteness Scale for DOM and Farkas's (2002a) discussion thereof.)

The next section presents the basics of the Kazym Khanty adnominal possessive construction, before proceeding to the next five sections where I will present evidence of determinacy presuppositions (Farkas 2002a; Roberts 2003; Farkas & Brasoveanu 2019) in Kazym Khanty Unpossessives. Specifically, I will make use of the presuppositions of existence, familiarity, uniqueness, and partitive specificity, as well as the presupposition of salience (cf. Barlew 2014) and intensional rigidity (Kripke 1980; Muñoz 2019).

3.2. *Kazym Khanty adnominal possessive construction*

Example (91) illustrates the basic adnominal possessive construction of the Kazym dialect of Northern Khanty. The possessor is unmarked and precedes the head noun which

between plurals and singulars, see (Nouwen 2016) for discussion. We needn't concern ourselves with these details since plurality will not play a role anywhere (besides partitive specificity) below.

⁸⁸ However, the Kazym Khanty Partitive Article is limited to **proper** partitive specificity, when $g(i) \subset g(k)$, as we will see in §3.5. So, I leave it open whether Enç's definition provides a cross-linguistically valid understanding of partitive specificity or if we should instead opt for narrower proper partitive specificity and see the Turkish Accusative Case as requiring one of two statuses, partitive specificity or familiarity.

is marked with a possessive agreement suffix indexing the possessor's person-number features. The possessor may also be expressed as a null pronoun, *pro*.

- (91) *nǎŋ kǎt'-en moś-λ*
 you.SG cat-POSS.2SG purr-NPST[3SG]
 'Your cat purrs.'

Possessive suffixes distinguish three persons and three numbers (singular, dual, and plural) of the possessor (Kaksin 2010), see Table 6.

Possessee Possessor	SG	DU	PL
1SG	-em -POSS.1SG	-ŋəλ-am -DU-POSS.1SG	-λ-am -PL-POSS.1SG
2SG	-en -POSS.2SG	-ŋəλ-an -DU-POSS.2SG	-λ-an -PL-POSS.2SG
3SG	-əλ -POSS.3SG	-ŋəλ -DU-POSS.3SG	-λ-aλ -PL-POSS.3SG
1DU	-emən -POSS.1DU	-ŋəλ-əmən -DU-POSS.1DU	-λ-əmən -PL-POSS.1DU
2DU	-ən -POSS.2NSG	-ŋəλ-ən (-ŋəλ-an) -DU-POSS.2NSG	-λ-ən -PL-POSS.2NSG
3DU	-ən -POSS.3DU	-ŋəλ-ən (-ŋəλ) -DU-POSS.3DU	-λ-ən (-λ-aλ) -PL-POSS.3DU
1PL	-ew -POSS.1PL	-ŋəλ-əw -DU-POSS.1PL	-λ-əw -PL-POSS.1PL
2PL	-ən -POSS.2NSG	-ŋəλ-ən (-ŋəλ-an) -DU-POSS.2NSG	-λ-ən -PL-POSS.2NSG
3PL	-eλ -POSS.3PL	-ŋəλ -DU-POSS.3PL	-λ-aλ -PL-POSS.3PL

Table 6. Possessive suffixes of Kazym Khanty (field data)⁸⁹

⁸⁹ The parenthesized allomorphs are preferred by a small number of my consultants. This interspeaker variation is ignored in what follows and only the more widely used allomorphs (without parentheses) are considered. I thank Daniil Burov (p.c.) for verifying and correcting some datapoints in this paradigm and for helpful discussion.

My field data differs significantly from the possessive paradigm provided in Kaksin's (2010) grammar sketch and not all differences are due to the different transcriptions adopted. My data is very close to the paradigms provided by (Sipos 2022). There seem to be minor differences with Sipos's consonant stem paradigm.

With possessed nouns instead of the regular dual and plural number markers (*-ŋən* [DU] and *-ət* [PL]), special possessed number allomorphs appear (*-ŋəλ* [DU] and *-λ* [PL]) and they trigger morphophonological alternations in possessive markers.

Possessive markers cannot be stacked, even if the second marker is used in a non-possessive function, for example, **amp-em-en* [dog-POSS.1SG-POSS.2SG] (int.) ‘my aforementioned dog’.⁹⁰

With lexical (*i.e.*, non-pronominal) possessors, possessive marking may be absent: *waśaj-en amp / amp-λ* [V.-POSS.2SG dog / dog-POSS.3SG] ‘Vasya’s dog’. According to preliminary data (Muravyev et al. 2023; Muravyev et al. 2024), **the presence of possessive marking in such constructions is dependent on the status of the possessor as Proximate or Obviative** (*cf.* (Nikolaeva & Bárány 2019))⁹¹. In what follows, I only consider instances of pronominal or null *pro* possessors which obligatorily trigger possessive agreement: *λ_{HW} / Ø_{3SG} amp-*(λ)* [(s)he / pro.3SG dog-POSS.3SG] ‘his/her dog’.

In §4.1.1, I suggest that the possessive agreement probe in Kazym Khanty requires Proximate targets only. When there is no possessive marker in a possessive construction, agreement has failed (Preminger 2014) because the possessor was not Proximate and no features were copied onto the probe.

3.3. Proper Possessive

The Proper Possessive expresses inherent possessive relations (Vikner & Jensen 2002; Partee & Borschev 2003; Karvovskaya 2018) (*cf.* the Associative Possessive, §3.4). The Proper Possessive has the full possessive paradigm from §3.2 and it expones different person-number features depending on the features of the Possessor DP (see Chapter 4 for a model of how this comes about).

The Proper Possessive does not require uniqueness of the referent, which is shown in example (92) using the negation test (Löbner 2011; Coppock & Beaver 2015) (see §3.1).

⁹⁰ Compare Meadow Mari *ūdər-em-že* [daughter-POSS.1SG-POSS.3SG] ‘my daughter (topicalized)’, adapted from (Kuznetsova 2003), where a POSS.3SG in a non-possessive function attaches to an already possessed noun resulting in ‘possessive stacking’.

⁹¹ In languages with obviation (*e.g.*, (Aissen 2001)), third-person nominals within a clause are subdivided into the Proximate and Obviatives. The Proximate is a syntactically privileged, prominent third person participant, as compared to the Obviatives. (Muravyev et al. 2023) hypothesize that only Proximate lexical possessors trigger possessive agreement in Kazym Khanty.

(92) [A child made a mess in the kitchen. Their parents tell them: “What a mess you made!”]

<i>păsan</i>	<i>əxtij-ən</i>	<i>năŋ</i>	<i>juntut-en</i>	<i>uλ,</i>
table	on-LOC	you.SG	toy-POSS.2SG	lie[NPST.3SG]
<i>kər_λaŋəλ-ən</i>	<i>năŋ</i>	<i>juntut-en</i>	<i>uλ,</i>	
stove-LOC	you.SG	toy-POSS.2SG	lie[NPST.3SG]	
<i>păsan</i>	<i>ilpij-ən</i>	<i>năŋ</i>	<i>juntut-λ-an</i>	<i>kerət'λ'-əλ-ət</i>
table	under-LOC	you.SG	toy-PL-POSS.2SG	lie.around-NPST-3PL

‘There’s a toy of yours on the table, [a toy of yours] on the stove, under the table your toys are scattered.’

A DP carrying the presupposition of uniqueness is expected to be unacceptable in such a context, as is the case, for example, with the English Definite Article or the Saxon Genitive in argument positions (§3.1.1; Coppock & Beaver 2015). If the Proper Possessive required uniqueness, example (92) would be contradictory, as the first sentence with the description *năŋ juntut-en* ‘your toy’ would attribute to the unique toy belonging to the addressee the property of being on the table and the second sentence would attribute it the property of being on the stove, while the third one would imply that there is a unique plural individual consisting of addressee’s toys each of which is under the table. This is clearly contradictory. Since (92) is felicitous, the conclusion is that the Proper POSS.2SG descriptions in (92) do not presuppose uniqueness.

This is quite unlike the behavior of the English Saxon Genitive which presupposes uniqueness in argument position but does not do so in predicate position. As discussed in §3.1.1, Coppock and Beaver (2015) attributed this shifting behavior of the Saxon Genitive to the IOTA-shift (81) applying in argument position but not in predicate position.

Given this difference between Kazym Khanty Proper Possessives and English Saxon Genitives, I suggest that Proper Possessives are not interpreted via IOTA but are shifted via EX instead. I unpack this argument below in §3.3.1 where I propose that Kazym Khanty does not have IOTA in its inventory of type-shifts, following (Mikhailov 2021c; Golosov 2024; Tiutiunnikova 2024; Tiutiunnikova et al. 2024).

This is supported by the observation that the Proper Possessive freely combines both with indefinite and with definite determiners. In (93), it combines with the

epistemically nonspecific pronoun *mul̥sər* ‘some’.⁹² In (94), it combines with the proximal demonstrative *tām* ‘this’.

(93) [The speaker is at a friend’s place.]

mul̥sər an-en mij-a
 some.EN cup-POSS.2SG give-IMP[SG]
 ‘Give me any cup of yours.’

(94) (adapted from Pleshak 2018: (6))

tām ma xələm puχ-λ-am armija-ja măn-s-ət
 this I three son-PL-POSS.1SG army-DAT go-PST-3PL
 ‘These three sons of mine joined the army.’

The Proper Possessive in (93) functions as a modifier that narrows down the extension of the NP ‘cup’ to the set of ‘cups that belong to you’, but it does not determine the NP. As a modifier, it can be freely combined with the indefinite pronoun *mul̥sər*.

However, the Proper Possessive also carries the presupposition of existence of the NP referent, which is confirmed by the following examples. The Proper Possessive is not used in the predicative possessive construction with the verb *tăjti* ‘to have’ in the existential meaning and cannot have a narrow scope in relation to negation (95a).⁹³ Adding a Possessive leads to a change in the interpretation of the verb to ‘temporary possession’ (95b).

- (95) a. *năŋ welik ănt tăj-λ-ən*
 you.SG bicycle NEG have-NPST-2SG
 ‘You don’t have a bike.’
- b. *năŋ welik-en ănt tăj-λ-ən*
 you.SG bicycle-POSS.2SG NEG have-NPST-2SG
 ‘You don’t have your bike [currently; e.g., it’s broken].’
 Int.: #You don’t have a bike.

⁹² Epistemic specificity is concerned with the question of “whether the speaker (or another cognitive agent) has a particular entity in mind or not” (Farkas & Brasoveanu 2019: 12), cf. the Russian pronoun *kakoy-to* ‘some (epistemically nonspecific)’ (Aloni & Degano 2022).

⁹³ That is in the sense ‘the subject referent owns an entity from the extension of the direct object’.

Simonenko (2017) shows that possessive agreement markers come in two types depending on the availability of narrow scope with respect to negation. Moksha (< Mordvin < Uralic), Meadow Mari (< Mari < Uralic), and Turkish possessives freely take narrow scope with respect to negation in a construction like (95a), whereas Buryat (< Mongolic) and Izhma Komi (< Permic < Uralic) possessives can only have wide scope in such constructions (as in (95b)) (Ibid.: 431-432). Simonenko links this difference to the presence of an existence presupposition in the possessive. If the possessive triggers such a presupposition, then its use in the scope of negation would lead to a contradiction: the presupposition ‘there is an x such that x stands in the possession relation to y (the possessor)’ contradicts the assertion ‘there is no x such that x stands in the possession relation to y (the subject)’ (where y is the same referent). In languages where the use of the possessive in the scope of negation is possible, the possessive does not carry the presupposition of existence.

Applying Simonenko's reasoning to Kazym Khanty data in (95), I propose that the Proper Possessive carries an existence presupposition, as do the possessives of Buryat and Izhma Komi. Thus, the Proper Possessive has the denotation in (96). It is unmarked for definiteness but derives referential NPs (due to its presupposition of existence).

(96) DENOTATION OF THE PROPER POSSESSIVE

$$\llbracket \text{Poss} \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda x_e \lambda y_e: \exists z [\text{POSS}(x, z) \wedge P(z)]. \text{POSS}(x, y) \wedge P(y)$$

where POSS is an inherent possessive relation in the sense of (Vikner & Jensen 2002; Partee & Borschev 2003; Karvovskaya 2018)

3.3.1. Interpreting Kazym Khanty Proper Possessives as bare nouns

An NP with a Proper Possessive such as *năn juntut-en* ‘your toy’ from (92), thus, has the predicative denotation in (97). This is essentially a determiner-less denotation which makes sense given that the Proper Possessive cooccurs with determiners of different types (93)-(94) but it may also occur ‘bare’, that is, without a determiner.

$$(97) \llbracket năn juntut-en \rrbracket^{g,c} = \lambda y_e: \exists y [\text{POSS}(a_c, y) \wedge \text{toy}'(y)]. \text{POSS}(a_c, y) \wedge \text{toy}'(y)$$

In order to occur bare in argument position, Proper Possessive descriptions require type-shifting via IOTA or EX (Coppock & Beaver 2015), and we have seen that, unlike Saxon Genitive DPs interpreted via IOTA, Proper Possessive descriptions are felicitous in the negation test (98) which entails that they do not have a type e denotation: they cannot be interpreted via IOTA.

(98) [A child made a mess in the kitchen. Their parents tell them: “What a mess you made!”]

păsan	əxtij-ən	năŋ	juntut-en	uλ,
table	on-LOC	you.SG	toy-POSS.2SG	lie[NPST.3SG]
kər_λaŋəλ-ən	năŋ	juntut-en	uλ,	
stove-LOC	you.SG	toy-POSS.2SG	lie[NPST.3SG]	
păsan	iłpij-ən	năŋ	juntut-λ-an	kərət'λ'-əλ-ət
table	under-LOC	you.SG	toy-PL-POSS.2SG	lie.around-NPST-3PL

‘There’s a toy of yours on the table, [a toy of yours] on the stove, under the table your toys are scattered.’

This is puzzling on both approaches that make use of IOTA, Coppock and Beaver’s (2015) approach to English DPs and the neo-Carlsonian approach (Chierchia 1998; Dayal 2004) to bare nominals, because both approaches assume that IOTA is preferred over EX in most (if not all) contexts where type-shifting is needed.⁹⁴ This suggests that IOTA is unavailable in Kazym Khanty and all bare NPs are interpreted via EX. This is what (Golosov 2024a; Tiutiunnikova 2024; Tiutiunnikova et al. 2024) have argued for in detail, focusing on singular and dual nominals (without possessives).

Consider, for instance, the bare singular *amerikanets* ‘American’ in the negation test in (99). In a language whose type-shift inventory contains IOTA, the negation test necessarily leads to infelicity because IOTA is preferred over EX and IOTA’s uniqueness presupposition fails such a context.⁹⁵ In Kazym Khanty, however, bare nouns are felicitous in the negation test suggesting that they are interpreted with EX and not with IOTA.

⁹⁴ For C&B, this is due to the principle of Type Simplicity (Coppock & Beaver 2015: (98)) which forces the addressee to parse a predicative nominal as interpreted via IOTA and, thus, of type *e*, wherever consistent with the common ground, instead of the more complex type $\langle\langle e, t \rangle, t\rangle$ that it would arrive at via EX. Under the neo-Carlsonian approach, this is due to the principle of Revised Meaning Preservation proposed by Dayal (2004: (39c)), based on (Chierchia 1998), which also prefers IOTA over EX (specifically, in the interpretation of bare, article-less nouns). See the original papers for details.

⁹⁵ This is the case at least in Hindi (< Indo-Aryan < Indo-European; Dayal 2004; 2018) and in Teotitlan del Valle Zapotec (< Otomanguean; Oaxaca, Mexico; Deal & Nee 2018), see the Hindi example (i).

(i) #**kuttā** so-rahā-he or **kuttā** bhōk-rahā-he
 dog sleep-PROG-PRS and dog bark-PROG-PRS
 ‘The dog is sleeping and the dog is barking.’ (adapted from Dayal 2018: (6))

(99) [‘There were several participants from different countries in a race.’]

amerikanets met siri juχt-əs pa
 american most front come-PST[3SG] ADD
 met juλta χaśəs i_śi **amerikanets**
 most behind stay-PST[3SG] ADD american

‘An American came first and an(other) American came last.’ (Tiutiunnikova et al. 2024)

Thus, I assume that Proper Possessive descriptions and (other) bare nouns in Kazym Khanty are interpreted via EX as existential generalized quantifiers (100). See (Tiutiunnikova 2024; Tiutiunnikova et al. 2024) for further arguments from scopal properties and other tests from (Dayal 2018) for this view.

(100) a. $EX(\llbracket n\check{a}n\ juntut-en \rrbracket^{g,c}) =$

$\lambda Q_{\langle e,t \rangle}. \exists x_e: \exists y[\text{POSS}(a_c, y) \wedge toy'(y)]. \text{POSS}(a_c, x) \wedge toy'(x) \wedge Q(x)$

the function from predicates Q to truth values, defined iff there is a toy owned by the addressee, when defined yields truth just in case there is such a toy and Q holds of it

b. $EX(\llbracket amerikanets \rrbracket^{g,c}) = \lambda Q_{\langle e,t \rangle}. \exists x_e. american'(x) \wedge Q(x)$

For example (98), this will give us the following Logical Form (LF) for the first clause (101) and truth conditions for the first two clauses (102).⁹⁶

(101) a. $[\text{PossP } you_e [\text{NP } toy_{\langle e,t \rangle}] \text{ Poss}_{\langle \langle e,t \rangle, \langle e, \langle e,t \rangle \rangle}]_{\langle e,t \rangle} \xRightarrow{EX} \exists x[\text{PossP } you [\text{NP } toy(x)] \text{ Poss}_{\langle \langle e,t \rangle, t \rangle}]$

b. $[\text{CP } \exists x[\text{PossP } you [\text{NP } toy(x)] \text{ Poss}_{\langle \langle e,t \rangle, t \rangle}]_{\langle e,t \rangle} [\text{VP } lie-on-the-table]_{\langle e,t \rangle}]_t$

(102) $\llbracket (98) \rrbracket^{g,c} = 1$ iff $\exists y[\text{POSS}(a_c, y) \wedge toy'(y)]. \exists x[\text{POSS}(a_c, x) \wedge toy'(x) \wedge on-the-table(x)] \wedge \exists z[\text{POSS}(a_c, z) \wedge toy'(z) \wedge on-the-stove(z)]$

In prose, the first two clauses of (98) are defined just in case there is a toy-entity that stands in an inherent possessive relation to the addressee (presupposition), they are

⁹⁶ I write only one existential possessive presupposition here since two instances of this presupposition triggered by two Proper Possessives are semantically equivalent to one instance.

In the LF, I use $\exists x[\dots(x)]$ to designate a nominal that EX has applied to. I assume that syntactically the Proper Possessive heads its own projection, the Possessive Phrase. See Chapter 4 for details.

true iff there is such an entity on the table and there is such an entity on the stove (assertion). Clearly, this is not contradictory and provides the intuitively correct truth conditions for (98).

Thus, I argue that Proper Possessive is not inherently definite and is not a determiner in any sense. Nominals with the Proper Possessive (Proper Possessive descriptions) and without other determiners are interpreted in the same way Kazym Khanty bare nouns are, as existential generalized quantifiers shifted via EX.

3.4. Associative Possessive

According to Nikolaeva (2003), who examines the associative uses of a number of Uralic Possessives, ASSOCIATIVE POSSESSIVES denote relations that are not based on the possessive prototype (possession, part-whole, social relations) (Taylor 1996) or on the semantics of the head noun (Partee & Borschev 2003), i.e., these are not inherent relations. Associative relations are temporary and context-dependent, in Partee's terminology these are free possessive relations.

The use of the Associative Possessive carries the presupposition that there is a contextually accessible relation between the DP referent and the Possessor. Like the Proper Possessive, the Associative Possessive has the full possessive paradigm from §3.2 with the same exponents dependent on the person-number features of the Possessor as in the Proper Possessive. (In Chapter 4, I provide a Distributed Morphology analysis that accounts for this.)

In (103), the cup marked with the Associative POSS.2SG does not belong to the speaker's friend and is not related to her in any way beyond the fact that it is next to the friend. Nevertheless, possessive marking is possible and even obligatory here. (A bare noun would be infelicitous here, which is marked as *an-#(en)* with the infelicity sign (#) outside the brackets.)

(103) [A friend is over at the speaker's place. There's one cup on the table.]

an-#(en) mij-e
cup-POSS.2SG give-IMP.SG>SG
'Give me the cup.'

On the other hand, if the entity does not uniquely fit the DP description, an Associative Possessive is infelicitous (104).

(104) [Context from (103) with several cups.]

an-#en) mij-a
cup-POSS.2SG give-IMP[SG]
‘Give me a cup.’

Consultant’s comment on *-en*: “[the addressee] will then ask ‘Which cup do you mean?’”.

The same is true of other person-numbers and is exemplified for POSS.1PL in (105)-(107). Example (107) shows that POSS.1PL is obligatory with the sun presumably via association with the speech community (‘the sun that shines over us’).

(105) [At the family dinner. There’s a fish pie and other dishes on the table.]

ολᾱη-ᾱλ-ᾱν χυλ-ᾱη **ńań-#(ew)** λɛ-λ-ew
beginning-POSS.3SG-LOC fish-PROP bread-POSS.1PL eat-NPST-1PL>SG
‘Let’s eat the fish pie first.’

(106) [Context from (105) with several fish pies.]

ολᾱη-ᾱλ-ᾱν χυλ-ᾱη **ńań-#(ew)** λɛ-λ-ᾱw
beginning-POSS.3SG-LOC fish-PROP bread-POSS.1PL eat-NPST-1PL
‘Let’s eat a fish pie first.’

(107) **χătl-#(ew)** εtm-ᾱs
sun-POSS.1PL appear-PST[3SG]
‘The sun came out.’

The obligatoriness of Associative Possessives with unique referents can be accounted for if one assumes that the uniqueness implication is a presupposition and the pragmatic principle *Maximize presupposition!* (Heim 1991) enforces the use of presupposing expressions such as the Associative Possessive in all contexts where their presuppositions are satisfied. Using the presupposition-less alternative to the presupposing expression in such a context, in our case a bare noun, triggers the antipresupposition that the referent is not unique (Percus 2006; and §2.1). I suggest that this is the reason for the infelicity of the bare noun in examples like (103), (105), and (107). The bare noun implies anti-uniqueness which clashes with the common ground resulting in a contradiction (cf. Magri 2009). As discussed in §2.1, this is the strategy used to explain the English Definite Article’s obligatoriness in uniqueness-implying

contexts in (Heim 1991; Percus 2006; Coppock & Beaver 2015), ditto for the other types of definites discussed in Chapter 2.

Based on these observations, I claim that the Associative Possessive presupposes uniqueness.⁹⁷ This, in turn, suggests that the Associative Possessive may present yet another definite marker type that differs from the other known types in also being a possessive marker, requiring an associative relation salient in the context and a possessor (as discussed in Chapter 4). Thus, this answers the challenge put forward by Beaver and Coppock (2016) who suggested that there may be no inherently-definite possessives in the world's languages (cf. §3.1.1)

The Associative Possessive also presupposes existence/familiarity of the referent, as the examples (108) and (109) show. In these examples, the speaker asks the addressee to make a bonfire in different contexts: in (109) the wood is already stacked, and the fire just needs to be lit, while in (108) a campfire has to be made from scratch. In (108) the referent is novel due to non-existence, while in (109) the referent is familiar due to its physical presence in the utterance situation. Associative Possessive marking is obligatory when the referent is familiar (and unique) and impossible when it is unknown.

(108) **tut-#(en)** wer-a
 fire-POSS.2SG do-IMP[SG]
 ‘Make a fire’.

Consultant's comment on *-en*: “[if the marked form is used] it's a bonfire in the place where we usually have bonfires, [with the bare noun] it's a bonfire in a new place”.

(109) **tut-#(en)** āλλ’-e
 fire-POSS.2SG light-IMP.SG>SG
 ‘Light the fire’.

Consultant's comment on *-∅*: “[only the marked form can be used,] if the firewood is already stacked and prepared”.

These observations bring the Associative Possessive even closer to Roberts’ (2003) analysis of English *the*. The Associative Possessive presupposes uniqueness and weak

⁹⁷ This observation was first made in the proceedings paper (Mikhailov 2021a).

familiarity, in addition to its ‘associative relation’ presupposition, making it a bona fide definite determiner.

This is further confirmed by its incompatibility with the epistemically nonspecific determiner *mul̥sər* (110), discussed above as cooccurring with the Proper Possessive (93). Here, just like in examples (103)-(104), the addressee and the DP referent are connected only by the contextual relation of physical proximity. The Associative Possessive is barred in the presence of the indefinite *mul̥sər*.

(110) [Context from (104) with several cups.]

mul̥sər	an-(#en)	mij-a
some.EN	cup-POSS.2SG	give-IMP[SG]

‘Give me some cup [no matter which].’

This fact is expected given the analysis proposed above. The Associative Possessive does not cooccur with the indefinite pronoun *mul̥sər* because it is definite: the pronoun and the Possessive carry mutually contradictory implications. This distinguishes the Associative Possessive from the Proper Possessive, which is only a possessive modifier, limited to inherent possessive relations and unmarked for definiteness according to the semantics in (96), see (111) (repeated from (93)).

(111) [The speaker is at a friend’s place.]

mul̥sər	an-en	mij-a
some.EN	cup-POSS.2SG	give-IMP[SG]

‘Give me any cup of yours.’

Example (112) shows the Associative Possessive used anaphorically in an elicited narrative. Here, the Associative Possessive has 1SG features, because it marks an associative relation between the DP referent and the speaker that was introduced in the preceding context.

(112) [“I found a passport on the street. I went to the administration office, where I met my friend and we talked.”]

nəm_nəpek-#(ɛm)	suwet-ən	χǎj-s-ɛm
passport-POSS.1SG	council-LOC	leave-PST-1SG>SG

‘[And then] I left the (lit. my) passport at the administration. [Let them find the owner and return it to him.]’

Other examples with the Associative Possessive appear in the discussion of the other Unpossessives below and in the following chapters as the Associative Possessive often provides an alternative marking option in contexts where the other Unpossessives are used.

One such example is (113) where a dog is, on the one hand, associated with the protagonist, Masha, and, on the other hand, it is salient. Thus, it may be marked with the Associative POSS.3SG *-λ* due to the association, but it may also be marked with the Salient Article *-en*, glossed as POSS.2SG here, due to its salience (see §3.6).

(113) [“Masha was walking down the street when she saw a big angry dog.”]

amp-λ/-en/#-Ø λ_{uw} pελ-aλ-a χurət-ti pit-əs
 dog-POSS.3SG/-POSS.2SG (s)he at-POSS.3SG-DAT bark-NFIN.NPST become-PST[3SG]
 ‘The dog started barking at her.’

This example, together with (107), also shows that the Associative Possessive may appear on subjects, which is important because other examples above were all of direct objects.

Example (114) provides a minimal pair to (113) where the protagonists are now 1DU. As expected, Associative POSS.1DU is an option here, along with the Salient Article as above.

(114) [“We (two) were walking down the street when we saw a big angry dog.”]

amp-εmən/-en min pελ-amən-a
 dog-POSS.1DU/-POSS.2SG we.DU at-POSS.1DU-DAT
 χurət-ti pit-əs
 bark-NFIN.NPST become-PST[3SG]
 ‘The dog started barking at us.’

It should be noted that in both these examples the Salient Article is more often provided first, but the Associative Possessive is still generally accepted as expected. (One type of contexts where there is a strong preference for the Associative Possessive over the Salient Article is the case of direct objects with third person subjects, see (142) and the discussion in §§5.2.4, 5.2.5, 5.4.)

One last examples is given in (115). Here, the ball is marked with Associative POSS.2NSG due to an association with the (dual) addressees of the speaker. The association

is based on the fact that the speaker knows that her addressees like to play with this ball. This is an example where the Associative Possessive is used with a promoted subject in a passive sentence.⁹⁸

(115) [Vasya and Katya are looking for a ball to play with at the kindergarten in the box with toys. The teacher, knowing that they love to play with the ball, says to them:]

m' ačok-ən	pet' aj-ən-ən	wu-s-i
ball-POSS.2NSG	P.-POSS.2SG-LOC	take-PST-PASS[3SG]

‘Petya’s already taken the ball.’

3.4.1. Analysis

The difference between the Proper Possessive and the Associative Possessive is fleshed out in the denotation of the Associative Possessive in (116), where a free possessive semantics (Vikner & Jensen 2002; Partee & Borschev 2003; Karvovskaya 2018) is combined with the semantics for WEAK DEFINITE ARTICLES using ι (see (19)). Compare the Proper Possessive denotation from (96) which is restricted to inherent relations and is not definite.

(116) DENOTATION OF THE ASSOCIATIVE POSSESSIVE

$$\llbracket \text{Assoc}_i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda x_e \iota y_e. R_i(x, y) \wedge P(y)$$

where R_i is a familiar free relation picked up from the context⁹⁹

In prose, the Associative Possessive applies to a nominal as a combined modifier and determiner: (i) it adds information about a context-dependent relation (R_i) and a possessor (x) that stands in this relation to the DP referent (y) and (ii) it returns this referent y while presupposing that that y is (weakly) familiar and uniquely satisfies $R_i(x, y)$ and $P(y)$ (which is achieved by ι). After it composes with the NP denotation and the Possessor denotation it yields a denotation of type e , just like other definites discussed in Chapter 2 and unlike the Proper Possessive.

⁹⁸ POSS.2SG on Petya in this example is the Proprial Article, see §3.7.

⁹⁹ $R_i(x, y)$ is a more intuitive shorthand in the spirit of (Partee 2011/1983) for the proper notation for a contextually-accessible relation supplied by the assignment function $g(i)(x, y)$. A fully explicit denotation for the Associative Possessive, using Heim & Kratzer’s (1998: 292) notation for indices of non-individual types (not type e), is as follows, where the complex index also represents the type of denotation retrieved from the context:

(i) $\llbracket \text{Assoc}_{\langle i, \langle e, \langle e, t \rangle \rangle} \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda x_e \iota y_e. g(\langle i, \langle e, \langle e, t \rangle \rangle \rangle)(x, y) \wedge P(y)$

Below I keep on using the shorthand above.

For the sake of illustration, let me analyze (115). Assume that it has the Logical Form in (117). The Associative Possessive-marked DP has a silent second-person dual *pro* representing the possessor/addressee. This gives us the truth conditions in (118a) with the contextually supplied relation in (118b) and the presupposition in (118c).

(117) $[_{TP} [_{PossP} pro.2DU_e [_{NP} ball_{\langle e,t \rangle}] Assoc_{i, \langle \langle e,t \rangle, \langle e,e \rangle} }]_e [_{VP} Petya_e take_{\langle e, \langle e,t \rangle \rangle}]]$

(118) a. $\llbracket (115) \rrbracket^{g,c} = 1$ iff $take'(p, \iota x. [R_i(a_c, y) \wedge ball'(x)])$

b. $R_i = \lambda x \lambda y. x$ likes to play with y

c. Presupposition: there is a unique ball that the addressee likes to play with

In prose, (115) is true if and only if Petya took the unique boy that the addressee (of the current context) likes to play with.

Furthermore, we can now account for the incompatibility of the Associative Possessive with the epistemically nonspecific determiner *muḷsər*. Assume the simplest indefinite denotation for *muḷsər* (119) (cf. (24)).¹⁰⁰ A DP like *muḷsər an-en* [some.EN cup-POSS.2SG] from (100) with the logical form in (120a) would be unable to compose (120b), as *muḷsər* requires a predicate $\langle e, t \rangle$, but the Associative Possessive yields a type e PossP.

(119) $\llbracket muḷsər \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle}. \lambda Q_{\langle e,t \rangle}. \exists x. P(x) \wedge Q(x)$

(120) a. $[_{DP} muḷsər_{\langle \langle e,t \rangle, \langle \langle e,t \rangle, t \rangle}] [_{PossP} pro.2SG_e [_{NP} cup_{\langle e,t \rangle}] Assoc_{i, \langle \langle e,t \rangle, \langle e,e \rangle} }]_e]$

b. $[\lambda P_{\langle e,t \rangle}. \lambda Q_{\langle e,t \rangle}. \exists x. P(x) \wedge Q(x)](\iota x. R_i(a_c, y) \wedge cup'(x)) \Rightarrow \#$, due to type clash

Thus, the proposed analysis covers the core facts of the Associative Possessive's distribution.

Let me note that the predictions that (116) makes for the availability of the Associative Possessive in specific contexts depend on the assumptions we make regarding what counts as an associative relation and what makes such a relation available for anaphoric pick-up by the Associative Possessive. In most examples discussed here, an associative relation arises as a result of the two referents interacting in a single situation

¹⁰⁰ A more serious analysis of *muḷsər* is that it requires the discourse referent of the DP to be unidentified in the current context in the sense of (Farkas 2002b). This roughly means that the speaker cannot match a specific actual individual to the discourse referent in question, cf. (Aloni & Degano 2022).

in the preceding contexts, and this is the case also for the examples discussed by Nikolaeva (2003) for the Ostyak dialect and in other research on Uralic unpossessives (see the references in Chapter 4). There were also cases where the association seems to be due to proximity in the speech situation (the cup examples) or due to background knowledge in the common ground (the ball example (115)).

My focus in this section has been on the Associative Possessive as a definite determiner, but the problem of the associative relation is also clearly an important matter for the semantics of this marker. What are the constraints and the rules regulating the associative relation? Are there parallels in the possessive constructions of other languages when it comes to free / associative relations (e.g., Storto 2005)? These questions will have to be left for future research, as my current data on the Associative Possessive is limited and does not shed more light on these issues.

3.5. Partitive Article

The Partitive Article (grammaticalized from POSS.3SG) marks nominals referring to part of a contextually specified set. In other words, it marks partitive-specific DPs (Enç 1991; Farkas 2002a; von Heusinger 2002; Farkas & Brasoveanu 2019).

Unlike the two Possessives discussed above, the Partitive Article only realizes the POSS.3SG cell of the possessive paradigm from §3.2. I argue in Chapter 4, following (Mikhailov 2021b), that this is because it grammaticalized out of a possessive agreement marker but it no longer has the morphosyntax of one. Thus, it presents our first truly clear example of an UNPOSSESSIVE marker.

An example of the use of the Partitive Article is given in (121). Here, the Partitive Article with the numeral *i* ‘one’ is used to refer to one of the two previously introduced dogs.

(121) [“I was walking down the street when I saw two dogs.”]

i	amp-#(əλ)	ma	pɛλ-am-a
one	dog-POSS.3SG	I	at-POSS.1SG-DAT
χurət-ti	pit-əs		
bark-NFIN.NPST	become-PST[3SG]		
‘One of the dogs started barking at me.’			

Using a bare noun in a context where the Partitive Article is appropriate leads to infelicity, which is a *Maximize presupposition!*-signature just like we’ve seen above for

the Associative Possessive (examples (103), (105), (107), and (109) and the other definites discussed in Chapter 2. Therefore, I assume that the Partitive Article triggers a partitive presupposition (‘the DP referent is a subset of a set of familiar referents’, cf. (89)).

The Partitive Article is impossible without a strongly familiar antecedent (122).

(122) [“What's going on outside?”]

amp-(#əλ) λəw pur-λ

dog-POSS.3SG bone gnaw-npst[3sg]

‘A dog is chewing on a bone.’ (Mikhailov & Tiutiunnikova 2024)

Additional examples with the Partitive Article from (Mikhailov 2021b) are provided below.

(123) **i** **an-əλ** mij-e

one cup-POSS.3SG give-IMP.SG>SG

‘[– Can I help you? – See the two cups on the table?] Give me one cup.’

(124) **kāt-ŋəλ** / **kāt** **eweŋəλ** ma wəλ-λam¹⁰¹

two-DU.POSS.3SG / two girl.DU.POSS.3SG I know[NPST]-1SG>NSG

‘[Some girls entered the classroom.] I knew two of them / two of the girls.’

At the same time, the Partitive Article is infelicitous when the DP referent is identical to the familiar referent from the preceding context (125). The Salient Article should be used instead, grammaticalized from the POSS.2SG (see the next section).

(125) [“I was walking down the street when I saw a dog.”]

#amp-əλ ma pəλ-am-a χurət-ti pit-əs

dog-POSS.3SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]

Int.: #The dog started barking at me.

The infelicity of (125) shows that the Kazym Khanty Partitive Article differs from some other partitive-specific markers such as the Turkish Accusative (§3.1.2, (Enç 1991)) or the Izhma Komi 3SG Unpossessive (Simonenko 2017), because it marks proper partitivity between the DP referent and its antecedent superset referent and it is

¹⁰¹ The suffix -ŋəλ [DU.POSS.3SG] is a portmanteau of the possessive dual -ŋəλ and POSS.3SG -əλ, see the paradigm in §3.2 and the morphological analysis sketched in Chapter 4.

infelicitous in the case of identity.¹⁰² Note that proper partitivity entails non-uniqueness which makes the Partitive Article somewhat complementary to the other Unpossessives which are all definite.

3.5.1. Analysis

I propose the following semantics for the Partitive Article. According to (126), the Partitive Article is an indefinite determiner which presupposes that the individual whose existence is asserted is a proper part of the strongly familiar antecedent referent $g(i)$. Excluding the partitive presupposition, this is equivalent to the basic semantics for the English Indefinite Article from Chapter 2, (24).¹⁰³

(126) DENOTATION OF THE PARTITIVE ARTICLE

$$\llbracket \text{Part}_i \rrbracket^g = \lambda P_{\langle e, t \rangle} \lambda Q_{\langle e, t \rangle} \exists x_e: x < g(i). P(x) \wedge Q(x)$$

An interesting observation is that the Partitive Article is possible with a conjunctive antecedent, like ‘meat and eggs’ in (127), which looks like an identity context, apparently contradicting the proper partitivity presupposition of (126). In terms of indices, it seems reasonable to assume that $\acute{n}u\chi\acute{\alpha}\lambda$ ‘meat’ in the target sentence of (127) carries the index i , same as the conjunct ‘meat’ in the preceding sentence, which would make this an anaphoric identity context violating the partitivity presupposition.

(127) [“— Why does the house smell like that?

— I left meat_i and eggs_k on the balcony for three days.”]

$\acute{n}u\chi\acute{\alpha}\lambda$ $\alpha\lambda pa$ $\acute{s}ak-s$

meat-POSS.3SG maybe spoil-PST[3SG]

‘Maybe the meat’s gone bad.’ (Mikhailov & Tiutiunnikova 2024)

This seems puzzling given the outright infelicity of the Partitive Article in identity contexts with a non-conjunctive antecedent.¹⁰⁴

¹⁰² According to (Simonenko 2017), the Meadow Mari 3SG Unpossessive in the partitive use is subject to similar restrictions as the Kazym Khanty Article. For a discussion of partitive uses of possessives in other Uralic languages, see (Nikolaeva 2003; Simonenko 2017; É. Kiss & Tánzos 2018) and Chapter 4.

¹⁰³ Alternatively, we could assume a modificational semantics for the Partitive Article which would yield a type $\langle e, t \rangle$ denotation (cf. the Proper Possessive (96)) and let ex introduce existential quantificational semantics. As far as I know, the Partitive Article does not combine with other determiners, unlike the Proper Possessive, which speaks in favor of hard-coding quantificational semantics in its denotation.

¹⁰⁴ This is also reported by Varvara Tiutiunnikova (p.c.) who is currently investigating the licensing conditions of the Partitive Article in more detail.

I assume that a conjunctive DP, apart from introducing/mentioning indices of the conjuncts, also introduces a complex index for the conjoined DP referent which we could represent as $\{i, k\}$. In the case of (127), this index would be mapped by the assignment function to a sum individual consisting of (this particular) meat and (these particular) eggs: $g(\{i, k\}) = \text{meat} \sqcup \text{egg-a} \sqcup \text{egg-b} \sqcup \text{egg-c}$, where $\sqcup$ is the mereological sum operation which forms sums of individuals (see Nouwen 2016), in this case the sum of meat and three different eggs. This gives us the denotation for the DP $\acute{n}\acute{u}\chi\acute{\alpha}\lambda$ ‘meat’ in (127).

$$(128) \llbracket \acute{n}\acute{u}\chi\acute{\alpha}\lambda_{\{i, k\}} \rrbracket^g = \lambda Q_{\langle e, t \rangle} \exists x_e : x < g(\{i, k\}). \text{meat}'(x) \wedge Q(x)$$

a generalized existential quantifier of ‘meat’ individuals that are a proper part of $g(\{i, k\})$ which maps every predicate Q to truth just in case there is such an individual of which Q also holds

So, according to my analysis, (127) is in fact also a proper partitivity context relative to the complex index of the conjunction $\{i, k\}$. Naturally, (127) may also be construed as a familiarity context (with the single index i ; see example (147)) which requires the use of a different Unpossessive, the Salient Article to which we now turn.

3.6. Salient Article

In (Mikhailov 2021b), I argued that the Salient Article is a definite determiner just like the Associative Possessive. There I called it a “topic marker” and suggested that it marks aboutness topics in the sense of (Strawson 1964; Reinhart 1981; Nikolaeva 2001). Here and especially in Chapter 5, I provide a more detailed investigation of the semantics of this marker. In Chapter 5, I will also show that the Salient Article is not limited to aboutness topics but rather it requires salience. (Of course, topichood and salience are correlated. See Chapter 5 and below.)

The Salient Article is formally equivalent to POSS.2SG, but, unlike the Proper and Associative Possessives, it never varies in person-number. Based on this, and several other arguments developed in Chapter 4, I contend that the Salient Article is a grammaticalized Unpossessive determiner.

As a definite determiner, the Salient Article is barred with novel referents (129) and is obligatory with (some) familiar (and unique) referents (130). (The following three examples form a single narrative, based on (Nikolaeva 2003: 139–140).)

(129) ma χot-a λuŋ-s-əm. šāta **sāldat-#(en)** oməs-λ.
 I house-DAT enter-PST-1SG there.LOC soldier-POSS.2SG sit-NPST[3SG]

‘I entered a house. There was a soldier sitting there.’

(130) ma **sāldat-(%en)** χuś-a wana mǎn-s-əm, puškan-ən šāš-s-əm
 I soldier-POSS.2SG at-DAT closer go-PST-1SG gun-LOC show-PST-1SG>SG

‘I came closer to the soldier and raised my gun at him.’

(131) **sāldat-#(en)** pakn-əs.
 soldier-POSS.2SG get.frightened-PST[3SG]

‘The soldier got frightened.’

Similarly to the Associative Possessive, using a bare noun in a context where the Salient Article is required triggers an anti-familiarity implication (132) (based on (125) with the Partitive Article). I attribute this to *Maximize presupposition!* (Heim 1991) enforcing the use of the Salient Article as a presuppositional item (cf. §3.4 and §2.1).

(132) [“I was walking down the street when I saw a dog.”]

amp-#(en) ma pɛλ-am-a χurət-ti pit-əs
 dog-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]

‘The dog started barking at me.’

Consultant’s comment on Ø: “Then it’s some other dog, not clear which”.

To these examples, we can add the following example, which shows that the Salient Article also requires uniqueness. Example (133a) shows the Salient Article used with a maximal plural (dual) referent introduced in prior context.¹⁰⁵ In the same context, a non-maximal DP does not admit the Salient Article (133b) (repeated with additions from (121)). The Partitive Article POSS.3SG must be used instead (see §3.5).

(133) [“I was walking along the street when I saw two dogs.”]

a. **amp-ηəλ-an** ma pɛλ-am-a χurət-ti pit-s-əŋən
 dog-DU-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST-3DU

‘The dogs started barking at me.’

b. **i amp-əλ/#-en** ma pɛλ-am-a
 one dog-POSS.3SG/-POSS.2SG I at-POSS.1SG-DAT

¹⁰⁵ The reader interested in the semantics of the Kazym Khanty dual is referred to (Golosov 2024b).

χurət-ti pit-əs
bark-NFIN.NPST become-PST[3SG]
‘One of the dogs started barking at me.’

Example (134) shows that uniqueness, however, is not a sufficient condition for the Salient Article’s use. The example is modeled on the English example (135), repeated from (20), where the English Definite Article is felicitous.

(134) [“I recently found a box on my cupboard.”]

ma nuχ puš-s-εm, λaraś λipij-ən
I up open-PST.1SG>SG box inside-LOC
wəλ-əm χur-(#en) nuχ wu-s-εm
be-NFIN.PST image-POSS.2SG up take-PST-1SG>SG

‘I opened it and took out the photo that was inside the box. [In the photo was my father in youth.]’

(135) I found a box in my attic the other day. I opened the lid and pushed **the button I found inside**. You won’t believe what happened. (adapted from Roberts 2003: 209)

Roberts has argued that in cases like (135) the English Definite Article can be used thanks to accommodation of semantic uniqueness and existence of the described individual (‘there was one and only one button inside the box’) which satisfies the Definite Article’s informational uniqueness presupposition (‘the DP referent is unique among familiar discourse referents in satisfying this description’). The fact that the Salient Article cannot be used in such a context (134) suggests that accommodation for some reason is not available here. The natural explanation is that the Salient Article’s presupposition is in fact stronger than informational uniqueness (among weakly familiar discourse referents), and hence it cannot be so easily accommodated. Thus, uniqueness is not a sufficient condition for the Salient Article.

Example (136), repeated with adjustments from (112), shows that strong familiarity is not enough for the Salient Article as well. Here a passport was introduced in the preceding context, which makes it strongly familiar (even if the story digresses between the two mentions of the passport). Nevertheless, the Salient Article is excluded here, and the Associative POSS.1SG must be used instead (see §3.4). Thus, strong familiarity and uniqueness cannot be all there is to Salient Article’s conditions of use.

(136) [“I found somebody’s passport in the street. I went to the town administration. Met a friend there and talked to her for some time.”]

nɛm nɛpek-ɛm/#-en suwet-ən χǎj-s-ɛm
 name paper-POSS.1SG/-POSS.2SG council-LOC leave-PST-1SG>SG
 ‘[Then] I left the passport at the administration. [Let them find the owner.]’

At first glance, these two examples support the assumption from (Mikhailov 2021b) that the Salient Article requires the referent to be topical. Indeed, if topicality is the property a referent has if the utterance communicates new information **about it** (aboutness topic (Reinhart 1981; Strawson 1964; Lambrecht 1994; Nikolaeva 2001)), the passport in (136) is clearly not topical. Specifically, we may say that a topical referent must be given in the QUD (question under discussion) that the utterance addresses (Roberts 2012; 2019).¹⁰⁶ Since (136) presents a continuation of the narrative with the speaker as its single actor, this utterance answers a question such as “What did you do next?” or “What happened next?”. From this we conclude that the passport in (136) is not topical because it is not part of the QUD which (136) answers. (And similar reasoning could be applied to (134).) This explains the infelicity of the Salient Article in this example under the assumption from (Mikhailov 2021b).

However, not all uses of the Salient Article fit into the logic described above, as there are cases in which the DP marked with the Salient Article is nevertheless in narrow focus, e.g., as in (137). In this example, a DP with a relative clause and a Salient Article refers to one of the two girls under discussion as part of an answer to the question “Who will play the sunshine?”. According to the above reasoning, the answer part to such a question, this very DP, bears narrow focus. The use of the Salient Article, however, is obligatory here, contra the topicality proposal of (Mikhailov 2021b).

(137) [We are putting on a play at the school theatre. Two girls have not yet been assigned a role. We discuss their strengths to decide who will play the sun: “— The girl on the left is tall. — The girl on the right is a good singer. So, who will play the sun?]

jǎm pɛlək-ən oməs-ti ewen / #ewi at junt-λ

¹⁰⁶ In Roberts’ (2012) QUD model each utterance in a discourse answers a (possibly implicit) question. Answers are required to be congruent to QUDs and the constituent which corresponds to the question part of the QUD must be focused. See Chapter 5 for details.

good side-LOC sit-NFIN.NPST girl.POSS.2SG / girl OPT play-NPST[3SG]
 ‘Let the girl on the right play [the sun].’

Instead, I propose that the Salient Article marks the relatively maximally salient referent in the sense of (Roberts 2011; Barlew 2014) as discussed in Chapter 2. This is motivated by the examples below. A referent is at the center of the interlocutors’ attention and is therefore salient (§2.3) in the following cases: if it is relevant to the interlocutors’ goals, or it has been recently mentioned, or it is perceptually salient in the extralinguistic context. Any topical referents are salient by definition, but this is not true only for them.

3.6.1. *The requirement of salience*

In (137) above the girls under discussion are salient because they are relevant to the interlocutors’ goals. Another case of referent salience occurs with DPs with deictic demonstratives, as in (138). In the deictic use of a demonstrative, which is accompanied by a pointing gesture, the speaker directs the addressee’s attention to a certain referent (Diessel 2006; Reisinger & Huijsmans 2021; Stojnić 2021: Ch. 4; Huijsmans & Reisinger 2022). As a result, the referent is salient, and the conditions of use of the Salient Article are met, as in (138).

(138) [“Who hit you?”]

ma	ši	aj_iken-ən	χăts-əs-ij-əm
I	DEM	boy.POSS.2SG-LOC	hit-PST-PASS-1SG

‘I was hit by that boy.’

It should be noted that uses with DPs with deictic demonstratives are extremely common for the Salient Article (e.g., see examples with demonstratives from Marinina 2024). Many such cases clearly indicate the irrelevance of information-structural distinctions for its use (contra Mikhailov 2021b). For instance, in (138), the DP with the demonstrative is under narrow focus, but the Salient Article is used with it despite this. (See also the more detailed discussion and analysis in Chapter 5.)

The fact that the Kazym Khanty Salient Article requires salience brings it close to the Bulu Salient Article (Barlew 2014), as well as to anaphoric/salience-based demonstratives (Reisinger & Huijsmans 2021; Huijsmans & Reisinger 2022; Marinina 2024). There is a crucial difference with the Bulu Salient Article, however, which is apparent in Roberts’ ‘salience without uniqueness’ contexts.

Recall from Chapter 2 that definite determiners crosslinguistically, excluding demonstratives but including the Bulu Salient Article, are barred in ‘salience without uniqueness’ contexts because these contexts incur a violation of informational uniqueness. This is not a problem for personal pronouns, which are the preferred form in these contexts crosslinguistically, because pronouns presuppose relative maximal salience on Roberts’ approach.¹⁰⁷

As it turns out, the Kazym Khanty Salient Article does **not** require informational uniqueness, at least not merely among salient discourse referents, as can be seen from its felicitous use in the ‘salience without uniqueness’ example (139).

(139) [A theatrical play for children: “A girl_k appeared from the room on the left. Another girl_k appeared from the room on the right.”]

λəw_{k/*i} / ^{OK}ewen_{k/*i} još-ən lipət təj-əs
 (s)he / girl.POSS.2SG hand-LOC flower have-PST[3SG]
 ‘She_{k/*i} [lit.: the girl] was holding flowers.’

As expected, using the personal pronoun is felicitous in (139) and is also the first choice for most of my consultants. However, unlike the definite determiners in English, Akan, and Bulu, the Salient Article in (139) is also acceptable (accepted by all my consultants). This presents a striking difference between the Kazym Khanty Salient Article and the definite determiners of other languages, especially the Bulu Salient Article, showing that the Kazym Khanty Salient Article indeed requires (relative) maximal salience as proposed above.

3.6.2. Analysis

Based on this, I adopt Roberts’ (2003: (49)) analysis of personal pronouns (Chapter 2, (42)) for the Kazym Khanty Salient Article in (140).

(140) DENOTATION FOR THE SALIENT ARTICLE

¹⁰⁷ A moment’s reflection tells me that demonstratives are also expected to be available in these contexts (i), in agreement with (Reisinger & Huijsmans 2021; Huijsmans & Reisinger 2022) salience-based analysis. However, neither Huijsmans and Reisinger, nor Marinina (2024) report such data.

(i) RUSSIAN (own knowledge)
 Sleva na scenu vyšla ženšina. Sprava vyšla eščë odna ženšina. U neě [by her.GEN] / u etoy ženšiny [by this.GEN woman.GEN] v rukax byli cvety.
 ‘A woman entered from stage left. Another woman entered from stage right. She / this woman had flowers in her hands.’

$$\llbracket \text{Sal}_i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle}: \text{sal}(i, c) \wedge \forall j \in \text{Dom}(g)[P(g(j)) \wedge j \geq_{\text{Sal}(c)} i \rightarrow j = i]. g(i)$$

The function in (140) takes a predicate P and is defined iff the index i is salient in the context c and it is the **most** salient discourse referent among strongly familiar P -discourse referents (presupposition), when defined, it yields the individual corresponding to the discourse referent i (at-issue content). An additional piece of notation, that was introduced in Chapter 2 following Roberts, is the partial order over salient discourse referents $\geq_{\text{Sal}(c)}: j \geq_{\text{Sal}(c)} i$ is true just in case j is at least as salient as in c as i .

This denotation derives the correct predictions for the ‘salience without uniqueness’ example in (139) as only the most recently mentioned referent is considered the maximally salient discourse referent, while the other, less salient portions of discourse are ‘screened out’ in Roberts’ terms (for a formal account of this effect and a demonstration of how the analysis works, see §5.5.2).

Thus, I propose that the Kazym Khanty Salient Article is a definite determiner that requires relative maximal salience, like personal pronouns and ʔayʔajuθəm Salience Demonstratives from Chapter 2. This is the strongest requirement on the Salience-Familiarity Scale (repeated from (61)) excluding absolute maximal salience which is not attested for determiners.

(141) THE SALIENCE-FAMILIARITY SCALE (main version)¹⁰⁸

weak familiarity > strong familiarity > salience > relative maximal salience >
absolute maximal salience

3.6.3. A note on differences with (Mikhailov 2024)

Before moving on, I need to comment briefly on the differences with the analysis proposed for the Salient Article in (Mikhailov 2024), the paper which forms the basis for Chapter 5. The concern of (Mikhailov 2024), and of Chapter 5, is with figuring out the conditions of use of the Salient Article and, specifically, with the question what factors do **not** play a role contrary to previously voiced suggestions. As summarized above, I argue against the Salient Article being sensitive to information-structural distinctions, and

¹⁰⁸ The reason why (61) is called the main version of the SFS is that I suggest a possible extension including partitive specificity in §3.1.2, which, however, is not very important for this study.

I also critically investigate the obviation hypothesis first proposed in the report (Muravyev 2022).

Muravyev showed that Kazym Khanty grammar features properties of obviation systems as found among others in Algonquian languages (Goddard 1990; Aissen 1997; 2001). He floated the conjecture that the Salient Article is a proximate marker, signaling prominent third person participants and subject to the proximate uniqueness constraint: there may be at most one proximate third person per obviation span (e.g., per clause).

The motivation for this is found in examples such as (142). It appears that only one third person DP may be marked with the Salient Article within a single clause, as (142) shows. The less topical of the two DPs must be marked with the Associative POSS.3SG instead (cf. §3.4), a second Salient Article is unavailable even though it appears that the presuppositions of (140) are not violated, and the boy is the secondary topic in (142) in the sense of (Nikolaeva 2001). (The Associative Possessive is licensed because the boy is associated with the girl via the preceding context.)

(142) [“I was walking along the street when I saw a boy and a girl.”]

ewen **aj_ikeλ** / #**aj_iken** χătś-əs-λe
 girl.POSS.2SG boy.POSS.3SG / boy.POSS.2SG hit-PST-3SG>SG
 ‘The girl hit the boy.’

In (Mikhailov 2024), I argued for a semantic treatment of the proximate uniqueness effects observed with the Salient Article in examples like (142). There I proposed a presupposition of maximal salience among discourse referents simpliciter (‘global maximal salience’, see §5.4.1). In the case of (142), the second Salient Article is barred because there may be at most one maximally salient discourse referent per utterance.

The arguments for that analysis as well as the connection with the demonstratives are unpacked in Chapter 5. In §5.4.2, I discuss the reasons for eschewing the analysis of (Mikhailov 2024) in favor of (140). The main reason is that the (Mikhailov 2024) analysis severely undergenerates as there are numerous examples where proximate uniqueness effects are **not** observed with the Salient Article. Here is just one example like this (143) (see §5.2.5 for more). In (143) which is a minimal pair to (142), a second Salient Article is just as acceptable as the Associative POSS.3SG. The only relevant difference is the syntactic role of the second DP which is a complement of a postposition in (143) as compared to the direct object in (142).

(143) [“I was walking along the street when I saw a boy and a girl.”]

aj_iken	eweλ/ewen	χuś-a	măn-əs
boy.POSS.2SG	girl.POSS.3SG /girl.POSS.2SG	at-DAT	go-PST[3SG]
pa	lipt-ən	mă-s-λe	
add	flower-LOC	give-PST-3SG>SG	

‘The boy came closer to the girl and gave her a flower.’

My conclusion in §5.4.2 is somewhat less optimistic than that of (Mikhailov 2024). I believe the proximate uniqueness effects of the Salient Article are not a semantic phenomenon and are thus beyond the scope of this work. See (Muravyev et al. 2024) and §5.4.2 for some suggestions for solutions.

3.6.4. *Salient Article vs. other Unpossessives*

Having proposed analyses for three of the four Kazym Khanty Unpossessives, it is time to consider if and how they compete. We saw in Chapter 2 that in determiner systems with several determiners they may be in complementary distribution. I will now show that the Salient Article does not compete with either the Partitive Article or the Associative Possessive. That is, there are contexts where they are freely interchangeable, as, indeed, is to be expected based on the analyses proposed above for the three determiners.

Example (143) already shows a context where both the Associative Possessive and the Salient Article are viable marking options for the same DP, in this case a direct object. Example (144) shows the same in the subject position. The thief is clearly very salient here so the Salient Article is available, but he is also associated with the computer that he stole, so an Associative POSS.3SG is also an option here, preferred by some consultants.¹⁰⁹

(144) [“— A computer was stolen from our museum. — Did they find the thief yet?”]

pa	muj,	mulχatəλ	λoλmax-en/-əλ/%-Ø	wəjət-s-a
ADD	what	yesterday	thief-POSS.2SG/-POSS.3SG	find-PST-PASS[3SG]

‘You betcha! Yesterday, the thief was found [and the computer was returned].’

Thus, the Salient Article and the Associative Possessive do not seem to compete in a way like, say, the English Definite and Indefinite Articles compete via *Maximize presupposition!* (MP; cf. §2.1). To see why this is expected, consider again the

¹⁰⁹ More variations on this context are discussed in §5.2.2ff.

denotations proposed above for the two Kazym Khanty determiners, repeated from (116) and (140), respectively:

$$(145) \llbracket \text{Assoc}_i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda x_e \iota y_e. R_i(x, y) \wedge P(y)$$

where R_i is a familiar free relation picked up from the context

$$(146) \llbracket \text{Sal}_i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} : \text{sal}(i, c) \wedge \forall j \in \text{Dom}(g) [P(g(j)) \wedge j \geq_{\text{Sal}(c)} i \rightarrow j = i]. g(i)$$

The Associative Possessive presupposes uniqueness, familiarity, and the existence of an associative relation R_i between the DP referent and the possessor argument. If these presuppositions are satisfied for a given referent, the Associative Possessive may be used for it. The Salient Article presupposes salience of the referent and relative maximal salience among familiar discourse referents satisfying the DP description. It is easy to imagine contexts, where a referent is maximally salient but it is not associated with any other familiar referent (there is no R_i), and thus the Salient Article is available but the Associative Possessive is not (see the many examples at the beginning of §3.6).¹¹⁰ And vice versa, there are contexts where a referent is associated with another familiar referent, but it is not salient, so the Associative Possessive is available, but the Salient Article is not (e.g., (137)). Thus, it is clear that neither determiner's presupposition entails the presupposition of the other determiner, that is, neither determiner has a strictly stronger presupposition than the other one. This means that they are not subject to MP and do not compete.

The same holds for the Salient Article and the Partitive Article. Example (147) is an expanded version of (121) which shows that the Salient Article can also be used with

¹¹⁰ To be sure, this reasoning depends on what counts as an associative relation and what determines whether an associative relation is available for the Associative Possessive to pick it up. I have not provided a proper discussion of this matter, in part, because I do not have enough data to deal with this issue. One thing I can say is that sometimes when the 'protagonist' of a narrative-like sequence of examples is the speaker, as in many examples at the beginning of §3.6, the Associative POSS.1SG seems to be dispreferred (i) (based on (132)). (This is so even though the Associative POSS.1SG is in principle possible, as (136)/(112) shows.)

(i) **amp-en/%-εm** ma peλ-am-a χurət-ti pit-əs
 dog-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]
 'I was walking down the street when I saw a dog.] The dog started barking at me.'

This is not the case for third-person protagonists however (ii) (repeated from (113)), but the reasons for this are unclear and have to be left for future research.

(ii) ["Masha was walking down the street when she saw a big angry dog."]
amp-λ/-en/#-∅ λuw peλ-aλ-a χurət-ti pit-əs
 dog-POSS.3SG/-POSS.2SG (s)he at-POSS.3SG-DAT bark-NFIN.NPST become-PST[3SG]
 'The dog started barking at her.'

the familiar (and unique) ‘meat’ introduced by the conjunctive antecedent in the preceding context. This shows that, in some contexts, the Salient Article and the Partitive Article are not in complementary distribution, which is to be expected given that neither has a strictly stronger presupposition than the other according to the analysis proposed for them in (126) (repeated in (148)) and (140)/(146).

(147) [“— Why does the house smell like that?

— I left meat_i and eggs_k on the balcony for three days.”]

ńux-eni/-əλ_{i, k} aλpa šak-s

meat-POSS.2SG/-POSS.3SG maybe spoil-PST[3SG]

‘Maybe the meat’s gone bad.’¹¹¹

(148) $[[\text{Part}_i]]^{g,c} = \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle} \exists x_e: x < g(i). P(x) \wedge Q(x)$

With the Partitive Article, we could expect the two markers to compete, given its similarity to the unmarked INDEFINITE ARTICLE denotation, but again, the data clearly show that the two markers do not compete.

As for the Associative Possessive, we saw in §3.4 that it too requires uniqueness/maximality, so it is excluded in properly partitive contexts. Given that the Partitive Article is available with conjunctive antecedents as in (147), I expect that one can construct a similar context for the Associative Possessive and the Partitive Article, where both markers will be available, because neither marker’s presupposition is stronger than the other one’s. An example like this has to be collected in the future, as I do not have a suitable context in the currently available data.

Having described and discussed the semantics of three Unpossessives, we are ready to turn to the final one, the Proprial Article. The comparisons between it and the other determiners will be made below.

3.7. Proprial Article

The last Unpossessive for us to discuss is the Proprial Article. It is also formally equivalent to POSS.2SG like the Salient Article and it marks human names in argument

¹¹¹ The version of this example with the Salient Article is due to Varvara Tiutiunnikova (p.c.). The indices on the two suffixes indicate that they have different antecedents: the Salient Article *-eni* has as antecedent only the ‘meat’ conjunct, indexed *i*, of the preceding conjoined DP, while the Partitive Article *-əλ_{i, k}* has as antecedent the whole conjunction as signified by the complex index $\{i, k\}$.

positions.¹¹² Unlike the other Unpossessives discussed here, the motivation for treating the Proprial Article as a marker that is synchronically independent from the Salient Article is purely semantic, and so, I will not be delaying its presentation until Chapter 4 like I did with the other Unpossessives (§3.7.3). Thus, I will compare the Proprial Article and the Salient Article in detail here. (A brief comparison with the Associative Possessive is made in §3.8, as for the Partitive Article, I unfortunately do not have data on its cooccurrence with proper names, and so a comparison with it is a task for future research.)

Below (§3.7.2), I show that the Proprial Article fits the profile of PROPRIAL ARTICLES formulated by Muñoz (2019). According to Muñoz, a PROPRIAL ARTICLE takes names of predicate type $\langle e, t \rangle$ and derives DPs of the individual type e which refer to a unique and familiar individual that bears this name. A PROPRIAL ARTICLE differs from the other types of definites surveyed in Chapter 2 in that it derives **intensionally rigid** descriptions, rigid designators in the sense of (Kripke 1980): PROPRIAL descriptions cannot covary in reference with a modal/temporal quantifier unlike other types of definites. A PROPRIAL description (a proper name) being a rigid designator picks out the same individual for any circumstance of evaluation, the same individual at whatever past or future time we talk about and in whatever non-actual/counterfactual world we consider (Kripke 1980: 48–49).¹¹³

Here is Muñoz’s story for English.

3.7.1. Muñoz (2019) on English

Muñoz (2019) has argued that English names are marked with an unpronounced proprial article akin to those of the languages that use a dedicated morpheme for it (see the references on Tagalog, Māori, Catalan, Icelandic, etc. in the original paper). Muñoz

¹¹² Other types of proper names such as toponyms are not marked with the Proprial Article.

(i) **maskwaj-#en-ən** atəm tərəm
M.-POSS.2SG-LOC bad sky
‘In Moscow the weather is bad’.

¹¹³ Kripke was mostly concerned with rigidity vis-à-vis counterfactual worlds or situations, i.e., situations that somehow differ from the actual world/situation, but the rigidity of proper names also holds vis-à-vis different times, past or future, as we will see below. This is so because a nonrigid designator, such as a definite description, might pick out a different referent depending on the time of evaluation, just as well as depending on the situation of evaluation. But a proper name picks out the same individual at all times and situations. For instance, *the fastest Formula One driver* (nonrigid description) picks out different drivers at different years (perhaps, for 1991 it was Ayrton Senna, and for 1994 it was Michael Schumacher), but the name *Michael Schumacher* (rigid designator) always picks out the same individual, regardless of the circumstances of evaluation.

adopts the hypothesis that names denote predicates (or intensions of predicates, i.e. properties) like common nouns do (type $\langle e, t \rangle$) and, thus, require a PROPRIAL ARTICLE to become arguments (cf. Matushansky 2008). This does not happen in vocative and predicative positions since they do not require type e DPs, unlike argument positions.

The argument for a covert proprial article in English is based on the observation that proper names pass some of the crucial tests for the semantic type of predicates $\langle e, t \rangle$. For example, proper names occur in the vocative position, unlike NPs with determiners (149).

- (149) a. Dinner is ready, (*most) friends / (*the) guest / (*some) brothers.
 b. It's the economy, stupid.
 c. Ede, the pizza is ready. (Eckardt 2014: (1), (6), (15))

It would be quite counterintuitive if all expressions in vocative position were of type $\langle e, t \rangle$ with the exception of proper names which are usually assumed to be of type e .¹¹⁴

Muñoz develops an analysis in which the PROPRIAL ARTICLE only combines with unmodified proper names (of type $\langle e, t \rangle$) to yield the individual that bears that name. Naturally, this only happens in argument positions, since only in argument positions is a proper name required to be of type e .

It is predicted that if a language has an overt PROPRIAL ARTICLE, it will be missing in non-referring non-argument positions. This prediction is borne out in my Kazym Khanty data.

Muñoz shows that PROPRIAL ARTICLE-marked names, unlike names marked with a DEFINITE ARTICLE, **cannot covary with a (modal or temporal) quantifier to denote different referents in different circumstances** (they are intensionally rigid). Consider (150) which is uttered in the context of a discussion among teachers: in (150a), the speaker claims that a particular student named Smith always cheats (the same student in different circumstances), while in (150b) the claim is that for every exam situation, the student named Smith cheats (possibly different students in different circumstances). This

¹¹⁴ Eckardt actually maintains this latter assumption introducing a type-shifting operation which allows proper names to be predicates in vocative position (Eckardt 2014: 277). However, this is clearly a stipulation and — as Muñoz shows — the view of proper names as predicates is superior in accounting for the data of languages with PROPRIAL ARTICLES, as well as for English.

is because the Definite Article derives nonrigid DPs (Kripke 1980), while the proprial article derives rigid ones.

(150) a. **Smith** always cheats.

b. **The Smith** always cheats. (Muñoz 2019: 7)

3.7.2. Back to Kazym Khanty data

The Kazym Khanty Proprial Article is obligatory with human names in argument positions (151)-(152). Unlike with the other Unpossessives, using a bare noun is simply never an option here. This already indicates a difference with the Salient Article (132).¹¹⁵

(151) **wəntɾ-*(en)** jʊχ šəp sewr-əs

A.-POSS.2SG wood piece cleave-PST[3SG]

‘Andrej cleaved a log.’

(152) [Vasya and Katya are looking for a ball at the kindergarten. The teacher says to them:]

mʼačok-ən **petʼaj-*(en)-ən** wʉ-s-i

ball-POSS.2NSG P.-POSS.2SG-LOC take-PST-PASS[3SG]

‘Petya’s already taken the ball.’ (repeated with adjustments from (115))¹¹⁶

The Proprial Article is barred in vocative and predicative positions (153)-(155). This is expected under Muñoz’s (2019) analysis because these positions require predicative nominals of type $\langle e, t \rangle$.

(153) **maša** / ***mašaj-en**, ow-en pʉnš-e

M. / M.-POSS.2SG door-POSS.2SG open-IMP.SG>SG

‘Masha, open the door.’

(154) ma λəχs-ɛm nɛm-əl **lʼoša** / ***lʼošaj-en**

I friend-POSS.1SG name-POSS.3SG L. / L.-POSS.2SG

‘My friend’s name is Liosha.’

(155) [My friend recently gave birth to a boy.]

¹¹⁵ Some of my consultants freely allow bare proper names in argument positions, which may indicate that the Proprial Article is not grammaticalized in their idiolects. In this section, I consider only the judgments of speakers who do not accept bare argumental proper names.

¹¹⁶ The POSS.2NSG on the ball in this example is an Associative Possessive as in ‘the ball you’re looking for’ or ‘the ball you usually play with’. I specifically indicated during elicitation that the ball does not belong to the two children.

ma λəχs-εm λəwət nεmt-s-əλλε **miša-ja/*-jen-a**
 my friend-POSS.1SG 3SG.ACC name-PST-3SG.SG m.-DAT/-POSS.2SG-DAT
 ‘My friend named him Misha.’

The second important difference from the Salient Article is that the Proprial Article does not appear to have salience restrictions, since it does mark human names even when their referents are not salient. This is demonstrated in (156) where the context sets up an (implicit) question under discussion along the lines of “Who will look after the children while the parents are away?”. Andrej is not salient here, as he was not mentioned in the preceding context, but he is still marked with the Proprial Article.

(156) [Andrej is a caretaker in the kindergarten. The parents are going away for the weekend and they tell their children.]

ńawrεm-ət, nin **wεntr-en-ən** λawəλ-aj-ti
 child-PL, you.PL A.-POSS.2SG-LOC baby.sit[NPST]-PASS-2PL
 ‘Children, Andrej will look after you.’

Thus, we can suggest an explanation for fact that the Salient Article may be substituted for a bare noun which triggers an anti-familiarity inference, but the Proprial Article simply cannot be substituted for a bare noun. The Salient Article requires strong familiarity and salience while the Proprial Article does not. Its presuppositions are a lot less restrictive than those of the Salient Article which provides for a much wider distribution of the Proprial Article with much less contexts where a bare name may surface.

The third, most important distinguishing property of the Proprial Article is that it derives rigid DPs, which is not the case for the Salient Article.

Thus, a Salient Article-marked DP such as the ‘dog’ in (157) may refer to different dogs in different circumstances of evaluation. This is confirmed by the acceptability of the continuation of (157) given in brackets that implies different dogs.

(157) kašəŋ śos amp šiwaλ-t-εm-ən
 every hour dog see-NFIN.NPST-1SG-LOC
amp-en ma pελ-am-a χurət-λ
 dog-POSS.2SG I at-POSS.1SG-DAT bark-NPST[3SG]

‘Every time I meet a dog, the dog barks at me. [Sometimes it is a big dog, sometimes it is a smaller dog.]’

On the other hand, a Proprial Article-marked name cannot refer to different referents in different circumstances (158a). Instead, a bare name must be used to get a nonrigid reading of the DP. (Another option provided by one of my consultants was to use a common NP like ‘boy named Vasya’ (158b).)

(158) [“Every year we give a present to the 4th year student who gets the best grades.”]

a. kaʃəŋ ol mojləpsi wəjl’ **waśa** / #**waśaj-en**¹¹⁷

every year present take[NPST.3SG] V. / V.-POSS.2SG

‘Every year the present is taken by a Vasya. [Last year it was Vasya Tas’manov and this year it’s Vasya Tarlin.]’

Consultant’s comment on *-en*: “then it’s the same Vasya each year / some particular Vasya”.

b. kaʃəŋ ol **waśa nəm-əp** **ńawrəm** śi mojləpsi wəjl’

every year V. name-PROP child DEM present take[NPST.3SG]

(Same as above.)

Thus, the Kazym Khanty Proprial Article derives rigid DPs, while the Salient Article derives nonrigid, intensionally variable DPs, exactly as in English according to Muñoz. Based on this difference and the fact that the Proprial Article has no salience restrictions (see above), I claim that the Kazym Khanty Salient Article and the Kazym Khanty Proprial Article are mutually independent and must be treated as distinct morphemes.

We also see that the context where bare names become an option is the context of intensional nonrigidity such as (230). This suggests that the Proprial Article presupposes

¹¹⁷ An anonymous reviewer for *Ural-Altaic Studies* asks whether a focused agent is expected to require a Passive construction in an example like (65a). In the Kazym dialect, a focused agent does not necessarily require a Passive construction as documented by Muravyev, e.g., in (Muravyev 2023). In (65), the fact that the agent is animate, and the patient is inanimate is enough to maintain an active construction.

The reviewer also wonders about the validity of this example. This example was approved by three consultants, one of whom used the Passive of the verb *māti* ‘to give’ instead. Another example demonstrating that Proprial Article-marked names cannot refer to different referents in different circumstances received the judgements of four more consultants. Overall, the judgements of seven different speakers of the Kazym dialect show that Proprial Article-marked names are indeed rigid as (230) is intended to show.

intensional rigidity, and its use is barred in contexts of nonrigidity. Muñoz proposes a denotation for PROPRIAL ARTICLES that includes this presupposition, and I adopt it for the Kazym Khanty Proprial Article (with a selectional restriction to human names) in (160). But first, a few additional assumptions need to be incorporated into our framework to account for intensional rigidity.

3.7.3. A situation-semantic rendition of (Muñoz 2019)

First of all, we will now additionally assign denotations relative to situations (type s) in a Kratzer-style possibilistic Situation Semantics, adding a situation parameter of interpretation to our interpretation function: $\llbracket \cdot \rrbracket^{g,s,c}$.¹¹⁸ Situations should be understood roughly as specific parts of worlds at particular times. Situations provide a handy way to model different circumstances of evaluation, including both worlds and times. Worlds are understood as maximal situations and for every situation s' , there is a world $w_{s'}$ which is the world of this situation: $s' \leq w_{s'}$. Predicates are now true in situations and sentences denote sets of situations, true in a situation s' iff s' is included in the set of situations denoted by this sentence.¹¹⁹

In non-intensional contexts, the situation parameter is interpreted as the actual world of the current context, w_c .¹²⁰ In intensional contexts, an intensional quantifier shifts the situation parameter to the situations/worlds that it quantifies over, along the lines of (von Fintel & Heim 1997-2023). Examples of intensional quantifiers include a modal like *might* and an adverb of quantification like *kašəŋ šos* ‘every time’ or *kašəŋ ol* ‘every year’. Instead of the extension of an expression α , namely, $\llbracket \alpha \rrbracket^{g,s,c}$, intensional quantifiers require the **intension** of the expression they compose with, namely, $\llbracket \alpha \rrbracket_{\mathfrak{c}}^{g,c}$, which is a

¹¹⁸ For this variety of Situation Semantics, see (Portner 2009: §4.4.2; Schwarz 2009; Kratzer 2012: Ch. 5; 2023; Elbourne 2013). I opt for a situation-semantic framework instead of Muñoz’s system with worlds only, because situations are more handy for dealing with adverbs of quantification *kašəŋ šos* ‘every time’ and *kašəŋ ol* ‘every year’ in (157)-(158). As we’ve seen in Chapter 2, Situation Semantics is also the favored framework of a range of studies dealing with definiteness. The system I use in this section is something of a mix between Kratzer-style Situation Semantics and Muñoz’s system because of the use of an interpretation parameter instead of syntactically-represented situation pronouns, but this is merely a matter of simplification. I try to keep the formal details to a minimum just to illustrate the main benefits of the proposed analysis.

¹¹⁹ The set of situations denoted by a sentence is the set of situations which exemplify it, see (Kratzer 2023).

¹²⁰ Alternatively, we could take it to be the Austinian topic situation, i.e., the specific situation that the current utterance is about (Austin 1950/1979; Elbourne 2013; Kratzer 2023).

function that maps a situation argument to the extension of this expression in this situation.¹²¹

Intensions are defined as follows.

(159) For any expression α , the intension of α is defined as

$$\llbracket \alpha \rrbracket_{\mathfrak{c}}^{g,c} = \lambda s_s. \llbracket \alpha \rrbracket^{g,s,c}$$

- ‘the function which, given an assignment and a context, maps every situation to the extension of α in this situation (at this assignment and in this context)’
- If $\llbracket \alpha \rrbracket^{g,s,c}$ has the type σ , then $\llbracket \alpha \rrbracket_{\mathfrak{c}}^{g,c}$ has the type $\langle s, \sigma \rangle$.

Now I can present my analysis of the Kazym Khanty Proprial Article, based on (Muñoz 2019: (26)).

(160) DENOTATION OF PROPRIAL ARTICLE

$$\begin{aligned} \llbracket \text{Prop}_i \rrbracket^{g,s,c} &= \lambda P_{\langle s, \langle e, t \rangle \rangle}: \forall s', x \\ &\left[P(s', x) \leftrightarrow \exists g'' \text{ in } s'. g''(i) \text{ is conventionally allowed to map to } x \wedge \right. \\ &\quad \left. i \text{ is a human name index} \right]. g(i) \end{aligned}$$

The Proprial Article bears an index i and takes the intension of a name predicate as an argument, presupposing that an individual bears this name in a situation iff (i) there is a convention in that situation (the world of that situation) of using this index to refer to this individual and (ii) this index is a human name index.^{122,123} This ensures both rigidity and that the Proprial Article combines only with (human) proper names. If this presupposition is satisfied, the Proprial Article returns the individual which the current assignment maps i to.

Intensional rigidity that the Proprial Article provides is encoded in the first part of the presupposition (i) and the fact that the situation argument of the name predicate is

¹²¹ The $\mathfrak{c}$ notation for intensions is taken from (von Stechow & Heim 1997-2023: (26)).

¹²² The purpose of this ‘convention’ presupposition is in part to ensure that proper names with the PROPRIAL ARTICLE do not admit restrictive modification. See the original paper for details on this and on the technicalities of how name-bearing is formalized using assignments.

¹²³ Muñoz takes the set of indices $\mathbb{N}$ to have disjointed subsets for each name of a given language. Thus, for Khanty we may assume that there are such sets of indices as $\mathcal{N}_{waša} \subset \mathbb{N}$ and $\mathcal{N}_{maša} \subset \mathbb{N}$ which are respectively the set of indices associated with the name *waša* ‘Vasya’ and the set of indices associated with the name *maša* ‘Masha’ and so on. To be a human name index then means to belong to one of such sets of indices associated with a human name.

Another option suggested by Alexander Podobryaev (p.c.) is to instead make the human name requirement a contextual restriction on the Sense Insertion rule (in the sense of Distributed Morphology) for the Proprial Article, as proposed in Chapter 4. See fn. 128.

bound by the universal quantifier in the presupposition, $\forall s', \dots [P(s', \dots) \leftrightarrow \dots]$ (see also fn. 122 and the original paper for the role that assignments play here). Thus, the Proprial Article ensures a rigid denotation for the whole DP, which is insensitive to the situation parameter s . The second part of the presupposition (ii) is responsible for selecting only human names. If both parts are satisfied, the Proprial Article simply returns the individual intended by the speaker, just like the Salient Article and several types of definites discussed in Chapter 2.

Let us see how this analysis can account for the difference between the Salient Article and the Proprial Article. We also need to amend the denotations of proper names and common nouns from §1.2 to account for the import of the situation parameter (161). Specifically, names like *waśa* ‘Vasya’ (230) now denote name-bearing predicates of individuals in the sense of (Muñoz 2019: (20), (22-23)) true in situations, and nouns like *amp* ‘dog’ (157) denote predicates of individuals simpliciter true in situations. (For the sake of illustration, I also provide the intensions (159) of these expressions.)

(161) SITUATION SEMANTICS-FRIENDLY DENOTATIONS FOR NAMES AND NOUNS

- a. $\llbracket \text{waśa} \rrbracket^{g,s,c} = \lambda x_e. x \text{ bears the name Vasya in } s = \lambda x. \text{vasya}'(x, s)$
 $\llbracket \text{waśa} \rrbracket_{\mathfrak{c}}^{g,c} = \lambda s' \lambda x. \text{vasya}'(x, s')$
 b. $\llbracket \text{amp} \rrbracket^{g,s,c} = \lambda x_e. x \text{ is a dog in } s = \lambda x. \text{dog}'(x, s)$
 $\llbracket \text{amp} \rrbracket_{\mathfrak{c}}^{g,c} = \lambda s' \lambda x. \text{dog}'(x, s')$

First, consider the example with the Salient Article which derives a nonrigid DP ‘the dog’, repeated from (157).

(162) *kašəŋ śos amp šiwaλ-t-əm-ən*
 every hour dog see-NFIN.NPST-1SG-LOC
amp-en *ma pəλ-am-a χurət-λ*
 dog-POSS.2SG I at-POSS.1SG-DAT bark-NPST[3SG]
 ‘Every time I meet a dog, the dog barks at me. [Sometimes it is a big dog, sometimes it is a smaller dog.]’

The extension of the target DP is given in (163) with a rough paraphrase.

- (163) $\llbracket \text{amp-Sal}_i \rrbracket^{g,s,c} =$
 $[\text{sal}(i, c) \wedge \forall j \in \text{Dom}(g)[g(j) \text{ is a dog in } s \wedge j \geq_{\text{Sal}(c)} i \rightarrow j = i]]. g(i)$
 ‘the most salient individual of c which is a dog in s ’

The extension in (163) is given relative to the situation parameter but in the larger context of (162) the DP will be interpreted relative to the situations introduced and quantified over by *kašəŋ šos* ‘every time’ according to a denotation for this quantifier like (164) (based on (Portner 2009: (269)) with simplifications). The Logical Form in (165) provides a representation of the effect the quantifier has on the interpretation of the predicates in (162).

$$(164) \llbracket \text{kašəŋ šos} \rrbracket^{g,s,c} = \lambda p_{\langle s,t \rangle} \lambda q_{\langle s,t \rangle}.$$

$$\forall s' \leq s. [\llbracket p(s') \rrbracket = 1] \rightarrow \exists s'' [s' \leq s'' \wedge \llbracket q(s'') \rrbracket = 1]$$

‘the universal quantifier over situations such that it takes two propositions (intensions of sentences) p, q and yields truth iff for all situations s' , that are a part of the situation parameter s , in which p is true, there is an extended situation s'' which includes s' where q is also true’

Based on (164), *kašəŋ šos* ‘every time’ will apply to the intensions of the two clauses in its sentence (162), thus shifting their situation parameter, and, therefore, it will shift the situation relative to which all the predicates in the sentence are interpreted. This is schematized in rough form in (165) with English glosses for Khanty expressions, where TP1 represents the first clause, TP2 the second clause, and CP the whole sentence with the quantifier. Note the use of $\lambda s'. in the two TPs in (165) which symbolizes that the two TPs are not interpreted relative to the matrix situation parameter but rather their situation parameter is bound by the quantifier $\llbracket \text{every time} \rrbracket^{g,s,c}$.¹²⁴$

(165) LOGICAL FORM OF (162)

$$[\text{CP } \llbracket \text{every time} \rrbracket^{g,s,c} [\text{TP1 } \lambda s'. \llbracket \text{I meet a dog}_i \text{ in } s' \rrbracket^{g,s',c}]$$

¹²⁴ Strictly speaking, given that the Salient Article makes use of an index, we need a dynamic-semantic mechanism of context change to properly model the donkey sentence-like example (164). In particular, we need the bare noun *amp* ‘dog’ in the first clause to locally introduce an index relative to the quantification done by *kašəŋ šos* ‘every time’ that can then be dynamically picked up by the Salient Article in the second clause: the resulting effect will be that the dog individual covaries with situations quantified over by *kašəŋ šos* ‘every time’.

The same point is made in a related situation-semantic framework by Schwarz (Schwarz 2009: §6.2.4) regarding the German Strong Article which incorporates an index (32b)/(34). With Schwarz, I avoid a fully explicit model of this sort for our index-bearing definite not to overcomplicate presentation. Various dynamic solutions available on the market can easily account for this effect (Muskens 1996; Beaver & Coppock 2015; Köpping 2020; cf. Nouwen et al. 2022). (See Coppock & Champollion 2024 for a user-friendly introduction.)

$$[\text{TP}_2 \lambda s'. \llbracket \text{the dog}_i \text{ in } s' \text{ barks at me} \rrbracket^{g,s',c}]]$$

The quantifier takes the intensions of the two clauses respectively as its arguments p and q and returns truth just in case the truth conditions in (166a) hold given that the presupposition contributed by the Salient Article is satisfied (166b), undefined otherwise.
(166) SEMANTICS OF (162)

- a. $\llbracket (162) \rrbracket^{g,s,c} = 1$ iff $\forall s' \leq s [\llbracket \text{I meet a dog}_i \rrbracket_{\text{c}}^{g,c}(s') = 1] \rightarrow$
 $\exists s'' [s' \leq s'' \wedge \llbracket \text{the dog}_i \text{ barks at me} \rrbracket_{\text{c}}^{g,c}(s'') = 1]$
- b. defined iff i is salient in c and **i is the most salient dog in s'' in c**

Note that the quantifier over situations *kašəŋ šos* ‘every time’ ‘binds into’ the presupposition of the Salient Article, so the discourse referent i is presupposed to be a dog in s'' , the (extension of a) situation quantified over by the quantifier. This ensures that dogs covary with situations, so for every situation that verifies (166a) we are speaking of **the dog in this situation** barking at me. This is the intensionally nonrigid reading of the Salient Article-marked DP that we have been working towards.

Now consider the Proprial Article example repeated from (230a) and split here into the felicitous bare name variant (167a) and the infelicitous Proprial Article-marked variant (167b).

(167) [“Every year we give a present to the 4th year student who gets the best grades.”]

- a. *kašəŋ ol mojləpsi wəjl' waša*
 every year present take[NPST.3SG] V.

‘Every year the present is taken by a Vasya. [Last year it was Vasya Tas’manow and this year it’s Vasya Tarlin.]’

- b. *#kašəŋ ol mojləpsi wəjl' wašaj-en*
 every year present take[NPST.3SG] V.-POSS.2SG

Int.: Same as above.

Consultant’s comment on *-en*: “then it’s the same Vasya each year / some particular Vasya”.

I assume that *kašəŋ ol* ‘every year’ has approximately the same denotation as *kašəŋ šos* ‘every time’ (164) and keep the assumption from §3.3.1 that bare nouns are interpreted as existential quantifiers via EX in Kazym Khanty. With this, we get the

Logical Form in (168) and the semantics in (169) for the bare name example variant (167a).¹²⁵

(168) LOGICAL FORM OF (167a) AFTER EX APPLIED^{126,127}

$$\begin{aligned} & [\text{CP } \llbracket \text{every year } C \rrbracket^{g,s,c} \\ & \quad [\text{TP } \lambda s'. \exists x \llbracket [\text{NP } \text{vasya}(x) \text{ in } s'] \rrbracket^{g,s',c} \\ & \quad \quad \exists y \llbracket [\text{NP } \text{present}(y) \text{ in } s'] \rrbracket^{g,s',c} [\text{VP } \llbracket x \text{ takes } y \text{ in } s' \rrbracket^{g,s',c}]]] \end{aligned}$$

(169) SEMANTICS OF (167a)

$$\begin{aligned} & \llbracket (167a) \rrbracket^{g,s,c} = 1 \text{ iff } \forall s' \leq s [\text{year}'(s') \wedge C(s')] \rightarrow \\ & \exists s'' [s' \leq s'' \wedge \exists x \exists y [\text{vasya}'(x, s'') \wedge \text{present}'(y, s'') \wedge \text{take}'(x, y, s'')]] \\ & \text{‘for all contextually relevant year-situations } s' \text{ there is an extended situation } s'' \text{ in} \\ & \text{which there is an individual } x \text{ that bears the name Vasya in } s'' \text{ and there is a present-} \\ & \text{individual } y \text{ in } s'' \text{ and } x \text{ takes } y \text{ in } s'' \text{’} \end{aligned}$$

These truth conditions are intuitively accurate. In particular, ‘the Vasya’ and the present end up being (potentially) different each year, covarying with the situations quantified over by *kašəŋ ol* ‘every year’.

Now consider the infelicitous Proprial Article-marked variant (167b): specifically, the LF (170), denotation of the target DP (171), and the semantics of the whole sentence (172).

(170) LOGICAL FORM OF (167b)

$$\begin{aligned} & [\text{CP } \llbracket \text{every year } C \rrbracket^{g,s,c} \\ & \quad [\text{TP } \lambda s'. \exists y \llbracket [\text{NP } \text{present}(y) \text{ in } s'] [\text{VP } \text{Vasya-Prop}_7 \text{ takes } y \text{ in } s'] \rrbracket^{g,s',c}]] \end{aligned}$$

(171) $\llbracket \text{Vasya-Prop}_7 \rrbracket^{g,s,c} = \forall s', x$

¹²⁵ Here is a Situation Semantics-friendly EX is just like the standard version (82) except that the predicates that it applies to are now also interpreted relative to situations.

Of course, to interpret a sentence with a generalized quantifier in object position (derived by EX here) we need Quantifier Raising or an analogous mechanism to alleviate the arising type clashes. See (Heim & Kratzer 1998: §7.3).

¹²⁶ *every year C* symbolizes that the quantifier is additionally implicitly restricted to some set of situations *C* (see von Stechow 1994; Schwarz 2009: §3.2.1), in this case roughly $\{s': s' \text{ is a situation of giving presents to the best students at our school}\}$. This is needed because (167), unlike the Salient Article example (162), does not have a ‘TP1’ so to speak which would restrict the situation quantifier *kašəŋ ol* ‘every year’ to some relevant set of situations.

¹²⁷ Recall from fn. 96 that I use $\exists x[\dots(x)]$ in LFs to designate a nominal that EX has applied to.

$$\left[\begin{array}{c} vasya'(s', x) \leftrightarrow \\ \exists g'' \text{ in } s'. g''(7) \text{ is conventionally allowed to map to } x \wedge \\ i \text{ is a human name index} \end{array} \right]. g(7)$$

‘the individual that g maps 7 to, if 7 is a human name index that is conventionally associated with this individual (the individual bears the name Vasya in all relevant situations), undefined otherwise’

(172) SEMANTICS OF (167b)

$$\begin{aligned} \llbracket (167b) \rrbracket^{g,s,c} = 1 \text{ iff } \forall s' \leq s [year'(s') \wedge C(s')] \rightarrow \\ \exists s'' [s' \leq s'' \wedge \exists y [present'(y, s'') \wedge take'(g(7), y, s'')]] \end{aligned}$$

‘for all contextually relevant year-situations s' there is an extended situation s'' such that there is a present-individual y in s'' and $g(7)$ takes y in s'' ; defined iff the presupposition of (171) is satisfied’

It is apparent from these truth conditions why (167b) does not have a covarying interpretation unlike (167a): the Proprial Article DP is essentially interpreted relative to the matrix assignment function g , and the situations quantified over by *kašəŋ ol* ‘every year’ play no role whatsoever in its denotation (171). This is how the Proprial Article derives rigidly denoting DPs, which differentiates it from bare nouns (168)-(169) and Salient Article DPs (165)-(166). (Compare specifically the role of situations in the denotation of a Salient Article DP (163) and the lack of it in (171).)

This concludes our discussion of the Proprial Article.

3.8. Summary

In this Chapter, we discussed the semantics of the four Kazym Khanty Unpossessives, in comparison with the Proper Possessive and the bare nouns of the language, against the typological background set up in Chapter 2 and §3.1.

I showed that the semantics of these Unpossessives can be precisely characterized in terms of well-known determinacy presuppositions such as familiarity, uniqueness, partitive specificity, salience, and intensional rigidity. I have argued that these Unpossessives function as full-fledged determiners, opening the path for semantic studies that treat Unpossessives on an equal footing with determiners of other language families and use tests from the extensive literature on determinacy as the present study does.

Using such tests, I demonstrated in §3.3 that the Kazym Khanty Proper Possessive is not inherently definite. Rather, it is a possessive modifier, same as the English Saxon

Genitive (Coppock & Beaver 2015; Beaver & Coppock 2016), the Russian adnominal possessive constructions, and other known possessive constructions (Partee & Borschev 2003; Beaver & Coppock 2016). This distinguishes the Proper Possessive from the four other Unpossessive determiners of Kazym Khanty.

The mutual distribution of the four Unpossessives and the bare nouns is presented in Figure 7 against the background of the Salience-Familiarity Scale typology from Chapter 2.

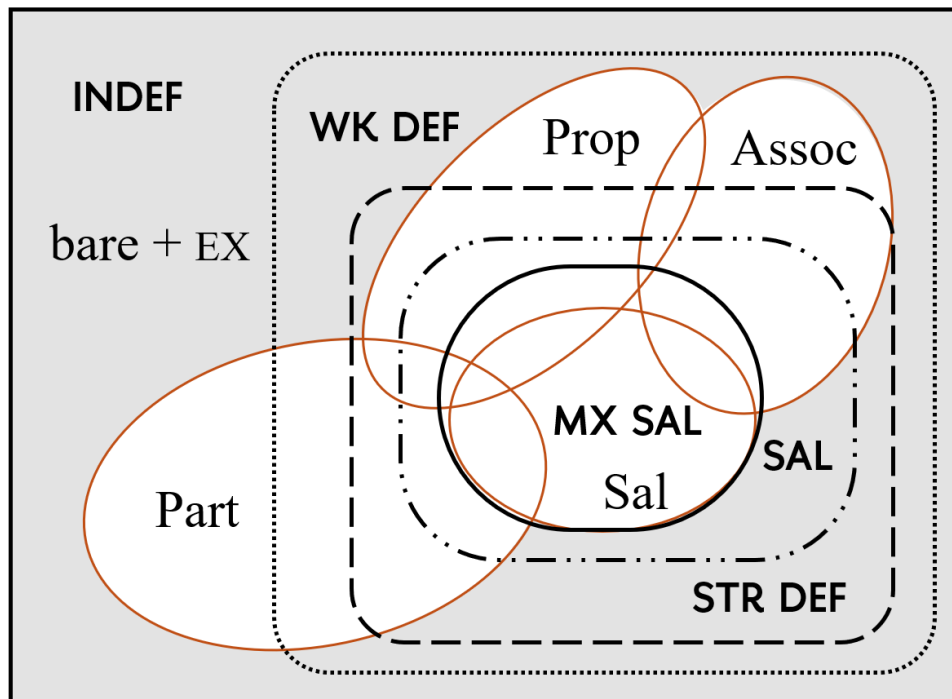


Figure 7. The determinacy system of Kazym Khanty against the background of the Salience-Familiarity Scale typology of definites (Figure 6). Figure legend: (i) crosslinguistic definite types are represented by rectangles with black outlines and subscripted in **BOLD SMALL CAPS SANS SERIF**: INDEF – INDEFINITE; WK DEF – WEAK DEFINITE; STR DEF – STRONG DEFINITE; SAL – SALIENT ARTICLE; MX SAL – RELATIVE MAXIMAL SALIENCE ARTICLE; (ii) Kazym Khanty Unpossessive determiners are represented by ovals with light outlines and subscripted in Capitalized Regular Serif: Assoc – Associative Possessive; Part – Partitive Article; Sal – Salient Article; Prop – Proprial Article; (iii) Shading indicates the contexts where bare nouns ('bare + EX') can be used.

There are several observations we can make regarding Figure 7. As mentioned at the beginning of the chapter, Kazym Khanty Unpossessives do not fit neatly into the types

of definites defined in Chapter 2. The Associative Possessive, the Partitive Article, and the Proprial Article all cross boundaries of the types of definites in Figure 7. For instance, the Associative Possessive is usable in any contexts, where the referent is at least weakly familiar (cf. the Salience-Familiarity Scale (90)), as long as there is a familiar relation with another referent in the common ground. See, e.g., (143) where it can be used with a salient referent as an alternative option to the Salient Article; or (173) where both the Associative Possessive and the Proprial Article are available with a proper name, since its referent is also associated with the other referent.

(173) [Petya is Masha's husband.]

mašaj-en	petr-əλ/-en	šiwaλə-s-λe
M.-POSS.2SG	P.-POSS.3SG/-POSS.2SG	see-PST-3SG>SG
'Masha saw Petya.'		

Incidentally, this example shows that these two determiners also do not compete via *Maximize presupposition!* (MP). In Chapter 2, following the work of my predecessors, I explained cases of complementary distribution of determiners/bare nouns by appealing to MP. MP standardly plays a role in cases where the two alternative expressions differ in terms of presuppositional strength: the presuppositions of one expression entail the presuppositions of the other (Heim 1991; Percus 2006; Magri 2009; Coppock & Beaver 2015; Marty & Romoli 2021). The fact that the Associative Possessive and the Proprial Article are not in a complementary distribution, as shown by (173), is expected given the denotations proposed for them in their respective subsections of the current chapter.

As discussed in §3.6.4 for three of the markers, neither Kazym Khanty Unpossessive has strictly stronger presuppositions than the other ones. This is also true for the Proprial Article. These four determiners, the Associative Possessive, the Partitive Article, the Salient Article, and the Proprial Article, are tuned to different kinds of properties of the current context / situation under discussion / head nominal, and their requirements intersect in examples like (143) and (173). This gives us the observed

distribution where in some contexts, two (or more) of the four determiners are available as marking options for a given DP. (See the other examples in §3.6.4.¹²⁸)

The properly weaker presuppositional alternative for all Unpossessives is the bare noun, which is interpreted as a presupposition-less existential generalized quantifier under the proposal from §3.3.1 (Tiutiunnikova et al. 2024). And, thus, we observe that in contexts where an Unpossessive is licensed, bare nouns are usually infelicitous, just like MP predicts. This is depicted in Figure 7.

Taking into account the findings of Marinina (2024), who investigated the three demonstrative determiners of Kazym Khanty (as discussed in §2.4.3), the Kazym Khanty determinacy system appears to be an unusually rich and diverse system which incorporates at least five uncommon types of definites (three of the Unpossessives and two of the Demonstratives), a Partitive Article, and bare nouns that are interpreted existentially across the board, contrary to the predictions of (Chierchia 1998; Dayal 2004). (See the concluding chapter for some discussion.)

In the next chapter, I provide a model of the morphosyntax of the Unpossessives discussed above, arguing for their mutual independence. In Chapter 5, I take a more careful look at the distribution and semantics of the Salient Article.

¹²⁸ As mentioned above, I do not currently have examples which show the overlapping distribution of the Partitive Article with the Associative Possessive or the Proprial Article, but the prediction is that they will also not compete with respect to MP.

Another issue, raised by Alexander Podobryaev (p.c.), is the potential competition of the Proprial Article with the Salient Article, which was not covered in §3.7.3. It is complicated by the full homonymy of the two markers so, given a human name marked with a POSS.2SG ‘lookalike’ marker *-en/-an*, it is hard to be sure exactly which of the two Articles it is. Regarding MP-based competition, the prediction is the same as for the other pairs of markers, the two Articles are expected to have overlapping distributions. In Chapter 4, I argue that the two Articles are in fact Senses (in the sense of Distributed Morphology) of the same syntactic head $\text{Poss}_{[\text{Sal/Prop}]}$, which are freely inserted at Logical Form, whichever is appropriate. Alexander Podobryaev (p.c.) suggests that we may instead make the Sense Insertion rules for the two Articles sensitive to the head noun, so that with human names, only the Proprial Article Sense could be inserted (see fn. 155). This is an interesting proposal and it is worthwhile to explore in the future, but the challenge, again, is to devise a way to tease apart the two Articles, given a human name marked with a POSS.2SG lookalike. One benefit of this proposal is that it would allow us to simplify the Proprial Article’s presupposition to make it exactly as proposed by Muñoz, maintaining the strong claim that proprial articles have the same denotation crosslinguistically.

4. Diagnosing the Unpossessives as synchronically independent markers¹²⁹

In the foregoing chapters, I assumed that the four non-possessive functions of possessives in Kazym Khanty discussed in this thesis are synchronically distinct Unpossessive markers that differ from the Proper Possessive. In the previous chapter, I even provided independent semantic denotations for each Unpossessive, are distinct from the Proper Possessive denotation and from each other. Here are the base examples from Chapter 1 for each function including the Proper Possessive.

(174) PROPER POSSESSIVE

nǎŋ kǎt'-en moś-λ
you.SG cat-POSS.2SG purr-NPST[3SG]
'Your cat purrs.'

(175) ASSOCIATIVE POSSESSIVE

[A friend is over at the speaker's place. There's one cup on the table.]
an-en mij-e
cup-POSS.2SG give-IMP.SG>SG
'Give me the cup.'

(176) SALIENT ARTICLE

['I was walking along the street when I saw a dog.']
amp-en ma peλ-am-a χurət-ti pit-əs
dog-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]
'The dog started barking at me.'

(177) PROPRIAL ARTICLE

wənt-r-en jʉχ šəp sewr-əs
A.-POSS.2SG wood piece cleave-PST[3SG]
'Andrej cleaved a log.'

(178) PARTITIVE ARTICLE

['I was walking down the street when I saw two dogs.']
i **amp-əλ** ma peλ-am-a
one dog-POSS.3SG I towards-POSS.1SG-DAT

¹²⁹ This chapter closely reproduces (Mikhailov 2024c) with the added discussion of the Partitive Article based on the earlier proceedings paper (Mikhailov 2021b).

χurət-ti	pit-əs
bark-NFIN.NPST	become-PST[3SG]

‘One of the dogs started barking at me.’

In this chapter, I make good on the assumption of independence made above and provide a fully spelled-out argument for the unpossessivehood of the four non-possessive functions. I follow the lead of five recent studies (É. Kiss & Tánczos 2018; É. Kiss 2018; Halm 2018; Serdobolskaya et al. 2019; and Logvinova 2019 on Chuvash (< Turkic)) have taken a polysemic stance on the the non-possessive functions they investigated, arguing that they are in fact **distinct markers grammaticalized from proper possessive suffixes**.

The central claim of this chapter is that the non-possessive functions of Kazym Khanty possessives introduced above must be treated polysemically, as mutually independent markers. Building on the work of my predecessors, I develop diagnostics that aim to test whether a non-possessive function shows the same morphophonological, morphosyntactic, semantic, and pragmatic behavior as the proper possessive function, as would be expected under a monosemic approach. Using these diagnostics, I show that Associative Possessive, the Salient Article, the Proprial Article, and the Partitive Article **observe different behavior than the Proper Possessive** (1) with respect to several parameters. I develop an analysis within the framework of Distributed Morphology (Halle & Marantz 1993) to model this behavior, which allows us to cover the differences between these functions while also accounting for their similarities. Thus, I argue that these functions are UNPOSSESSIVES that are homonymous with the proper possessive but are synchronically distinct from it (see the definition in (7)).

To set up the context for the argument, I extend the analysis of possessives’ semantics from §3.3 with a model of the syntax and morphology of possessive agreement markers in §4.1. In §4.2, I discuss some monosemic approaches to data of other Uralic varieties and contrast them with the polysemic treatments by É. Kiss & Tánczos, Halm, Serdobolskaya et al., and by Logvinova on Chuvash. This discussion leads to the development of diagnostics for UNPOSSESSIVE MARKERS in §4.3.

In sections §4.4-4.6, I apply the unpossessive diagnostics to the Associative Possessive, the Partitive Article, the Salient Article, and the Proprial Article, arguing that these functions qualify as UNPOSSESSIVES.

In §4.7.1, I present an analysis of the four markers and the Proper Possessive within DM using the mechanism of allosemy (Wood 2015; Myler 2016, a.o.), and in §4.7.2, I follow it up with an argument for allosemy from the normal workings of grammaticalization based on (Bisang & Malchukov et al. 2020a).

The main results of the chapter are summarized in §4.8 with an outlook for future research into UNPOSSESSIVES beyond Kazym Khanty.

In the next section, I introduce my theoretical assumptions. I intend the framework sketched there to put the argument made in the following sections on a more solid footing, but I believe that the argument in its general form is relevant across theoretical frameworks (and object languages). For this reason, I advise readers more interested in the empirical side of my argument and less inclined to delve into the theoretical technicalities to skip ahead to section 4.2.

Regardless of the readers choice, I advise now to refresh the facts of the Kazym Khanty adnominal possessive construction as presented in the previous chapter in §3.2 as they will become directly relevant in what follows

4.1. Theoretical assumptions

My analysis is couched in Distributed Morphology (DM; (Halle & Marantz 1993)), a non-lexicalist approach to morphology based on the generative Y-model of grammar (see Figure 8 below). DM assumes a syntactic component that builds both word-level and phrase-level hierarchical structures, with Late Insertion of phonological and semantic content.

The derivation of a sentence starts in the narrow syntax where syntactic terminals (heads) from a lexicon called List 1 are assembled into structures via two operations: Merge, combining two syntactic units into a larger syntactic unit, and Agree, copying syntactic features from a unit onto another unit. The heads are devoid of phonological and semantic content (Bobaljik 2017). Other details of the syntax I assume are pretty much the same as in (Preminger 2014; Longenbaugh 2019; Newman 2021; Newman 2024). In particular, I assume that **syntactic operations are obligatory** but fallible (more on this below).

Merge is driven by Merge features. For instance, [$\cdot D \cdot$] on a head X triggers the Merge of a D element to X, e.g., a D(eterminer)P(hrase) (using the notation from (Heck & Müller 2007)). Agree is driven by agreement probes on heads, e.g., [<pers-num>] on a

head X triggers Agree with an element bearing person-number features (ϕ -features) in X's c-command domain, following Preminger (2014). Merge/Agree features that failed to be saturated at the head level (X) project to the phrasal level (X') and trigger Merge/Agree once more (Adger 2003; Béjar & Rezac 2009).

After a syntactic structure is assembled, it is spelled out to the semantic and (morpho)phonological levels (interfaces; Figure 8). Heads are mapped onto semantic denotations and phonological exponents via rules from two lists: the syntax-phonology mapping, Vocabulary (a.k.a. List 2), and the syntax-semantics mapping, Encyclopedia (a.k.a. List 3).

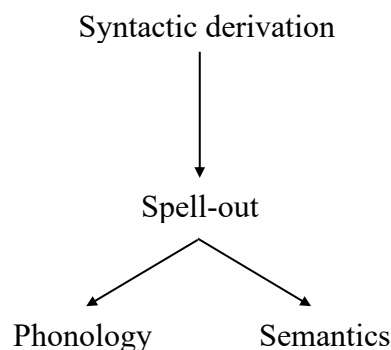


Figure 8. Y-model of grammar (based on (Bobaljik 2017))

Note that in this approach it does not make sense to speak of a single lexical entry for a morpheme or a word, since the lexicon is divided into three lists. This leads to the expectation of one-to-many correspondences between heads and phonological exponents, on the one hand, and between heads and denotations, on the other hand. The former case is allomorphy whereby a single morpheme is exponed by different allomorphs depending on its structural context. The latter case is allosemy, recently explored for functional heads in a DM-based setting by Wood (2015; 2023), Myler (2016), Wood & Marantz (2017), Kastner (2020), and Kasenov (2023), whereby a head is assigned different denotations depending on its (syntactic or semantic) structural context. Both types of syntax-interface correspondence play a significant role in my analysis.

4.1.1. *Syntax of possessives*

As is standard in syntactic literature on Uralic possessives (*e.g.*, (Dékány 2021) building on (Bartos 1999); and more generally, (Myler 2016)), I assume that possessives head a separate projection in the nominal functional sequence. This projection is called

Poss(essive)P(hrase), it takes another nominal projection (NumP) as its complement and introduces the possessor DP as its specifier.

Evidence from the order of nominal modifiers and the order of suffixes in the nominal wordform in Northern Khanty points to PossP being higher than Num(ber)P(hrase) and lower than another functional projection, which for the sake of convenience I will call DP (179). I assume that the DP from (179a) has the structure in (179b).

(179) a. (adapted from (Pleshak 2018: (6)))

tām	ma	χəλəm	puχ-λ-am	armiĵa-ĵa	mān-s-ət
this	I	three	son-PL-POSS.1SG	army-DAT	go-PST-3PL

‘These three sons of mine joined the army.’¹³⁰

b. [DP tām [PossP ma [NumP χəλəm [NP puχ] -λ] -am] D]

Poss attaches to NumP and takes a possessor DP as its specifier which is achieved with [$\cdot$ Num $\cdot$] and [$\cdot$ D $\cdot$] features on Poss (180)¹³¹. It also bears an agreement probe.

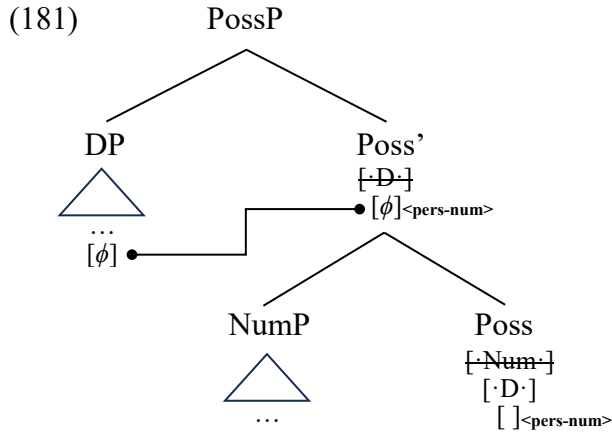
(180) FEATURE MAKEUP OF A PROPER POSS

Poss: [$\cdot$ Num $\cdot$][$\cdot$ D $\cdot$][$\cdot$ <pers-num>

This results in the following structure for the Northern Khanty PossP (181). Saturated Merge features are given with a strikethrough and omitted in higher nodes, Agree is represented with a line with bullets.

¹³⁰ An anonymous reviewer for *Ural-Altaic Studies* asks under what conditions plural marking is possible in a numeral-noun construction in Kazym Khanty, since this is generally considered impossible in Ugric languages. Pisarenko (2024: 46) notes that, although relatively uncommon, this pattern is observed with definite subjects. My impression is that (un)possessive marking is also needed for the head noun to be marked plural in the presence of a numeral, not just pragmatic definiteness.

¹³¹ For simplicity’s sake, I assume, following (Müller 2010), that Merge features are ordered so that (180) requires the NumP complement to be merged first and the DP specifier second. See (Newman 2021; Newman 2024) for pertinent discussion.



With these assumptions in place, we may already account for the fact that Kazym Khanty (un)possessives are in mutual complementary distribution (§3.2). This suggests that all (un)possessives occupy the same functional projection, Poss. The technical implementation goes as follows. Proper possessives cannot stack on top of other (un)possessives because they do not host [$\cdot$ Poss $\cdot$] (180). The same is assumed to hold for the other three Unpossessives. (See § 4.2.2 for possessive stacking in other Uralic languages.)

Finally, let me note that following (Preminger 2014), I assume that Agree may fail to find a suitable goal for the probe which, however, does not crash the derivation. This yields a default agreement marker (e.g., 3SG) or a zero exponent on the surface crosslinguistically, and this may occur with different kinds of agreement probes. I assume that this is exactly what happens in Kazym Khanty with an Obviative lexical possessor (as briefly noted in §3.2 and fn. 91): *waśaj-en^{obviative} amp* [V.-POSS.2SG^{obviative} dog] ‘Vasya’s dog’. The possessive probe requires a Proximate goal, but neither the head noun nor the Obviative possessor is Proximate. Hence, Agree fails, no features are copied onto the probe, and the head noun surfaces unmarked¹³².

4.1.2. Spelling out possessives

4.1.2.1. Morphophonology

One important feature of DM is underspecification of rules. Consider, for instance, the difference between the two number-conditioned POSS.2SG allomorphs:

¹³² For a detailed proposal to Kazym Khanty possessive agreement along these lines, see (Masliukov 2024).

(182) Poss[2SG] ↔ -an

Poss[2SG] ↔ -en / Num[SG]__

In (182), *-en* is specified as the allomorph that is inserted in the context of singular number (on the head noun), while *-an* is underspecified in terms of its context of insertion which makes it an elsewhere exponent of Poss_[2SG]. This is based on the principle that the rule that applies is the rule that matches the context of insertion the best (Bobaljik 2017: (11-12) and fn. 7). Rules in (182) thus reflect that *-an* will only be inserted with dual and plural head nouns (see Table 7, repeated from §3.2). The rest of the paradigm can be dealt with similarly. (Note the syncretisms in this paradigm.)

Possessee Possessor	SG	DU	PL
1SG	-em -POSS.1SG	-ηəλ-am -DU-POSS.1SG	-λ-am -PL-POSS.1SG
2SG	-en -POSS.2SG	-ηəλ-an -DU-POSS.2SG	-λ-an -PL-POSS.2SG
3SG	-əλ -POSS.3SG	-ηəλ -DU-POSS.3SG	-λ-aλ -PL-POSS.3SG
1DU	-emən -POSS.1DU	-ηəλ-əmən -DU-POSS.1DU	-λ-əmən -PL-POSS.1DU
2DU	-ən -POSS.2NSG	-ηəλ-ən (-ηəλ-an) -DU-POSS.2NSG	-λ-ən -PL-POSS.2NSG
3DU	-ən -POSS.3DU	-ηəλ-ən (-ηəλ) -DU-POSS.3DU	-λ-ən (-λ-aλ) -PL-POSS.3DU
1PL	-ew -POSS.1PL	-ηəλ-əw -DU-POSS.1PL	-λ-əw -PL-POSS.1PL
2PL	-ən	-ηəλ-ən (-ηəλ-an)	-λ-ən

	-POSS.2NSG	-DU-POSS.2NSG	-PL-POSS.2NSG
3PL	-eλ -POSS.3PL	-ηəλ -DU.POSS.3PL	-λ-aλ -PL-POSS.3PL

Table 7. Possessive suffixes of Kazym Khanty (field data)

In §3.2, I noted that with lexical possessors the head noun sometimes appears without a possessive. I assume that in such cases the possessor is Obviative and the possessive probe only “sees” Proximate possessors¹³³. So with Obviative possessors, the probe fails to find any suitable goal and gets sent to the interfaces unvalued: Poss[_i]. This in turn is spelled out simply as zero, leading to the assumption that the elsewhere exponent is $\emptyset$. See rule (183o) in the List 2 mapping (**Vocabulary Insertion**) rules for the Proper Possessive below.

(183) VOCABULARY INSERTION RULES FOR THE PROPER POSSESSIVE

- a. Poss[1SG] ↔ -am
- b. Poss[1SG] ↔ -εm / Num[SG]__
- c. Poss[2SG] ↔ -an
- d. Poss[2SG] ↔ -en / Num[SG]__
- e. Poss[1DU] ↔ -əmən
- f. Poss[1DU] ↔ -εmən / Num[SG]__
- g. Poss[1PL] ↔ -əw
- h. Poss[1PL] ↔ -ew / Num[SG]__
- i. Poss[3SG] ↔ -əλ / Num[SG]__
- j. Poss[3PL] ↔ -eλ / Num[SG]__
- k. Poss[3] ↔ -aλ / Num[PL]__
- l. Poss[3] ↔ $\emptyset$ / Num[DU]__
- m. Poss[3DU] ↔ -ən
- n. Poss[2] ↔ -ən

¹³³ The exact details of how this should be implemented must be developed on another occasion since the syntax of possessive agreement is not our direct concern.

o. Poss $\leftrightarrow \emptyset$

A few remarks are in order about the various decisions made in (183). Rules (183a–h) are entirely parallel to (182). Rules (183i–l) account for the syncretism found between POSS.3SG and POSS.3PL with dual and plural head nouns by underspecifying the conditions of insertion to simply [3]. (This rule will not apply with [3DU] since it has a more specific exponence rule (183m).) A similar simplification is made in (183n) for POSS.2DU and POSS.2PL which are fully syncretic (unlike POSS.2SG)¹³⁴.

I must note that I do not intend (183) as the best imaginable analysis of the Proper Possessive’s morphology. Rather, this is an illustration of what a DM analysis might look like, with a level of sophistication sufficient for our purposes. Perhaps, a more involved analysis could account for more syncretisms in the paradigm in Table 1, but I set it aside in the current thesis because morphology is not its focus.

4.1.2.2. *Semantics*

The semantics of the Kazym Khanty Proper Possessive is the same as in §3.3. See that section for the detailed arguments and explanations. Here, I simply repeat the denotation from (96) in the DM format of a List 3 mapping (**Sense Insertion**) rule.

(184) SENSE INSERTION RULE FOR THE PROPER POSSESSIVE

$$[\text{Poss}]^{g,s,c} \leftrightarrow \lambda P_{\langle e,t \rangle} \lambda x_e \lambda y_e : \exists z [\text{POSS}(x, z) \wedge P(z)]. \text{POSS}(x, y) \wedge P(y)$$

where POSS is an inherent possessive relation in the sense of (Vikner & Jensen 2002; Partee & Borschev 2003; Karvovskaya 2018)

What is important is that (184) is more or less a basic possessive modifier denotation: it further restricts the extension of the nominal, but it does not determine its reference (*cf.* (Coppock & Beaver 2015); see §3.3 for evidence that this is so). On the other hand, the Kazym Khanty Unpossessives all determine the reference of the nominal they attach to (see Chapter 3).

4.2. *Previous research*

¹³⁴ I am grateful to Alexandra Shikunova (p.c.) for discussing possible analyses with me.

4.2.1. Against the “definiteness idea”: monosemic approaches

It has long been observed in the Uralic descriptive literature that Uralic possessives may figure in contexts reminiscent of definite articles (*e.g.*, (Collinder 1957); see (Nikolaeva 2003; Gerland 2014) for other references). The idea is essentially that in such functions a possessive is in fact a kind of definite article. Under standard conceptions of definiteness (Heim 1991; 2019; Roberts 2003; Coppock & Beaver 2015; König 2018; Schwarz 2019), this would mean that it presupposes uniqueness and familiarity of the DP referent and is obligatory where these presuppositions are satisfied.

More recently, however, several studies (Fraurud 2001; Nikolaeva 2003; Gerland 2014) have specifically argued against this idea and instead suggested that (some of) the non-possessive functions should be treated monosemically, as stemming from some basic possessive meaning.

Nikolaeva (2003) observes that apart from POSS.3SG and POSS.2SG (185), POSS.1SG markers may also be used non-possessively, specifically, as ASSOCIATIVE POSSESSIVES. This is problematic for the definiteness account as it is not clear how the choice of person-number is conditioned. If a possessive truly grammaticalized into a definite article of the kind familiar from European languages, it would not be expected to vary in person-number features.

(185) OBDORSK DIALECT OF NORTHERN KHANTY (adapted from (Nikolaeva 2003: 137))

wanta tām mašinaj-en jowra mǎn-əs
look this car-POSS.2SG awry go-PST[3SG]

‘Look, that car went awry.’

(186) (adapted from (Nikolaeva 1999: 83))

ma iši taxa:j-e:m-na il ko:ri-s-əm
me same place-POSS.1SG-LOC down fall-PST-1SG

‘I fell down in the same place [lit.: at the same my place].’

Instead, Nikolaeva (2003) proposes that Uralic possessives may describe a presupposed **associative relation** between two entities. This results in many familiar referents being marked with possessives as they may frequently be associated either with another referent in the narrative, as in (186) where the speaker has previously been said to have fallen in this place, or to the utterance situation via the addressee, as in (185) where “the car is ‘yours’ because I am talking to you about it” (Ibid.: 137). The

possessives are not obligatory in these functions, unlike European definite articles, indicating a lower degree of grammaticalization, if any at all.

Furthermore, according to (Fraurud 2001: 254), some work on Proto-Uralic suggests that definite article-like functions were present already at that stage¹³⁵, which also speaks against these functions being a result of grammaticalization.

Gerland (2014) takes an even stronger stance, arguing that no grammaticalization process has taken or is taking place, but rather the meaning responsible for these definite article-like functions is inherent to Uralic possessives and was present already in Proto-Uralic. On both possessive and non-possessive functions, the referents of the marked nominals are definite and are related to some entity which may be another referent or a situation available in the context, the utterance context itself or even the whole world.

While a detailed critical overview of the existing monosemic approaches available on the market is certainly a worthwhile enterprise, for the purposes of the present work it is enough to discuss the general **monosemic prediction** that all of them make (given reasonable background assumptions)¹³⁶.

(187) MONOSEMIC PREDICTION

Non-possessive functions of a possessive must observe the same morphophonological, syntactic, semantic, and pragmatic properties as the PROPER POSSESSIVE function of this marker. Any differences must receive an independent explanation.

If the monosemic approaches are on the right track, then it should be the case that the possessive markers they aim to account for are consistent in their morphosyntax, semantics, and pragmatics across functions. In terms of our framework (§4.1), this means that the functions correspond to a single syntactic terminal (Poss), semantic denotation ([Poss]), and set of exponents (Vocabulary Insertion rules like (183)) — this is the deeper reason for the same surface properties observed according to (187).

Consider, for example, the article-like function of the Komi POSS.3SG marker in (188) (Izhma dialect; Permic group). In this example, it would appear that the marker

¹³⁵ However, see Fraurud's paper for reasonable skepticism and references for this claim.

¹³⁶ Some other monosemic works include (Körtvély 2010; Janda 2015; Siegl 2015; Simonenko 2017). See also the discussion of monosemic studies in (É. Kiss & Tánecz 2018: 239-240).

does not carry its possessive meaning but rather seems, like the English definite article, to be marking a globally unique entity.

(188) IZHMA DIALECT OF KOMI (adapted from (Simonenko 2017: 428))¹³⁷

šondy-(*)*(ys)* *žeb-s* '-i-s.
 sun-POSS.3SG set-DETR-PST-3SG
 'The sun has set.'

The monosemic approach must claim that this seemingly divergent semantic behavior is derivable from the basic possessive meaning. For instance, the possessive marker links the referent to the whole world which acts in place of a more prototypical possessor (Fraurud 2001, cf. Gerland 2014). Fraurud suggests that possessives differ crosslinguistically in whether they admit abstract possessors of this kind. However, it seems that none of the monosemic proposals on the market provide independent evidence for the availability of such abstract possessors or, for that matter, an explicit account of how such an abstract possessor enters the derivation to compose with a possessive deriving the correct reading.

There is, I believe, yet a tougher challenge for monosemic approaches, namely, the morphophonological and morphosyntactic divergence of non-possessive functions from the PROPER POSSESSIVE one. As far as I am aware, the proponents of monosemic approaches have not yet considered data of this kind.

In what follows, we will see that the challenge offered by such data proves insurmountable to monosemic approaches as it stands in stark contradiction to the monosemic prediction (187).

4.2.2. *For polysemic approaches*

Recently, five papers (É. Kiss 2018; É. Kiss & Tánčzos 2018; Halm 2018; Logvinova 2019; Serdobolskaya et al. 2019) have adduced data from morphosyntax, taking a polysemic stance on several possessive markers and investigating their synchronic distribution and diachronic development.

Thus, É. Kiss and Tánčzos (2018) analyze the three functions of the Udmurt *-jez* marker: cross-referencing a possessor, encoding partitive specificity, and marking

¹³⁷ An anonymous reviewer for *Ural-Altaic Studies* suggested some corrections to the Izhma dialect transcription in (188) which I have adopted.

specific objects. They argue that the latter two functions are synchronically independent from the Proper Possessive function of the marker and represent subsequent stages of a grammaticalization path. In another paper (É. Kiss 2018), it has been shown that grammaticalization along the same path had taken place in Hungarian with two PARTITIVE MARKERS developing from a POSS.3PL and a POSS.3SG possessive.

The crucial arguments for the independence of these non-possessive functions in Udmurt are based on two observations. Firstly, the Partitive *-jez*, used as a nominalizer, **does not agree in person-number features** with the supposed possessor. Thus, in (189b) with a plural possessor, it is expected that the POSS.3PL marker *-zy* will be used, but in the case of an elided head noun the *-jez* marker is used instead, clearly not agreeing in number with the possessor ('boys'). Similarly, in (190), there is an agreement mismatch with a first-person singular possessor, suggesting that *-jez* here is not a possessive agreement marker.

(189) UDMURT (adapted from É. Kiss & Tánčzos 2018: 736)

- | | | | | |
|----|---------------------|------------------|---|----------------------------|
| a. | <i>Ivan-len</i> | <i>gurt-ez</i> | → | <i>Ivan-len-ez</i> |
| | Ivan-GEN | village-POSS.3SG | | Ivan-GEN-POSS.3SG |
| | 'Ivan's village' | | | 'that of Ivan' |
| b. | <i>pi-os-len</i> | <i>gurt-zy</i> | → | <i>pi-os-len-ez</i> |
| | boy-PL-GEN | village-POSS.3PL | | boy-PL-GEN-POSS.3SG |
| | 'the boys' village' | | | 'that of the boys' |

(190) (adapted from (Ibid.))

<i>Ta-iz</i>	<i>solen</i>	<i>mašina-jez,</i>	<i>noš ta-iz</i>	<i>mynam— ojdo</i>
this- POSS.3SG	that.GEN	car-POSS.3SG	but this-POSS.3SG	I.GEN PTCL
<i>mynom</i>	<i>mynam-en-yz.</i>			
go.FUT.1PL	I.GEN-INS-POSS.3SG			
'This is his car, and that is my one—let's go with my one!'				

Secondly, a restricted class of nouns triggers the *-yz* allomorph of the Proper Possessive *-jez* (*Saša-len puny-jez* [S.-GEN dog-POSS.3SG] 'Sasha's dog', but *Saša-len ki- yz* [S.-GEN hand-POSS.3SG] 'Sasha's hand'), but **the Accusative *-jez* does not use the special allomorph with these nouns** (although it does use it with plurals just like the

Proper Possessive)¹³⁸. Thus, in (191) an Accusative-marked ‘hand’ receives the *-jez* allomorph, while a POSS.3SG-marked ‘hand’ receives the *-yz* allomorph. This clearly shows that the two markers are morphophonologically distinct. In our terms from §4.1, they have different Vocabulary Insertion rules, as sketched in (192).

(191) (adapted from (ibid.: 738))

- a. *Šekyt ki-jez operirovat' karyny.*
 hard hand-ACC to.operate make.INF
 ‘It is hard to operate the hand.’
- b. *Solen ki-yz čeber.*
 she/he:GEN hand-POSS.3SG nice
 ‘His/her hand is nice.’

(192) a. Poss[3SG] ↔ *-jez*

b. Poss[3SG] ↔ *-yz* / { $\sqrt{\text{HAND}}$, ...} __

c. Case[ACC] ↔ *-jez*

These observations lead É. Kiss and Tánčzos to treat these functions as independent from the Proper Possessive *-jez*, since if the markers used in (189), (190), and (191a) were instances of the Proper Possessive, they would be expected to observe the same morphosyntactic and morphophonological properties as in the Proper Possessive function, which is not the case.

Finally, it is interesting to observe that the Udmurt Accusative Case Unpossessive may combine with Proper Possessive morphemes, including the POSS.3SG *-jez* that it grammaticalized from.

(193) a. (adapted from (ibid.: 735))

- Mon Saša-leš agaj-z-e utčaj.*¹³⁹
 I Sasha-ABL brother-POSS.3SG-ACC search.PST.1SG
 ‘I searched for Sasha’s brother.’

¹³⁸ Apart from plurals, É. Kiss and Tánčzos (2018: 734) also list the instrumental, elative, translative, and egressive case endings as contexts triggering the *-yz* allomorph. Presumably, this is the reason for the Partitive *-jez* surfacing as *-yz* in (190).

¹³⁹ The possessor of the direct object is assigned ablative case instead of genitive (É. Kiss & Tánčzos 2018: 735).

b. [CaseP [PossP Saša-leś [NP agaj] -z] -e]

Here, both the POSS.3SG Possessive and the Accusative undergo morphophonological alternations, so that the former surfaces as *-z* instead of *-jez* and the latter as *-e* instead of *-jez*. (Thus, we have another instance where the Accusative Unpossessive shows distinct allomorphy.) The fact that **possessive stacking** of this sort is possible with the Accusative Unpossessive further shows that it is independent from the proper POSS.3SG: it occupies a separate slot of the nominal morphological template. In present terms, this suggests that the accusative *-jez* occupies a distinct nominal projection above Poss (*cf.* §4.1.1), for instance, Case as sketched in (193b), and it is subject to Poss-conditioned allomorphy. See (Logvinova 2019) for similar observations on the third-person possessive in the Maloe Karachkino dialect of Chuvash (< Turkic) which also shows different allomorphy, allows for possessive stacking, and does not exhibit agreement in non-possessive functions (with nominalized adjectives and partitive-specific/anaphorically accessible referents).

Another paper proposing a polysemic view on a non-possessive function of a possessive is (Halm 2018) who investigates the negative affective demonstrative (NAD) function of the Hungarian POSS.3SG (194).

(194) HUNGARIAN (adapted from (*ibid.*: 362))

A hülyé-je!
the stupid-POSS.3SG
'That total idiot' (lit. 'its stupid')

According to Halm, NAD is not possessive since **there is no possessor present**¹⁴⁰. Even a null *pro* possessor cannot be assumed here since there is no anaphorically or deictically available antecedent for it (*ibid.*: 362).

Furthermore, NAD behaves syntactically like a demonstrative in that (i) **it is in complementary distribution with the other demonstratives** (195) and (ii) it obligatorily requires a definite article like the other demonstratives.

(195) (adapted from (*ibid.*: 364))

Hallottad, hogy {az a marha / a marhá-ja /

¹⁴⁰ This argument was also used in (Logvinova 2019: 109–110).

heard that_{COMP} that_{DEM} the cow / the cow-POSS.3SG /
 *az a *marhá-ja*} már megint mit csinált!/?
 that_{DEM} the cow-POSS.3SG yet again what did
 ‘Have you heard what that fool has done again?’

Based on these observations, Halm (ibid.: 387) proposes that NAD was reanalyzed as a Det(erminer) head. This explains its complementary distribution with other demonstratives, which are assumed to initially Merge in the same projection (*cf.* Dékány 2021: §3.2.2).

Morphosyntactic evidence of the kind adduced in (É. Kiss 2018; É. Kiss & Tanczos 2018; Halm 2018; Logvinova 2019; Serdobolskaya et al. 2019) presents very clear arguments in favor of treating non-possessive functions of several possessives as independent UNPOSSESSIVE MARKERS. Building on these arguments, I formulate a set of unpossessive diagnostics that aim to test the monosemic prediction (187) in the following section¹⁴¹.

4.3. Unpossessive diagnostics

In this section, I formulate diagnostics for unpossessivehood, i.e., tests that diagnose a non-possessive function of a possessive marker as a synchronically independent marker. The diagnostics probe the similarity of morphophonological, morphosyntactic, semantic, and pragmatic properties of a non-possessive function to the proper possessive function of a marker. In case an instance of differing behavior of the former is found, it can be used to argue that this function is an independent unpossessive. Note that not having a proper possessive reading (at least, intuitively) is a pre-condition for being a candidate unpossessive. It is the correlation between a non-possessive function and a positive result on one of the diagnostics that provides grounds for arguing that it is an independent marker¹⁴².

¹⁴¹ The current work is limited to definiteness-based UNPOSSESSIVES, thus largely neglecting UNPOSSESSIVES with syntactic functions and potentially other types. For examples of syntactic UNPOSSESSIVES, see studies on the Chuvash (Logvinova 2019) and Udmurt (É. Kiss & Tanczos 2018; Serdobolskaya et al. 2019) POSS.3SG markers used for nominalizing modifiers in the absence of nominal heads.

¹⁴² See § 4.7 for a discussion of how negative results in the presence of some positive results should be treated.

In some diagnostics below I use *might* to show that an UNPOSSESSIVE does not have to differ from the PROPER POSSESSIVE in all its properties to be considered independent. Where applicable, I mention in parentheses relevant examples from the above discussion. Together with the diagnostics, I state the implications of a positive result for the analysis in the current system (from § 4.1).

(196) POSSESSIVE ALLOMORPHY

- a. An UNPOSSESSIVE might have different morphophonology than the PROPER POSSESSIVE, for example, in not having the full range of allomorphs available to the proper possessive or in having special allomorphs of its own. (*cf.* (191), (193))
- b. *Analysis*: Different set of Vocabulary Insertion rules for the unpossessive (*e.g.*, (192)) which likely also means that it is represented by a distinct head in the syntax.

(197) POSSESSIVE STACKING

- a. An UNPOSSESSIVE might have a different order in the nominal morphological template, meaning that:
 - i. An UNPOSSESSIVE might stack on a PROPER POSSESSIVE or another UNPOSSESSIVE and vice versa (*cf.* (193));
 - ii. An UNPOSSESSIVE might be in a complementary distribution with another nominal modifier category (*e.g.*, demonstratives) (*cf.* (195)).
- b. *Analysis*: Distinct syntactic category (*e.g.*, Det) for the unpossessive. If it attaches outside the proper possessive, it appears higher in the nominal functional sequence. Complementary distribution with another nominal category helps establish the category of the unpossessive (*cf.* (Halm 2018: § 5.2)).

(198) AGREEMENT MISMATCH

- a. An UNPOSSESSIVE might never vary in person-number features despite the actual features of the supposed possessor. (*cf.* (189), (190))
- b. *Analysis*: No agreement probe on the unpossessive.

(199) EXPLICIT POSSESSOR

- a. An UNPOSSESSIVE might not allow an explicit DP-internal possessor under any conditions. (*cf.* (194))

- b. *Analysis*: No [$\cdot D \cdot$] Merge feature on the unpossessive¹⁴³.

Furthermore, I add two diagnostics that did not figure in the discussion above.

(200) SEMANTIC IMPLICATIONS

- a. An UNPOSSESSIVE might differ from the PROPER POSSESSIVE w.r.t. to its presuppositions¹⁴⁴, for example, the unpossessive may trigger a uniqueness presupposition whereas the proper possessive does not. (*cf.* Hungarian NAD that is necessarily definite (Halm 2018: 364))
- b. *Analysis*: Different semantic denotation for the unpossessive that includes a presupposition not present in the proper possessive denotation.

I also add a pragmatic diagnostic that has not been considered in the studies surveyed above. It is relevant specifically in the case of POSS.2SG UNPOSSESSIVES considered in this thesis.

The diagnostic is based on the idea that a possessive construction with a 2SG possessor must pragmatically compete with the same construction with a 1DU (1PL) possessor. An utterance of ‘your (SG) dog’ in a context where ‘our dog’ is not ruled out must generate a Q-implicature (Horn 2006) that ‘the dog is yours, but not ours’.

While this is expected of proper possessives, a nominal marked with an unpossessive POSS.2SG lookalike will not generate any such implicatures as it is not marked with a possessive affix and hence does not pragmatically compete with possessive-marked nominals.

(201) PRAGMATIC COMPETITION

¹⁴³ An anonymous reviewer for *Ural-Altaic Studies* wonders if this claim is too general, since the [$\cdot D \cdot$] feature may be saturated not by a possessor DP but by some determiner, for instance, by a demonstrative. In a system like (Newman, E. S. B. 2021), this is indeed technically possible. This, then, makes it possible for an UNPOSSESSIVE to bear a [$\cdot D \cdot$] feature even though it does not admit a possessor DP, which might be due to semantic reasons. The upshot is that the analytical suggestion in (199b) might need to be revised to state instead that the UNPOSSESSIVE in question has a different semantic type than the PROPER POSSESSIVE, a semantic type which is one type e argument short: *e.g.*, $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ instead of $\langle\langle e, t \rangle, \langle e, \langle e, t \rangle \rangle\rangle$ (the type of the Proper Possessive in (184)). See § 4.7.1 for a specific proposal of this kind. There, however, I still assume that the unpossessive lacks [$\cdot D \cdot$], so its syntactic feature makeup and semantic type work in tandem to exclude possessor DPs. Pursuing here the interesting idea proposed by the reviewer would take us too far afield.

¹⁴⁴ Other implication types may also be relevant, for a taxonomy and relevant tests see (Tonhauser et al. 2013).

- a. A 2SG UNPOSSESSIVE must not pragmatically compete with a 1PL POSSESSIVE so that an utterance of *N*-POSS.2SG ‘your *N*’ with the unpossessive will never give rise to a Q-implicature that the alternative with the POSS.1PL is false (i.e., ‘your *N*, but not ours’).
- b. *Analysis*: Different semantic denotation for the unpossessive, making it irrelevant for competition of possessive-marked nominals.

I intend the diagnostics presented above to serve as universal tests for establishing which approach should be taken with a particular non-possessive function. I expect that with the addition of new data from unpossessive diagnostics, many of the existing monosemic treatments will have to be abandoned in favor of polysemic ones.

Below, I offer four case studies of non-possessive functions of the Kazym Khanty POSS.2SG exponents *-en/-an* and POSS.3SG exponents *-əλ/-aλ* where the unpossessive diagnostics provide crucial evidence for a polysemic treatment of the Associative Possessive, the Partitive Article, the Salient Article, and the Proprial Article.

4.4. *Applying the diagnostics*

In Chapter 1, I introduced the five functions of the Kazym Khanty possessives: the Proper Possessive function, the Associative Possessive function, the Partitive Article function, the Salient Article function, and the Proprial Article function. Equipped with the unpossessive diagnostics from §4.3, we can now test the monosemic prediction (§4.2.1) for each function. Do they observe the same behavior as the Proper Possessive function? Below I argue that they do not.

For convenience, the predictions of a monosemic approach for each diagnostic are sketched in Table 8. The differences in semantic implications were discussed in Chapter 3 so below I only briefly review them. See Table 9 in §4.5 for a summary of the actual results for each diagnostic, as discussed in the following sections.

	(i) Proper Possessive	(ii) Associative Possessive	(iii) Partitive Article	(iv) Salient Article	(v) Proprial Article

A. Shows allomorphy of (i) ¹⁴⁵	—	Yes	Yes	Yes	Yes
B. Allows possessive stacking	No	No	No	No	No
C. Agrees with the Addressee/superset antecedent in number	Yes	Yes	Yes	Yes	Yes
D. Admits an explicit possessor	Yes	Yes	Yes	Yes	Yes
E. Requires uniqueness (or other implications)	No	Same as (Ei)	Same as (Ei)	Same as (Ei)	Same as (Ei)
F. Competes with <i>-ew</i> [POSS.1PL]	Yes	Yes	n/a	Yes	Yes

Table 8. Predictions of a monosemic approach to each of the Kazym Khanty POSS.2SG and POSS.3SG functions w. r. t. the unpossessive diagnostics (§4.3)

4.4.1. Morphological diagnostics

The morphological diagnostics from (É. Kiss & Tánzos 2018; Halm 2018; Logvinova 2019), possessive stacking and different allomorphy, are not relevant for Kazym Khanty.

As mentioned in §3.2, possessive stacking is impossible regardless of the marker's function. Furthermore, all the morphophonological facts of the POSS.2SG exponent *-en* and the POSS.3SG exponent *-əλ* that I know of are consistent across all their functions. For instance, in all the functions of POSS.2SG the *-an* allomorph is used with plural (or dual) number marked nouns:

(202) a. *an-λ-an*

cup-PL-POSS.2SG

‘your cups / the cups [next to you]’ (Proper/Associative Possessive)

b. *amp-λ-an*

¹⁴⁵ I use narrower descriptions of diagnostics in Table 8 for clarity, but the order of diagnostics is the same as in §4.3. I suppress irrelevant parts of diagnostics that do not play a role in the following discussion.

dog-PL-POSS.2SG

‘your dogs / the dogs’ (Proper Possessive/Salient Article)

c. andrej-ηəl-an

A.-DU-POSS.2SG

‘your (two) Andrejs / Andrejs’ (Proper Possessive/Proprial Article)

Similarly for POSS.3SG, regardless of its function, when it cooccurs with dual number, the two markers are expounded as the portmanteau *-ηəl* [-DU.POSS.3SG] instead of the expected sequence **-ηəl-əl* [-DU-POSS.3SG].

(203) *ńawrɛm-ηəl*

child-DU.POSS.3SG

‘his/her two children / two of the children’ (Proper Possessive/Partitive Article)

These morphological facts support the monosemic view: the markers behave the same in the non-possessive functions as in the Proper Possessive function. However, other diagnostics suggest that this view is not correct for Kazym Khanty.

4.4.2. Agreement mismatch

The Proper Possessive agrees in person-number features with the possessor. In the case of a plural (or dual) addressee, the POSS.2NSG marker *-ən* is used instead of *-en* [POSS.2SG]. In the case of plural or dual third person possessor, the respective markers *-el* and *-ən* from the possessive paradigm in Table 7 are used instead of *-əl* [POSS.3SG].

The same holds for the Associative Possessive (204).

(204) a. *waśa, soχλ-en mɵηχ-e*

V. board-POSS.2SG wipe-IMP.SG>SG

‘Vasya, wipe the blackboard.’

b. *ńawrɛm-ət, soχλ-ən mɵηχ-a-λən*

child-PL board-POSS.2NSG wipe-IMP-NSG>(N)SG

‘Children, wipe the blackboard.’

These data suggest that even if the Associative Possessive is synchronically independent from the Proper Possessive, it is still a possessive agreement morpheme. The same is not true of the Salient Article and the Proprial Article, however.

Examples (205)–(206) show that even with a plural addressee the two markers still appear as *-en*. The “agreeing” form with the POSS.2NSG marker *-ən* is infelicitous here: it results in a Proper Possessive interpretation as if the dog belongs to the children.

(205) [A mother is telling her children: “I was walking along the street when I saw a dog.”]

amp-en/#-ən *ma pɛλ-am-a* *χurət-ti*
 dog-POSS.2SG/-POSS.2NSG I at-POSS.1SG-DAT bark-NFIN.NPST
pit-əs
 become-PST[3SG]
 ‘The dog started barking at me.’

(206) [Andrej is a caretaker in the kindergarten. The parents are going away for the weekend and they tell their children.]

ńawrɛm-ət, nin *wɔntr-en-ən/#-ən-ən* *lawəλ-aj-ti*
 child-PL, you.PL A.-POSS.2SG-LOC/-POSS.2NSG-LOC baby.sit[NPST]-PASS-2PL
 ‘Children, Andrej will look after you.’

The same is true of the Partitive Article (207). Neither POSS.3DU, nor POSS.3PL are felicitous in (207) under the desired reading despite the fact that the supposed possessor (namely, the superset antecedent) expressed by a null pronoun *pro*_[3DU] by hypothesis would be dual here.¹⁴⁶

(207) *i* *an-əλ/#-ən/#-eλ* *mij-e*
 one cup-POSS.3SG/-POSS.3DU/-POSS.3PL give-IMP.SG>SG
 ‘[– Can I help you? – See the two cups on the table?] Give me one cup.’
 Consultant’s comment on *-ən/-eλ*: “this is [the form used] if the cup belongs to the two of them [some third person]”.

These data suggest that the Partitive Article, the Salient Article, and the Proprial Article are not possessive agreement morphemes themselves which distinguishes them from the two Possessives.

¹⁴⁶ The reason I mention POSS.3PL here as well is that sometimes speakers might not use a dual form where it is expected because it seems that the dual is being lost in some idiolects.

4.4.3. Explicit possessor

The Proper Possessive admits an explicit DP-internal possessor (see §3.2 and §4.1.1). This is, however, much inhibited for the Associative Possessive and impossible for the other three functions.

In (208) with an Associative POSS.2SG marked cup, an explicit possessor is infelicitous. Adding it triggers a Proper Possessive interpretation (*i.e.*, ownership). (Although, see the next example for a caveat.)

(208) [A friend is over at the speaker's place. There's one cup on the table.]

(#*nǎŋ*) *an-en* *mij-e*
 you.SG cup-POSS.2SG give-IMP.SG>SG
 'Give me the cup.'

Consultant's comment on *nǎŋ*: "It's like 'give me your cup, don't touch grandma's cup', it should really be your cup".

The Associative Possessive does admit explicit possessors given sufficient context, *e.g.*, introducing a contrast on the possessor, as in (209) where two dogs are contrasted with respect to which possessor stands in an associative relation with them.

(209) ["Both Petya and I have been attacked by a dog recently."]

ma amp-em *wera* *pǎltap* *wθ-s*
 I dog-POSS.1SG very scary be-PST[3SG]
 'My dog was very scary. [But Petya's dog even turned out to be rabid.]'

The proper description of the Associative Possessive's syntax then is that it **disprefers** explicit possessors but does not rule them out entirely.¹⁴⁷

Examples (210), (211), and (212) show the same for the Salient Article, the Proprial Article, and the Partitive Article, respectively. Neither admits an explicit possessor. Given an explicit possessor, NPs with these markers are reinterpreted as Proper Possessive-marked ('your dog', 'your Andrej', 'one of his cups').

(210) [Context from (205).]

¹⁴⁷ This is probably due to information-structural restrictions on activated referents since Khanty is an extensive *pro*-drop language. Since the relation described by the Associative Possessive is strongly familiar, the possessor argument is strongly familiar too, if not salient, which would provide sufficient reason for *pro*-drop. I leave further exploration of this idea for another occasion.

(#nǎŋ) *amp-en* *ma* *pɛλ-am-a* *χurət-ti* *pit-əs*
 your dog-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]

‘The dog started barking at me.’

- (211) (#nǎŋ) *wontr-en* *jux* *šəp* *sewr-əs*
 you A.-POSS.2SG wood piece cleave-PST[3SG]
 ‘Andrej cleaved a log.’

- (212) [Context from (207).]

#*λuw i* *an-əλ* *mij-e*
 (s)he one cup-POSS.3SG give-IMP.SG>SG
 ‘Give me one of his cups.’
 Int.: #Give me one of the cups.¹⁴⁸

This suggests that the markers used in (210)–(212) have different selectional restrictions than the Proper Possessive, a different syntax.

4.4.4. Pragmatic competition for POSS.2SG lookalikes

Given standard neo-Gricean pragmatics (*e.g.*, (Horn 2006)), a second-person singular possessive is expected to compete in certain contexts with the first-person dual/plural possessive (201).

For the Proper Possessive function, the infelicity of POSS.2SG in a context where POSS.1PL is appropriate is demonstrated in (213). The use of POSS.2SG triggers the ‘yours, but not ours’ implicature just as expected.

- (213) [Vasya tells his wife:]

χot_λaŋλ-ew/#-en *pos-ijəλ*
 house_roof-POSS.1PL/-POSS.2SG drip-FREQ[NPST.3SG]

‘(Our) roof is leaking.’

Consultant’s comment on *-en*: “Then it’s only his wife’s roof, this is wrong”.

The same is shown for the Associative Possessive in (214). The fact that the speaker and their friend are waiting for the kettle to boil licenses an associative POSS.1DU. An

¹⁴⁸ The opposite order of possessor and numeral is also infelicitous in this example under the intended reading.

associative POSS.2SG, however, is degraded here as it suggests that only the addressee is associated with the kettle.

(214) [The speaker and their friend are sitting in the speaker’s kitchen, tired after a bath. They just put the kettle on fire and they wait for it to boil in silence. The speaker says:]

šajput-ɛmən/%-en *sora* *kawərm-əλ*¹⁴⁹
 kettle-POSS.1DU/-POSS.2SG quickly boil-NPST[3SG]
 ‘The kettle is boiling quickly!’

Consultant’s comment on *-en*: “Is it the case that only he [the addressee] needs the kettle or was it only him who put the kettle on the stove?”.

The picture is reversed for the Salient Article and the Proprial Article.

With the Salient Article (215), the POSS.2SG exponent is appropriate and does not trigger the ‘yours, but not ours’ implicature and a POSS.1PL is infelicitous as there are no grounds for claiming that the dog is ‘ours’. This suggests that the marker used in (215) is some semantically distinct marker that does not compete with other possessives.

(215) [Context from (205).]

amp-en/#-ew *ma* *peλ-am-a* *χurət-ti*
 dog-POSS.2SG/-POSS.1PL I at-POSS.1SG-DAT bark-NFIN.NPST
pit-əs
 become-PST[3SG]
 ‘The dog started barking at me.’

With the Proprial Article (216), a POSS.1PL marker, accepted by some of my consultants, suggests that Andrej is a relative or friend of ours, *i.e.*, has a proper possessive meaning. The same is not the case for POSS.2SG used as a Proprial Article: in (216) Andrej need not be ‘your friend’ to receive POSS.2SG marking. And here again no ‘yours, but not ours’ implicature is observed.

(216) *wəntɾ-en/#-ew* *jɯχ* *šəp* *sewr-əs*
 A.-POSS.2SG/-POSS.1PL wood piece cleave-PST[3SG]

¹⁴⁹ In this example, a Proper Possessive POSS.1SG is also appropriate since the kettle belongs to the speaker. I thank Alexey Kozlov (p.c.) for suggesting this context.

‘Andrej cleaved a log.’

Since the Salient Article and the Proprial Article do not pragmatically compete with other possessives, and since the Salient Article, the Proprial Article, and the Partitive Article do not agree with the supposed possessor (addressee or superset antecedent) in number, and do not admit an explicit possessor, it is safe to assume that they should be viewed as UNPOSSESSIVE markers.

What’s left, is to show that the two pairs of the Proper Possessive with the Associative Possessive and the Salient Article with the Proprial Article are also mutually independent. The issue is that the two Possessives both introduce a possessor and have the full possessive paradigm of exponents (i.e., they covary in ϕ -features with a possessor). At the same time, the Salient and the Proprial Articles both do not vary in ϕ -features, do not introduce a possessor, but are homonymous with *-en/-an* [POSS.2SG]. To tease apart the two problematic pairs, I review the semantic evidence from Chapter 3.

The Partitive Article does not share both formal and morphosyntactic properties with any other marker, because it is homonymous with *-əλ/-aλ* [POSS.3SG] instead and it does not vary in ϕ -features, and so it is obviously distinct from the four other markers already at this stage. Here is the denotation I assigned the Partitive Article in (126) presented in Sense Insertion rule-form (see §3.5 for details):

$$(217) \llbracket \text{Part}_i \rrbracket^{g,s,c} = \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle} \exists x_e: x < g(i). P(x) \wedge Q(x)$$

4.5. *The Proper Possessive vs. the Associative Possessive*

The Proper Possessive is a possessive modifier that is compatible with indefinite determiners such as *mułsər* ‘some (epistemically nonspecific)’ and it does not presuppose uniqueness. These two properties are illustrated with the following two examples repeated from §3.3.

(218) [The speaker is at a friend’s place.]

mułsər	an-en	mij-a
some.EN	cup-POSS.2SG	give-IMP[SG]

‘Give me any cup of yours.’

(219) [A child made a mess in the kitchen. Their parents tell them: “What a mess you made!”]

păsan	əxtij-ən	năŋ	juntut-en	uλ,
-------	----------	------------	------------------	-----

table on-LOC you.SG toy-POSS.2SG lie[NPST.3SG]
 kər_λaŋəλ-ən **nǎŋ** **juntut-en** uλ,

stove-LOC you.SG toy-POSS.2SG lie[NPST.3SG]
 pǎsan iλpij-ən **nǎŋ** **juntut-λ-an** kərət'λ'-əλ-ət

table under-LOC you.SG toy-PL-POSS.2SG lie.around-NPST-3PL

‘There’s a toy of yours on the table, [a toy of yours] on the stove, under the table
 your toys are scattered.’

If the Proper Possessive was inherently definite, e.g., by requiring uniqueness, both above examples would be infelicitous. Given that this is not so, I contend that the Proper Possessive is not inherently definite, as captured in its Sense Insertion rule repeated from (184).

(220) $[[\text{Poss}]]^{g,s,c} \leftrightarrow \lambda P_{\langle e,t \rangle} \lambda x_e \lambda y_e: \exists z [\text{POSS}(x, z) \wedge P(z)]. \text{POSS}(x, y) \wedge P(y)$

where POSS is an inherent possessive relation in the sense of (Vikner & Jensen 2002; Partee & Borschev 2003; Karvovskaya 2018)

The Associative Possessive, on the other hand, requires uniqueness (221)-(222) and is incompatible with the indefinite determiner *muləsər* (223).

(221) [A friend is over at the speaker’s place. There’s one cup on the table.]

an-#(en) mij-e
 cup-POSS.2SG give-IMP.SG>SG
 ‘Give me the cup.’

(222) [Context from (221) with several cups.]

an-#(en) mij-a
 cup-POSS.2SG give-IMP[SG]
 ‘Give me a cup.’

Consultant’s comment on *-en*: “[the addressee] will then ask ‘Which cup do you mean?’”.

(223) [Context from (222) with several cups.]

muləsər **an-#(en)** mij-a
 some.EN cup-POSS.2SG give-IMP[SG]
 ‘Give me some cup [no matter which].’

These properties are captured in the denotation of the Associative Possessive from (116) restated as a Sense Insertion rule in (224) (see §3.4 for comments and explanations).

$$(224) \llbracket \text{Assoc}_i \rrbracket^{g,s,c} \leftrightarrow \lambda P_{\langle e,t \rangle} \lambda x_e t y_e. R_i(x, y) \wedge P(y)$$

where R_i is a familiar free relation picked up from the context

The difference between the two Sense Insertion rules (220) and (224), provided that they represent accurate semantic analyses of the two Possessives, clearly shows that the Associative Possessive is semantically distinct from the Proper Possessive (see §4.7 for a discussion of their morphosyntax).

4.6. *The Salient Article vs. the Proprial Article*

The Salient and the Proprial Articles both do not vary in ϕ -features, do not introduce a possessor, but are homonymous with *-en/-an* [POSS.2SG]. It seems natural to try to analyze them monosemically, but the semantic differences between them described in Chapter 3 also force us to treat them separately. Here's a review of the relevant facts.

The Salient Article marks maximally salient referents. This is apparent from Roberts' 'salience without uniqueness' context (225) in which only definites that require maximal salience can be used (e.g., personal pronouns; §2.3); and from a non-salient strongly familiar context such as (226) where the passport is not a part of the current question under discussion, as I argued in §3.6, which makes it not salient enough. (Both examples are repeated from §3.6.)

(225) [A theatrical play for children: "A girl_i appeared from the room on the left. Another girl_k appeared from the room on the right."]

$\lambda u w_{k/*i} /$ ^{OK} **ewen**_{k/*i} još-ən lipət täj-əs
 (s)he / girl.POSS.2SG hand-LOC flower have-PST[3SG]
 'She_{k/*i} [lit.: the girl] was holding flowers.'

(226) ["I found somebody's passport in the street. I went to the town administration. Met a friend there and talked to her for some time."]

nəm **nəpek-əm/#-en** suwet-ən χăj-s-əm
 name paper-POSS.1SG/-POSS.2SG council-LOC leave-PST-1SG>SG
 '[Then] I left the passport at the administration. [Let them find the owner.]'

Compare the Proprial Article which is easily used in a clearly non-salient weak familiarity context such as (227).

(227) [Context from (206): Andrej is a caretaker in the kindergarten. The parents are going away for the weekend and they tell their children.]

ńawrɛm-ət, nin **wɔntr-en-ən** λawəλ-aj-ti
 child-PL, you.PL A.-POSS.2SG-LOC baby.sit[NPST]-PASS-2PL
 ‘Children, Andrej will look after you.’

Most importantly, the Salient Article derives simple, intensionally nonrigid definite descriptions (Kripke 1980; Muñoz 2019). A Salient Article-marked DP such as the ‘dog’ in (228) may refer to different dogs in different circumstances in the scope of the situation quantifier *kašəŋ śos* ‘every time’. This is confirmed by the acceptability of the continuation of (228) given in brackets that implies different dogs.

(228) *kašəŋ śos amp šiwaλ-t-ɛm-ən*
 every hour dog see-NFIN.NPST-1SG-LOC
amp-en ma pɛλ-am-a χurət-λ
 dog-POSS.2SG I at-POSS.1SG-DAT bark-NPST[3SG]
 ‘Every time I meet a dog, the dog barks at me. [Sometimes it is a big dog, sometimes it is a smaller dog.]’

This was explicitly modelled in §3.7.3 in a situation-semantic framework using the denotation of the Salient Article that I proposed in (140). I repeat it here for convenience as a Sense Insertion rule. (Intensional variability is achieved thanks to the predicate denotation’s *P* dependence on situations which is not apparent from (229), see §3.7.3.)

(229) $[[\text{Sal}_i]]^{g,s,c} \leftrightarrow \lambda P_{\langle e,t \rangle}: \text{sal}(i,c) \wedge \forall j \in \text{Dom}(g)[P(g(j)) \wedge j \geq_{\text{sal}(c)} i \rightarrow j = i]. g(i)$

As predicted by Muñoz’s (2019) crosslinguistic analysis of PROPRIAL ARTICLES, the Kazym Khanty Proprial Article derives rigid designators (Kripke 1980). A Proprial Article-marked name cannot refer to different referents in different circumstances (230) (repeated from (158a)). Instead, an unmarked name must be used to get a nonrigid reading of the DP.

(230) [“Every year we give a present to the 4th year student who gets the best grades.”]

kašəŋ ol mojləpsi wɔjl’ waśa / #waśaj-en
 every year present take[NPST.3SG] V. / V.-POSS.2SG

‘Every year the present is taken by a Vasya. [Last year it was Vasya Tas’manow and this year it’s Vasya Tarlin.]’

Both the felicitous bare noun variant of (230) and the infelicitous Proprial Article variant of (230) were analyzed in §3.7.3 using the Proprial Article denotation repeated here as the Sense Insertion rule (231).¹⁵⁰

$$(231) \llbracket \text{Prop}_i \rrbracket^{g,s,c} \leftrightarrow \lambda P_{\langle \langle s, \langle e, t \rangle \rangle} : \forall s', x \\ \left[P(s', x) \leftrightarrow \exists g'' \text{ in } s'. g''(i) \text{ is conventionally allowed to map to } x \wedge \right. \\ \left. i \text{ is a human name index} \right]. g(i)$$

Thus, exactly as in English under Muñoz’s analysis, the Kazym Khanty Proprial Article derives rigid DPs, while the Salient Article derives intensionally variable DPs, which is correctly captured by the denotations (229) and (231).

Based on this difference and the fact that the Proprial Article has no salience restrictions (see above), I claim that the Kazym Khanty Salient Article and the Kazym Khanty Proprial Article are mutually independent and must be treated as distinct morphemes. The differences between all the five markers are summarized in Table 9 in the next section.

4.7. Capturing the facts in DM

4.7.1. The analysis

So far I have made the following claims regarding the differences and similarities between the markers under discussion (see Table 9 for a summary of the empirical generalizations):

1. The two Possessives (the Proper Possessive and the Associative Possessive) are semantically distinct but morphosyntactically largely equivalent.
2. Two Articles (the Salient Article and the Proprial Article) are morphosyntactically different from the two Possessives but similar to each other.¹⁵¹
3. The Salient Article and the Proprial Article **are** different from each other semantically.

¹⁵⁰ But see fn. 155 for a slight alternative to the Proprial Article’s Sense Insertion rule.

¹⁵¹ See Chapter 3 (§§3.6.4, 3.8) for a semantic comparison of the four markers.

4. The Partitive Article differs from all four other markers both morphosyntactically and semantically.

	(i) Proper Possessive	(ii) Associative Possessive	(iii) Partitive Article	(iv) Salient Article	(v) Proprial Article
Abbreviation	Poss	Assoc	Part	Top	Prop
Allomorphs	εm/əm/am [1sg], en/an [2sg], əλ/ελ/aλ [3sg], ...	εm/əm/am [1sg], en/an [2sg], əλ/ελ/aλ [3sg], ...	əλ/ελ/aλ [~3sg] ¹⁵²	en/an [~2sg]	en/an [~2sg]
A. Shows allomorphy of (i)	—	Yes	Yes	Yes	Yes
B. Allows possessive stacking	No	No	No	No	No
C. Agrees with the Addressee/ superset antecedent in number	Yes	Yes	No	No	No
D. Admits an explicit possessor	Yes	Heavily restricted*	No	No	No
E. Requires uniqueness / partitivity (^U / _P)	No	Yes ^U	Yes ^P	Yes ^U	Yes ^U
F. Competes with -ew [POSS.1PL]	Yes	Yes	n/a	No	No
G. Admits interpretations beyond prototypical possessive ones	No	Yes	Yes	Yes	Yes
H. Requires salience	— [†]	— [†]	— [†]	Yes	No
I. Derives rigid noun phrases	— [†]	— [†]	— [†]	No	Yes

Table 9. Unpossessive diagnostics (§4.3), results for each of the Kazym Khanty possessives' five functions

* — explicit possessor only available under contrast (209)

¹⁵² [~3sg] means 'this marker looks like POSS.3SG but is not one'.

† — in the interest of space, these data were not included in this thesis¹⁵³

The simplest possible story would claim that each marker is represented with a distinct Poss head subscripted with a distinguishing feature (*e.g.*, Poss_[Poss], Poss_[Assoc], Poss_[Part], Poss_[Sal], Poss_[Prop]) and, possibly, with further differences in the feature makeup. Each head would have a dedicated set of spell-out rules for both interfaces and that's that.

However, such a story is not very convincing as it ignores two facts. First, the Proper Possessive and the Associative Possessive have fully equivalent paradigms. Second, the Salient Article and the Proprial Article have the same allomorphs *-en/-an* as the 2SG exponents of the proper and Associative Possessives. Do we really have to assume that a Kazym Khanty speaker's language competence contains redundant rules for these markers?

I believe that we do not. There is a more minimal account one can formulate granted Distributed Morphology's (DM) mechanism of allosemy. Recent work by (Wood 2015; 2023; Myler 2016; Wood & Marantz 2017; Kasenov 2023) has fruitfully explored allosemy of functional heads, whereby one head may map onto several distinct senses. These authors argued that this provides insightful accounts of predicative possession, argument introducing heads, modal ambiguities, and other phenomena.¹⁵⁴

Thus, we are enabled to account for the morphosyntactic similarity between the Proper and the Associative Possessives by saying that they simply **are** the same head, Poss_[Poss/Assoc]. This head has a single set of Vocabulary Insertion (List 2) rules, discussed in §4.1.2.1 which gives the full possessive paradigm, and two Sense Insertion (List 3) rules: one mapping to the Proper Possessive semantics (220), the other to the Associative Possessive semantics (224). The choice between the two List 3 rules is free. Under this account there is no question as to why the Proper and the Associative Possessives have equivalent paradigms — it is **the same** paradigm, *i.e.*, the same set of List 2 rules.

The same reasoning goes for the pair of the Salient and the Proprial Articles. Neither of the two markers selects for a possessor or Agrees with one. Both have the two allomorphs *-en/-an* (which are not dependent on ϕ -features). This is because they simply

¹⁵³ The facts are that the Proper, the Associative Possessive, and the Partitive Article do not require salience and do not derive rigid DPs.

¹⁵⁴ See also the references in (Wood 2023: fn. 21).

are the same head, POSS_[Sal/Prop], with a single set of List 2 rules and with two List 3 rules chosen freely.

And the Partitive Article is genuinely fully independent from the others. It has a dedicated syntactic head POSS_[Part], and dedicated List 2 and List 3 rules. Note that it has to be distinct from POSS_[Sal/Prop] because they are mapped to different exponents at the same time as they are mapped to different denotations. This correlation of exponents and denotations is the paradigm indication that different syntactic terminals are required, because in DM syntactic terminals are the only thing linking semantics to morphophonology.

Thus, I propose the following syntactic terminals and spell-out rules for our five markers (232)–(234). (The syntactic and morphophonological details are as in §4.1. The semantic system is §1.2 with the situation-semantic extension from §3.7.3 for the Proprial Article. The denotations are cross-referenced below.)

(232) SYNTACTIC TERMINALS

- a. POSS: [Poss/Assoc][·Num·][·D·][]_{<pers-num>}
- b. POSS: [Sal/Prop][·Num·]
- c. POSS: [Part][·Num·]

(233) VOCABULARY INSERTION RULES

- a. POSS_[Poss/Assoc]_[ϕ] is spelled out depending on the ϕ -features that it bears after Agree.
- b. POSS_[Sal/Prop] $\leftrightarrow$ -an
- c. POSS_[Sal/Prop] $\leftrightarrow$ -en / Num[SG]__
- d. POSS_[Part] $\leftrightarrow$ -a λ
- e. POSS_[Part] $\leftrightarrow$ $\emptyset$ / Num[DU]__
- f. POSS_[Part] $\leftrightarrow$ -ə λ / Num[SG]__

(234) SENSE INSERTION RULES

- a. POSS_[Poss/Assoc] $\leftrightarrow$ [[Poss]]^{*g,s,c*} (see (220))
- b. POSS_[Poss/Assoc] $\leftrightarrow$ [[Assoc]_{*i*}]]^{*g,s,c*} (see (224))
- c. POSS_[Part] $\leftrightarrow$ [[Part]_{*i*}]]^{*g,s,c*} (see (217))
- d. POSS_[Sal/Prop] $\leftrightarrow$ [[Sal]_{*i*}]]^{*g,s,c*} (see (229))

4.7.2. *Is this the best we can do?*

Before we conclude there are two concerns to address. First of all, why not say that there is only **one** head which maps onto a Possessive (234a, b) if it bears ϕ -features and onto a POSS.2SG-lookalike Article (234c, d) if it does not? Secondly, what about the full equivalence of the three Articles' allomorphs and the two possessives' 2SG and 3SG allomorphs *-en/-an* and *-əλ/-aλ/-∅*? Surely, this is a mystical situation! As it turns out, the two questions are connected. Let us turn to the first one.

There are two reasons why one cannot assume a single Poss head. (Assume that we ignore the Partitive Article for now.)¹⁵⁶ First, the two POSS.2SG-lookalike Articles do not select or admit a possessor DP. Under the obligatory operations view of syntax that I committed myself to in §4.1.1 (following (Preminger 2014) and others), there is no way for a Merge feature [$\cdot D \cdot$] to not trigger the Merge of a possessor DP. And if a possessor DP **is** Merged it must somehow contribute to the meaning of the whole possessive nominal, resulting in a type mismatch, since the two Articles' denotations are not suited to compose with a possessor DP. Additionally, such a possessor DP must always stay implicit as we never find overt possessor DPs with the two Articles. However, there is no evidence even for an implicit one. Thus, it seems that a [$\cdot D \cdot$] feature with the two Articles runs us into theoretical and empirical problems that require us to abandon the well-motivated obligatory operations view and/or to include ad hoc assumptions. I maintain that the Poss head behind the Salient and the Proprial Articles does not bear a [$\cdot D \cdot$] feature. (The same argument is easily made for the Partitive Article, if we consider mapping Poss to it in the absence of ϕ -features.)

The second reason to not assume one Poss head is that the Salient and the Proprial Articles **look** like 2SG possessives but show no traces of bearing [2SG] ϕ -features and, therefore, cannot be spelled out with the rules from (183) that map Poss_[2SG] onto *-en/-an*. One also cannot assume that *-en/-an* is the most underspecified possessive exponent since, as discussed in §4.1.2.1, the absence of ϕ -features is expounded as $\emptyset$. Ditto for the Partitive Article as looking like a 3SG possessive. To sum up, the Salient Article, the Proprial Article, and the Partitive Article cannot be assumed to bear [$\cdot D \cdot$], [$\cdot \cdot \cdot$]_{<pers-num>}, or

¹⁵⁶ Even given that the Partitive Article still gets a separate Poss head as in the account that I proposed.

any ϕ -features in the syntax which is why they need two dedicated syntactic terminals and Vocabulary Insertion rules. One terminal for the Salient Article and the Proprial Article with one set of List 2 rules; and one terminal for the Partitive Article with another set of List 2 rules, as proposed above.

So how about their allomorphy? Do we just ignore that the allomorphs are the same as for 2SG/3SG possessives? Synchronically, I believe, we have no choice, for the reasons just discussed. But I do not think that that is a weakness of the present account. There is a natural diachronic explanation for this fact.

It seems reasonable to hypothesize that the Salient Article and the Proprial Article grammaticalized from 2SG possessives and the Partitive Article grammaticalized from POSS.3SG.¹⁵⁷ And this is why they inherited their allomorphs. In fact, this is something amply documented for grammaticalization cases across the world's languages. Roughly speaking, grammaticalization processes involve semantic reanalysis of a form/construction in a specific context where its base meaning and its goal meaning overlap. Whether a change in phonological form follows the change in meaning is a separate issue. It is by no means a necessity.

Based on a statistical analysis of 1003 grammaticalization paths, Bisang, Malchukov and colleagues (Bisang & Malchukov et al. 2020a) show that in numerous cases **changes in Lehmann's (2002/2015) parameters of grammaticalization do not cooccur** (Lehmann 2002/2015). That is, the semantic, morphological, phonological, and syntactic changes that can take place during grammaticalization do not necessarily go hand in hand.

For instance, one might have semantic erosion (loss and generalization of meaning/semantic integrity) without loss of phonological material. Indeed, (Ibid.: 41 and §3.4) observe that there is no correlation between semantic erosion and phonetic reduction. The overall conclusion that Bisang, Malchukov et al. reach is:

¹⁵⁷ Arguably, this is a very slight step in grammaticalization since both the source marker and the target marker are bound, abstract in meaning, etc. Nevertheless, this is still genuine grammaticalization according to Lehmann's (2002/2015) parameters of grammaticalization, as discussed below.

As for the Partitive Article, this is in fact what (Fraurud 2001; É. Kiss 2018; É. Kiss & Tánčoz 2018) have proposed for Udmurt and Hungarian Partitive Unpossessives.

«Semantic Integrity seems to involve stronger interactions with function-related parameters such as Paradigmaticity and Syntagmatic Variability, while Phonetic Reduction only shows a strong correlation with Bondedness» (ibid.: 47)¹⁵⁸.

In the case of the Salient Article, the Proprial Article, and the Partitive Article along with the change in meaning, we have a reduction of paradigmatic complexity (paradigmaticity), since only “one row” of the source paradigm is preserved, but the allomorphy stays the same.

It seems quite natural that the implications of this **partial grammaticalization** for the synchronic competence of a speaker are also in a way partial, and allosemy (as in (234)) provides a good way to model this. This is especially so in the case of the Proper Possessive and the Associative Possessive. The only thing that they differ in is their semantics. They have the same syntax and morphology, and this is completely in line with what we know about grammaticalization.

Thus, I contend that the present account provides an interesting take on what grammaticalization means for formal theories of synchronic grammar. Lehmann’s (2002/2015) parameters of grammaticalization may be seen as resulting from changes to one of the three Lists, and they do not have to happen in parallel. Semantic reanalysis without phonological or paradigmatic change simply means that a new Sense Insertion rule becomes available. Phonological or paradigmatic change further implies changes in the syntactic terminal and in List 2 rules (compare the case of the Udmurt accusative *-jez* discussed in §4.2.2 where changes to all three Lists presumably occurred).

In the case of Kazym Khanty POSS.2SG Unpossessives, we have four instances of grammaticalization along the following lines (step 4 is independent from steps 2 and 3):

1. The Proper Possessive develops a new Sense (234b). → The Associative Possessive appears.
2. One of the possessives develops a new Sense (234d) and loses syntactic features along with paradigmatic complexity (232b)–(233b–c). → The Salient Article appears.

¹⁵⁸ Paradigmaticity concerns the size of the paradigm and its degree of formal homogeneity, and syntagmatic variability concerns freedom of linear order. See (Bisang et al. 2020b) for details.

3. The Salient Article develops a new Sense (234e). → The Proprial Article appears.
4. The Proper Possessive develops a new Sense (234c) and loses syntactic features along with paradigmatic complexity (232c)–(233d–f). → The Associative Possessive appears.

Aspects of a target marker that were not affected in the above steps remain the same as in the source marker, in particular, its allomorphy.¹⁵⁹

The grammaticalization chain just sketched serves as an illustration for the DM–grammaticalization connection that I wanted to draw. Of course, this is not the only imaginable scenario. A serious grammaticalization proposal requires much more (*cf.* (É. Kiss & Tánzos 2018; Halm 2018)), but this is not the place to make one. My goal in this subsection was to argue that the apparent weaknesses of the proposed analysis are a natural consequence of how grammaticalization works.¹⁶⁰

4.8. *Summary*

In this chapter, I argued that the non-possessive functions of the Kazym Khanty POSS.2SG and POSS.3SG markers must be treated polysemically, as UNPOSSESSIVE MARKERS homonymous with, but synchronically independent from the Proper Possessive. This was done using the **unpossessive diagnostics** developed specifically for this purpose.

I showed that the behavior of the Associative Possessive, the Partitive Article, the Salient Article, and the Proprial Article does not conform to the monosemic prediction (discussed in §4.2.1) with respect to multiple properties. Thus, **these markers must be diagnosed as UNPOSSESSIVE**.

I presented an analysis of the four Unpossessives within DM using the mechanism of (free) allosemy. I argued that the Proper Possessive and the Associative Possessive are **different Senses spelling out the same syntactic terminal** (and I claimed the same for the Salient Article and the Proprial Article) which explains why they show identical

¹⁵⁹ Also, for instance, its syntactic category, Poss, which also shows no evidence of change for any of the four grammaticalized markers.

¹⁶⁰ The analysis presented in this section was influenced by Daniar Kasenov, who introduced me to DM and gave insightful comments, and Timur Maisak, who taught me grammaticalization theory. I am deeply grateful to them both. Naturally, they should not be held responsible for any errors or mistakes here.

allomorphy and syntactic behavior, while the Partitive Article was argued to be the most independent from the other four markers. I further argued that allosemy is a natural consequence of the normal workings of grammaticalization and, thus, a DM architecture of the sort assumed here provides a good framework for modeling the synchronic results of partial grammaticalization.

The unpossessive diagnostics presented in this chapter can and should be applied to data from other Uralic varieties and languages that have non-possessive functions of possessives. I expect that this will lead to the discovery of additional UNPOSSESSIVE markers, thus, opening the route to detailed investigations into their synchronic semantics and diachronic development paths, as exemplified by (É. Kiss 2018; É. Kiss & Tánczos 2018; Halm 2018; Logvinova 2019; Serdobolskaya et al. 2019). Ultimately, this could enrich the typology of definiteness and specificity markers with previously unattested types, as exemplified by Chapter 3 of this thesis.

5. Semantics of the Salient Article¹⁶¹

The Salient Article was first described in (Mikhailov 2021b). In the following examples, it is used with a topical referent (235) and with a demonstrated referent (236).

(235) [“I was walking along the street when I saw a dog.”]

amp-en ma pɛλ-am-a χurət-ti pit-əs
dog-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]
‘The dog started barking at me.’

(236) [A mother is walking with her child. The child points at a flower:]

ma **tum** **lipt-en** mənəm-ti λaηχα-λ-əm
I that flower-POSS.2SG pick-NFIN.NPST want-NPST-1SG
‘I want to pick that flower.’

In both examples there does not seem to be any discourse-internal relation between the addressee and the referent of the highlighted noun phrase. Intuitively, this suggests that we are not dealing with the Proper 2SG Possessive here, but rather with some Unpossessive, as argued in detail in the preceding chapter. I call this Unpossessive the ‘Salient Article’ in this thesis but I called it the ‘Topic Marker’ in (Mikhailov 2021b) because I hypothesized that it marks topical referents. In what follows I will reject this hypothesis (§5.2.2); hence a different name is needed.

The goals of this chapter are three-fold. Firstly, to provide a detailed empirical study of a single Uralic unpossessive, namely, the Salient Article POSS.2SG of Kazym Khanty. Secondly, to give a motivated semantic analysis that correctly reflects the Salient Article’s conditions of use. And lastly, to situate the Salient Article in the typology of definites.

Thus, this chapter attempts to do for the Salient Article with more care and scrutiny, what was done in broader strokes in Chapter 3 for five Kazym Khanty (Un)possessives.

I will argue for the following claims:

- 1) Noun phrases with the Salient Article are definite.
- 2) The Salient Article does not directly encode syntactic role or information-structural distinctions.

¹⁶¹ This chapter closely reproduces the narrative from (Mikhailov 2024a) with important refinements and significant additions in §§5.2.3, 5.4.2, 5.5.

- 3) The Salient Article marks the most salient discourse referent satisfying the DP description in the given context.

Thus, I argue that the Kazym Khanty Salient Article instantiates a previously unattested definite determiner type. Even though it is similar to other topic-related phenomena in the world's languages such as the Obdorsk Khanty Agreeing Objects (Nikolaeva 2001) or the Blackfoot Proximate Marking (Bliss 2017), the Kazym Khanty Salient Article does not directly encode either topichood (contra (Mikhailov 2021b)) or proximate status (contra (Muravyev 2022)) but is used exclusively as a definite determiner.

The rest of the chapter is structured as follows. Section 5.1 introduces the theoretical notions to be employed in what follows, alongside cross-linguistic 'standards of comparison' for our description of the Salient Article. In Section 5.2, I describe the Salient Article in detail and argue against a syntactic account of its distribution. In Section 5.4, I outline and motivate a semantic analysis of the Salient Article, based on Roberts' (2003) semantics for definite DPs, and in Section 5.5, I implement it in a dynamic semantic system, based on Stojnić's (2021) Attention-Coherence approach, modelling the Salient Article's use with demonstratives and in 'salience without uniqueness' contexts.

5.1. Basic theoretical notions

5.1.1. Definiteness and salience

In what follows, I will be referring to the relevant semantic notions from the Salience-Familiarity Scale typology of definites explored in Chapter 2, so it is useful to repeat here some definitions.

I understand definiteness following (Roberts 2003) as involving two presuppositions: weak familiarity and informational uniqueness (§2.1).

(237) WEAK FAMILIARITY AND INFORMATIONAL UNIQUENESS OF DEFINITE DPs (informal)
(Roberts 2003: 308)

Given a context C , use of a definite DP_i presupposes that it has as antecedent¹⁶² a discourse referent x_i which is:

- a. weakly familiar in C , and
- b. unique among discourse referents in C in being contextually entailed to satisfy the descriptive content of DP_i .

It's also useful to remember how I proposed to distinguish weak familiarity from strong familiarity in §2.3.2.

(238) WEAK VS. STRONG FAMILIARITY

For the context c' prior to the current utterance and the context c_0 of the current utterance (as in (54)),

- a. A strongly familiar discourse referent i is included in the domain of the assignment function prior to its mention in the current utterance.

For a given assignment $g_{c'}$, i is strongly familiar iff $i \in Dom(g_{c'})$

- b. A weakly familiar discourse referent i is entailed to exist by the common ground but is not necessarily in $Dom(g_{c'})$ prior to its mention in the current utterance.

Thus, weak familiarity includes two cases:

- i. 'Narrow' weak familiarity: $i \notin Dom(g_{c'})$ and $i \in Dom(g_{c_0})$
- ii. Strong familiarity: $i \in Dom(g_{c'})$

(239) REFERENT INTRODUCTION

Referents may be introduced:

- a. linguistically,
- b. by non-linguistic events in the utterance situation,
- c. by entailment from the common ground when relevant (Roberts 2003: 305), or
- d. by accommodation.

The difference of (238)-(239) from Roberts' proposal and with the subsequent literature is in where the line is drawn in (239) with respect to how the discourse referent needs to be introduced in order to be strongly familiar (cf. Roberts 2003: (29)). I assume

¹⁶² Since, as we saw above, the discourse referent need not be introduced by any explicit linguistic material, the term 'antecedent' should be taken to refer to the corresponding discourse referent here, and not to any explicit DP.

that extralinguistic events may have the same effect on the assignment, and be treated the same as, proper linguistic introduction of discourse referents. The crucial difference is then spelled out as whether the discourse referent was in $Dom(g_{c'})$ prior to the utterance (strong familiarity) or not (weak familiarity). I suggested in §2.3.2 that this interpretation of strong familiarity may provide the correct boundary to delimit STRONG ARTICLE definites. (See (Schwarz 2019: §4.2) for evidence of subtle crosslinguistic variation in the use of STRONG ARTICLE definites.)

Following Barlew (2014) on the Bulu Salient Article and Roberts (2003, 2011) on personal pronouns, I take **salience** to be a function of **attention capture**. A referent that is salient to a given individual is a referent that she is paying attention to. Just like attention, salience is a gradient notion so that a referent may be salient to a higher or lesser degree. The possible sources of salience are summarized in (240), repeated from (43).

(240) SOURCES OF SALIENCE

a. **Top-down attention capture:**

- The referent is relevant to the interlocutors' current goals, and so they attend to it;
- The referent is a part of the current question under discussion, and so the interlocutors attend to it;

b. **Bottom-up attention capture:**

- The referent is perceptually prominent in the speech situation due to movement, sudden change, standing out from the surroundings, or other factors, and so the interlocutors attend to it;
- The speaker directs the addressee's attention to the referent via a demonstrative accompanied by a pointing gesture;
- The referent is mentioned by the speaker and thus comes to the addressee's attention.

Using this and following Barlew (2014), I gave the definition of discourse referent salience in the current context, notated as $sal(i, c)$, in (44) repeated as (241). Salience is defined in terms of an attention relation: $att(i, a_c, t_c) \leftrightarrow a_c$ is attending to i at t_c .

(241) Given a context c with speaker s_c and addressee a_c and a discourse referent $i \in D_c$:
 $sal(i, c) \leftrightarrow att(i, s_c, t_c) \wedge C_c$ entails that $att(i, a_c, t_c)$

A discourse referent *i* is salient in *c* iff the speaker is attending to *i* at t_c and the common ground entails that the addressee is too attending to *i* at t_c . (Barlew 2014: 632–633)

I further introduced a degree-based formalization of salience in (62) which helped us distinguish relative maximal salience, associated in (Roberts 2003) with personal pronouns, and absolute maximal salience, which, I suggested building on (Gundel et al. 1993), is associated with unstressed and zero pronouns. Together, these types of conditions on discourse referents, imposed by different kinds of definite expressions, form the Salience-Familiarity Scale (repeated from (61)):

(242) THE SALIENCE-FAMILIARITY SCALE

weak familiarity > strong familiarity > salience > relative maximal salience > absolute maximal salience

See the relevant examples of WEAK, STRONG, and SALIENT ARTICLES in Chapter 2. Below, I will pay special attention to the distinction between salience simpliciter and relative maximal salience which is based on ‘salience without uniqueness’ contexts. In contexts like this, the SALIENT ARTICLE gram type (exemplified by the Bulu Article) and other article types discussed in Ch. 2 are ruled out, while relative maximal salience-presupposing expressions (pronouns and demonstratives) are felicitous (§§2.2.1, 2.3, and fn. 107). This is illustrated for English below:

(243) A woman entered from stage left. Another woman entered from stage right. **She** / **#the woman** was carrying a basket of flowers. (adapted from (Roberts 2003: 324))

I discuss ‘salience without uniqueness’ contexts for Kazym Khanty in §5.2.3, arguing that the Kazym Khanty Salient Article presupposes relative maximal salience as it is felicitous in these contexts.

5.1.2. Information structure and topicality

Another dimension that is relevant to our discussion of the Salient Article is information structure. After (Lambrecht 1994) and (Nikolaeva 2001), I understand information structure as creating a pragmatically-structured proposition that mediates between the meaning of an utterance and its form.

One part of the proposition corresponds to information assumed by the speaker to be known to the addressee (Lambrecht’s pragmatic presupposition, cf. common ground

(Stalnaker 1974)). I will refer to the corresponding part of the utterance as **theme**, following one terminological tradition (Halliday 1985), in order to avoid confusion with the notion of topic that is used in another sense below.¹⁶³

The other part of the pragmatically structured proposition is defined via Lambrecht's notion of pragmatic assertion. Pragmatic assertion is that information which the addressee is expected to know upon hearing the utterance. The **rheme** (focus in Lambrecht's terminology) is that part of the utterance where the pragmatic assertion differs from the pragmatic presupposition.

To illustrate, consider the example (244) below from the Obdorsk dialect of Northern Khanty. Following standard practice in research on information structure, the theme-rheme division of a given sentence can be inferred from question-answer pairs. The theme of the answer is taken to be the part that is congruent with the question and the rheme is the part that corresponds to the question word (in content questions). Thus, in the answer in (244), the theme is the open proposition "X killed the/a reindeer", and the rheme is "X = John", with the latter supplying a value for the variable X in the theme (cf. (Nikolaeva 2001: 4ff.)).

(244) OBDORSK DIALECT OF NORTHERN KHANTY

kalaŋ xoj-na we:l-s-a? Juwan-na we:l-s-a
 reindeer who-loc kill-pst-pass[3sg] J.-loc kill-pst-pass[3sg]¹⁶⁴
 'Who killed the/a reindeer? John did.' (adapted from (Nikolaeva 2001: 25))

The reason that I illustrate these notions with data from the Obdorsk dialect is that these data provide a standard of comparison for our discussion of the Kazym dialect Salient Article in the following sections.

In her seminal article (Nikolaeva 2001), Irina Nikolaeva has argued that information-structural notions of primary topic and secondary topic are grammaticalized in the Obdorsk dialect as the syntactic roles of Subject and Agreeing Object (the latter is marked as O2 in the glosses).

¹⁶³ See (Roberts 2019) for a discussion of the terminological matters and the references there. Question under discussion can be taken to refer roughly to the same notion as theme. See below for details.

¹⁶⁴ Where nothing hinges on it, I adopt for Obdorsk examples the glossing conventions I use for the Kazym dialect.

Following a long tradition, Nikolaeva defines (primary¹⁶⁵) **topic** as whatever the proposition is ABOUT ((Reinhart 1981; Strawson 1964; Lambrecht 1994) and other works). The **aboutness relation** holds between a discourse referent and a proposition if the speaker assumes that the discourse referent is a center of current interest about which the assertion is made (Nikolaeva 2001: 4). By virtue of this definition, the topic is necessarily a part of the theme and is subject to a presupposition of salience (cf. the discussion in §2.3, §3.6, and §5.1.1 above).

I assume following Roberts (2012; 2019) that each utterance in a discourse answers a (possibly implicit) question under discussion (QUD) and that its theme is defined as that part of the utterance which is congruent with the QUD that it addresses. Given this, the discourse referents figuring in the theme are guaranteed to be relevant and, therefore, salient. Thus, we might add to the definition of the aboutness relation that the referent that the proposition is about figures in the QUD addressed by this proposition, which (at least) partly guarantees its salience.

As for the Obdorsk dialect data, Nikolaeva observes that the constituent that refers to the (primary) topic is required to be the Subject. This leads to extensive use of Passivization to promote topical non-agents to the Subject role. And, conversely, rhematic agents are demoted via the Passive to the Oblique role marked with the Locative *-na*, given the presence of another topical referent.¹⁶⁶ Thus, in (244) the Passive in the answer is **obligatory** because, based on the question, the patient argument is the topic in this sentence, and the agent argument is in the narrow argument rheme.¹⁶⁷

Similarly, in a predicate rheme sentence with the recipient argument as the topic, the recipient is **obligatorily** promoted to Subject via Passivization, while the agent and patient arguments (in the predicate rheme) are demoted to Obliques.

(245) [“What about Peter?”]

(luw)	Juwan-na	ke:si-na	ma-s-a
he	J.-LOC	knife-LOC	give-PST-PASS[3SG]

¹⁶⁵ For a discussion of secondary topic see below.

¹⁶⁶ Sentence rheme utterances maintain Active Voice since there is no topical constituent that would need to be promoted to subject.

¹⁶⁷ I adapt Lambrecht’s taxonomy of foci that Nikolaeva uses (Nikolaeva 2001: 3–4) to the terminology used here. Thus, instead of “argument focus”, I will speak of argument rheme and similarly for Lambrecht’s predicate focus and sentence focus.

‘John gave him a knife.’ (adapted from (Nikolaeva 2001: 25))

Now, Nikolaeva associates another syntactic role with the secondary topic. The **secondary topic** is defined as “[a]n entity such that the utterance is construed to be ABOUT the relationship between it and the primary topic” (Ibid.: 26). The two referents are presupposed to stand in a relationship. In the Obdorsk dialect, the secondary topic necessarily triggers Object Agreement on the verb, which is the central property of the Agreeing Object (O2) recognized by Nikolaeva as a distinct syntactic role.

As the secondary topic, the O2 is barred from occurring in the rheme, not given by the current QUD. Thus, O2 is excluded from sentences with narrow patient rheme (246), with predicate rheme including the patient (247), or with sentence rheme (248). (The rheme is highlighted in each of these examples.) A Non-agreeing Object (O1) is used in such sentences instead.

(246) a. [“Whom did John hit?”]

luw/Juwan	Pe:tra	re:sk-əs	/	*re:sk-əs-li
he/J.	P.	hit-PST[3SG.O1]	/	*hit-PST-3SG.O2 ¹⁶⁸

‘He/John hit Peter.’

b. Juwan **man-e:m** wa:nt-əs / *wa:nt-əs-li anta naŋ-e:n
 J. I-acc see-pst[3sg.o1] / see-pst-3sg.o2 not you-acc
 ‘John saw me, not you.’ (adapted from (Ibid.: 29))

(247) [“What did John do?”]

luw	Pe:tra re:sk-əs	/	*re:sk-əs-li
he	Peter hit-pst[3sg.o1]/	/	*hit-pst-3sg.o2

‘He hit Peter.’ (adapted from (Ibid.: 30))

(248) [“What happened?”]

ma	tam	kalaŋ	we:l-s-əm	/	*we:l-s-ə:m
I this	reindeer	kill-PST-1SG.O1	/	kill-PST-1SG.O2	

‘I killed this reindeer.’ (adapted from (Ibid.: 28))

¹⁶⁸ Since O1 and O2 are only morphologically distinguished in verbal agreement, Nikolaeva marks the verb as either O1 or O2 depending on whether it agrees with the Object.

Where O2 does occur, it is given in the QUD, so the conditions on secondary topic outlined above obtain. Compare (249) to (247) above. According to Nikolaeva, both examples contain predicate rheme structures, but only in (27) is the Object a secondary topic which ensures obligatory Object Agreement.

(249) [“What did John do to Peter?”]

luw Pe:tra / luw-e:lre:sk-əs-li / *re:sk-əs
 he P. / he-acc hit-pst-3sg.o2 / hit-pst[3sg.o1]

‘He hit Peter/him.’ (adapted from (Ibid.))

A similar argument for the role of information structure in the Kazym dialect grammar has been put forward by (Koshkareva 2002) (with important dialectal differences from the Obdorsk dialect). However, in a recent paper, Muravyev (Muravyev 2023) has shown that at least in the contemporary Kazym dialect grammar, information-structure does not wholly account for the distribution of the Passive and the Object Agreement (see also (Muravyev & Zhornik 2023)).

Rather, topicality hierarchies (person, animacy, definiteness, cf. (Zúñiga 2006: 21; Givón 2001b: 156; 2001a: 200)) play an important role in determining which noun phrase is the Subject or is promoted to Subject and whether the Object is agreed with. This likens the Kazym dialect (and specifically its Passive) to languages with direct/inverse systems (Zúñiga 2006). Thus, according to Muravyev, only the Passive voice is available in (250) regardless of information structure because the patient *mašaj-en* ‘Masha’ outranks the agent *wot-en* ‘wind’ on the animacy hierarchy (animate > inanimate¹⁶⁹).

(250) KAZYM DIALECT OF NORTHERN KHANTY

- a. wot-en mašaj-en iλ **pāwət-s-a**
 wind-LOC M.-POSS.2SG down fell-PST-PASS[3SG]
 b. *wot-en mašaj-en iλ **pāwt-əs**
 wind-POSS.2SG M.-POSS.2SG down fell-PST[3SG]

‘The wind knocked Masha off her feet.’ (adapted with from (Muravyev 2023))

¹⁶⁹ Here and below, I quote only the relevant parts of the hierarchies in question, following Muravyev. For details see the works cited above and the references there.

Similarly, the person hierarchy constrains the availability of Passive with 1st and 2nd person pronoun agents (speech act participant > third person). As it turns out, these pronouns do not form the Locative Case at all, which is the case assigned to demoted participants, so that they may never be demoted regardless of any other considerations (251). This might be seen as an instance of grammaticalization of a person hierarchy effect: the most prominent noun phrases on the hierarchy cannot be demoted.

- (251) a. ma waśaj-en χătśə-s-əm
 I V.-POSS.2SG hit-PST-1SG
 b. *waśaj-en <mănemən>¹⁷⁰ χătśə-s-i
 V.-POSS.2Sg I.LOC hit-PST-PASS[3SG]
 ‘I hit Vasya.’ (adapted from (Ibid.)

Another trait common to languages with a direct/inverse system is obviation. Based on the similarities of Kazym Khanty Passive to the Inverse of these languages, Muravyev proposes in an unpublished report (2022) that the Kazym dialect of Kazym Khanty also has an obviation system which, crucially for us, includes the Salient Article. I elaborate on this in the next section.

5.1.3. Obviation and proximate status

One more parallel that I will consider for the Salient Article is with proximate markers in languages with obviation systems such as the Algonquian languages.

Obviation refers to a separate tier of grammar that distinguishes a privileged third-person argument, called the **proximate**, from the other third persons, called **obviatives**. Crucial features of obviation systems (Aissen 2001) involve the condition of ‘proximate uniqueness’ and the condition that only third persons are eligible for proximate/obviative marking, with the first- and second-person pronouns excluded from consideration. **Proximate uniqueness** states that there must be one and only one proximate in an obviation span (which might consist in a single clause or a whole episode of a narrative) (Aissen 2001: 12).

Obviation systems are a common trait of Algonquian languages but are also found in the other language families of the Americas, as well as beyond that region (see (Ibid.)

¹⁷⁰ Since personal pronouns do not form Locative, a constructed form is provided here in angle brackets.

for a preliminary typology). In Algonquian languages, assignment of proximate status is conditioned on the following hierarchies (Ibid.: 6–7, cf. Zúñiga 2006; Givón 2001b; 2001a):

- animacy (Anim > Inan, i.e., animates are preferred as proximates);
- semantic role (Agent > Patient);
- topichood (Top > Non-top);
- possessorhood (Possessor > Possessee, in a possessive construction the Possessor is preferred as proximate over the Possessee).

Beyond that language family, mostly the same constraints are relevant, although particular languages may diverge from this set to varying degrees, and some systems also include pronominality (Pron > Common N) as a relevant hierarchy.

As mentioned above, Muravyev (2022) hypothesizes that the Kazym dialect also has an obviation system which includes the following grammatical devices distinguishing the proximate from obviatives:

- 1) The Passive marks sentences with proximate patients (250);
- 2) The Salient Article marks proximate status;
- 3) Possessive agreement is triggered by proximate lexical Possessors.¹⁷¹

For the purposes of this chapter, I forgo a deeper discussion of the Passive and the Possessive agreement, focusing on the Salient Article (see Muravyev et al. 2024 for more on obviation in Kazym Khanty). In the remainder of this section, I introduce Heather Bliss' (2017) analysis of obviation in Blackfoot (< Algonquian) to serve as a standard of comparison when we consider Muravyev's hypothesis regarding the Salient Article in §§5.2.4–5.3.

In Blackfoot, both Proximate and Obviative receive special morphological marking (252). As expected, proximate uniqueness constrains occurrence of more than one proximate per obviation span (here, clause) (253).

(252) Ann-wa Leo íhpok-inihkim-yii-wa ann-yi n-Itán-yi
 dem-PROX L. ACCOM-sing.TA-DIR-PROX DEM-OBV 1-daughter-OBV
 'Leo sang with my daughter.' (adapted from (Bliss 2017: 4))

¹⁷¹ Muravyev cites (Nikolaeva & Bárány 2019) with a similar proposal that Tundra Nenets (< Uralic) possessive agreement with lexical possessors is conditioned on their proximate status.

- (253) *Ann-wa Leo íihpok-inihkim-yii-wa ann-**wa** n-Itán-**wa**
 dem-PROX L. ACCOM-sing.TA-DIR-PROX DEM-PROX 1-daughter-OBV
 Intended: ‘Leo sang with my daughter.’ (adapted from (Ibid.))

Bliss’ paper is concerned with the question of whether obviation in Blackfoot is a primarily syntactic or discourse phenomenon, which has been the subject of intense debate in Algonquian studies (see the summary and references in (Ibid.)). Bliss argues that Blackfoot Proximate and Obviative categories code primarily syntactic distinctions, but the nature of these distinctions makes these categories suitable for “recruitment” for discourse purposes such as main character *vs.* peripheral character marking in narratives (Goddard 1990).

According to Bliss, the core syntactic property coded by Blackfoot Proximate/Obviative marking is syntactic (in)dependence. Proximate nominals exhibit free word order (254) and may be omitted (255). (Examples below are adapted from (Ibid.: 7–8).)

- (254) a. Ann-wa n-ínsst-innaan-wa nít-sspommo-a-wa
 DEM-PROX 1-sister-1PL-PROX 1-help.TA-DIR-PROX
 ‘I helped our sister.’
 b. Nítsspommoawa **anna nínsstsinaana**¹⁷²
 (Ibid.)

- (255) a. A’páwaawahkaa-wa **ann-wa** **Pióhkomiaaki**
 walk.around.AI-PROX DEM-PROX far.sounding.woman
 ‘Far Sounding Woman is walking around.’
 b. A’páwaawahkaawa
 ‘S/he is walking around.’
 (Ibid.)

On the other hand, the linear position of Obviative nominals is constrained: they cannot occur preverbally (unless resumed by an enclitic pronoun):

- (256) a. Áókataki-yini ann-yi w-inssts-yi

¹⁷² Bliss does not provide glosses for this example but note that *anna nínsstsinaana* is the same expression as the morphologically segmented *ann-wa n-ínsst-innaan-wa* [DEM-PROX 1-sister-1PL-PROX], namely, ‘our sister (proximate)’. The same remarks apply to examples below.

bead.AI-OBV DEM-OBV 3-sister-OBV
 ‘His sister does beadwork.’

b. ***Anni ónssts** áókatakiyini
 Ibid.

Furthermore, Obviatives cannot be freely omitted (similarly, unless resumed by an enclitic pronoun):

- (257) a. Áísokssta-yini ann-yi ot-issítsimaan-yi
 nurse.well.AI-OBV DEM-OBV 3-baby-OBV
 ‘Her baby is nursing well.’
 b. Áísokssta-yini-**áyi**
 nurse.well.AI-OBV-3SG.PRN
 ‘S/he is nursing well.’
 c. *Áísoksstayini.
 (Ibid.)

Bliss suggests that the reason for these distributional differences between Proximates and Obviatives in Blackfoot is that Proximate nominals are adjuncts to the clause, while Obviative nominals are arguments. This directly reflects the intuition that Proximates are syntactically independent, and vice versa for Obviatives.

This claim is supported by the fact that Proximate-marked verbs and nouns can function as independent clauses, while Obviative nominals necessarily function as arguments (dependent on the verb), and the same is true of Obviative-marked verbs which must appear as dependent clauses. See the examples in (Bliss 2017: 10–12).

Having argued for the syntactic nature of obviation in Blackfoot, Bliss also discusses its discourse function. She considers a narrative and shows that Proximate marks the main character in this narrative (Ibid.: 12–15).¹⁷³ Later in the story, when the main character of the story changes, a different noun phrase receives Proximate marking, while peripheral characters are marked Obviative (cf. (Goddard 1990)). This clearly shows that obviation devices can be intentionally deployed by narrators to highlight

¹⁷³ Bliss uses the terms “topic/non-topic” for “main/peripheral character” which I don’t use here in order to avoid confusion with the information-structural topic in Nikolaeva’s sense.

referents that are central to their story. As mentioned above, Bliss views this as an instance of a recruitment process whereby a syntactic device is utilized for a discourse function.

Furthermore, Bliss suggests that the “Proximate as independent/Obviative as dependent” analysis extends to other obviation systems in Algonquian, citing evidence from a range of other Algonquian languages. For instance, Bliss cites Benjamin Bruening’s analysis of obviation in Passamaquoddy (Bruening 2001; 2009). Bruening proposes that the Proximate/Obviative distinction works similarly to the Nominative/Accusative distinction in Dependent Case Theory (Marantz 1992) where Accusative is a **dependent case** assigned to a noun phrase in the presence of another structurally prioritized DP (see §5.3 for details). The Obviative is similarly dependent in Passamaquoddy, as it is assigned in the presence of a Proximate DP. Thus, according to Bliss, obviation in Algonquian is a syntactic phenomenon that rests on syntactic (in)dependence, realized differently in different languages.

When we discuss Kazym Khanty data, I will argue against a syntactic account of the Salient Article’s distribution (§5.3) despite its similarities to proximate markers in Algonquian.

5.2. The Salient Article as a definite determiner

Having introduced the relevant theoretical notions and crosslinguistic ‘standards of comparison’, we are now prepared to consider the Salient Article in detail. In Section 5.2.1, I show that the Salient Article requires both uniqueness and familiarity — the defining property of definite determiners (Chapter 2) — but these conditions are not sufficient. In Section 5.2.2, I argue that the availability of the Salient Article does not depend on information-structural distinctions or on syntactic role of the DP. Section 5.2.3 discusses ‘salience without uniqueness’ contexts and points to a limitation in the view of the Salient Article advanced in this chapter. In Sections 5.2.4 and 5.3, I consider the Salient Article as a proximate marker and argue against a syntactic analysis of its distribution.

5.2.1. Uniqueness and familiarity

The Salient Article demonstrates some of the basic properties of definite determiners. Consider the marking possibilities of noun phrases referring to the soldier in a small narrative-like sequence in the following three examples (based on the example from (Nikolaeva 2003: 139–140)).

(258) ma χot-a λuŋ-s-əm. šāta **sāldat-#(en)** oməs-λ.
 I house-DAT enter-PST-1SG there.LOC soldier-POSS.2SG sit-NPST[3SG]

‘I entered a house. There was a soldier sitting there.’

(259) ma **sāldat-(%en)** χuś-a wana mǎn-s-əm, puškan-ən šāš-s-əm
 I soldier-POSS.2SG at-DAT closer go-PST-1SG gun-LOC show-PST-1SG>SG
 ‘I came closer to the soldier and raised my gun at him.’

(260) **sāldat-#(en)** pakn-əs.
 soldier-POSS.2SG get.frightened-PST[3SG]
 ‘The soldier got frightened.’

Upon first mention of the soldier (258), the Salient Article is barred. Upon the second mention (259), where the Subject is still the narrator, and the soldier noun phrase is in an Oblique position (complement of a postposition), the bare form is preferred, although the Salient Article is at least acceptable for some speakers (signified by the percent sign). Finally, in (260), where the soldier is the Subject, the Salient Article is obligatory for all the speakers I consulted.

These data show that the Salient Article is unacceptable with novel referents and that it can be used with familiar referents. Based on this, we may propose that the Salient Article requires familiarity, although it does not seem to be a sufficient condition, as (259) shows.

In the case of novel unique referents, however, the Salient Article is unacceptable. For instance, it is generally not accepted with superlatives (261), unlike the English Definite Article (see Ch. 2). This suggests that uniqueness is not a sufficient condition for the Salient Article to be used. (See also (264).)

(261) [There is a fishing contest on Fisherman’s Day. I come home from the contest and tell my family about it.]

tām ol mēt wōn χuλ-#(en) wet-χuś-jǎŋ kila-kem
 this year very big fish-POSS.2SG five-DEC-ten kg-APPR
 ‘This year, the biggest fish was 15 kilograms.’

Let us demonstrate that the Salient Article indeed requires informational uniqueness / maximality. Example (262a) shows that the Salient Article is acceptable with maximal plural referents that were introduced in prior context. In the same context,

a non-maximal DP does not admit the Salient Article (262b). The Partitive Article POSS.3SG must be used instead.¹⁷⁴

(262) [“I was walking along the street when I saw two dogs.”]

a. **amp-ηəλ-an** ma pɛλ-am-a χurət-ti pit-s-əŋən
 dog-DU-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST-3DU

‘The dogs started barking at me.’

b. **i amp-əλ/#-en** ma pɛλ-am-a
 one dog-POSS.3SG/-POSS.2SG I at-POSS.1SG-DAT

χurət-ti pit-əs

bark-NFIN.NPST become-PST[3SG]

‘One of the dogs started barking at me.’

This suggests that the Salient Article presupposes uniqueness which gives rise to infelicity in (262b).

Further support for this proposal comes from the obligatory use of the Salient Article in examples like (263). Here, the bare form is barred because, judging by my consultants’ comments, it gives rise to an anti-familiarity/anti-uniqueness inference, i.e., the dog referred to is not the unique familiar dog introduced in prior context. (Cf. the obligatoriness of the other types of definites in Chapter 2.)

(263) [“I was walking along the street when I saw a dog.”]

amp-#(en) ma pɛλ-am-a χurət-ti pit-əs
 dog-POSS.2SG I at-POSS.1SG-DAT bark-NFIN.NPST become-PST[3SG]

‘The dog started barking at me.’

Consultant’s comment on Ø: “then it’s some other dog, not clear which”.

One way to account for such inferences is by appealing to Heim’s pragmatic principle *Maximize Presupposition!* (Heim 1991) which enforces the use of definite articles in contexts where their (uniqueness and familiarity) presuppositions are satisfied

¹⁷⁴ The partitive article function is quite often found for 3SG possessives in Uralic languages, e.g., in Udmurt (É. Kiss & Tánzos 2018), Komi Izhma, and Meadow Mari (Simonenko 2017). É. Kiss and Tánzos argue that the partitive article in Udmurt is not merely a function of the POSS.3SG but is an independent marker. The same argument was made for the Kazym Khanty Partitive Article in Chapter 4. See §3.5 on its semantics.

(cf. Chapter 3). According to my proposal, presented in §5.4, the Salient Article presupposes salience and informational uniqueness among salient referents (i.e., relative maximal salience) (in Roberts’ sense §5.1.1). Failing to use the Salient Article in a context where its presuppositions are satisfied leads the addressee to infer that its use would not be felicitous in this context, e.g., because the intended referent is **not** unique/familiar/salient.

So far it seems that the Salient Article is much like the English Definite Article or, perhaps, the narrower Akan Definite Article. Is our job here done? No, because the Salient Article is still narrower than both these markers. It differs from uniqueness-based definites in that it is not used with superlatives (261) and it differs from familiarity-based definites in that strong familiarity is not a sufficient condition on its use (e.g., in (259) the bare form is preferred with a strongly familiar referent).

Another example which highlights the narrower distribution of the Salient Article is (264), modeled after Roberts’ (2003) example (14)/(20), discussed in Chapter 2. Here, the Salient Article is infelicitous with a novel and unique referent.

(264) [“I recently found a box on my cupboard.”]

ma	nuχ	puš-s-εm,	λaraś	λipij-ən
I	up	open-PST.1SG>SG	box	inside-LOC
wəλ-əm	χur-(#en)	nuχ	wu-s-εm	
be-NFIN.PST	image-POSS.2SG	up	take-PST-1SG>SG	

‘I opened it and took out the photo that was inside the box. [In the photo was my father in youth.]’

Consider first the English gloss of (264). Roberts argues that the use of *the* with DPs like *the photo that was inside the box* in examples like these is possible thanks to accommodation. The English Definite Article requires informational uniqueness which is not satisfied for *the photo* here. But accommodating semantic uniqueness and existence (‘there was one and only one photo inside the box’) gives informational uniqueness. Thus, in examples of this kind speakers expect the addressees to accommodate the use of the Definite Article and this allows to convey the uniqueness of the referent.¹⁷⁵

¹⁷⁵ This reasoning mirrors Roberts’ discussion of example (21).

Now, the fact that the Salient Article is not felicitous in (264) suggests that its presuppositions are stronger than the English Definite Article's presuppositions, and they cannot be so easily accommodated. As I suggest below, the Salient Article presupposes relative maximal salience. To be able to accommodate such a high status of the Salience-Familiarity Scale (61) would require the referent to at least be strongly familiar. Since *the photo* is not at all familiar in (264), the Salient Article is infelicitous for it.

Example (265), repeated from (136), also shows that strong familiarity is not enough. Here a passport was introduced in the preceding context, which makes it strongly familiar (even if the story digresses between the two mentions of the passport). Nevertheless, the Salient Article is excluded here, and the Associative POSS.1SG must be used instead (cf. §3.4). Thus, strong familiarity and uniqueness cannot be all there is to Salient Article's conditions of use.

(265) [“I found somebody's passport in the street. I went to the town administration. Met a friend there and talked to her for some time.”]

nɛm	nɛpek-ɛm/#-ɛn	suwet-ən	χǎj-s-ɛm
name	paper-POSS.1SG/-POSS.2SG	council-LOC	leave-PST-1SG>SG

‘[Then] I left the passport at the administration. [Let them find the owner.]’

One candidate for another condition on the Salient Article is the requirement of salience, as proposed by Barlew (2014) for the Salient Article of Bulu (§2.3.1). For example, it seems reasonable to assume that one important difference between (265), where the Kazym Khanty Salient Article is barred, and (263), where it is obligatory, is that in the latter the referent is salient to a high degree but in the former it is not.

However, a requirement of salience (of some positive degree) by itself, which was enough for the Bulu Salient Article, still wouldn't be sufficient for the Kazym Khanty Salient Article. Recall that according to Roberts and Barlew a factor determining salience is relevance of the referent to the current question under discussion (QUD). Even though the passport cannot be assumed to be at the center of the addressee's attention in the context immediately preceding (265), the QUD ‘What did you do with the passport?’ presumably still looms somewhere in the stack of relevant questions under discussion at that point, which ensures that the passport is salient to some positive degree, even if not to a high one. The same applies to the narrative examples (258)–(260): the soldier referent arguably becomes relevant to the story upon his introduction in (258) which guarantees

that he is salient to some degree. Nevertheless, the Kazym Khanty Salient Article is still dispreferred upon the second mention of the soldier in (259) and is barred in (265), which one may attribute to the fact that the soldier has not yet attained a **sufficiently high** degree of salience and the passport has already lost it by the time it is mentioned for the second time (cf. (276) discussed below).

What this discussion suggests is that, perhaps, unlike the Salient Article of Bulu (Barlew 2014: 630–631), the Kazym Khanty Salient Article does care to which degree the DP referent is salient. Before we develop this idea in some detail, let us consider some further possible factors.

One candidate suggested in (Mikhailov 2021b) is information structure. There, I proposed that the Kazym Khanty marker is a ‘topic marker’. Following this suggestion, it is reasonable to check if perhaps it marks topics in Nikolaeva’s (2003) sense. Another candidate is syntactic role. In all the examples above, except the one where the Salient Article is unavailable (265), the DP with the Salient Article is the Subject. I address these two possibilities in the following section.

5.2.2. *Information structure and syntactic role*

In the preceding section I argued that the Salient Article is a definite determiner which must have some additional requirements besides informational uniqueness and existence. In this section I show that neither syntactic role nor information structure provide the needed requirements of the Salient Article.

Let us consider a set of examples from an elicited dialogue about a thief stealing a computer from the local museum. Examples in (266) show that the Salient Article is used with both Subjects and Objects. (Note that the context in these examples ensures that the thief is topical and in the theme of the utterance, as defined in §5.1.2.)

(266) [“— A computer was stolen from our museum. — Did they find the thief yet?”]

- a. pa muj, muλχatəλ **λoλmaχ-en** wəjət-s-a,
 ADD what yesterday thief-POSS.2SG find-PST-PASS[3SG]
 kampjuter-ew juχli mā-s-i
 computer-poss.1pl back give-pst-pass[3sg]
 ‘You betcha! Yesterday, the thief was found, and the computer has been returned.’
- b. pa muj, ma muλχatəλ **λoλmaχ-en** wəjət-s-em
 ADD what I yesterday thief-POSS.2SG find-PST-1SG>SG

‘You betcha! Yesterday, I found the thief.’

The Salient Article may also be used with non-core syntactic roles such as Demoted Agent in the Locative (267) and Possessor (268).

(267) [“— A computer was stolen from our museum. — Did they return the computer yet?”]

kampjuter-ew **λoλmaχ-%(en)-ən** tinij-əm
computer-POSS.1PL thief-POSS.2SG-LOC sell-NFIN.PST[PASS.3SG]

‘It turns out that the computer was sold somewhere by the thief.’^{176, 177}

(268) [“— A computer was stolen from our museum. — Did they find the thief yet? — You betcha! Yesterday, I found the thief. — What did they do to him in the end?]

λoλmaχ-en mašinaj-əλ milcaj-ən nuχ wə-s-i
thief-POSS.2SG car-POSS.3SG police-LOC up take-PST-PASS[3SG]

‘The thief’s car was taken by the police. [And he himself was put to jail.]’

I conclude from these data that the Salient Article has no special restrictions on syntactic role.

Now we turn to (the lack of) information-structural restrictions. Recall from §5.1.2 that in the Obdorsk dialect of Northern Khanty there is a strict link between primary topics and the Subject position and secondary topics and the Agreeing Object (O2) position as argued by (Nikolaeva 2001). Thus, in agent rheme sentences primary topic patients are obligatorily promoted to Subject via Passivization (and rhematic agents are obligatorily demoted). Narrow argument rheme on the Subject or the O2 is impossible. In other words, in utterances that do have an explicit theme, the Subject DP is obligatorily a part of the theme (cf. footnote 166). If another DP is also topical, it is obligatorily in the O2 position.

Is there such a strict correspondence between information structure and the Salient Article in the Kazym dialect? I argue no.

¹⁷⁶ The non-finite form used here functions as a Mirative Past Tense.

¹⁷⁷ Note that here the question under discussion (“What happened to the computer?”) ensures that the thief is in the rheme of the utterance, anticipating the discussion below that the Salient Article is not restricted to utterance theme. On the other hand, I assume that the thief is still salient in (267) by virtue of bridging. A robbery entails the existence of a thief. Since the robbery is the topic of the dialogue, the thief is salient as its agent.

Even though the computer is also salient here, it cannot be marked with the Salient Article since it is marked with a Proper Possessive which is in complementary distribution with the Salient Article, see Chapter 4.

In most of the examples considered so far, the DP with the Salient Article was indeed either a primary or a secondary topic. However, example (267) already shows that the Salient Article may be strongly preferred even with a rhematic DP.

To this we may add example (269) where a DP is under narrow argument rheme and is nevertheless obligatorily marked with the Salient Article.

- (269) [We are putting on a play at the school theatre. Two girls have not yet been assigned a role. We discuss their strengths to decide who will play the sun: “— The girl on the left is tall. — The girl on the right is a good singer. So, who will play the sun?]
- jām pelək-ən oməs-ti ewen / #ewi at junt-λ
 good side-LOC sit-NFIN.NPST girl.POSS.2SG / girl OPT play-NPST[3SG]
 ‘Let the girl on the right play [the sun].’

The same observation may be drawn from example (270) where a demonstrative DP is under narrow contrastive rheme and is also obligatorily marked with the Salient Article.

- (270) [I point at a boy and ask: “Who’s that?” My addressee answers: “This boy’s name is Petya.” Our friend corrects them.]
- ši aj_iken / #aj_iki χən pet’a,tum aj_iken pet’a
 DEM boy.POSS.2SG / boy NEG.EMPH P. that boy.POSS.2SG P.
 ‘This boy is not Petya, that one is.’

Table 10 summarizes the distribution of the Salient Article with respect to information structure as compared to Obdorsk dialect Subjects in argument or predicate rheme utterances (cf. Section 5.1.2).

Information-structural status	Kazym Salient Article	Obdorsk Subjects / Agreeing Objects
Narrow rheme	+	—
Narrow contrastive rheme	+	—
Predicate rheme, non-topical referent	+	—
Theme	+	+

Table 10. Independence of the Kazym dialect Salient Article from information-structural status as compared to the Obdorsk dialect Subjects and Agreeing Objects (O2) (Nikolaeva 2001)

Thus, I suggest that whether a DP is in the theme or in the rheme of the utterance is irrelevant for the Salient Article as long as its referent itself is salient in the given context. In all the examples in this section where the Salient Article is obligatory or strongly preferred, the referent of the marked DP is salient by virtue of the preceding context: the referent is either the primary topic of the target utterance, or it has been the primary topic of the preceding utterances (ensuring salience), or its salience is derived from bridging to another topic of the stretch of discourse in question.

The only example where I marked the Salient Article as possible, but less preferred than the bare form — the second sentence of the soldier example (259) — was an example where the context is underdetermined with respect to salience of the target referent and, thus, there is some ambiguity as to the speaker's intentions: whether the speaker intends the addressee to attend to the referent in question to a significant degree or not. I consider this to be the reason for the Salient Article's optionality in cases like this one.

5.2.3. *'Salience without uniqueness' contexts*

In Chapter 2, we discussed 'salience without uniqueness' contexts introduced by Roberts' to distinguish personal pronouns from definite determiners. Roberts' example is repeated in (271). The point was that the personal pronoun, unlike the Definite Article, requires relative maximal salience which allows it to 'screen out' less salient discourse referents and to pick up the unique most salient woman, the one which was most recently mentioned.

(271) A woman entered from stage left. Another woman entered from stage right. **She / #the woman** was carrying a basket of flowers. (adapted from (Roberts 2003: 324))

Roberts' analysis of pronouns is repeated in (272).

(272) PRESUPPOSITIONS OF PRONOUNS (informal) (Roberts 2003: (49))

Given a context C, use of pronoun Pro_i presupposes that it has as antecedent a discourse referent x_i which is:

- a. weakly familiar in C,
- b. salient in C, and
- c. unique in being the most salient discourse referent in C which is contextually entailed to satisfy the descriptive content suggested by the person, number and gender of Pro_i .

Crosslinguistic work by Arkoh & Matthewson and Barlew, among others, has shown that the same is true for narrower types of definites, like the Akan Definite Article (273a) or the Bulu Salient Article (274).¹⁷⁸ They are also unavailable in ‘salience without uniqueness’ contexts.

(273) (adapted from Arkoh & Matthewson 2013: (80))

[“A (certain) child came from Winneba and another child_i came from Swedru.”]

- a. #**Àbòfrá nòí** bá-ì árá nà kyìrèkyìrènyí nò hwí-ì **àbòfrá nòí**.
 child FAM come-PST just FOC teacher FAM cane-PST child FAM

Intended: Immediately the child_i came, the teacher caned the child_i.

- b. **Ói-bá-ì** árá nà kyìrèkyìrènyí nò hwí-ì **nòí**.
 3SG.SUBJ-come-PST just FOC teacher FAM cane-PST 3SG.ANIM.OBJ

‘Immediately s/he_i came, the teacher caned him/her_i.

(274) (adapted from Barlew 2014: (9))¹⁷⁹

- a. [I say to you: “Yesterday I saw two men and a woman at the farm. One of the man had black hair. The other man had white hair. The man that had black hair was white. The man that had white hair was black.” Then I continue:]

#**fâm tè** è mbó è kóbò fùlasi.
 man TE PN₉ AUX PN₉ speak French

Intended: The [black haired] man was speaking French.

- b. [I say to you: “Yesterday I saw two men and a woman at the farm. One of the man had black hair.” Then I continue:]

fâm tè è mbó è kóbò fùlasi.
 man TE PN₉ AUX PN₉ speak French

‘The [black haired] man was speaking French.’

The same explanation is available here. Neither the Akan Definite Article, nor the Bulu Salient Article require **maximal** salience which is why when they are used in ‘salience without uniqueness’ contexts, there is no unique choice of discourse referent fitting the DP description and this results in a violation of the articles’ uniqueness

¹⁷⁸ This is relevant regardless of whether we accept Bombi’s (2018) and Owusu’s (2022) criticisms of Arkoh and Matthewson’s analysis.

¹⁷⁹ I corrected minor inconsistencies in glosses between (47a) and (47b).

presuppositions. This is in contrast to pronouns in the three languages ((271), (273b), and (Barlew 2014: 630–631)) which require maximal salience and simply pick up the last mentioned referent as it is the maximally salient one. (The same likely holds for demonstratives, see fn. 107.)

Thus, ‘salience without uniqueness’ contexts work as a diagnostic to single-out definites that require maximal salience.

An important observation is that the Kazym Khanty Salient Article passes this diagnostic as (275) shows.

(275) [A theatrical play for children: “A girl_i appeared from the room on the left. Another girl_k appeared from the room on the right.”]

λuW_{k/*i} / ^{OK}ewen_{k/*i} još-ən lipət təj-əs
 (s)he / girl.POSS.2SG hand-LOC flower have-PST[3SG]
 ‘She_{k/*i} [lit.: the girl] was holding flowers.’

As expected, using the personal pronoun is felicitous in (275) and is also the first choice for most of my consultants. However, unlike the definite determiners in English, Akan, and Bulu, the Salient Article in (275) is also accepted by all my consultants. This presents a striking difference between the Kazym Khanty Salient Article and the definite determiners of other languages, especially the Bulu Salient Article, showing that the Kazym Khanty Salient Article indeed requires maximal salience as proposed above. This makes it unique among non-demonstrative determiners as the only non-demonstrative determiner that requires maximal salience in the typology of Chapter 2. Any analysis of the Salient Article must reflect this requirement.

In Chapter 2 (§2.3.3.1), we discussed that there is a distinction to be made between relative maximal salience and absolute maximal salience. The relatively maximally salient referent only has to be more salient than the other referents (if any) satisfying the DP description. The absolutely salient referent has to be at the very center of current attention, it has to have the highest possible degree of salience. Roberts’ ‘salience without uniqueness’ contexts diagnose relative maximal salience. Given that the Salient Article passes this test, it is still open whether it makes a stronger requirement than this.

Indeed, there is evidence suggesting that relative maximal salience might not be enough for the Salient Article, which I discuss here briefly before moving on to focus on other pertinent issues.

Example (276) presents a sentence out of a small narrative in which the speaker first met a certain woman but then couldn't find her for some time. Then the speaker comes to Kazym village and finds the woman there which the target sentence tells about. It is clear that the woman is a higher-order topic here. This is a narrative **about** meeting a woman who sings well in Khanty. Furthermore, there are no other women in this narrative. Thus, the woman in the target sentence is relatively maximally salient. However, we find that the Salient Article by itself is not sufficient here, but a specialized demonstrative *in* needs to be used alongside it.¹⁸⁰ Otherwise, the sentence is infelicitous.

(276) [“In Moscow, I had met a woman who sings well in Khanty. I didn't take her phone number then, and for two years, we didn't meet. Recently, I came to Kazym village.

I went to the museum, entered it, and...”]

#(in)	nəŋ-en	śata	om-əs	jontəs-man
DEM.AN	woman-POSS.2SG	there	sit-NPST[3SG]	sew-CVB

‘...the woman was sitting there sewing.’

The demonstrative *in* is mostly undescribed in the literature on Northern Khanty, and our consultants also do not appear to use it frequently in elicitation (Lera Marinina, p.c.).¹⁸¹ Fedorkiv (2018), as apparently the only source describing this marker, suggests that *in* serves an ‘actualizing’ function with familiar referents, and that it is the most frequent demonstrative to do so. Unfortunately, Fedorkiv does not provide contexts for the examples where *in* is used, so it is hard to judge what exactly she means.

Based on (276), we may suppose that *in* helps bring back into the center of attention a semi-active / deactivated topical referent. After all, the QUD for the target utterance of (276) seems to be ‘What happened next?’ or ‘What did you find there?’, i.e., the woman is not an immediate topic as she does not figure in the immediate QUD. There is a digression in the story which lowers the woman’s salience.

If my suggestion for the basis of why *in* is used here is correct, we may also suppose that the Salient Article by itself is not used with such semiactive referents as they are not

¹⁸⁰ Out of the three consultants who I elicited this narrative from, one used the unmarked demonstrative *śi* [DEM] (see §§2.4.3, 5.5). The other two used *in*.

¹⁸¹ In fact, L. Marinina (p.c.) reports that she could not elicit the demonstrative *in* consistently, with the speakers often parsing it as the homonymous adverb *in* ‘now’. For this reason, (Marinina 2024) does not discuss this demonstrative.

salient enough. And this is the purported reason why it is infelicitous in (276) without a demonstrative. In the felicitous version of (276), the demonstrative *in* reactivates the discourse referent for the target DP making it salient enough to satisfy the Salient Article's presupposition (cf. §5.5 on the Salient Article with deictic demonstratives). Or that is how the story behind (276) could go.

This picture, however, is puzzling if the Salient Article only requires relative maximal salience, since relative maximal salience is satisfied for the woman here. It seems that we are forced to suppose that the Salient Article requires something more than mere relative maximal salience.

Currently, I do not have enough data to shed further light on this issue and the suggestions made above, so it must be left as an area to be explored in future work. Figuring out better the conditions of use of *in* and the type of high degree of salience that the Salient Article requires could help us better understand how salience works and how best to model it. The model floated in Chapter 2 is not currently fit for the task, as it does not provide a natural way to accommodate an intermediate salience status in-between relative maximal salience and absolute maximal salience on the Salience-Familiarity Scale (61). Neither is Stojnić's (2021) stack-based dynamic model which I employ below in §5.5 to analyze the use of deictic demonstratives with the Salient Article.

In the rest of this chapter, we will keep working with the simplifying assumption that the Salient Article requires just relative maximal salience. Before proceeding to the possible analytical routes to take, we need to consider the Salient Article's similarity to proximate markers.

5.2.4. *Obviation*

Let us consider Muravyev's (2022) hypothesis that the Salient Article is a proximate marker much like those found in Algonquian languages (see §5.1.3).

Recall that two of the core properties of obviation systems is the proximate uniqueness constraint (Aissen 2001) and the condition that speech act participants are not considered for proximate/obviative marking. Both properties turn out to be observed by the Salient Article (although see the next subsection).

Firstly, only one third person DP may be marked with the Salient Article within a single clause, as (277) and (278) show. In both examples, the less topical of the two DPs

must be marked with the Associative POSS.3SG instead (cf. §3.4).¹⁸² In (277), the boy is associated with the girl via the preceding context which licenses the Associative marking on him. In (278), one may argue that the thief is associated with the computer under discussion since he stole it. What is crucial for our purposes is that the Salient Article is unavailable for both these referents, even though the boy is the secondary topic in (277) and the thief is salient in (278) (cf. footnote 177).¹⁸³

(277) [“I was walking along the street when I saw a boy and a girl.”]

ewen **aj_ike**λ / #**aj_iken** χᾱts̥-əs-λe
 girl.POSS.2SG boy.POSS.3SG / boy.POSS.2SG hit-PST-3SG>SG
 ‘The girl hit the boy.’

(278) [“A computer was stolen from our museum. But everything worked out.”]

waśaj-en **λoλmax-əλ/#-en/%-Ø** wəjət-s-əλe
 V.-POSS.2SG thief-POSS.3SG/-POSS.2SG find-PST-3SG>SG
 ‘Vasya found the thief [and returned the computer].’

Let’s compare (278) with the minimally different (279) where the other DP is an indefinite pronoun. An indefinite referent is obviously less salient than the thief, so the thief is marked with the Salient Article as the proximate third person.¹⁸⁴

(279) [“A computer was stolen from our museum. But everything worked out.”]

λoλmax-en χuj-at-ən wəjət-s-a,
 thief-POSS.2SG who-INDEF-LOC find-PST-PASS[3SG]
 ‘Somebody found the thief [and returned the computer].’

Secondly, if the first DP refers to an interlocutor in the thief examples, the second DP must be marked with the Salient Article (280) (repeated from (266)). Just as expected, if the Salient Article marks the proximate third person.

¹⁸² With the thief, some speakers also allow the unmarked option *λoλmax* [thief].

¹⁸³ According to my arguments from Chapters 3 and 4, *waśaj-en* ‘Vasya’ is marked with the Proprial Article here, and not with the Salient Article. Still, Vasya is arguably more salient and prominent than the thief, which would make him the proximate, and this excludes the Salient Article on the thief per Muravyev’s hypothesis.

¹⁸⁴ Note that in all the examples based on the thief story, Associative POSS.3SG is also a viable marking option for the DP referring to the thief, although, when the Salient Article is available, as in (279), it is preferred by most speakers.

(280) [“— A computer was stolen from our museum. — Did they find the thief yet?”]

pa muj, ma muλχatəλ λoλmaχ-en wəjət-s-em
 ADD what I yesterday thief-POSS.2SG find-PST-1SG>SG
 ‘You betcha! Yesterday, I found the thief.’

So far, the Salient Article fits the profile of proximate marking perfectly. The natural question arises: should the Salient Article be analyzed in a similar way to proximate markers in Algonquian languages (cf. §5.1.3), i.e., as a marker of syntactic independence (Bliss 2017)? In §5.3, I argue against such an analysis. To prepare for that, let me introduce another crucial set of data points.

5.2.5. Proximate uniqueness violations

As we’ve seen above, the Salient Article, at first sight, behaves similarly to a proximate marker. However, there are three types of cases where the Salient Article does not observe proximate uniqueness despite this. The first case is with demonstratives and it was discussed in (Mikhailov 2024a), which this chapter is based on. The second case is with oblique arguments, and the third case is with third person pronouns. These latter two cases were not discussed in (Mikhailov 2024a) and they require a substantive revision of the semantic analysis proposed there.¹⁸⁵ In particular, I will argue that we cannot provide a semantic account of proximate uniqueness effects with the Salient Article because of their gradient nature and because of the influence of morphosyntactic factors.

As we have seen above, the Salient Article is also used with DPs with demonstratives. It turns out that with demonstrative DPs, the Salient Article is no longer subject to proximate uniqueness. Thus, in (281) and (282), two demonstrative DPs with the Salient Article are perfectly felicitous within the same clause. (This observation is due to Nikita Muravyev (2022).¹⁸⁶)

(281) tām iken **tum** **imen** šiwəλ-əs
 this man.POSS.2SG that woman.POSS.2SG notice-PST[3SG]
 ‘This man saw that woman.’ (adapted from (Muravyev 2022: ex. 9))

(282) [“What happened here?”]

¹⁸⁵ They were first discussed in two talks (Mikhailov 2024b; Muravyev et al. 2024).

¹⁸⁶ Muravyev, however, does not provide any theoretical account but merely discusses preliminary observations.

tăm ewen tum aj_iken χăts'-əs
 this girl.POSS.2SG that boy.POSS.2SG hit-PST[3SG]
 'This girl hit that boy.'

Compare (283) repeated from (277) above where two DPs with the Salient Article are infelicitous without demonstratives.

(283) ["I was walking along the street when I saw a boy and a girl."]

ewen aj_ikeλ / #aj_iken χăts'-əs-λe
 girl.POSS.2SG boy.POSS.3SG / boy.POSS.2SG hit-PST-3SG>SG
 'The girl hit the boy.'

In what follows, I will argue that the demonstrative data present a problem for two possible lines of syntactic analyses. Before that, let us consider further minimal pairs to (283) where the second DP is not the direct object.

Another class of examples where proximate uniqueness is systematically violated has two lexical third-person DPs with one of them being in an oblique position, i.e., it is marked either with the dative case or the locative case or it is the complement of a postposition. This is opposed to examples considered in §5.2.4 where two DPs were in the subject and the direct object positions which are morphologically unmarked in Northern Khanty in the case of lexical DPs.

Thus, (284) shows that, when a subject is marked with the Salient Article, using a second Salient Article is **possible** with datives and complements of a postposition (284a, b), unlike with direct objects (283). I note, however, that two consultants barred the second Salient Article in (284a) and all consultants first volunteered POSS.3SG here, indicating at least a weak preference to observe proximate uniqueness in (a).

(284) [I was walking along the street when I saw a boy and a girl.]

- a. aj_iken eweλ-a/%ewen-a lipət mǎ-s
 boy.POSS.2SG girl.POSS.3SG-DAT/girl.POSS.2SG-DAT flower give-PST[3SG]
 'The boy gave a flower to the girl.'
- b. aj_iken eweλ / ewen χuśa mǎn-əs
 boy.POSS.2SG girl.POSS.3SG / girl.POSS.2SG to go-PST[3SG]
 pa lipt-ən mǎ-s-λe
 ADD flower-LOC give-PST-3SG>SG

‘The boy came closer to the girl and gave her a flower.’

Examples (285)-(286) present two minimal pairs demonstrating proximate uniqueness violations in Passive sentences, while in the Active pairs proximate uniqueness is observed. In the Active (a) sentences, Salient Article marking is unavailable for the direct object in the presence of a subject. In the Passive (b) sentences, where the Patient is promoted to subject position and the Agent is demoted to Locative, Salient Article marking becomes available for the Patient.

(285) [“A very scary big dog lives here. Recently the dog barked at Petya's daughter.”]

- a. pet'aj-en ewi **amp-əλ/#-en** λ'awətt-əs-λe
 P.-POSS.2SG girl dog-POSS.3SG/-POSS.2SG scold-PST-3SG>SG
- b. pet'aj-en ewij-ən **amp-en** λ'awətt-əs-i
 P.-POSS.2SG girl-LOC dog-POSS.2SG scold-PST-PASS[3SG]
- ‘Petya's daughter scolded the dog.’

(286) [“A computer was stolen from our museum, but it turned out okay.”]

- a. tām ik-en **λoλmax-(?'en)** kätλ-əs
 this man-POSS.2SG thief-POSS.2SG catch-PST[3SG]
- b. tām ik-en-ən **λoλmax-en** katλ-s-a
 this man-POSS.2SG-LOC thief-POSS.2SG catch-PST-PASS[3SG]
- ‘This man [*speaker points his finger at the man in the room*] caught the thief.’

Thus, we see that when one of the two co-argument DPs bears overt oblique marking (such as dative, locative, or a postposition), a Salient Article becomes available on the second DP, even though proximate uniqueness was expected to rule it out. (Incidentally, example (285a) also shows that a DP that cannot be marked with the Salient Article, because it already has a possessor which per arguments from Chapter 4 rules out other (un)possessive marking, is still a potential competitor for the purposes of Salient Article marking on the second DP, cf. example (278) and fn. 183.)

And now for the final case. Under the standard conception of proximate uniqueness (Aissen 2001), we expect that a third-person pronoun cannot cooccur with a Salient Article-marked DP, since personal pronouns are more prominent in terms of topicality hierarchies (Givón 2001a; 2001b; Zúñiga 2006), as discussed in §5.1.3. At the same time, the personal pronoun itself cannot be marked with the Salient Article, since Northern

Khanty personal pronouns generally do not combine with (un)possessives, e.g., a form like **λ_{HW}-en* [(s)he-POSS.2SG] is completely ungrammatical.¹⁸⁷

However, as examples (287), (289), and (290) show, the Salient Article quite freely cooccurs with third-person pronouns in the same clause, which distinguishes it rather sharply from prototypical proximate markers (Goddard 1990; Aissen 2001; Bliss 2017).

First, consider (287) which is a minimal pair to (278), repeated in (288). The majority of my consultants volunteered or accepted the Salient Article on the thief in (287) where the competing DP is a personal pronoun.¹⁸⁸ This is very different from the judgements given for (288) where the competing DP is lexical and the Salient Article is barred for the thief.

(287) [“A: A computer was stolen from our museum.

B: Did they find the thief?

A: Do you know Vasya?”]

λ_{uw} λ_oλ_max-əλ/%-en/%-Ø *wəjət-s-əλe*
 (s)he thief-POSS.3SG/-POSS.2SG find-PST-3SG>SG
 ‘He found the thief.’

(288) [“A computer was stolen from our museum. But everything worked out.”]

waśaj-en λ_oλ_max-əλ/#-en/%-Ø *wəjət-s-əλe*
 V.-POSS.2SG thief-POSS.3SG/-POSS.2SG find-PST-3SG>SG
 ‘Vasya found the thief [and returned the computer].’

Two other examples with personal pronouns are provided in (289) and (290). Here, the personal pronoun is found in the direct object position and as an oblique respectively, and the competing DP once again may receive Salient Article-marking despite the presence of a more prominent third-person pronoun.¹⁸⁹

¹⁸⁷ Apart from oblique forms like *λ_{HW}-eλ-a* [(s)he-POSS.3SG-DAT] and *năŋ-en-a* [you.SG-POSS.2SG-DAT] where the possessive marker has the same person-number as the pronoun and is essentially a fossilized part of the respective case form.

¹⁸⁸ The other option was to use the Associative POSS.3SG, accepted by all consultants and volunteered by some. Two more consultants volunteered a bare noun.

¹⁸⁹ In (270), again, most of my consultants preferred marking the dog with the Salient Article to Associative POSS.3SG marking.

(289) [“— Why is Petya so sad/unhappy? - Did you hear that the museum computer had been stolen? He was on guard duty that night.”]

λολmax-en λuwti sɛŋk-s-əλe pa χunt-əs
 thief-POSS.2SG (s)he.ACC beat.up-PST-3SG>SG ADD run-PST[3SG]
 ‘The thief beat him up and ran away..’

(290) [“Masha was walking down the street and saw a big angry dog.”]

amp-%en/%-əλ λuw pɛλ-aλ-a
 dog-POSS.2SG/-POSS.3SG (s)he toward-POSS.3SG-DAT
 χurət-ti pit-əs
 bark-NFIN.NPST become-PST[3SG]
 ‘The dog barked at her.’

All of these examples clearly contradict proximate uniqueness as the DP marked with the Salient Article is quite clearly less prominent than the respective third-person pronoun, therefore, it is a worse candidate for proximate status and we should expect the Salient Article to be barred which it isn’t.

To summarize, the problem that (283)-(287) and (289)-(290) present is that, under a proximate uniqueness approach, we expect the Salient Article to be barred in all of them on the second / less prominent DP, but it is available in (a part of) these examples and even preferred by some consultants. This presents a striking difference in the behavior of the Salient Article from Algonquian proximate markers, and presents a potentially recalcitrant set of data points.

Let us now discuss some possible accounts that cannot handle these data, and make suggestions toward a working account of the data surveyed in this section (from §5.2.1 to §5.2.5).

5.3. *Against a syntactic account of the Salient Article*

Recall from §5.1.3 that Heather Bliss (2017) proposed to analyze proximate markers in Algonquian languages as markers of syntactic independence (and obviative markers as markers of syntactic dependence). According to Bliss, this distinction is realized differently in different obviation systems across the language family, e.g., in Blackfoot syntactic independence corresponds to adjunct/independent constituent status, while in Passamaquoddy it roughly corresponds to the Nominative case as opposed to the Accusative (Bruening 2001) in Dependent Case Theory (Marantz 1992).

The question is whether we should account for the distribution of the Salient Article observed in the preceding section in a similar way. I would like to argue against this line of analysis.

First, consider again the generalizations about the Salient Article that we discussed so far. In §5.2.2, I have argued that the Salient Article is not restricted to any particular syntactic role(s). In general, I have found no special syntactic restrictions on DPs with the Salient Article as compared to DPs without it found within the same clause, except for the restriction on direct objects when the subject is a competing lexical DP (and not a pronoun). If the presence/absence of the Salient Article were to correspond to the Algonquian proximate/obviative distinction (under Bliss’ analysis), we should find an underlying syntactic distinction that would drive the salience effects observed in the preceding sections. In my current data, there is no evidence to that effect.

Second, there are data that would significantly complicate a syntactic account. As we have seen above, the Salient Article is also used with DPs with demonstratives. It turns out that with demonstrative DPs, the Salient Article is no longer subject to proximate uniqueness. Thus, in (291) and (292), two demonstrative DPs with the Salient Article are perfectly felicitous within the same clause. Compare (277) above where two DPs with the Salient Article are infelicitous without demonstratives.

(291) **tām** **iken** **tum** **im-en** **šiwəλ-əs**
 this man.POSS.2SG that woman-POSS.2SG notice-PST[3SG]
 ‘This man saw that woman.’ (adapted from (Muravyev 2022: ex. 9))

(292) [“What happened here?”]
 tām **ewen** **tum** **aj_iken** **χāts-əs**
 this girl.POSS.2SG that boy.POSS.2SG hit-PST[3SG]
 ‘This girl hit that boy.’

To see why this observation presents a problem for syntactic accounts, let us consider two possible lines of analyses in more detail.

The literature presents at least two syntactic approaches to proximate uniqueness. The first one is the optimality-theoretic (OT) account by Judith Aissen (2001).¹⁹⁰

¹⁹⁰ For an introduction to Optimality Theory see (McCarthy 2008).

According to Aissen, obviation can be modeled via a set of markedness constraints which ban the occurrence of either obviative third person DPs (*Obviative), proximate third person DPs (*Proximate) or two (or more) proximates (Proximate Uniqueness) within a single obviation span (Ibid.: 10–12). The generator component of Aissen’s model generates a candidate set that contains tuples of third persons DPs each assigned either proximate or obviative status. Next, the evaluator component applies to the candidates the hierarchy of constraints to find the optimal candidate. The hierarchy of constraints that correctly rules out two proximates within a single obviation span is given in (293).

(293) AN OPTIMALITY-THEORETIC HIERARCHY OF CONSTRAINTS FOR OBVIATION (based on (Aissen 2001))

PROXIMATE UNIQUENESS $\gg$ *OBVIATIVE $\gg$ *PROXIMATE

To see how this works, consider Tableau 1 that models a basic example of proximate uniqueness effects, e.g., our example (277) above.

Input: <x, y> x, y = 3 rd persons	PROXIMATE UNIQUENESS	*OBVIATIVE	*PROXIMATE
(a) P _x P _y	*!		**
(b) O _x O _y		*!*	
(c)  P _x O _y		*	*

Tableau 1. Proximate Uniqueness in Optimality Theory (based on (Aissen 2001: 12))

The generator component supplies three candidates: (a) with two proximates, (b) with two obviatives, and (c) with one proximate and one obviative. In Kazym Khanty terms, this would, roughly speaking, correspond to (a) a clause with two DPs with the Salient Article, (b) a clause with no DPs with the Salient Article, (c) a clause with one DP with the Salient Article.¹⁹¹ The (a) candidate is ruled out since it violates Proximate Uniqueness, the highest ranked constraint. And the optimal candidate is (c), which, unlike (b), violates the second highest ranked constraint only once. Thus, Aissen’s account

¹⁹¹ This is an oversimplification since at least proper nouns also appear to compete for proximate status in Northern Khanty, even though they are not marked with the Salient Article (cf. footnote 183).

correctly derives the observed pattern in cases which concern subject vs. direct object DPs **without** demonstratives.

It is not clear, however, how this analysis could be extended to cases **with** demonstratives like (291) and (292) and to cases with obliques and personal pronouns from §5.2.5. Focusing on demonstratives, one possibility is to say that DPs with demonstratives form an independent obviation span. This proposal seems problematic. There does not seem to be any reason why demonstrative DPs would form a separate obviation span. After all, obviation spans are clauses or larger units of discourse (as per Aissen's definition (1997: 713–714) which is based on much preceding work on Algonquian). There do not seem to be cases where an obviation span is smaller than a clause. Aissen even states explicitly that “it seems impossible to initiate a new obviation span within a clause” (Ibid.). The other examples are problematic for the same reason.

Therefore, it seems that an Aissen-style OT account of obviation effects in the Salient Article would require a modification of the notion of obviation span that is both ad hoc and contradicts the Algonquian data, which is why I will not pursue this line of analysis any further.

The second syntactic approach to proximate uniqueness effects was proposed by Bruening (2001) for Passamaquoddy within a Minimalist syntactic framework and later applied to Tundra Nenets (< Uralic) data by (Bárány & Nikolaeva 2021) (cf. footnote 171).

In Bruening's system (Bruening 2001: 119ff.), all DPs bear a syntactic proximate feature [P] which may be either valued [+P] or unvalued [P]. First and second persons are inherently [+P]. Third persons are not inherently valued but may become valued [+P] from context and in opposition to another third person DP. If two third person DPs cooccur, one will become [+P], while the other will remain unvalued [P] and receive an additional obviative feature [Obv] which is spelled out as obviative morphology. (Note that in Passamaquoddy there is no proximate morphology, unlike Blackfoot.)

Both variants of the proximate feature [+P]/[P] are uninterpretable and need to be checked off by functional heads in the syntax (which is a natural set of assumptions within the 2000s Minimalist syntax): [P] is checked off by the *v* (voice) head, while [+P] must be checked off by some higher head which Bruening calls H^0 . This essentially means that all DPs need to be licensed in a particular structural configuration. In Passamaquoddy,

this setup allows to model movement of the proximate to the left of the verb and agreement of the verb with the proximate.

More formal detail than that need not concern us here, but the general idea is that proximate status of a third person DP may only be received in competition with another third person DP. This entails that the only third person that co-occurs with a second or first person will be neither proximate nor obviative in Bruening’s model. (Recall that the proximate does not receive any overt morphological marking in Passamaquoddy.) And there is a good reason for this assumption with respect to Passamaquoddy data.

In a Passamaquoddy sentence with a first or second person and a third person, only the former is indexed by an agreement prefix on the verb and appears to the left of it. In a sentence with two third persons, either one may be indexed with the same agreement prefix and appear to the left of the verb. Thus, it seems that the bare third person DP in the first case indeed is **not** proximate as it cannot be indexed by the agreement prefix and appear to the left of the verb. On the other hand, it is also not obviative, as it does not bear obviative morphology. This pattern may be summarized in terms of features as follows:

(294) PASSAMAQUODDY OBVIATION IN BRUENING’S MODEL: DP FEATURES (Bruening 2001: 119–124)

- (a) 1st/2nd person proximate & 3rd person non-proximate: [1/2, +P] ... [3, P]
- (b) 3rd person proximate & 3rd person obviative: [3, +P] ... [3, P, Obv]

At this point, it is already obvious that Bruening’s approach does not derive correct predictions for the Salient Article, since the Salient Article freely appears in the (a) case, see (295) repeated from (280).

(295) [“— A computer was stolen from our museum. — Did they find the thief yet?”]

pa muj, ma muλχatəλ λoλmaχ-en wəjət-s-em
ADD what I yesterday thief-POSS.2SG find-PST-1SG>SG
‘You betcha! Yesterday, I found the thief.’

If the Salient Article were associated with proximate status, which corresponds to the [+P] feature in Bruening’s model, then it must have been barred in (295) according to (294), which is not the case. If we were to allow [+P] to be freely assigned to any third person, we would lose Bruening’s account of proximate uniqueness which is wholly grounded in competition between two local third persons.

On the other hand, it seems easy to account for violations of proximate uniqueness with demonstrative DPs (291) and (292) in Bruening’s model. One has only to assume that demonstratives are inherently valued [+P], just like first and second person pronouns. This, of course, is stipulative but at least does not require any additional assumptions beyond the stipulations already made by Bruening himself.

What is another problem, is the case with third-person pronouns (296), repeated from (287). We would expect that the Salient Article on the direct object is barred with a subject third-person pronoun, assuming that the proximate argument is preferably promoted to subject (cf. the Passamaquoddy data above), and instead Bruening’s system predicts that the direct object must be obviative. However, Salient Article-marking on the direct object is a viable and even preferred option for some of my consultants in (296).

(296) [“A: A computer was stolen from our museum.

B: Did they find the thief?

A: Do you know Vasya?”]

λuw λoλmax-əλ/%-en/%-Ø wəjət-s-əle

(s)he thief-POSS.3SG/-POSS.2SG find-PST-3SG>SG

‘He found the thief.’

The other data from §5.2.5 also present a problem for Bruening’s system.

Thus, Bruening’s approach to proximate uniqueness makes the wrong predictions for the Kazym Khanty data, which is why I also reject it as a model of the Salient Article.

Another problem with both the OT and the Minimalist approaches sketched above is that they have nothing to say about the Salient Article’s status as a definite determiner, which I argued for in §5.2.1. In the next section I discuss the semantic analysis of the Salient Article from (Mikhailov 2024a) that both reflects the Salient Article’s status as a definite and attempts to model proximate uniqueness without unmotivated stipulations. Despite its merits, I will show that it too has a problem handling some of the data from §5.2.5.

5.4. *Toward a semantic analysis*

Let us sum up what generalizations an analysis of the Salient Article must account for:

- 1) The Salient Article is a definite determiner as it requires (at least) informational uniqueness and familiarity in the sense of (Roberts 2003), but it does not fit the general profiles of uniqueness and familiarity-based determiners (§5.2.1).
- 2) The Salient Article is used with demonstratives (examples (236), (270), cf. (Marinina 2024)).
- 3) The Salient Article is subject to a kind of proximate uniqueness constraint (Aissen 2001) such that there may be at most one DP with the Salient Article per clause (§5.2.4).
- 4) If a DP with the Salient Article has a demonstrative, it is no longer subject to proximate uniqueness (§5.3).
- 5) Similarly, if a DP with the Salient Article is oblique or its competitor is a personal pronoun, it is no longer subject to a (categorical) proximate uniqueness constraint. (New generalization, not discussed in (Mikhailov 2024a).)

In the preceding section, I argued that neither of the two existing syntactic accounts of proximate uniqueness work for the Salient Article. Thus, the Salient Article cannot be reasonably said to mark the syntactically proximate argument. Instead, I propose a semantic account based on the intuition voiced in the preceding sections that the Salient Article requires salience of the discourse referent in question.

5.4.1. The proposal from (Mikhailov 2024a)

Recall from §§5.1.1–5.1.2 that we understand salience as a function of attention capture (Roberts 2011; Barlew 2014). The salient discourse referents are those that the interlocutors attend to, be that in virtue of relevance to the interlocutors' goals or due to the referents' perceptual prominence. Assuming that the Salient Article requires salience as well as informational uniqueness and weak familiarity, this already accounts for the data in §§5.2.1–5.2.2.

Furthermore, as discussed in §2.3 and following general consensus in the work on demonstratives, in particular (Diessel 2006),¹⁹² I assume that demonstratives serve to coordinate the interlocutors' joint attention. Speakers use demonstratives and deictic pointing gestures to focus the addressee's attention on a particular referent (see also

¹⁹² I thank Lera Marinina (p.c.) for introducing me to Diessel's paper. See (Marinina 2024) for evidence that Diessel's point holds of Northern Khanty demonstratives as well.

Reisinger & Huijsmans 2021; Stojnić 2021: Ch. 4; Huijsmans & Reisinger 2022; Marinina 2024). Given that the demonstrated referent comes into addressee’s attention, this ensures that the referent is salient. And this invites the use of the Salient Article as the demonstrative fulfils the Salient Article’s salience requirement on the discourse referent.

Thus far we have said nothing to account for the obviation facts. In (Mikhailov 2024a), I proposed that the Salient Article requires **global maximal salience** of the discourse referent among the salient discourse referents in a given context as summarized in the excerpt in (297). This is different from the presuppositions of relative maximal salience and absolute maximal salience discussed in §2.3.3.1 and repeated in (298).

(297) ‘GLOBAL’ MAXIMAL SALIENCE PRESUPPOSITION OF THE KAZYM KHANTY SALIENT ARTICLE (according to Mikhailov 2024a: (59c))

Given a context C and time t , DP_i with the Salient Article presupposes that its antecedent discourse referent x_i is the most salient discourse referent in C at t excluding interlocutors (where t may be either the utterance time t_{UT} or the time of the pointing gesture t_{DEM} when a demonstrative is present)

(298) DEGREE-BASED SALIENCE

Given a context c with speaker s_c and addressee a_c and a discourse referent $i \in Dom(g_c)$:

a. **Salience measure function:** $m_{sal(c)}(i) = d$, the degree of salience of discourse referent i in context c is equal to d , where d is a real number in $[0; 1]$.

b. **Salience:** $sal(i, c) = [m_{sal(c)}(i) > 0]$

c. **Relative maximal salience** for given descriptive content P (e.g., features of a pronoun):

$$rel.max.sal(P, i, c) = \forall j \in Dom(g_c)[P(g_c(j)) \wedge m_{sal(c)}(j) \geq m_{sal(c)}(i) \rightarrow j = i]$$

d. **Absolute maximal salience:** $abs.max.sal(i, c) = [m_{sal(c)}(i) = 1]$

Global maximal salience (297) differs from absolute maximal salience (298d) in that there is no requirement that the degree of salience of the referent is maximal, and it differs from relative maximal salience (298c) in that global maximal salience compares

the referent to all other familiar discourse referents without relativizing to the descriptive content of the DP (this is why I call it ‘global’ here).

The proposal in (297) from (Mikhailov 2024a) follows in spirit the discourse-based approaches to the Algonquian proximate category which states that the proximate marks the more salient discourse referent among the referents under discussion (see the references in (Aissen 1997; Bliss 2017)). Its benefits include that it accounts for the obviation-like distribution facts of the Salient Article from §5.2.4 and for the exceptional proximate uniqueness violation with demonstratives (ignoring for the moment the other facts from §5.2.5).

First of all, global maximal salience ensures that at a single moment, say the utterance time of (277) repeated as (299), only one discourse referent may be maximally salient and thus eligible for Salient Article marking. This models the basic proximate uniqueness signature. In (299), only the Subject DP *ewen* ‘girl’ may be marked with the Salient Article even though the preceding context does not make either the girl or the boy more salient.¹⁹³

(299) [“I was walking along the street when I saw a boy and a girl.”]

ewen **aj_ikeλ** / #**aj_iken** χătś-əs-λe
 girl.POSS.2SG boy.POSS.3SG / boy.POSS.2SG hit-PST-3SG>SG
 ‘The girl hit the boy.’

I suggest that in cases like this, the speaker intends one referent (the girl) to be more salient than the other one and uses the Salient Article, which presupposes maximal salience of this referent, to convey this intention to the addressee. Seeing how the Salient Article’s presupposition is not satisfied in the context but is nevertheless plausible, the addressee accommodates it and shifts the focus of her attention to the referent marked with the Salient Article.¹⁹⁴

¹⁹³ An anonymous referee for *Voprosy Jazykoznanija* asks whether linear order affects the marking possibilities in (299). I have no evidence for the relevance of linear order, but syntactic role does have an effect here according to preliminary data. Thus, if the sentence is passivized and the girl demoted to Oblique, the preferred marking pattern is vice versa (given schematically as glosses): [boy.POSS.2SG] [girl.POSS.3SG-LOC] [hit-PST-PASS[3SG]]. I plan to investigate this further in future work.

¹⁹⁴ An anonymous referee for *Voprosy Jazykoznanija* notes that this reasoning might suggest that the salience of the referent is asserted rather than presupposed. I would disagree because that would predict the Salient Article to have a far wider distribution than it in fact has. The presuppositional view instead accounts for its narrow distribution, and the reasoning here fits neatly into the general workings of

Since only one referent may be maximally salient in a particular context at a particular point in time, marking the second referent (the boy) with the Salient Article would lead to presupposition failure. This gives the observed infelicity of using the Salient Article two times per clause in (299). Thus, my proposal accounts for the proximate uniqueness effects observed in the Salient Article.

Secondly, consider again the case with two demonstrative DPs with the Salient Article (repeated from (282)):

(300) [“What happened here?”]

tām	ewen	tum	aj_iken	χāts-əs
this	girl.POSS.2SG	that	boy.POSS.2SG	hit-PST[3SG]

‘This girl hit that boy.’

What would a typical utterance of a sentence like (300) look like? The speaker would first point at the girl, uttering *tām ewen* ‘this girl, and then at the boy, uttering the rest of the sentence, *tum aj_iken χātsəs* ‘hit that boy’. The addressee would follow with her gaze the first pointing gesture to the girl and then the second pointing gesture to the boy. Thus, it is fair to say that at different moments during the locutionary act of (300), two different referents are maximally salient, i.e., in focus of the interlocutors’ joint attention. And this is why the use of two Salient Articles per clause is possible with demonstratives according to (Mikhailov 2024a): demonstratives allow for the interlocutors’ joint attention to shift during the utterance.

The full proposal from (Mikhailov 2024a) is given in (301).

(301) SEMANTICS OF THE KAZYM KHANTY SALIENT ARTICLE (final version from Mikhailov 2024a; to be amended below)

Given a context *C* and time *t*, the use of a DP_i with the Salient Article presupposes that it has as antecedent a discourse referent x_i which is:

- a. weakly familiar in *C*,
- b. unique among discourse referents in *C* in being contextually entailed to satisfy the descriptive content of DP_i , and
- c. the most salient discourse referent in *C* at *t* excluding interlocutors

presupposition accommodation, see §2.1, §2.3.2 for examples from the other types of definites, and (Beaver & Zeevat 2007; von Stechow 2008) for a more general discussion of how accommodation works.

(where t may be either the utterance time t_{UT} or the time of the pointing gesture t_{DEM} when a demonstrative is present).

The final ingredient of this analysis is the clause excluding the interlocutors in (301c). This models the fact that the Salient Article is still available when the other DP is first or second person (cf. §5.2.4, §5.3), so Salient Article-marked DPs do not compete with interlocutor-denoting pronouns.

The rationale behind this clause may be found in the indexical approach to the semantics of first- and second-person pronouns¹⁹⁵ according to which these pronouns refer directly to the speaker and the addressee of the given context and not to the available discourse referents, i.e., reference is achieved without indices and without the assignment function g . Unlike third person pronouns and DPs that are interpreted as variables denoting discourse referents, the first and second person pronouns are interpreted as constants referring directly to the interlocutors. Thus, under the indexical view, the interlocutors are not even contained in the set of familiar discourse referents and will not be considered for the use of the Salient Article as it is already restricted to discourse referents. Strictly speaking, we can do without the addition to condition (301c) that excludes the interlocutors if we assume the indexical view, but it is left there for the sake of explicitness.¹⁹⁶

5.4.2. Corrections to (Mikhailov 2024a)

There are two corrections to the above analysis (301) that the full empirical picture from §5.2 requires us to make.

The first correction is a minor one and it concerns ‘salience without uniqueness’ contexts (302). The proposal from (301) incorporates Roberts’ informational uniqueness for definite DPs among weakly familiar discourse referents (301b) without relativizing to salient referents. Clearly, this makes the wrong prediction for (302) that the Salient Article should be barred: informational uniqueness among weakly familiar girls is violated here.

¹⁹⁵ See Kaplan’s seminal paper (Kaplan 1989) and much subsequent work, e.g., (Kratzer 2009).

¹⁹⁶ There are other views on the semantics of first and second persons which do not directly support this line of reasoning, in particular the proposal by (Sudo 2012) (see also Podobryaev 2017). Given that I am about to discard the analysis in (301), we will not need to further concern ourselves with the semantics of pronouns.

(302) [A theatrical play for children: “A girl_i appeared from the room on the left. Another girl_k appeared from the room on the right.”]

λ_{aw}_{k/*i} / ^{OK}**ewen**_{k/*i} još-ən lipət təj-əs
 (s)he / girl.POSS.2SG hand-LOC flower have-PST[3SG]
 ‘She_{k/*i} [lit.: the girl] was holding flowers.’

The solution is simple. Instead of informational uniqueness among the weakly familiar (301b), the Salient Article ought to presuppose relative maximal salience (298c), just like personal pronouns do according to Roberts, which accounts for the ‘salience without uniqueness’ contexts. For (302) this would mean that the Salient Article requires the *k* girl to be the most salient discourse referent among ‘girl’ discourse referents. This is satisfied in this context since *k* is the most recently mentioned girl and hence the most salient one. The less salient *i* girl is screened out thanks to the relative maximal salience presupposition.

The second correction is much larger in scale. Recall that in (Mikhailov 2024a) I proposed that the Salient Article presupposes global maximal salience (297). If we consider the full empirical picture from §5.2.5, it becomes clear that global maximal salience cannot be right as it undergenerates in the proximate uniqueness-violating examples like (303) and (304) repeated from (284) and (287) respectively. Recall that in these examples Salient Article-marking of the target DP is accepted by most of my consultants and even volunteered by some, alongside the second option, to mark it with an Associative POSS.3SG.

(303) [I was walking along the street when I saw a boy and a girl.]

c. aj_iken **eweλ-a/%ewen-a** lipət mə-s
 boy.POSS.2SG girl.POSS.3SG-DAT/girl.POSS.2SG-DAT flower give-PST[3SG]
 ‘The boy gave a flower to the girl.’

d. aj_iken **eweλ** / **ewen** χuśa mən-əs
 boy.POSS.2SG girl.POSS.3SG / girl.POSS.2SG to go-PST[3SG]
 pa lipt-ən mə-s-λe
 ADD flower-LOC give-PST-3SG>SG
 ‘The boy came closer to the girl and gave her a flower.’

(304) [“A: A computer was stolen from our museum.

B: Did they find the thief?

A: Do you know Vasya?"]

λuw λoλmax-əλ/%-en/%-Ø wəjət-s-əλe
(s)he thief-POSS.3SG/-POSS.2SG find-PST-3SG>SG
'He found the thief.'

In these examples, global maximal salience (297) would rule out the Salient Article on the second DP, since the subject DP is more salient (either already marked with the Salient Article or is itself a pronoun) and the presupposition in question fails for the referent of the second DP.

It simply seems that the Salient Article's proximate uniqueness effects are not due to a semantic restriction (only). After all, examples (303) and (304) provide minimal pairs to (277)-(278), repeated below, up to syntactic role of the second DP / DP type of the subject, but the difference in proximate uniqueness effects is unequivocal.

(305) ["I was walking along the street when I saw a boy and a girl."]

ewen aj_ikeλ / #aj_iken χăts-əs-λe
girl.POSS.2SG boy.POSS.3SG / boy.POSS.2SG hit-PST-3SG>SG
'The girl hit the boy.'

(306) ["A computer was stolen from our museum. But everything worked out."]

waśaj-en λoλmax-əλ/#-en/%-Ø wəjət-s-əλe
V.-POSS.2SG thief-POSS.3SG/-POSS.2SG find-PST-3SG>SG
'Vasya found the thief [and returned the computer].'

It seems that we ought to abandon the promise of a semantic account of these data voiced in (Mikhailov 2024a). Some morphosyntactic or, perhaps, parsing factors are at play here, and a separate story needs to be told about them, as I am going to discuss shortly.

But before that, here is a stripped down proposal for the semantics of the Salient Article, in Roberts' semi-formal dynamic style (307) and in a static format (308). This proposal is based on Roberts' analysis of personal pronouns discussed in §2.3, which accounts for the availability of the Salient Article in 'salience without uniqueness' contexts §5.2.3.

(307) SEMANTICS OF THE KAZYM KHANTY SALIENT ARTICLE (new version)

Given a context C , the use of a DP_i with the Salient Article presupposes that it has as antecedent a discourse referent x_i which is:

- a. familiar in C ,
- b. salient in C
- c. unique in being the most salient discourse referent in C which is contextually entailed to satisfy the descriptive content of DP_i .

(308) KAZYM KHANTY SALIENT ARTICLE: STATIC DENOTATION

$$\llbracket \text{SAL}_i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle}: \text{sal}(i, c) \wedge \forall j \in \text{Dom}(g) [P(g(j)) \wedge j \geq_{\text{sal}(c)} i \rightarrow j = i]. g(i)$$

According to (307)/(308) presupposes salience and relative maximal salience among referents satisfying the DP description. This should account for all of the data above (see the generalizations at the beginning of §5.4) except the proximate uniqueness patterns. Just like the analysis from (Mikhailov 2024a), this proposal correctly captures the fact that the Salient Article is a definite determiner (§5.2) which is what this chapter was aiming for. In the following and final subsection of this chapter, I provide a dynamic sketch of how deictic demonstratives invite the use of the Salient Article.

As for the proximate uniqueness signature exhibited by the Salient Article, I suggest, following (Muravyev et al. 2024), that it is rooted in parsing considerations. (Muravyev et al. 2024) propose that the proximate uniqueness effects observed in the Salient Article and beyond — in Kazym Khanty third-person Possessive Agreement, and in other obviation phenomena crosslinguistically — are functionally-driven.

The motivation for proximate uniqueness is to enhance resolution of syntactic and referential ambiguity by additional morphosyntactic marking (e.g., obviative marking in Blackfoot, §5.1.3) or by banning identical marking. The latter is proposed as the story behind the Salient Article’s behavior in the baseline example (305)-(306).

(309) FUNCTIONALLY-DRIVEN PROXIMATE UNIQUENESS (Muravyev et al. 2024)

- i. **Constraint:** Only one proximate argument per clause is possible within a single syntactic domain with formally and/or referentially ambiguous arguments, where:
 - b. distinct syntactic domains: are CP (clause) and DP (and possibly PP)
 - c. factors alleviating ambiguity include differences between competing DPs in animacy, syntactic role, or pronoun vs. full DP

- ii. **Motivation:** Efficient parsing during comprehension requires minimizing ambiguity. Proximate uniqueness is operative where ambiguity is not alleviated. It forces the speaker to use additional morphosyntactic means to disambiguate. Factors from (309i.c) may alleviate ambiguity which lifts the proximate uniqueness constraint.

According to this idea (309), the reason why the Salient Article is subject to proximate uniqueness only when the two DPs are subject and direct object is that this is the case where syntactic role assignment is ambiguous, because Northern Khanty does not distinguish Nominative and Accusative on lexical DPs. Where distinct syntactic roles are differentiated overtly by case (as in cases with obliques (303)) or by using a pronoun (304) which distinguishes Nominative and Accusative, proximate uniqueness is lifted as it is not needed to enforce disambiguation.

This is the current most promising story about proximate uniqueness effects in the Kazym Khanty Salient Article, but it is still in its infancy and this thesis is not the place to spell it out in more detail. What matters for us is that ‘global’ maximal salience is not a presupposition of the Salient Article as shown by (303) and (304), and so we should stick with the new Roberts-style analysis (307)/(308) that does not attempt to say anything about proximate uniqueness but succeeds in explicating the difference between the Kazym Khanty Salient Article on the one hand, and the Bulu Salient Article and other definites on the other. The Kazym Khanty Salient Article is a definite determiner that requires maximal salience among referents in the NP extension, just like personal pronouns. Thus, it makes the strongest requirement from the Salience-Familiarity Scale of Chapter 2, repeated here from (61), excluding absolute maximal salience which appears to never be presupposed by determiners (only by zero/unstressed pronouns):

(310) THE SALIENCE-FAMILIARITY SCALE

weak familiarity > strong familiarity > salience > relative maximal salience >
absolute maximal salience

Before concluding this chapter, I provide a dynamic sketch of how deictic demonstratives invite the use of the Salient Article.

5.5. Deictic demonstratives with the Salient Article

As discussed throughout this chapter, deictic demonstrative DPs are generally marked with the Salient Article.¹⁹⁷ I repeat here some examples from above (adding the co-speech pointing gesture to some individual *a* as a subscript “→*a*”):

(311) [A mother is walking with her child. The child points at a flower:]

ma **tum**→_b **lipt-en** mənəm-ti λaŋχa-λ-əm
 I that flower-POSS.2SG pick-NFIN.NPST want-NPST-1SG
 ‘I want to pick that flower.’ (= (236))

(312) [“Who hit you?”]

ma **śi**→_b **aj_iken-ən** χătś-əs-ij-əm
 I DEM boy-POSS.2SG-LOC hit-PST-PASS-1SG
 ‘I was hit by that boy.’ (= (138))

(313) [“What happened here?”]

tām→_b **ewen** **tum**→_d **aj_iken** χătś-əs
 this girl-POSS.2SG that boy-POSS.2SG hit-PST[3SG]
 ‘This girl hit that boy.’ (= (282))

The general answer for why the Salient Article is preferably used with deictic demonstratives as in (311)-(313) is that demonstratives and deictic pointing gestures are used to bring referents to salience (Diessel 2006; Reisinger & Huijsmans 2021; Stojnić 2021: Ch. 4; Huijsmans & Reisinger 2022; Marinina 2024) which fulfils the Salient Article’s salience presupposition.

To model this, we will take the denotations for Kazym Khanty demonstratives proposed by Marinina (2024; discussed in §2.4.3) and the Salient Article denotation (308) from above and embed them in a dynamic framework based on (Groenendijk & Stokhof 1991; AnderBois et al. 2015) and inspired by Ebert et al.’s (2020) treatment of demonstratives with pointing gestures. To this I also add dynamic salience building on Stojnić’s (2021) work.¹⁹⁸

¹⁹⁷ Anaphoric demonstratives are also used with the Salient Article (Marinina 2024) but they are of less interest because they do not introduce their discourse referent but rather pick it up from the context.

¹⁹⁸ Stojnić uses ‘prominence’ to talk about salience in the sense of the current work.

Here are Marinina’s denotations, without elaborating on how the anaphoric vs. deictic distinction is implemented, see §2.4.3 (l_c is the location of the speech act of context c):

(314) KAZYM KHANTY DEICTIC DEMONSTRATIVES (adapted from Marinina 2024: (11-13))

- a. $\llbracket \acute{s}i \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda R_{\langle e,t \rangle} . \iota x_e . P(x) \wedge R(x)$ (Unmarked)
- b. $\llbracket t\check{a}m \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda R_{\langle e,t \rangle} . \iota x_e . P(x) \wedge R(x) \wedge PROX(x)(P)$ (Proximal)
- c. $\llbracket tum \rrbracket^{g,c} = \lambda P_{\langle e,t \rangle} \lambda R_{\langle e,t \rangle} . \iota x_e . P(x) \wedge R(x) \wedge DIST(x)(P)$ (Distal)
- d. $PROX = \lambda x \lambda P . x$ is close in space to l_c
- e. $DIST = \lambda x \lambda P . x$ is far in space from l_c

5.5.1. A dynamic semantic model with demonstratives and salience

In standard static semantics of predicate logic (similar to what we have been working with so far), the interpretation of a formula is a set of assignment functions that verify this formula.¹⁹⁹ In their Dynamic Predicate Logic, Groenendijk & Stokhof (1991) propose that a formula ϕ is interpreted as a set of pairs of assignments, $\langle g, h \rangle$, such that h (the output assignment) is the result of updating g (the input assignment) with ϕ (cf. the discussion of different contexts in §2.3.2) or, notationally: $g \llbracket \phi \rrbracket h$. Thus, a dynamic formula is a context change potential (CCP; cf. fn. 59) if we understand contexts as assignments. Its effect on a context may be either adding of a new variable to h as compared to g (i.e., introduction of a new referent), or testing $\langle g, h \rangle$ to see whether the individuals that they assign variables to satisfy the predicates in the formula (i.e., predicating something of familiar referents). Predications on familiar referents function as tests on assignments that filter out those assignments which do not satisfy the predicate (with the individuals assigned to the variables of the formula).

I will make use of Stojnić’s (2021) Attention-Coherence account of anaphora resolution in a somewhat simplified form, i.e., ignoring the impact of coherence relations (Kehler 2002; Asher & Lascarides 2003; see also Stojnić & Stone & Lepore 2017). An important difference of the Attention-Coherence account from Groenendijk & Stokhof’s system is that stacks are used in the place of traditional assignments, so that the

¹⁹⁹ I.e., those assignment functions that make the formula true by virtue of which individuals they assign to the variables in the formula.

introduction of a new discourse referent pushes it on top of the stack (0-th position) demoting all the other discourse referents in the stack by 1. This is done via $\langle \alpha \rangle$ which is the equivalent of dynamic existential quantification in traditional dynamic systems, introducing a novel discourse referent. Anaphoric pick-up is done via $@p$, where p is some one-place predicate, e.g., she' (true of singular, female third persons): $@she'$ picks up the top referent of the current stack which satisfies she' . Naturally, anaphoric pick-up also promotes the referent on top of the stack, which is done via $\langle \pi d \rangle$ for some individual-denoting expression d . For instance, $\langle \pi @she' \rangle$ picks up the top she' -referent and pushes it on the very top of the stack. $\langle \pi \dots \rangle$ is also used with other definite-like expressions such as proper names or pointing gestures with demonstratives.

With this much, we can see how discourses with one-place predicates are interpreted, e.g., (315) is interpreted as (316), simplifying the way predicates get their arguments.

(315) A woman came in. She sat down.

(316) $\langle \alpha \rangle \wedge woman'(@) \wedge come.in'(@) \wedge \langle \pi @she' \rangle \wedge sit.down'(@)$

Introduce a woman referent to the stack; predicate that the top referent came in;²⁰⁰ pick up the top female referent and push her on top of the stack; predicate that she sat down.

The advantage of this approach is that it gives a unitary interpretation to anaphoric pronouns and other anaphoric expressions via $@$ by making use of stacks, while traditional approaches have to deal with ad hoc indexing on pronouns, possibly implying infinite lexical ambiguity, as well as other issues both conceptual and empirical (Haug 2014; Stojnić 2021). The use of stacks also gives us a natural model of salience at least in some instances. The most recently introduced referent is the most salient, since it is on the top of the stack making it the most accessible referent for anaphoric pick-up.

The framework I will be using works as follows. Natural language utterances are translated to dynamic formulas.²⁰¹ Formulas are assigned an interpretation by the

²⁰⁰ That is: filter out those candidate output stacks in which the top referent did not come in.

²⁰¹ Thus, we move here from a direct interpretation system like (Heim & Kratzer 1998) used so far to an indirect interpretation system where natural language expressions are translated to logical expressions which are then interpreted.

interpretation function $\llbracket \cdot \rrbracket^{\langle g, h \rangle, c, M}$ with respect to a pair of stacks $\langle g, h \rangle$ which are specified as part of a context c and relative to a model M .²⁰² A stack is defined as an ordered sequence of entities where a natural number starting with 0 corresponds to each entity in the sequence: g_0 is the top entity of the stack; g_m is the m th entity if there is one, otherwise it is the undefined value $\perp$.

Similarly to the proposal in §2.3.2 (54), a context c includes the common ground C_c , the interlocutors s_c and a_c , the utterance time t_c , and a pair of stacks $\langle g_c, h_c \rangle$ (the subscript is usually omitted): $c = \langle C_c, s_c, a_c, t_c, \langle g_c, h_c \rangle \rangle$. A formula ϕ uttered in a context c maps the stack of this context g_c to a new stack h_c (dynamic update): $g[\llbracket \phi \rrbracket^c] h$. More broadly, a formula ϕ uttered in a context c moves the conversation to a new context c' such that $h_c = g_{c'}$ (the output stack of c is the input stack of c'), including the addition of all the new referents in ϕ and having run all the tests of ϕ on c and so on. In notation: $c[\llbracket \phi \rrbracket] c'$.²⁰³

A formula ϕ is said to be **true** in c iff g and h are such that $g[\llbracket \phi \rrbracket^c] h$ (h is the result of updating g with $\llbracket \phi \rrbracket^c$).

To model the different salience of different arguments in English (and Bosnian-Croatian-Montenegrin-Serbian) and predication with two or more arguments, Stojnić make use of dedicated positions in the stack: 0 corresponds to the subject position since subjects are more discourse prominent (more salient) than objects in English; 1 corresponds to the direct object; 2 corresponds to the indirect object.²⁰⁴ Referent introduction refers to the specific position where the DP occurs which affects which

²⁰² I usually omit some of the parameters where they don't play a role, thus, the following are equivalent under this simplification: $\llbracket \cdot \rrbracket^{\langle g, h \rangle, c, M}$, $\llbracket \cdot \rrbracket^{\langle g, h \rangle}$, $\llbracket \cdot \rrbracket^c$, and $\llbracket \cdot \rrbracket$.

See (Stojnić 2021; Stojnić & Stone & Lepore 2017) for details on how exactly such a logical system is defined (without the simplifications adopted here). I'm also incorporating elements of (AnderBois et al. 2015) as a matter of convenience (cf. Ebert, C. et al. 2020; Groenendijk & Stokhof 1991).

²⁰³ We may also say that the common ground $C_{c'}$ of c' is the result of updating C_c of the utterance context with ϕ . Generally speaking, the output context $c' = c[\llbracket \phi \rrbracket]$ is the result of updating the utterance context c in all the relevant ways that $\llbracket \phi \rrbracket$ requires, including changes to the salience of referents on which see below (322).

Although, I include the common ground parameter here, I do not formally integrate it in the definitions below for the sake of simplicity.

²⁰⁴ On English see (Stojnić 2021: 59ff.). As far as I can see, the Northern Khanty facts are the same so that subjects are more discourse prominent than direct objects which are in turn more prominent than indirect object, so, a referent introduced in the object position is less salient than a referent introduced in the subject position. Although, there has been no systematic investigation of the relevant Khanty facts yet.

position of the stack is updated, e.g., $\langle \alpha 1 \rangle$ introduces a referent at the 1st position of the stack, keeping the 0th position intact and demoting the referent previously at the 1st position to the 2nd, the 2nd to the 3rd and so on. Predication is now done via variables like x_0 and x_1 which represent the 0th and 1st referents of the current stack respectively.

Here is how basic formulas are interpreted, including conjunction and presupposition (and omitting irrelevant details of the context):

(317) SEMANTICS OF SOME BASIC EXPRESSIONS

- a. **Novel referent introduction** at the k th position of the stack: $\llbracket \langle \alpha k \rangle \rrbracket^{(g,h)} = [1$
iff h is just like g except that some entity $b \in D_e$ occupies position k of h , all
positions n smaller than k are as in g , all positions m larger than k contain the
subsequence of entities of g starting with k in order] $= [1$ iff $h = g_{0,k} +$
 $b \cdot g_{k...}]$ ²⁰⁵
- b. **Anaphoric pick-up** of the top appropriate referent in g : $\llbracket @p \rrbracket^{(g,h)} = \perp$ if $g_0 = \perp$
(undefined value), otherwise g_0 if g_0 is a p , otherwise g_1 is a p and so on until
an appropriate entity is found, $\perp$ otherwise
- c. **Non-novel referent introduction**: $\llbracket \langle \pi kd \rangle \rrbracket^{(g,h)} = [1$ iff h differs from g at most
in that the entity b denoted by some entity-denoting expression d occupies k
with smaller positions as in g and entities from larger positions of g demoted by
one in h] $= [1$ iff $\llbracket d \rrbracket^{g,M} = b$ and $h = g_{0,k} + b \cdot g_{k...}]$
- d. **Predication**: $\llbracket P(x_k) \rrbracket^{(g,h)} = 1$ iff $[g = h$ and the k th position of h is in the
extension of $P]$ $= [g = h$ and $h_k \in \llbracket P \rrbracket^{(g,h)}]$ ²⁰⁶ (and similarly for 2-place
predicates)
- e. **Dynamic conjunction**: $\llbracket \phi \wedge \psi \rrbracket^{(g,h)} = 1$ iff [there is some stack i such that
 $\llbracket \phi \rrbracket^{(g,i)} = 1$ and $\llbracket \psi \rrbracket^{(i,h)} = 1]$ $= [\exists i. g \llbracket \phi \rrbracket i$ and $i \llbracket \psi \rrbracket h]$ ²⁰⁷
- f. **Presupposition**: $\llbracket \delta(\phi) \rrbracket^{(g,h)} = 1$ if $g \llbracket \phi \rrbracket g$, undefined ($\#$) otherwise²⁰⁸

²⁰⁵ See (Stojnić et al. 2017: 542-544; Stojnić 2021: 190-191).

²⁰⁶ Predication is a test which does not alter the assignments but filters out falsifying assignments by keeping only those pairs of assignments in the set of possible assignments for the current discourse that assign the variables in the predication to individuals that satisfy the given predicate.

²⁰⁷ There is a stack i which is the result of updating g with ϕ such that h is the result of updating i with ψ , i.e., this is the sequential update of g first with ϕ and then with ψ which leads to h .

²⁰⁸ In prose, $\delta(\phi)$ requires that the update of g with ϕ does not change g in any way, otherwise it yields undefinedness, or, more broadly, it requires that $c = c \llbracket \phi \rrbracket$ holds. The partial operator δ

g. **Identity:** $\llbracket x_k = x_n \rrbracket^{(g,h)} = 1$ iff $g = h$ and $h_k = h_n$

Example (317) makes use of some additional notation for stacks summarized in (318).

(318) NOTATION FOR STACKS

- a. g_k is the entity at the k th position of stack g
- b. $g_{0..k}$ is the subsequence of entities of g in order, starting with the 0th position and up to, but not including the k th position ($g_{0..1}$ is a sequence of entities which only contains the 0th element of g)
- c. $g_{k..}$ is the complete subsequence of entities of g starting with the k th position and until the end of g ($g_{0..}$ is equivalent to g)
- d. $g + h$ is a sequence of entities (a stack) which contains in order the elements of g followed by the elements of h
- e. $b.g$ is a stack which starts with the entity $b \in D_e$ and continues with the elements of g

Example (319) is an illustration of how Kazym Khanty expressions are translated ($\rightsquigarrow$) into formulas, including Ebert et al.’s (2020) treatment of demonstratives with co-speech pointing gestures (319d-e). A demonstrative with a pointing gesture introduces a discourse referent ($\langle \pi k I_b \rangle$) for the gesture referent (I_b), triggers its deictic presupposition, if any, and predicates the NP denotation of this referent. (Note that I avoid giving a fully compositional model here like (Muskens 1996; Coppock & Champollion 2024: Ch. 9) to simplify presentation, but it can easily be done using the fragment provided here if needed.)

(319) TRANSLATIONS OF SOME KAZYM KHANTY EXPRESSIONS AND DENOTATIONS OF CORRESPONDING LOGICAL CONSTANTS

- a. $ewi \rightsquigarrow girl'$
 $\llbracket girl' \rrbracket^{(g,h)} = \{x: x \text{ is a girl}\}$
- b. $aj_iki \rightsquigarrow boy'$
 $\llbracket boy' \rrbracket^{(g,h)} = \{x: x \text{ is a boy}\}$

(‘presupposing that...’) is borrowed from (Beaver 1992; Beaver & Krahmer 2001) following (Coppock & Champollion 2024).

c. $\chi\check{a}ts\acute{a}s \rightsquigarrow hit'$

$$\llbracket hit' \rrbracket^{\langle g, h \rangle} = \{ \langle x, y \rangle : x \text{ hit } y \}$$

d. $t\check{a}m \rightarrow_b ewi \rightsquigarrow \langle \pi k I_b \rangle \wedge \delta(PROX(x_k)) \wedge girl'(x_k)$

where k corresponds to 0, 1 or 2 depending on the syntactic position of the DP

$$\llbracket PROX(x_k) \rrbracket^{\langle g, h \rangle, c} = 1 \text{ iff } g = h \text{ and } h_k \text{ is close to } l_c \text{ (cf. (314d))}$$

$$\llbracket I_b \rrbracket^M = b, \text{ for some individual } b \text{ in the model } M$$

e. $tum \rightarrow_b ewi \rightsquigarrow \langle \pi k I_b \rangle \wedge \delta(DIST(x_k)) \wedge girl'(x_k)$

$$\llbracket DIST(x_k) \rrbracket^{\langle g, h \rangle, c} = 1 \text{ iff } g = h \text{ and } h_k \text{ is far from } l_c \text{ (cf. (314e))}$$

To illustrate how the system works, consider a simple sentence (320) without demonstratives or the Salient Article.

(320) *ewi aj_iki χḁts-əs*
 girl boy hit-PST[3SG]
 ‘A girl hit a boy.’

Assuming with (Tiutiunnikova et al. 2024) that Kazym Khanty bare nouns are interpreted as indefinites (cf. §3.3.1), we get the following translation and truth conditions for (320), along with an algorithmic clarification of the truth conditions:²⁰⁹

(321) $\llbracket \langle \alpha 0 \rangle \wedge girl'(x_0) \wedge \langle \alpha 1 \rangle \wedge boy'(x_1) \wedge hit'(x_0, x_1) \rrbracket^{\langle g, h \rangle} = 1 \text{ iff}$

$\exists i \exists b, d \in D_e [i = b.g \text{ and } i_0 \text{ is a girl and } h = i_{0.1} + d.i_{1...} \text{ and } h_1 \text{ is a boy and } h_0 \text{ hit } h_1]$

Push some girl b on top of g (0th position) resulting in stack i ; push some boy d on the 1st position of i resulting in stack h ; predicate that the girl hit the boy.²¹⁰

Now, in order to analyze the Salient Article, we need to say something about how salience works in the current system. Following Stojnić, I assume that the most recently

²⁰⁹ Again, note that I did not provide fully compositional rules for how predication is achieved or how bare nouns are to be interpreted using EX in a dynamic setting, see (Muskens 1996; Coppock & Champollion 2024) and on the latter see (Beaver & Coppock 2015).

Also, I simplify dynamic conjunctions (317c) by not introducing an additional assignment wherever the conjuncts entail that $g = h$.

²¹⁰ After this update, stack h has the form $\langle b, d, \dots \rangle$ with the girl on the 0th position and the boy on the 1st position.

introduced referents are the most salient (cf. §2.3/§5.1.1) which is reflected in the fact that referents are introduced to the top of the stack and anaphoric pick-up via @ proceeds from the top of the stack until an appropriate referent is found: the top-most referents are the most easily accessible for anaphoric. Among the salient referents, (referents introduced as) subjects are more salient than (referents introduced as) direct objects which are more salient than (referents introduced as) indirect objects. This is reflected in how referent introduction targets one of three specified positions in the stack (317a,b). In general, referents higher in the stack are more salient than referents lower in the stack. Using our degree-based definitions of salience (298)/(62) from §2.3.3.1, this view of salience in Stojnić’s Attention-Coherence approach with stacks is summarized in (322).

(322) DYNAMICS OF SALIENCE WITH STACKS

- a. Top three positions of the current stack g_c are salient: $sal(g_0, c)$, $sal(g_1, c)$, $sal(g_2, c)$. (The interlocutors attend to the most recently introduced referents.)
- b. Entities at higher²¹¹ positions of the current stack g_c are more salient than entities at lower positions: $\forall k, n \in \mathbb{N}[k < n \rightarrow m_{sal(c)}(g_k) > m_{sal(c)}(g_n)]$.
- c. Referent introduction pushes the referent to one of the top three positions of the current stack making this referent salient by (322a) (see (317)).
- d. Beyond that, salient referents are the ones the interlocutors attend to, as discussed in §2.3/§5.1.1.

Note that this formalization by necessity simplifies some of the aspects of salience discussed in §2.3, e.g., attention is not directly represented here, and it also does not give us a clear handle on degrees of salience. Nevertheless, this is a clear step forward as it allows us to operationalize salience in order to model the Salient Article’s co-occurrence with deictic demonstratives, to which I now turn.

I adapt the new semantics of the Salient Article proposed in (308) as follows:

(323) TRANSLATION AND INTERPRETATION OF THE SALIENT ARTICLE

Given some noun phrase N whose logical translation is P ,

- a. $N-en \rightsquigarrow \delta(SAL(@P)) \wedge \delta(REL.MAX.SAL(@P, P))$

²¹¹ Higher positions have the smaller number, i.e., 0 is the top position, 1 is the second highest position, and so on.

- b. $\llbracket \text{SAL}(@P) \rrbracket^{(g,h),c} = 1$ iff $g = h$ and it holds of the top P -referent x of g that $\text{sal}(x, c)$
- c. $\llbracket \text{REL.MAX.SAL}(@P, P) \rrbracket^{(g,h),c} = 1$ iff $g = h$ and it holds of the top P -referent x of g that

$$\forall n \in \text{Dom}(g)[g_n \in \llbracket P \rrbracket^{(g,h),c} \wedge m_{\text{sal}(c)}(g_n) \geq m_{\text{sal}(c)}(x) \rightarrow g_n = x]$$

According to (323), the Salient Article presupposes salience of its discourse referent and maximal salience relative to the predicate denoted by its noun phrase. This is a more complicated translation than one can imagine using Stojnić’s $@$ variable. Let us consider the simpler alternative to (323) to see why (323) should be preferred.

Recall that the basic translation of anaphoric expressions in this system is simply $@p$ where p is a one-place predicate summarizing the information encoded by the given anaphoric expression. One can imagine translating the Salient Article simply as $@P$ where P is the logical translation of the marked DP. This ensures relative maximal salience, since $@$ searches for an appropriate referent starting from the salient top of the stack; which would automatically give us an account of salience without uniqueness contexts as discussed in §5.4.2. However, $@P$ by itself does not ensure salience since it could in principle search deep down in the stack down to non-salient strongly familiar referents. As we saw throughout this chapter, this is unsatisfactory. Salience needs to be hard-coded as a presupposition of the Salient Article to rule out reference to non-salient familiar referents, which (323) achieves.

There is another benefit to (323) which is less apparent since we are not working in a fully compositional system here. In Stojnić’s system, anaphoric expressions contribute their meaning as part of non-novel referent introduction (317c) done with $\langle \pi k \dots \rangle$. I believe that this is a suitable analysis for the Salient Article. When we consider the Salient Article’s uses with demonstratives below, I will make use of the idea that the deictic demonstrative (specifically, its pointing gesture) contributes the referent of the DP (319d,e) (Diessel 2006; Ebert et al. 2020; Stojnić 2021: Ch. 4). Making the Salient Article introduce a referent for the same DP seems like an overkill. I will discuss this aspect of (323) more below as we work through the central examples, although ultimately, this has to be justified in a compositional fragment based on the current system to be developed in future work.

5.5.2. *The Salient Article with demonstratives and in ‘salience without uniqueness’ contexts*

First, consider the example with two demonstratives (324). The two DPs with a demonstrative, a pointing gesture, and a Salient Article are translated as (325i-ii), and the whole sentence is translated as (325iii).

(324) [“What happened here?”]

tăm_{→b} ewen tum_{→d} aj_iken χăts-əs
 this girl.POSS.2SG that boy.POSS.2SG hit-PST[3SG]
 ‘This girl hit that boy.’ (= (282))

Using the rules for dynamics of salience (322), we eliminate the presuppositions introduced by the Salient Article for each DP (325iv) since they are trivially satisfied by virtue of the respective deictic demonstratives, which introduce a referent, making it salient and promoting it to maximal relative salience (see (319d,e)). The sentence as a whole receives the truth conditions in (325v) along with its presuppositions and an algorithmic explanation of the truth conditions.

- (325) i. $tăm_{→b} ewen \rightsquigarrow \langle \pi 0I_b \rangle \wedge \delta(PROX(x_0)) \wedge girl'(x_0) \wedge \delta(SAL(@girl')) \wedge \delta(REL.MAX.SAL(@girl', girl'))$ (by (319d))
- ii. $tum_{→d} aj_iken \rightsquigarrow \langle \pi 1I_d \rangle \wedge \delta(DIST(x_1)) \wedge boy'(x_1) \wedge \delta(SAL(@boy')) \wedge \delta(REL.MAX.SAL(@boy', boy'))$ (by (319e))
- iii. (324) $\rightsquigarrow \langle \pi 0I_b \rangle \wedge \delta(PROX(x_0)) \wedge girl'(x_0) \wedge \delta(SAL(@girl')) \wedge \delta(REL.MAX.SAL(@girl', girl')) \wedge \langle \pi 1I_d \rangle \wedge \delta(DIST(x_1)) \wedge boy'(x_1) \wedge \delta(SAL(@boy')) \wedge \delta(REL.MAX.SAL(@boy', boy')) \wedge hit'(x_0, x_1)$
- iv. Formula (325iii) is equivalent to
- $\langle \pi 0I_b \rangle \wedge \delta(PROX(x_0)) \wedge girl'(x_0) \wedge \langle \pi 1I_d \rangle \wedge \delta(DIST(x_1)) \wedge boy'(x_1) \wedge hit'(x_0, x_1)$ (by (322c,a))
- v. $\llbracket \langle \pi 0I_b \rangle \wedge \delta(PROX(x_0)) \wedge girl'(x_0) \wedge \langle \pi 1I_d \rangle \wedge \delta(DIST(x_1)) \wedge boy'(x_1) \wedge hit'(x_0, x_1) \rrbracket^{(g,h),c} = 1$ iff $\exists i. [i = b. g \text{ and } i_0 \text{ is a girl and } h = i_{0.1} + d. i_{1...}$
 and h_1 is a boy and h_0 hit h_1];
 defined iff i_0 is close to l_c and h_1 is far from l_c (by (317))

Store (the pointed at) girl *b* at the 0th position of the current stack; (check that the girl is close to the speech act location); store (the pointed at) boy *d* at the 1st position of the current stack; (check that the boy is far from the speech act location;) predicate that the girl hit the boy.

Thus, we get the intuitively correct truth-conditions for (324). It is true iff the pointed out individual *b*, which is close to the speech situation, is a girl and the pointed out individual *d*, which is far from the speech situation, is a boy and the girl hit the boy. The Salient Article's presuppositions are trivially satisfied by the deictic demonstratives and hence do not affect the global presuppositions of (324): the demonstratives raise *b* and *d* to salience and to maximal relative salience among girls and boys respectively, fulfilling both presuppositions of the respective Salient Articles locally.

Note that it is crucial for the above argument to stipulate that the demonstrative and the deictic gesture are evaluated before the Salient Article.²¹² For non-demonstrative DPs we must assume that the Salient Article's presuppositions are evaluated prior to the content of the rest of the DP. I believe it is a reasonable assumption overall that the presuppositions of a constituent XP are evaluated prior to all other semantic content of this XP (except the special case of deictic gestures with demonstratives). This is needed to ensure that non-demonstrative DPs with the Salient Article such as *ewen* are infelicitous in a context where the Salient Article's presuppositions are not satisfied. (Otherwise the system so far would allow for *ewen* to receive an existential interpretation as above (320) and satisfy the Salient Article's presuppositions locally.)

²¹² A further note about order of composition is needed here. The fact that the dynamic fragment we're working with currently is not compositional allows me to side-step the issue of how exactly a demonstrative DP is composed and, in particular, the question of what types the demonstratives and the Salient Article have.

An attentive reader might have noticed that the static denotations from above actually result in a type clash. The demonstratives are of type $\langle\langle e, t \rangle, \langle\langle e, t \rangle, e \rangle\rangle$ in (314) while the Salient Article is of type $\langle\langle e, t \rangle, e \rangle$ in (308). The solution is to give either one of them a predicative denotation, so that they add their presuppositions to the meaning of the DP and return a predicate $\langle e, t \rangle$. According to Lera Marinina's recent data on Kazym Khanty demonstratives (specifically, their incompatibility with adnominal quantifiers), it seems that demonstratives really do yield a denotation of type *e*, meaning that the Salient Article must actually have the predicative type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ (cf. Coppock & Beaver 2015).

I leave the solution of this issue for another occasion.

I demonstrate how infelicity comes about with a non-demonstrative DP like this for example (326), repeated from (261). The crucial detail is that the context of (326) neither introduces a strongly familiar fish referent for the target DP nor makes it salient.

(326) [There is a fishing contest on Fisherman's Day. I come home from the contest and tell my family about it.]

#tām ol mæt wən χuλ-en wet-χuś-jǎŋ kila-kəm
 this year very big fish-POSS.2SG five-DEC-ten kg-APPR
 'This year, the biggest fish was 15 kilograms.'

(327) $\delta(\text{SAL}(@\text{biggest.fish}')) \wedge$
 $\delta(\text{REL.MAX.SAL}(@\text{biggest.fish}', \text{biggest.fish}')) \wedge$
 $\langle \pi 0 @ \text{biggest.fish}' \rangle \wedge \text{weigh.15.kg}'(x_0)$

Example (326) receives the translation in (327) (ignoring irrelevant details).²¹³ The translation involves two presuppositions for a referent x ($= @\text{biggest.fish}'$, relative to the input stack g_c) which are evaluated in a context where x is neither salient nor (by assumption) familiar and this leads to infelicity (#) of the Salient Article in this example.

Finally, let us look at how the current system models the felicity of the Salient Article in Roberts' 'salience without uniqueness' contexts which served as the crucial piece of data for the analysis of the Salient Article proposed in §5.4.2.

Here is the full example, first given in §5.2.3, including each sentence of the context:

(328) [A theatrical play for children:]

- a. pawa pələk χot_jit ewəlt ewi^x ɛt-əs
 left side room from girl appear-PST[3SG]
 'A girl^x appeared from the room on the left.'
- b. i ewi^y pa jām pələk χot_jit ewəlt ɛt-əs
 one girl ADD good side room from appear-PST[3SG]

²¹³ In line with the above discussion, I assume that the DP under the intended definite reading contributes non-novel referent introduction of the anaphoric kind via $\langle \pi 0 @ P \rangle$ picking up the topmost P referent of the current stack and pushing it to the 0th position (since the DP is in the subject position). Again, developing a fully compositional fragment which models how a non-demonstrative DP composes with the Salient Article is a task for future work.

‘Another girl^y appeared from the room on the right.’

- c. ewen_y još-ən lipət tāj-əs
 girl.POSS.2SG hand-LOC flower have-PST[3SG]
 ‘She_{y/*x} [lit.: the girl] was holding flowers.’

Ignoring irrelevant details, here is the translation of each sentence of (328) alongside its output stack after the updates it encodes are carried out.²¹⁴

(329) i. (328a) $\rightsquigarrow \langle \alpha 0 \rangle \wedge \text{girl}'(x_0) \wedge \text{appear.from.left}'(x_0)$

- 1) Output stack: $h = \langle x, \dots \rangle$, where x is a girl and she appeared from the room on the left

ii. (328b) $\rightsquigarrow \langle \alpha 0 \rangle \wedge \text{girl}'(x_0) \wedge \text{appear.from.right}'(x_0)$

- 1) Output stack: $h = \langle y, x, \dots \rangle$, where y is a girl and she appeared from the room on the right

iii. (328c) $\rightsquigarrow \delta(\text{SAL}(@\text{girl}')) \wedge \delta(\text{REL.MAX.SAL}(@\text{girl}', \text{girl}')) \wedge$

$\langle \pi 0 @ \text{girl}' \rangle \wedge \langle \alpha 1 \rangle \wedge \text{flowers}'(x_1) \wedge \text{hold}'(x_0, x_1)$

- 1) Output stack: $h = \langle y, f, x, \dots \rangle$, where y is a girl, f is flowers, and y is holding f

- 2) Truth conditions: $\llbracket (329\text{iii}) \rrbracket^{\langle g, h \rangle, c} = 1$ iff $\exists i. i = g^{215}$ and $h = i_{0.1} + f \cdot i_{1\dots}$ for some flowers f and h_0 holds h_1 ; defined iff the presuppositions are satisfied in c

- 3) Evaluating the presuppositions:²¹⁶

$c = c \llbracket \text{SAL}(@\text{girl}') \rrbracket$ because $\llbracket @\text{girl}' \rrbracket^{\langle g_c, h_c \rangle} = g_{c_0}$ and $\text{sal}(g_{c_0}, c)$ holds (by (329ii.1) and (322));

$c = c \llbracket \text{REL.MAX.SAL}(@\text{girl}', \text{girl}') \rrbracket$ because the following holds:

$\forall n \in \text{Dom}(g_c) [g_{c_n} \in \llbracket \text{girl}' \rrbracket^{\langle g_c, h_c \rangle} \wedge m_{\text{Sal}(c)}(g_{c_n}) \geq$

$m_{\text{Sal}(c)}(g_{c_0}) \rightarrow g_{c_n} = g_{c_0}]$ (by (329ii.1) and (322))

²¹⁴ This format of presentation is inspired by (Stojnić & Stone & Lepore 2017: 545).

²¹⁵ Pushing the topmost girl to the 0th position does not change the stack as she already occupies it: given that the topmost girl is g_0 per (329ii.1), $g_0 \cdot g = g$, which is equivalent to the output stack of (329ii) $\langle y, x, \dots \rangle$.

²¹⁶ g_{c_0} should be read as ‘the referent in the 0th position of g_c ’. The (admittedly awkward) underscore (‘_’) is used for readability.

Specifically, $m_{Sal(c)}(g_{c,0}) > m_{Sal(c)}(g_{c,1})$ (by (329ii.1) and (322)), i.e., girl y is more salient in c than girl x due to being the more recently introduced subject (higher in the stack)

As (329iii.3) shows, the presuppositions of the Salient Article are satisfied thanks to the preceding context which makes girl y more salient than girl x by virtue of our rules of referent introduction (317) and the dynamics of salience (322).

To conclude, in this section I have shown how the Roberts-style analysis from §5.4.2 can be embedded in a dynamic semantic framework based on Stojnić's (2021) Attention-Coherence approach with a treatment of deictic demonstratives and salience to properly model why the Salient Article is used with deictic demonstratives and why it is appropriate in 'salience without uniqueness' contexts.

5.6. *Summary*

I have argued that the Salient Article functions as a definite determiner, marking the relatively maximally salient discourse referent within a given context. The Salient Article represents a previously unattested type of definite, distinct from both uniqueness-based and familiarity-based definites, as described by Schwarz (2009) and Arkoh & Matthewson (2013), as well as from the Salient Article of Bulu (Barlew 2014).

To account for the crucial examples discussed in this chapter, I implemented the proposed analysis within a dynamic semantic framework, specifically utilizing the Attention-Coherence approach developed by Stojnić (2021) and Stojnić et al. (2017). This framework offers an effective—albeit somewhat limited—method for operationalizing salience, and its application to Northern Khanty data has proven fruitful. Additionally, the analysis has revealed important questions for future research, particularly regarding the order of composition and evaluation in demonstrative and non-demonstrative determiner phrases (DPs).

In pursuing the proposed analysis, I have considered and rejected syntactic alternatives based on the similarity of the Salient Article to the proximate markers of the Algonquian languages. The argument rested on applying to the Salient Article two existing syntactic analyses of proximate uniqueness effects (Aissen 2001; Bruening 2001). This yielded wrong predictions, so the analyses were rejected. It would be interesting to see whether the semantic analysis of the Salient Article proposed here may

be productively adapted and applied to (any of) the Algonquian data, which is, however, a task for future work.

If the analysis proposed here is on the right track, it lends further support to the view that salience, understood as a function of the interlocutors' attentional states, is a crucial dimension for the semantics of definite expressions, as argued by (Gundel et al. 1993; Roberts 2022; Barlew 2014; Huijsmans & Reisinger 2022; Marinina 2024), among others.

6. Conclusions and outlook

6.1. Summary of results

This thesis was concerned with the determiner system of one Uralic language, the Kazym dialect of Northern Khanty. Like many other Uralic varieties, Kazym Khanty features possessive agreement markers with non-possessive functions. I investigated four such functions and argued that they have in fact already grammaticalized into Unpossessive determiners, independent from the Proper Possessive, namely: the Associative Possessive, the Partitive Article, the Salient Article, and the Proprial Article.

The main goal of this study was to describe the semantics of these four markers against the background of the broader definite determiner typology that formal semantic research of the past two decades has accumulated (Farkas & Brasoveanu 2019; Schwarz 2019; Reisinger et al. 2021).

In Chapter 2, I gave an overview of the current typology of definites that arises from language-specific semantic studies based on the work of Heim (1982), Roberts (2003), Schwarz (2009), and Löbner (1985; 2011). I also attempted at unifying some of the results of my predecessors in a more comprehensive picture, which was summarized as (330), repeated from (73).

(330) THE SCHWARZ-LÖBNER DEFINITE TYPOLOGY²¹⁷

- a. **Salience-Familiarity dimension:** A definite determiner may presuppose one of the four statuses of the discourse referent along the Salience-Familiarity Scale (61)/(331) (Roberts 2003; Barlew 2014; Schwarz 2019; this work, Ch. 3 and 5; among many others).
- b. **Concept type dimension:** A definite determiner may be restricted to one of the four Löbnerian concept types (Wespel 2008; Löbner 2011; Albers 2024, among many others).

The Salience-Familiarity dimension is based on the Salience-Familiarity Scale (SFS) from §§2.3.2-2.3.3.1, whose driving idea is that the four statuses of discourse referents discussed in Chapter 2 stand in an asymmetric entailment relation, $>$, from the

²¹⁷ See the many references collected in (Schwarz 2019; Coppock 2022: fn. 1; Albers 2024: fn. 1) for other crosslinguistic work on the semantics of definite determiners in terms of these two dimension.

bottom to the top of the scale (excluding absolute maximal salience which is not relevant for determiners):²¹⁸

(331) THE SALIENCE-FAMILIARITY SCALE

weak familiarity > strong familiarity > salience > relative maximal salience > absolute maximal salience

I suggested that this perspective on weak vs. strong familiarity and salience could clarify some outstanding issues in the SFS-based typology of definites as discussed by Schwarz (2019) and Coppock (2022). I also discussed in §2.4 some definites that do not neatly fit the typology in (330), including demonstratives (following Ahn, Huijsmans & Reisinger, and Marinina, a.o.).

Building on that, Chapter 3 presented a semantic investigation of the four Kazym Khanty Unpossessive determiners. I argued that the Proper Possessive is semantically a simple possessive modifier and is not inherently definite, unlike the four determiners that developed from it. I showed that the semantics of these Unpossessives can be precisely characterized in terms of well-known determinacy presuppositions such as familiarity, uniqueness, partitive specificity, salience, and intensional rigidity. However, with the exclusion of the Salient Article, they do not fit neatly into the types of definites defined in Chapter 2.

The Associative Possessive was argued to be a free (associative) possessive definite which presupposes that the unique referent of this DP stands in a contextually derived strongly familiar relation to another familiar referent. This answers the challenge put forward by Beaver and Coppock (2016) who suggested that there may be no inherently-definite possessives in the world's languages, by providing an example of such a marker.

I analyzed the Partitive Article as an indefinite determiner which presupposes that the individual whose existence is asserted is a proper part of another strongly familiar antecedent referent. Interestingly, the Partitive Article presupposes proper partitivity, unlike Turkish Accusative Case (Enç 1991), but this presupposition can be satisfied by a conjunctive antecedent that includes only two individuals such that the Proper Partitive DP referent is identical to one of them.

²¹⁸ Read as ' $x > y$ ' should be read as ' x delineates a set of contexts which is a superset of the contexts delineated by y '; or 'all discourse referents that have status y also have status x '.

My analysis of the Proprial Article was heavily based on (Muñoz 2019). The Proprial Article applies to unmodified human names to derive intensionally rigid DPs. I provided a formal, situation-semantic demonstration of its intensional rigidity, as compared to the Salient Article and to bare nouns in Kazym Khanty, which are both nonrigid.

Finally, the Salient Article was argued to be a RELATIVE MAXIMAL SALIENCE ARTICLE, similar to personal pronouns (Roberts 2003) and salience-based demonstratives (Huijsmans & Reisinger 2022; Marinina 2024), which imposes additional restrictions similar to proximate uniqueness of languages with obviation systems (Aissen 2001). Following (Muravyev et al. 2024), I suggest in Chapter 5 that these additional restrictions are rooted in morphosyntactic parsing considerations. I showed that a semantic account of these restrictions like the one in (Mikhailov 2024a) is unlikely.

Taking into account the findings of Marinina (2024), who investigated the three demonstrative determiners of Kazym Khanty (as discussed in §2.4.3), the Kazym Khanty determinacy system appears to be an unusually rich and diverse system which incorporates at least five uncommon types of definites (three of the Unpossessives and two of the Demonstratives), a Partitive Article, and bare nouns that are interpreted existentially across the board, contrary to the predictions of (Chierchia 1998; Dayal 2004), as argued by (Golosov 2024a; Tiutiunnikova 2024; Tiutiunnikova et al. 2024).

In Chapter 4, I argued at length that these four possessive-lookalike determiners are indeed UNPOSSESSIVE MARKERS that are homonymous with the Proper Possessive but are synchronically distinct from it (see the definition (7) from the introduction). To put the argument on a more solid footing, I formulated an explicit model of the syntax, morphophonology, and semantics of possessive agreement markers formulated in the frameworks of current mainstream generative grammar (Preminger 2014; Newman 2024), Distributed Morphology (Halle & Marantz 1993), and formal semantics (Heim & Kratzer 1998). The argument was based on UNPOSSESSIVE diagnostics that I developed based on the work of my predecessors. These diagnostics aim to test whether a non-possessive function shows the same morphophonological, morphosyntactic, semantic, and pragmatic behavior as the proper possessive function, as would be expected under a monosemic approach.

I developed an analysis in DM (Halle & Marantz 1993) to model the behavior of Kazym Khanty Unpossessives making crucial use of the mechanism of allosemy from the recent work by Wood (2015), Myler (2016), and others. And I also addressed some potential weaknesses of this analysis by linking the analytical freedom of DM to recent results from grammaticalization theory (Bisang & Malchukov et al. 2020a). That is, the dissociation of syntax from semantics and morphophonology that DM provides with its three Lists makes for a good model of how grammaticalization processes normally work. Changes in parameters of grammaticalization of some construction do not necessarily proceed hand-in-hand with semantics possibly running ahead of syntax and both of them running ahead of morphophonology / allomorphy. This is apparent when we observe that the Salient and the Proprial Articles on the one hand, and the Proper and Associative Possessives on the other hand completely share paradigms within their pairs, even though they differ semantically (semantics running ahead of morphosyntax). Adding the Partitive Article to the mix, the three Articles share their allomorphs with the Possessives, although the Articles' paradigms are severely reduced to just one cell of the original possessive paradigm (syntax running ahead of allomorphy).

The unpossessive diagnostics presented in Chapter 4 can and should be applied to data from other Uralic varieties and languages that have non-possessive functions of possessives. I expect that this will lead to the discovery of additional UNPOSSESSIVE markers, thus, opening the route to detailed investigations into their synchronic semantics and diachronic development paths, as exemplified by (É. Kiss 2018; É. Kiss & Tánzos 2018; Halm 2018; Logvinova 2019; Serdobolskaya et al. 2019) and the present work. Ultimately, this could enrich the typology of definiteness and specificity markers with previously unattested types, as was the case in Chapter 3 of this thesis.

Finally, Chapter 5 was concerned with figuring out the conditions of use of the Salient Article and, specifically, with the question of what factors do **not** play a role, contrary to previously voiced suggestions. I presented a broader and more detailed inquiry into the distribution of the Salient Article, as compared to Chapter 3, and I argued against the Salient Article being sensitive to information-structural distinctions (contra Mikhailov 2021b), and against Muravyev's (2022) hypothesis that the Salient Article is a PROXIMATE MARKER. In §5.5, I implemented the analysis of the Salient Article as a RELATIVE MAXIMAL SALIENCE DEFINITE in a dynamic semantic system, based on Stojnić's (2021)

Attention-Coherence approach. This allowed me to model the Salient Article's use with demonstratives and in 'salience without uniqueness' contexts. The Attention-Coherence approach has proven effective in operationalizing salience to make sense of the Salient Article's distribution, and I believe it shows good promise for further crosslinguistic application.

6.2. *Directions for future studies*

In Chapters 3 and 4, we saw that Kazym Khanty bare nominals trigger indefiniteness antipresuppositions when used in utterances where an Unpossessive could have been used instead. So, a natural question is what is the place of bare nouns in the Kazym Khanty determinacy system? More specifically, what possible readings do they have (Chierchia 1998; Dayal 2004)?

The picture presented above would seem to suggest that they are basically indefinites, used when indefinites are used in languages with binary definite-indefinite determiner systems by virtue of *Maximize presupposition!*-based competition (see §2.1) with the Unpossessives. After all, I did not present any examples of Kazym Khanty bare nouns in definite contexts.²¹⁹ Yet there are definite contexts which are not covered by the Unpossessives. Compare Figure 9 (repeated from Figure 7).

²¹⁹ Almost, the second clause of the soldier examples is one such context (130) / (259).

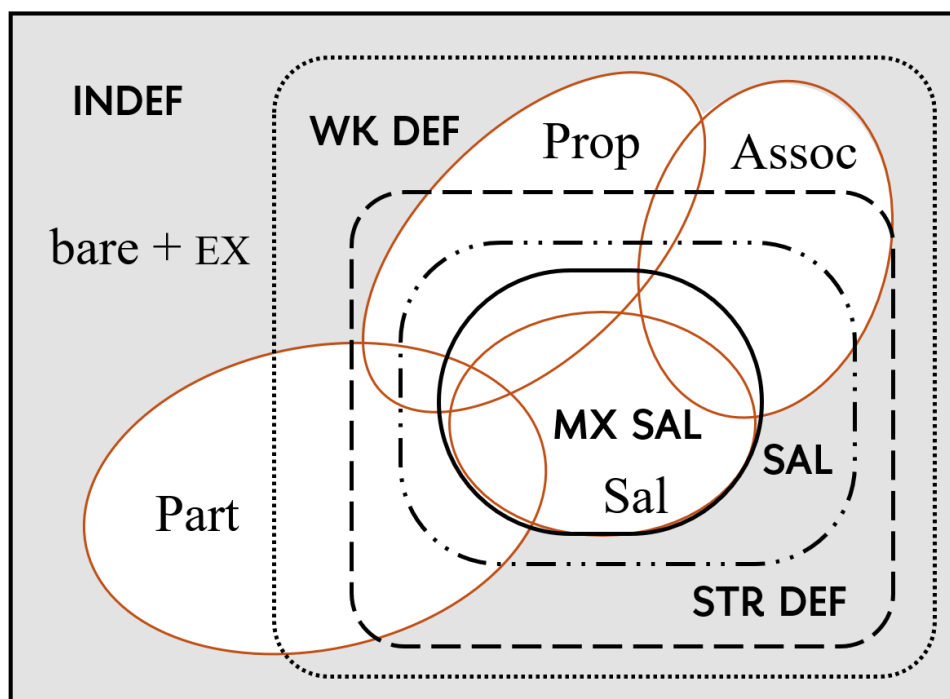


Figure 9. The determinacy system of Kazym Khanty against the background of the Salience-Familiarity Scale typology of definites (Figure 6). Figure legend: (i) crosslinguistic definite types are represented by rectangles with black outlines and subscripted in **BOLD SMALL CAPS SANS SERIF**: INDEF – INDEFINITE; WK DEF – WEAK DEFINITE; STR DEF – STRONG DEFINITE; SAL – SALIENT ARTICLE; MX SAL – RELATIVE MAXIMAL SALIENCE ARTICLE; (ii) Kazym Khanty Unpossessive determiners are represented by ovals with light outlines and subscripted in Capitalized Regular Serif: Assoc – Associative Possessive; Part – Partitive Article; Sal – Salient Article; Prop – Proprietary Article; (iii) Shading indicates the contexts where bare nouns ('bare + EX') can be used.

(Tiutiunnikova 2024) has shown that this picture is too simplistic, and Kazym Khanty bare nouns **are** used in many canonical definite contexts, e.g., the larger situation uniqueness context in (332).

(332) tāmχātł tiliwisar xuwat wan-s-əm sərχanł rajon kəša
 today television on see-PST-1SG S. district head
 ‘I saw the head of Surgut district on TV today.’

The question then arises what exactly their distribution is. (Goloso 2024a; Tiutiunnikova 2024; Tiutiunnikova et al. 2024) argue that bare nouns are existentials in

contexts such as (332) (cf. §3.3.1), supporting the possibility floated in Chapter 3 that Kazym Khanty altogether lacks the IOTA type-shift which derives definite readings of bare nouns. Drawing on this, I would like to maintain the hypothesis, based on (Heim 1991; 2019), that the bare nouns' distribution is constrained by competition with the Unpossessives. And this means that we need to develop an even clearer picture of the Unpossessives' distribution and the constraints on their use.

In Chapter 5, I attempted to clarify the distribution and the constraints on the use of the Salient Article. This led me to propose a dynamic semantic analysis of the Salient Article which raised further questions, regarding the order of composition and evaluation in demonstrative and non-demonstrative determiner phrases (DPs). These need to be addressed in future work. At the same time, we observed that the semantic analysis by itself might not be enough, and morphosyntactic / parsing constraints might play a non-trivial role as well, as suggested by (Muravyev et al. 2024). This suggestion too needs to be developed in more detail.

Doubtless, future inquiries into other questions that the Unpossessives raise, such as the question of possible antecedents for the Associative Possessive's presupposed free possessive relation or the Partitive Article's superset presupposition, will similarly yield some needed progress and some unexpected results.

This is the path towards a comprehensive view of the Kazym Khanty determinacy system envisioned in this study. Hopefully, this thesis provides a solid foundation and a useful semantic framework for achieving this goal for Kazym Khanty data and, potentially, for determiner systems of other languages.

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